United Nations Development Programme

Project Title: Combating Climate Change through the Promotion and Application of Sustainable Biomass Energy Technologies in Pakistan (PASBET)		
Country(ies): Pakistan	**Implementing Partner (GEF Executing Entity):** Ministry of Climate Change & Environmental Coordination (MoCC&EC)	**Execution Modality**: Assisted National Implementation Modality (NIM)
Contributing Outcome (UNDAF/CPD, RPD, GPD): CPD Outcome 6: Enhanced resilience and socioeconomic development of communities **UNDP Strategic Plan Output:** Energy Gap Closed		
UNDP Social and Environmental Screening Category: *Substantial*	**UNDP Gender Marker: *2***	
Quantum Award ID: 00123744.1	**Quantum Project ID:** 00123744	
UNDP-GEF PIMS ID number: PIMS 6542	**GEF Project ID number:** 10720	
LPAC meeting date: 17 August 2023		
Last possible date to submit to GEF: 10 October 2022		
Latest possible CEO endorsement date: 31 December 2022		
Project duration in months: 60		
Planned start date: 16 November2023	**Planned completion date**: 15 November 2028	
Expected date of Mid-Term Review (MTR) submission to the GEF: 16 May 2026	**Expected date of Terminal evaluation (TE) submission to the GEF:** 16 May 2028	
Expected Operational Closure Date: 15 August 2029	**Expected Financial Closure Date:** 15 February 2030	
Brief project description: The objective of the PASBET Project is the widespread application of sustainable biomass energy technologies for supporting socio-economic development of and reducing greenhouse gas (GHG) emissions from, the rural sector in Pakistan. The project's overall strategy is the removal of barriers to the achievement of this project objective. Component 1 of the project is comprised of technical assistance activities addressing the policy/regulatory and institutional barriers, and the expected outcome is the effective enforcement of policies and regulations on the sustainable production and use of woody biomass for energy generation and utilization in rural areas of Pakistan. Component 2 is composed of two outcomes: (a) in the form of technical assistance, enhanced woody biomass production on forested, non-forested, and on farmlands to cater local biomass energy needs, including for power generation and rural industry operations, and (b) be in the form of investment for demonstrating applications of technologies towards increased investments in the application of technologies for the production, and energy efficient utilization of woody biomass energy. Component 3 is regarding supporting financial requirements for Biomass Energy Technologies Initiatives so that		

there will be adequate amounts of financial resources available for woody biomass energy technology application projects in the country. Component 4 addresses the barriers related to capacity development and awareness in the areas of biomass energy technology capacity building and knowledge management and gender mainstreaming towards enhanced local capacity, skills and knowledge in the development, installation, and operation of biomass energy technology systems in rural Pakistan.

FINANCING PLAN	
GEF Trust Fund grant	**USD 0**
UNDP TRAC resources	USD 80,000
Confirmed cash co-financing to be administered by UNDP	USD 0
(1) Total Budget administered by UNDP	**USD 0**
CO-FINANCIERS THAT WILL DELIVER PROJECT RESULTS INCLUDED IN THE PROJECT RESULTS FRAMEWORK (FUNDS NOT ADMINISTERED THROUGH UNDP ACCOUNTS)	
Ministry of Climate Change & Environmental Coordination (MoCC&EC), Government of Pakistan (Public Investment)	USD 8,295,414
Ministry of Climate Change & Environmental Coordination (MoCC&EC), Government of Pakistan (In-Kind)	USD 2,198,853
Government of the Punjab (Forestry, Wildlife & Fisheries Department) (Grant)	USD 2,099,547*
Government of the Punjab (Forestry, Wildlife & Fisheries Department) (In-kind)	USD 742,081*
Government of Sindh (Forest & Wildlife Department) (Grant)	USD 135,747*
Government of Sindh (Forest & Wildlife Department) (In-kind)	USD 135,747*
Ten Billion Tree Tsunami Programme, Government of Khyber Pakhtunkhwa (Grant)	USD 5,696,832*
Government of Balochistan (Forests & Wildlife Department) (Grant)	USD 63,877*
Government of Balochistan (Forests & Wildlife Department) (In-kind)	USD 1,202,348*
UNDP (In-kind)	USD 500,000
(2) Total confirmed co-financing	**USD 21,070,446**
(3) Grand-Total Project Financing (1) + (2)	**USD 24,589,487**

SIGNATURES:		
Signature:	**Agreed by Government Development Coordination Authority**	**Date/Month/Year:**
Signature:	**Agreed by Implementing Partner**	**Date/Month/Year:**
Signature:	**Agreed by UNDP**	**Date/Month/Year:**

**Based on a PKR-USD Exchange Rate of PKR 221 per USD 1. The stated co-financing amount in the co-financing letter is in PKR.*

List of Abbreviations & Acronyms

AEDB	Alternative Energy Development Board
ARE	Alternative Renewable Energy
AWP	Annual Work Program
BEPS	Biomass Energy Portfolio Standards
DISCOs	Distribution companies
EAD	Economic Affairs Division
EC&EE	Energy conservation and energy efficiency
ESMP	Environment and Social Management Plan
EIA	Environmental Impact Analysis
FAO	Food and Agriculture Organization
GWh	Gigawatt-hours
GEB	Global environmental benefits
GEF	Global Environmental Facility
GoP	Government of Pakistan
GCF	Green Climate Fund
GHG	Greenhouse gas
GDP	Gross Domestic Product
INDC	intended nationally determined contribution
IPCC	Intergovernmental Panel on Climate Change
IRENA	International Renewable Energy Agency
KP	Khyber Pakhtunkhwa
MW	Megawatt
MTR	Mid-term Review
MoCC&EC	Ministry of Climate Change & Environmental Coordination
MoPDR	Ministry of Planning Development and Reforms
M&E	Monitoring and evaluation
MSW	Municipal Solid Waste
NBERM	National Biomass Energy Road Map
NEECA	National Energy Efficiency & Conservation Authority
NDC	Nationally determined commitment
PMU	Project Management Unit
PASBET	Promotion and Application of Sustainable Biomass Energy Technologies in Pakistan
SME	Small and medium enterprises
SDGs	Sustainable Development Goals
SBP	State Bank of Pakistan
TBTTP	Ten Billion Tree Tsunami Project
UNDP	United Nations Development Programme
UNFCCC	United Nations Framework Convention on Climate Change

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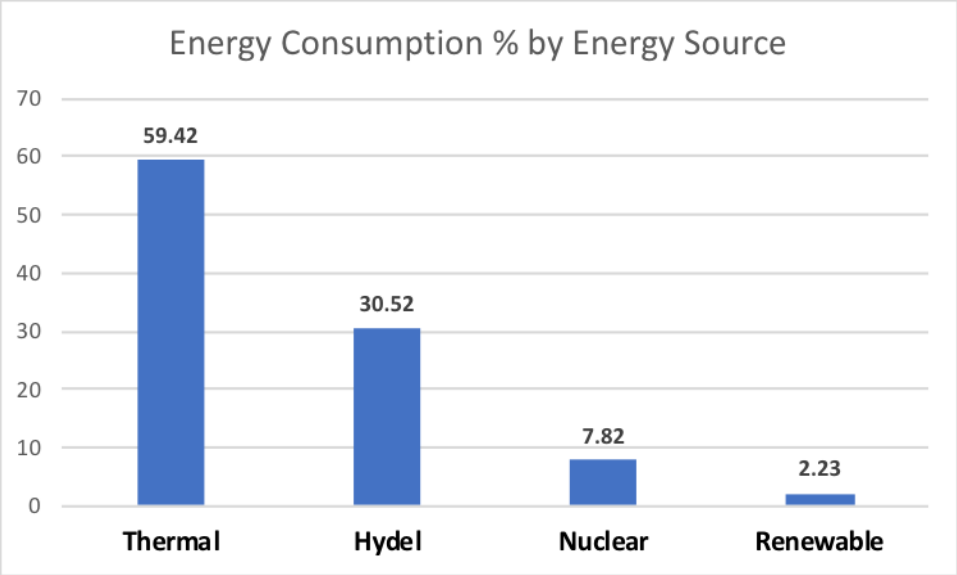
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II. DEVELOPMENT CHALLENGE

Development Challenge and Relevance to National Development Priorities, Global Environment and Sustainable Development Goals (SDGs)

Pakistan is the 6th largest country in the world having around 213 million population with annual growth rate of 2%. About 37% of these lives in cites, while a majority (63%) resides in rural areas (Pakistan Economic Survey, 2018-19)[1]. With ever increasing human population and recent development, energy demand has increased many folds. The country is one of the developing countries that are struggling hard to meet the demand for electric power generation. Pakistan currently relies on fossil fuel sources in the operation of its conventional thermal electricity generation assets, with adverse environmental impacts, particularly GHG emissions. In terms of energy-mix, in 2018 Pakistan's reliance on oil for energy production reduced to 31%, on gas to 34%, while share of hydro-based energy stood at 7.7% and renewable energy contributed 1.1% to the total energy mix of the country. Renewable energy's contribution increased to 2.1% in 2019 and is likely to increase further in coming years, as the Government of Pakistan is looking for cost effective, clean, and alternative energy sources to tackle the energy crises and to keep economic development on track. The government now plans to increase renewable energy share to 30% by the year 2030. Out of several available options of renewable energy technologies, biomass energy is considered promising especially in an agrarian country like Pakistan, where agriculture sector offers employment opportunities to around 38% of its population and contributes 18.5% to the country's GDP and contribution of forests to GDP is only around 0.4% (Pakistan Economic Survey, 2018-19). Pakistan's GDP stood at $82.02 billion in 2020 compared to $263.69 billion in 2020, according to official data from the World Bank.[2] However, GDP growth rate in FY2020 was -0.47%, and the provisional GDP growth rate for FY2021 is estimated at 3.94 percent, higher than the targeted growth of 2.1 percent, for the outgoing fiscal year. The GDP growth is based on 2.77, 3.57 and 4.43 percent growth in agriculture, industrial and services sector, respectively[3]



[1] Pakistan Economic Survey, 2018-19. Ministry of Finance, Islamabad. (finance.gov.pk): 185 pp.

[2] https://data.worldbank.org/indicator/NY.GDP.MKTP.CD?locations=PK

[3] https://www.finance.gov.pk/survey/chapters_21/Overview.pdf

Figure 1. Total Energy Consumption by Energy Source of Pakistan

Pakistan's energy demand is met primarily through fossil fuels with renewable resources only contributing about 2.3%. Oil and natural gas form the bulk of the energy source with hydropower and nuclear power responsible for approximately 39% of the energy production. Though the government has planned to increase the share of renewables to 30%, the reliance on fossil fuels and an increased demand in energy at the residential and industrial level has resulted in higher GG emissions over the last three decades.

- Pakistan's CO_2 emissions per capita have risen from 0.691 to 0.982 metric tons per capita from 1990 to 2018, an increase of 42%.
- In 2019, Pakistan was responsible for 0.68% of the global Co2 emissions. The corresponding share of CO2 emissions for the year 1990 was 0.3% which shows a rise of more than 100% over this time period. [4]
- The power sector is responsible for more than a quarter of the country's CO2 emissions and represents the most significant opportunity for decarbonization
- Bioenergy: Biomass is estimated to generate 50,000 GWh/ year in the country. 64,000 tons of Municipal Solid Waste (MSW) is produced daily by the urban areas in Pakistan. MSW is projected to generate 13,900 GWh of energy. Rural areas of Pakistan have huge biomass reserves and can produce 12 million m^3 of biogas per day.[5]
- Despite the slow growth rate in FY2020, the demand for energy in the country and renewable energy in particular is picking up because of:
    - Rapid population growth: By 2030, Pakistan's population at the current growth rate is expected to reach 263m.
    - Need for inexpensive electricity/power: 60% of electricity in Pakistan is generated through expensive imported fossil fuels and coal.
- Oil, natural gas, and hydropower are the primary sources of energy in the country with the share of renewables including solar and wind currently just over 2%. However, the Alternate Renewable Energy Policy of Pakistan aims to increase the share of renewable energy in the national power grid from 5% to 20% in 2025 and to 30% in 2030[6]
- Pakistan is a developing economy and has been facing an energy crisis for several years because of the insufficient energy production from different sources. The energy demand and consumption continue to grow as the population rises, as in other emerging economies. As a result of Pakistan's strong dependency on oil and gas, lower capacity ratings, distribution losses, cyclical debt, security and dangers of resources and poor governance, energy is one of the critical issues the country is facing and a shift to renewables is essential to help ease these issues. [7]
- The high cost of expanding the national grid to a majority of villages means that almost 59% of the population in rural areas in 2019 did not have electricity through this source. The majority of this population relies on traditional sources of power including firewood to meet their energy demands. Data on the amount of energy consumed by rural households as well as small industries using these sources, however, is limited due to a lack of research on the issue.[8]

[4] https://ourworldindata.org/co2/country/pakistan

[5] https://www.s-ge.com/en/article/export-knowhow/20213-c5-pakistan-renewable-energy

[6] https://www.s-ge.com/en/article/export-knowhow/20213-c5-pakistan-renewable-energy

[7] https://www.mdpi.com/2227-7390/9/17/2083/htm

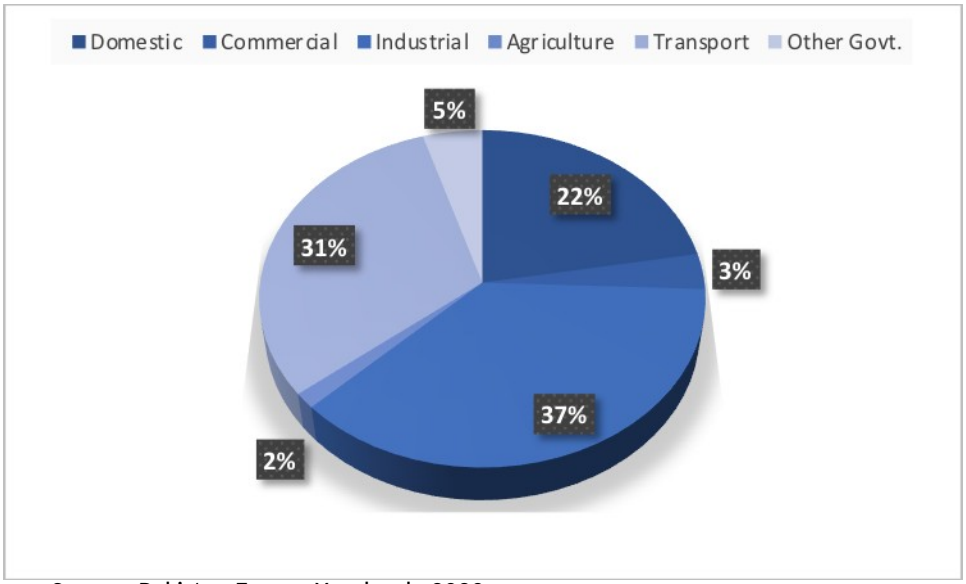
[8] https://data.worldbank.org/indicator/EG.ELC.ACCS.RU.ZS?locations=PK

- A regional comparison shows that Punjab accounts for the bulk of energy consumed (71,735 GWh) followed by Sindh (21,016 GWh), Khyber Pakhtunkhwa (10,677 GWh), Balochistan (4,778 GWh) and Azad Jammu and Kashmir (1,255 GWh).[9]
- According to the Pakistan economic Survey 2020, the share of electricity consumption in March 2020 was highest for households/residential (44.9%) and industries (29.45%) with commercial (7.72%) and agricultural activities (8.04%) making up for the rest of the consumption.

On the consumption side, as per Pakistan Energy Yearbook (2019), the overall energy consumption in Pakistan stood at 54,996,000 TOE (Tons of Oil Equivalent) for the year 2018-19 compared to 39,819,519 TOE in 2013-14, a 38% increase over the past 6 years or an average 6% per year increase.

Table 1: Energy Consumption by various Sectors (TOE)

Sector	2013-2014	2014-15	2015-16	2016-17	2017-18	2018-19
Domestic	9,881,983	10,286,929	10,513,575	11,343,757	11,660,210	12,215,000
Commercial	1,655,515	1,665,042	1,819,146	1,875,228	2,009,982	1,873,000
Industrial	13,772,078	14,844,302	15,928,956	17,927,974	20,601,909	20,419,000
Agriculture	723,704	693,012	709,513	764,194	839,993	815,000
Transport	12,934,115	13,584,670	15,287,332	17,007,968	18,637,229	17,073,000
Other Govt.	852,124	910,179	1,126,504	1,203,183	1,243,567	2,601,000
Total	**39,819,519**	**41,984,134**	**45,385,026**	**50,122,304**	**54,992,890**	**54,996,000**



Source: Pakistan Energy Yearbook, 2020

The 5th Assessment Report of the IPCC states that global anthropogenic GHG emissions have continued to increase from 1970 to 2010 with significant increases toward the end of this period and about half of cumulative anthropogenic CO_2 emissions between 1750 and 2010 have occurred during last 40 years of

[9] https://www.ceicdata.com/en/pakistan/electricity-generation-and-consumption/electricity-consumption-by-region-punjab

this period[10]. IPCC further states that bioenergy can play critical role for mitigation of GHG emissions[11]. Pakistan is party to the UNFCCC and a such has made a voluntary commitment of reducing GHG emissions up to 20% of its 2030 projected GHG emissions subject to international assistance to meet the total abatement cost[12].

Pakistan INDC refers to Land Use Change and Forestry as one of the pillars to achieve its GHG mitigation targets. It has been mentioned:

> *"Historical emissions from the land use change and forestry sector remained 2 to 3 percent of overall emissions. The projected increase in emissions is based on massive changes in land use and enormous deforestation which the country is currently suffering and potential use of biomass in energy and industrial processes. Large-scale tree plantation programmes in Khyber Pakhtunkhwa and Green Pakistan Programme are likely to increase forest cover from the current 5 percent to 6 percent, using domestic resources during the period 2016-2020. An approximate amount of US$ 936 million has been allocated for this purpose. An increase in the forest cover from 6 to 10 percent by the year 2030 requires an estimated US$ 3.74 billion. Projected emissions for the land use change and forestry sector for year 2030 are 29 MT CO2-equivalent."*[13]

Distributed grid, albeit based on solar and wind, as part of mitigation option in energy supply sector has also been mentioned.

Challenges in the energy sector in rural Pakistan:

- Given the lack of research on the issue, the level of understanding and knowledge on energy technology applications in the rural sectors of Pakistan is difficult to gauge.
- The high costs of extending the national grid to rural areas (some of which are far flung) means that a significant majority of villages in the country are not electrified. The government and other private stakeholders have been involved in the electrification of villages over the past few years through a decentralized approach: setting up renewable energy generation including solar and biomass. Energy through biomass (including animal dung and wood) is a popular/significant source given the availability of the raw material and low costs involved.
- The traditional use of biomass to produce energy in rural areas has long term implications on the health and safety of women who are primarily involved in the process as well as children.
- Women are more susceptible to indoor air pollution caused by the burning of traditional biomass, crop residues, untreated coal and other fossil fuels used for cooking. Similarly, a significant increase in the risk of stillbirths among those pregnant women exposed to smoke from inefficient biomass fuels and biomass smoke has also been found to be associated with low birth weight (LBW) deliveries
- Rural women are also vulnerable to harassment and other security threats when leaving their houses to collect biomass.
- Due to low literacy levels and lack of knowledge about the adverse impacts of these processes, the traditional use of biomass continues to be a major source of energy generation and affects the health and environment of users. More importantly, low-income levels of people residing in rural areas

[10] IPCC, WGIII, AR5, SPM pp 6 http://report.mitigation2014.org/spm/ipcc wg3 ar5 summary-for-policymakers approved.pdf

[11] IPCC, WGIII, AR5, SPM pp 26 http://report.mitigation2014.org/spm/ipcc_wg3_ar5_summary-for- policy makers approved. pdf.

[12] Pakistan's Nationally Determined Contribution (Pak-NDC). Ministry of Climate Change. 31 pp.

[13] https://www4.unfccc.int/sites/ndcstaging/PublishedDocuments/Pakistan%20First/Pak-INDC.pdf

means that those who are aware of more sustainable and environmentally friendly options cannot afford to switch to another form of energy. In most cases, traditional biomass is the only form of energy available unless the village becomes part of an electrification project of the government or some other organization.

- These limitations also affect the growth of industries within such rural areas and affect the development potential of the region. Switching to a more sustainable and efficient biomass plant requires high initial set up costs, which is a difficult step due to the lack of financing in these regions (partly due to long payback time to investors)

The energy sector in Pakistan has faced many difficulties over the past two decades, mainly because of high electricity generation costs, insufficient energy generation capacity, and highly inefficient energy generation, transmission, and distribution systems (GoP, 2013)[14]. In fact, biomass generated by the agriculture sector has massive potential for energy generation and can reduce the increasing power demand-supply gap. The biomass scoping exercise conducted jointly by the FAO and the World Bank suggested that biomass coming from the agriculture sector is mainly from crop residues, namely wheat straw, rice straw, and cotton stalk, and of food processing residues, i.e., bagasse and rice husk. In case of woody biomass harvesting and wood processing residues, the exercise could not draw any conclusion on the potential availability of woody biomass and wood residue, as there were large discrepancies in the reporting.

Current forest cover in the country is continuously decreasing with non-sustainable production and consumption of woody biomass. If current practices of non-sustainable production and consumption of woody biomass is not stopped then forest cover in the country will further diminish.

Further data collection and validation of existing data on wood biomass availability was recommended for complete assessment of availability of the woody biomass. The report recommended that both primary and secondary crop residues can be converted into electricity through combustion, gasification, and/or con-generation technologies. However, the selection of the most suitable biomass-energy conversion technology will depend on energy needs and the existing energy infrastructure available in the country (FAO 2014[15]). The renewable energy resource mapping conducted in 2016 provides both theoretical and technical potential of biomass energy generation from crop residues. The theoretical feedstock potential was estimated at about 25.3 million tons/year (62,785 GWh/year of thermal energy) for crop processing residues and 114 million tons/year (456,440 GWh/year of thermal energy) for crop harvesting residues. Based on the existing use of the residues, the technical potential was estimated to be around 25.1 million tons/year (96,890 GWh/year of thermal energy). The analysis also showed that bagasse offers the highest potential through their use as a fuel in cogeneration plants. The installed capacity of the cogeneration plants using bagasse generated by 86 existing sugar mills in Pakistan was estimated at about 1,840 MWe, whereas Municipal Solid Wastes (MSW) with a combine installed power capacity was estimated around 370 MWe (World Bank, 2016[16]). These estimates suggest that there is considerable information gap in woody biomass availability assessment in the country and its potential for generation of power.

Pakistan has made remarkable progress in connecting its towns and villages to the electric grid over the past few decades. The Global Tracking Framework report issued jointly by the World Bank and the International Energy Agency ranked Pakistan fourth in the world in terms of the number of people who

[14] National Power Policy, Government of Pakistan, 2013.

[15] FAO, 2014. Sustainable biomass production and biomass mapping for electricity in Pakistan (Scoping Phase).

[16] World Bank. 2016. Renewable Energy Resource Mapping: Final Report for Biomass Atlas for Pakistan. P. 76

gained access to electricity between 1990 and 2010 (after India, China, and Indonesia). Over this period, roughly 91 million people in Pakistan received electrical services for the first time. Although electrification has remained a priority of GoP since 1990s, still Pakistan has yet to achieve universal coverage. Estimates based on census data and the number of connections reported by utilities suggest that access to grid electricity was only about 74 percent in 2016 (90 percent in urban areas and 63 percent in rural areas in 2016). The household survey sponsored by IFC suggests that 35 percent of Pakistan's population lacked access to electricity in 2014 and that the access rate in parts of the country was still alarmingly low (almost half of the population in Sindh lived off grid, for example). Taken together, these estimates suggest that 5–54 million people in Pakistan may lack access to electricity[17]. Majority of this population (more than 90%) lives in rural areas. In any case, GoP acknowledges that more than 31 thousand villages are yet to be electrified in Pakistan. **Table 2** below shows there exists a substantial potential of rural electrification in Pakistan.

Table 2: Status of Electrified Villages in DISCOS of Pakistan

Year	Total Number of Villages in DISCO	Total Villages Electrified in DISCO	Remaining Villages in DISCO	Percentage of Total Electrified Villages in DISCOs
2016	165,998	133,732	32,266	80.56
2017	170,970	140,427	30,543	82.14
2018	178,592	150,812	27,780	84.44
2019	182,244	153,424	28,820	84.19
2020	188,592	157,203	31,389	83.36

Source: NEPRA State of Industry Report 2020

Overcoming challenges:

The issues around the use of biomass as a form of energy in rural areas can be significantly improved by the introduction of efficient and sustainable technologies including the use of processed, dried wood. These technologies can be made available to both residential households as well as small industries such as brick kilns. This will, however, require support in the form of easing access to finance for those who setup biogas plants as well as capacity building and training of individuals to ensure that the users are aware of the advantages of the technology and use it to the fullest.

Government initiatives that include setting up financing structures and subsidies for implementation of biogas plants/systems that use efficient and environmentally friendly technologies will go a long way in solving energy issues in rural areas. This will involve a more centralized approach as well as supportive policies by the provincial government.

Renewable and sustainable biomass has been shown to have the potential to generate 76% of the peak demand for electricity in Pakistan. It is estimated that agricultural waste materials could generate 56% of Pakistan's electricity, and woody biomass could sustainably generate 9.5% of the peak demand for electricity. Fuelwood has several advantages over fossil fuels i.e., it is a cheap source of energy generation, could mitigate environmental pollution and has less amount of sulfur and other poisonous gases. It has been estimated that 45,000 GWh annually can be added to the national grid if half of the total consumption of fuelwood is devoted to energy generation, which is 49% of the aggregate energy demand of 2018 (Pakistan Economic Survey, 2018-19).

[17] https://documents1.worldbank.org/curated/en/585231536778611429/pdf/WPS8582.pdf

The approved ARE policy[18] put forth an ambitious target of 20% RE by 2025 and 30% RE based electricity generation by 2030. The policy includes biomass and micro-grid based rural distributed generation as an integrated part of the achieving RE targets of Pakistan. However, it ought to be noted that when referring to biomass, the policy does not explicitly mention Woody biomass whereas it specifically mentions bagasse and other agricultural waste. Further, National Forest policy[19] talks about integrating forests with economic sectors. While energy efficient stoves have been explicitly mentioned as part of the Mitigation Options in Energy Demand Sector of Pakistan in the country's INDC[20], energy efficient wood-fired cook stoves are not covered under this project.

With the introduction of efficient woody biomass systems, the consequences of air pollution can be eradicated to a great extent in Pakistan. Currently, several kinds of energy equipment for utilizing biomass exist, which are based on the direct combustion process to obtain useful energy. In the industrial sector, these equipment have been utilized in sugar mill boilers and brick kilns to avoid harmful emissions. Wood fired systems are more energy efficient if the fuelwood is processed (dried, chipped, pelletized) before utilized

Potential of low/zero carbon development, i.e., woody biomass of rural towns and villages

Given the government's aim of increasing the share of renewable energy to 30% by the year 2030 (currently at 2.3%), energy through efficient and sustainable biomass can play a major role in achieving this target and meeting the country's nationally determined commitments (NDC) targets.

It ought to be noted that woody biomass is not part of the GoP radar for to be a viable source for meeting its energy targets therefore, whereas solar, wind, bagasse and MSW based targets are explicitly mentioned, there is no mention of agro-biomass, in general, and woody biomass, in particular. Some initiatives regarding woody biomass are restricted to firewood alone and do not target the power market. As of this time, there is minimal development nor success stories in this regard.

Success of micro-grid in rural sector has been limited to micro hydro-based alone in selected districts of KP and GB.

Among the options for Pakistan to reduce GHG emissions, particularly in the rural areas of the country, is the utilization of sustainable biomass energy resources, particularly woody biomass, to displace fossil fuel-fired energy systems, as well as the widespread application of energy efficient measures and practices in energy consuming operations and processes that support economic development. The sustainable production of woody biomass such as the establishment, operation and maintenance of dedicated fuelwood lots and the processing of wood wastes such as construction scrap wood, sawmill residues, and forest residues into woody biomass resources is an essential aspect of the promotion and widespread application of these renewable energy resources for displacing fossil fuel consumption. The processing of woody biomass into more efficient forms such as chips, pellets and briquets will enable more efficient and effective combustion, leading to lesser energy use and emissions. The application of appropriate woody biomass combustion technologies (e.g., gasification, fluidized bed, mass burning) would also facilitate more energy efficient utilization of this sustainable energy resource. Moreover, the production and

[18] https://www.aedb.org/images/ARE_Policy_2019_-_Gazette_Notified.pdf

[19] http://www.mocc.gov.pk/SiteImage/Policy/National%20Forest%20Policy%202015%20(9-1-17).pdf

[20] https://www4.unfccc.int/sites/ndcstaging/PublishedDocuments/Pakistan%20First/Pak-INDC.pdf

utilization of energy efficient wood burning or wood-fired equipment (e.g., industrial ovens/kilns/furnaces) will facilitate maximum benefits from the utilization of these sustainable renewable energy resources, The proper (i.e., wise, efficient, and effective) utilization of such equipment will further ensure the realization of the energy and environmental and socio-economic benefits of woody biomass energy resources. Though the Government of Pakistan is keen to enhance the share of renewable energy in the total energy mix of the country, especially from by utilizing the potential of biomass power energy, still there are many challenges which has to be overcome. Some of these challenges include: 1) lack of financing and high initial costs to set up a biomass power project; 2) lack of training, awareness, demonstration to farmers; 3) Lack of financial support and incentives; 4) lack of centralized approach and limited institutional support, especially from the provincial governments; and ineffective government policies; 5) long payback time to investors; and 6) undeveloped biomass market/lack of commercial services (Irfan et. al. 2019[21]).

PASBET Contributions to the achievement of several Sustainable Development Goals (SDGs)

- *Goal 7, 13*: The project will contribute to this goal directly through the production of clean and sustainable energy from biomass. The environmentally friendly biomass-based technologies introduced will ensure reduction in GHG emissions and counter climate change as opposed to the current systems in place. Given that the source of energy (fuelwood) is abundant in supply in the areas where several varieties are being planted and that it is renewable in nature, the energy production and utilization process is a sustainable one. Moreover, this will also promote the use of renewable energy sources over fossil fuels which is in line with the government's plans to combating climate change.

- *Goal 8*: The project will create employment opportunities for residents as well as ensuring a just division of labor through the mainstreaming of a gender action plan. This will directly lead to economic growth in the country i.e., higher incomes for those involved in the energy production process as well as growth in the project region due to increased multi-sectoral activities including forestry as well as small industries and commercial businesses.

- *Goal 11*: The environmentally friendly processes deployed in energy production in this project will have a positive result on the health of women and children i.e., traditional inefficient biomass energy production through wood has a damaging effect. Moreover, gender-balanced initiatives in the project will ensure inclusivity for women as well as safety measures and lessened drudgery for women who collect raw materials and perform other related manual tasks.

Root Causes and Threats

The major causes of low level of application of woody biomass-based energy generation both for electricity production and non-power applications in the rural areas of Pakistan are the following:

- *Unsustainable use of woody biomass*: Increase in human population has led to higher demand of limited forest resources (timber, fuelwood, and fodder) in the whole of the country to meet livelihood and household needs (e.g., building and energy needs). It is expected that unsustainable biomass use practices will further lead to deforestation and forest degradation.

[21] Irfan, M., Z. Yu Zhao, M. K. Panjani, F. H. Mangi, H. Li, A. Jan, M. Ahmed, and A. Rehman. 2019. Assess the energy dynamics of Pakistan: Prospects of biomass energy. Energy Report 6 (2020) pp. 80-93.

- *Use of inefficient methods/technologies for biomass energy conversion:* In Pakistan, rural industries that make use of woody biomass (e.g., tobacco curing) utilize energy inefficient equipment that consumes lots of woody biomass that often comes from unsustainable sources[22].

Pakistan is faced with the challenges in meeting the rural sector's energy needs, which has hampered the economic growth and socio-economic development efforts of the Government. Many villages and towns in rural areas are still either without electricity or experience frequent power outages due to load-shedding. In order to address these challenges, the government has shown its commitment for electricity generation through renewable energy sources. Biomass energy, such as woody biomass, is now being considered as a promising and alternate source of energy for generating electricity and of heating for various production processes in the remote towns/villages of the country where the resources are available and can be sustainably produced and utilized. In the case of woody biomass dual energy (electrical and/or thermal) generation, the developments in modern cogeneration technology that is already being used in the sugar industry, the sustainable commercial development, production, and utilization of this resource must consider the abovementioned issues to ensure that the country will fully benefit from its rational use.

Barrier Analysis

The overall project objective is the *widespread application of sustainable biomass energy technologies for supporting socio-economic development of, and reducing greenhouse gas (GHG) emissions from, the rural sector in Pakistan.* Despite a number of baseline initiatives (presented in the next section), a number of barriers still exist to realize this objective. These barriers: policy, regulatory and institutional; use of poor technologies; inadequate investments and access to financing, and limited capacity, awareness, and knowledge are listed and classified below:

Inadequate policy guidelines, regulatory framework, and institutional support for promoting sustainable woody biomass production for energy use and encouraging fuelwood production on farmlands:

- Woody biomass has not been explicitly mentioned as part of biomass and renewable energy in GoP policies
- Policy and regulatory uncertainties (e.g., pricing, tariff, complex government procedures, delays in clearance and government approvals) for woody biomass production and utilization for energy production
- Lack of clear strategy and economic incentives for promoting energy plantations and fuelwood production on farmlands and establishing woody biomass marketing channels
- Weak institutional support for market actors involved in promoting innovative practices and efficient biomass energy technologies
- Limited institutional capacity to provide technical support and regulatory enforcement for woody biomass energy production and use

[22] For example, the estimated firewood consumption in curing tobacco at Pakistan Tobacco Company's (PTC) operations in 2009 was about 2.2 kg of firewood per kg of cured tobacco. In 2009, about 16,505 tons of firewood was burned for tobacco curing in PTC owned barns resulting in 543 tCO_2e emissions. CH_4 and N_2O contributed 183 and 360 tCO_2e emissions, respectively. The high consumption of unsustainable wood in the tobacco curing industry contributed to deforestation and forest degradation in surrounding areas and produce higher quantity of GHG emissions.

- Lack of feasibility studies and economic analysis for promoting energy efficient biomass technologies.
- There is existing off-grid power generation and distribution strategy
- There is no GOP endorsed plan regarding micro-grids
- Developers of small-scale woody biomass projects would find it extremely difficult and cumbersome to go through such processes for getting various permits and clearances.

Limited woody biomass production on forested, non-forested, and on farmlands for use in power generation and rural industry operations

- General absence of guidelines on the commercial production of sustainable biomass in forested, non-forested, and farmlands for use as energy resources.
- Limited experience and knowledge on the utilization of woody biomass for energy purposes, both for power and non-power applications.
- Lack of understanding about the nature and capacity of the local sustainable biomass supply chain and market
- Limited information about the woody biomass demand and supply in the country.
- Limited number of established and operational farmers/communal forest nurseries in the forested areas of the country that can provide planting stock for raising woody biomass in provincial rural areas.
- Very few established and operational energy plantations of fast-growing native tree species at non-forested lands for woody biomass supply.
- Very few established and operational fuelwood production from trees of common fuelwood plant species on farmlands for use in woody biomass energy applications (power and non-power) in rural areas.
- Existing limited biomass-based energy applications in the country generally do not utilize sustainable woody biomass.
- Lack of comprehensive data about biomass (particularly woody biomass) energy resource endowment, economic and social development level, and energy use trends and patterns in rural Pakistan.
- Lack of baseline data on sustainable biomass (particularly woody biomass) resources present and available in rural areas.
- Lack of technical data about the physical, thermal, and chemical characteristics of woody biomass energy products from established wood lots and wood farms.
- Limited reliable information about the potential available sustainable biomass (particularly woody biomass) energy resources that can be developed and utilized in rural areas of the country.

Limited commercial applications of technologies for the production, and energy efficient utilization of woody biomass energy

- Absence of good examples of commercially operated woody biomass-based energy generation facilities that successfully and cost-effectively apply decentralized woody biomass-based electricity generation and distribution (through mini/micro-grids) schemes.
- Absence of good examples of commercially operated rural industries that successfully and cost-effectively apply decentralized woody biomass-based energy generation and utilization (through energy efficient wood-fired equipment) technologies.

- Absence of good examples of commercially operated production of woody biomass fuels such as wood chips, briquettes, and pellets for use in decentralized woody-biomass based power generation.
- Absence of good examples of commercially operated facilities that cost-effectively apply energy efficient woody biomass fired technologies and comprehensive energy management systems.
- Limited available published and disseminated information about the design, engineering, construction, operation, and maintenance of decentralized woody biomass-based energy production facilities, and the utilization of woody biomass for power and non-power applications.
- Low level of confidence by local governments (provincial, country, municipality/town, village) about the technical and economic viability, as well as the benefits of sustainable biomass (particularly woody biomass) energy technology projects as effective interventions for sustainable rural development.
- Low level of confidence by local governments about the cost-effectiveness of the application of EC&EE technologies and comprehensive energy management systems in selected energy end-use sectors in promoting rural socio-economic development.
- Absence of published and disseminated guidance and information about sustainable biomass (particularly woody biomass) energy technologies that are in line with the country's rural development program.
- Lack of technical capacity in the rural towns and villages on enabling sustainable biomass (particularly woody biomass) energy generation and supply, efficient energy utilization and environmental quality improvement.
- Limited knowledge and skills in the application of biomass (particularly woody biomass) energy technologies within the local governments (provincial, country, municipality/town).
- Limited knowledge and capacity of the local governments in the identification, design, and application of EE and RE technologies that are appropriate and feasible for rural development.
- Lack capacity of the local governments in carrying out energy-integrated development planning.
- In general, the private sector's technical capacity is lacking in regards the development and implementation of sustainable biomass (particularly woody biomass) energy and energy efficiency technology projects.
- Adoption of biomass (particularly woody biomass) and EE technologies in rural areas is generally not properly screened and not integrated with major development factors.
- Previous and current biomass energy projects have not yet led to significant rural energy system transformation.
- Limited data in rural areas relating to socio-economic development, in general, and in sustainable energy development, particular.
- Low level of adoption of applicable sustainable biomass (particularly woody biomass) energy technologies to support sustainable rural energy development and utilization.
- Renewable energy technology development in Pakistan is relatively weak, characterized by inadequate systematic and properly set processes, standards, and normative systems.
- Currently, the implementation of sustainable biomass (particularly woody biomass) energy technologies in rural areas often do not have the benefit of proper screening and integration of other key aspects such as local natural, environmental, and ecological conditions, including economic conditions.
- There have been no major breakthroughs in scientific research and development of energy efficient wood-fired industrial equipment for many years.

- There are no locally made heat process equipment (e.g., boilers, kilns) for the efficient combustion of available biomass energy forms.
- Energy technology applications in rural areas are not supported with systematic and properly set processes, standards, and normative systems.
- The development of decentralized biomass (particularly woody biomass)-based energy generation projects has not yet led to significant rural energy system transformation model that is suitable for different regional economic, climatic conditions and resources distribution in the country.
- Limited and inadequate promotion and implementation of sustainable biomass (particularly woody biomass) technology applications.
- Lack of technical data about the physical, thermal, and chemical characteristics of sustainable biomass (particularly woody biomass) energy resources present and available in rural areas.
- Low level of technology adoption in rural energy development plans and initiatives.
- Lack of successful demonstration and promotion of relevant sustainable biomass (particularly woody biomass) technology application models.
- Previous efforts on demonstrating and promoting RE technologies, in general, and particularly sustainable biomass energy technologies have not been effective.

Limited financial investment and economic incentives for woody biomass production and application of energy efficient technologies and adoption of innovative and sustainable approaches for woody biomass energy conversion and use:

- Lack of sustainable supply of woody biomass and fuelwood to meet the current and future demands of woody biomass energy production
- Lack of economic incentives or financial matching between supply and demand for sustainably growing woody biomass resulting in poor confidence for local energy market
- Lack of investment by the private sector for raising energy plantations and promoting fuelwood production on farmlands mainly because of limited incentives for the private sector and financial resources available to farmers and SMEs
- Poor technology: The current wood-fired equipment used in industries is energy inefficient and utilizes substantial amounts of fuelwood that often results in high rate of local deforestation and increased GHG emissions
- Limited private sector involvement in biomass energy production: So far, private sector involvement is limited to sugar mills that are utilizing bagasse for cogeneration of electricity (for own use and outside sales) and thermal energy (for own use); and,
- The most common biomass energy conversion practice is direct burning which is both inefficient and polluting leading to greater CO2 emissions.

Limited capacity, awareness, and knowledge about the environmental, social, and economic benefits of using efficient technologies.

- Limited capacity in raising energy plantations in wider landscapes and in developing/using efficient technologies for biomass energy conversion
- Limited knowledge and low priority on creating public awareness on role of forests for rural socio-economic development

- Lack of awareness and knowledge about modern biomass energy conversion technologies, especially for their industrial applications
- Lack of transparent trade markets at the local level for the buyers and sellers of woody biomass. As a result, fuelwood prices vary from vendor to vendor. Entrepreneurs must come forward to exploit this opportunity, which could be explored by understanding the value/supply chain of woody biomass enterprises.

The potential impact of the barriers discussed above on the implementation front has been assessed for Pakistan as shown in the following table[23].

Parameters	Off-Grid rural electrification using Woody Biomass	Independent Power Producers using Woody Biomass	Captive power generation using Woody Biomass
Resources (Woody Biomass)			
• Availability & certainty	High	High	High
Technology			
• Availability	High	Low	High
• Service	High	Low	Medium
• Business models			
• Price of electricity/tariff	High	High	Low
• Cash flow	High	Medium	Low
Policy & regulatory			
• Processes	High	High	Low
• Regulations	Low	High	Low
• Access to finance	High	High	Medium

Source: UNIDO Project: Promoting Sustainable Energy Production and Use from Biomass in Pakistan

SUMMARY OF BARRIER ANALYSIS

Policy, Regulatory and Institutional Barriers: The main problems are: the absence of policies on mainstreaming woody biomass production in the agriculture and forestry sectors; limited and not fully implemented policies and market-based regulatory framework for supporting woody biomass production and use; inadequate technical, management and operational standards in renewable energy (specifically biomass) development and utilization, energy storage, energy conservation and energy efficiency (EC&EE) and comprehensive energy management; absence of energy-integrated development planning in rural towns and villages and absence of trained and qualified provincial government personnel that provide technical support on sustainable production of woody biomass and enforcement of regulatory framework for biomass use for energy production and utilization. The tariff issue applicable to biomass-based power is the single most important regulatory barrier. While placing much emphasis on renewable energy development in the Alternative Renewable Energy (ARE) Policy, there are no specific targets on biomass in general and woody biomass in particular. Although biomass energy use has a departmental 'home' at Alternative Energy Development Board (AEDB), Ministry of Energy, biomass production and supply need to be coordinated with the varied stakeholders in government (e.g., Forest Department, Agriculture Department, NGOs) and with private sector players. AEDB Act 2010 does not provide enough

[23] UNIDO Project - Promoting Sustainable Energy Production and Use from Biomass in Pakistan: Draft recommendations- policy on biomass gasification and combustion technology

administrative and financial power to the Board for promoting ARE technologies. AEDB can at best play the role of a facilitator.

Technical Barriers: The main problems are grouped into two: (a) limited woody biomass production on forested, non-forested, and on farmlands for use in power generation and rural industry operations and (b) limited commercial applications of technologies for the production, and energy efficient utilization of woody biomass energy. On the biomass production side, the main problems include: general absence of guidelines on the commercial production of sustainable biomass in forested, non-forested, and farm lands for use as energy resource; limited information about the woody biomass demand and supply in the country; limited number of established and operational farmers/communal forest nurseries in the forested areas of the country that can provide planting stock for raising woody biomass in provincial rural areas; and very few established and operational fuelwood production from trees of common fuelwood plant species on farmlands for use in woody biomass energy applications (power and non-power) in rural areas. On the technologies for the production and energy efficient utilization of woody biomass energy, the following problems need to be addressed: absence of good examples of commercially operated woody biomass-based energy generation facilities that successfully and cost-effectively apply decentralized woody biomass-based electricity generation and distribution (through mini/micro-grids) schemes; absence of good examples of commercially operated production of woody biomass fuels such as wood chips, briquettes, and pellets for use in decentralized woody-biomass based power generation; low level of confidence by local governments about the cost-effectiveness of the application of EC&EE technologies and comprehensive energy management systems in selected energy end-use sectors in promoting rural socio-economic development; and limited available published and disseminated information about the design, engineering, construction, operation, and maintenance of decentralized woody biomass-based energy production facilities, and the utilization of woody biomass for power and non-power applications.

Financial Barriers: The key barrier is the absence of financial mechanisms. Considering the high initial costs of setting up biomass projects, the government lacks better and more investment-friendly financial mechanisms. There are no established schemes of subsidies and monetary benefits to local developers and investors during the development stage that would alleviate the cash flow problems arising from the low-income profile at the initial years of the project lifecycle especially in new areas where production of biomass fuels is still lagging the demand. Electricity pricing for biomass-based electricity is still not established that would allow competitive rates in the form of feed-in tariffs when the power is fed to the grid. On the other hand, because of the risk-averse attitude of the banks to cater to the financing needs of the prospective biomass investors, the cost of money for loans tends to be high. Another factor is the high transmission and distribution cost in a situation of dispersed location of the relatively low volume consumers and intermittent demand from domestic/residential and small and medium scale industries and commercial establishments. The full-scale development of the biomass-based projects is hindered by a lot of factors such that private sector is not yet attracted to participate. If this situation prevails, biomass-based projects will continue to be dependent on the limited government funding for such projects.

Awareness and Capacity Barriers: These barriers come in many forms as they relate to the challenges in introducing the new biomass resources development and energy utilization technologies in the rural setting. Local workforce base is definitely insufficient in various aspects of the biomass fuel chain and the generation aspects as well. To a certain degree, similar awareness, and technical knowledge in the bagasse (also a biomass fuel) utilization for power has remained in the private sector operating the sugar mills and have not been passed on for woody biomass to develop local technical experts. Given the lack of technologies for sustainable woody biomass energy systems, the number of local technicians is therefore

low. There is no capacity building program that includes training of technical experts who can trickle down the knowledge and skills for operating and maintaining biomass-based projects. There is no programmed capacity building for government officials regarding the importance of environmental protection/emissions associated with energy sources and the application of, sustainable biomass (particularly woody biomass) energy and EE technologies including the advantages, disadvantages, and costs of such technologies. The energy and environmental awareness campaigns have not yet significantly reached these local regions, such that this could lead to lack of support for any projects that will be initiated. Aside from the fuel development and energy generation side, there is no program existing for energy users and consumers as well, particularly in the rural areas, regarding the benefits associated with biomass energy used in a sustainable manner, using new technologies. The problems become significant as the projects will necessarily involve district participation through village committees or similar groups. Given that most of the workforce are in the fields and the energy end-users are less educated, there is a big gap in the awareness levels regarding the health (particularly for women and children) and environmental benefits of biomass growing and usage from the standpoint of their values and culture in the different provinces to be covered. At present, there is lack of knowledgeable human resources and of a system how women could productively be involved in wood collection and use of biomass energy as well as in decision making and ensuring mechanisms to protect their health and safety in the design, construction, operation, and maintenance of woody biomass-based energy production systems. The lack of data on rural energy use and biomass makes it difficult to assess energy requirements of different regions. There is a lack of research on this topic on a provincial-district level and how the knowledge could assist in planning for local biomass projects given the differences in requirements and use of energy in various regions. R&D is insufficient thus limiting an essential value-adding part of the biomass industry's value chain, investment to improve intelligent and innovative and technologies to reduce operational costs. There is also lack of biomass technology in the country and limited access to international cooperation and technical assistance to affect the technology transfer process and training of local professionals.

The abovementioned barriers, if not properly and adequately addressed/removed, will continue to prevent the successful transition towards zero/low-carbon strategy producing and using biomass in rural villages and towns.

Business-As-Usual (BAU) Scenario

Prior to the PASBET interventions, the BAU scenario with the array of existing barriers will persist in terms of the usual situation of high dependency on petroleum fuels for rural energy, slow growth and high GHG emissions.

Baseline Initiatives Pakistan is currently relying heavily on fossil fuels to meet its energy needs and unable to make transition to Renewable Energy (RE) despite have tremendous potential for renewable energy technology development and application. It is estimated that about 27% of the population still lacks electricity access and those who do experience frequent blackouts. The total power generation capacity in the country is around 25,000 MW, out of this RE-based electricity production accounts only 1,558 MW[24]. Over-reliance on fossil fuel for power generation will worsen climate change impacts on the country, which already stands at number eight of countries affected by the climate change. By utilizing the potential of RE production, Pakistan could not only meet its growing energy needs, but also help in

[24] WWEA and HBS. 2017. Impact of 18th Constitutional Amendment on Governance of Renewable Energy in Pakistan. WWEA Policy Paper Series (PP-02-17): 31 pp.

reducing GHG emissions. To achieve the production of RE, actions are required both at the national and provincial levels. However, the situation has changed with the 18th constitutional amendment in 2010 when the subject of energy was decentralized by giving autonomy to provinces to regulate policy measures and take initiatives for energy production, including linked to power production from renewable resources like biomass. Preliminary analysis of baseline scenario shows that the provinces have taken steps and launched number projects and programs for RE production, including biomass conversion to energy which provide a solid baseline for the proposed project.

The major on-going and planned baseline projects/programs being undertaken by the federal and provincial governments of Punjab, KP, Sindh, and Punjab are listed in **Table 3** below:

Table 3. Ongoing and Planned Biomass-related Baseline Projects and Initiatives

Project/Program	Implementing Entity	Status	Source of Funding	Total Cost (In US$ million)	Links to PASBET
Federal Ministries					
Distributed Generation Schemes through Renewable Energy	Ministry of Energy (Power)	Planned	Government of Pakistan	6.329	RE-based distributed power generation will get traction in Pakistan
Ten Billion Tree Tsunami Program (Federal Component)	MoCC&EC	On-going	Government of Pakistan	41.76	Woody biomass supply will increase as result of increase of forest cover
Punjab					
Feasibility Study Consultancy for Biomass based Power Plant setup near wheat and rice production	Energy Dept. GoPb	On-going	Government of Punjab	1.028	Biomass-based residue will be introduced as a viable and bankable option for power generation
Piloting, waste to energy projects and closure of selected solid waste dump sites and trapping of methane emission	Local Govt. & Comm. Dev. Dept.	Planned	Government of Punjab	1.266	MSW-based fuel will be introduced as a viable and bankable option for power generation in Pakistan
Ten Billion Tree Tsunami Program (Punjab Component)	Forest Dept.	On-going	Government of Punjab	84.05	Woody biomass supply will increase as result of increase of forest cover
Khyber Pakhtunkhwa (KP)					
Electrification of 100 Villages through Solar/Alternate Energy	Ener. & Power Dept. KP	On-going	Government of KP	1.484	RE-based distributed power generation will get traction in Pakistan
Electrification of Un-Electrified villages through Solar/Alternate Energy in remaining Districts of Khyber	Ener. & Power Dept. KP	On-going	Government of KP	0.814	RE-based distributed power generation will get traction in Pakistan

Project/Program	Implementing Entity	Status	Source of Funding	Total Cost (In US$ million)	Links to PASBET
Pakhtunkhwa Phase-II.					
Sustainable Energy & Economic Development Programme (SEED)	Ener. & Power Dept. KP	Planned	DFID	43.599	RE-based distributed power generation will get traction in Pakistan
Raising of Fuel wood species plantation in Swabi (where demo location is in Kalam)	Forest Dept. KP	On-going	Government of KP	1.026	Improved, sustainable and reliable fuel wood supply for power projects
Promotion of Energy Plantation in District Kurram	Forest Dept. KP	On-going	Government of KP	0.506	Improved, sustainable and reliable fuel wood supply for power projects
Energy Plantation in Sub-Division, D.I Khan.	Forest Dept. KP	On-going	Government of KP	0.152	Improved, sustainable and reliable fuel wood supply for power projects
Billion Tree Afforestation Project KP (Phase-III)	Forest Dept. KP	On-going	Government of KP	45.89	Woody biomass supply will increase as result of increase of forest cover
Ten Billion Tree Tsunami Program (KP Component)	Forest Dept. KP	On-going	Government of KP	81.36	Woody biomass supply will increase as result of increase of forest cover
Sindh					
Ten Billion Tree Tsunami Program (Sindh Component)	Forest Dept. Sindh	On-going	Government of Sindh	16.43	Woody biomass supply will increase as result of increase of forest cover
Enhancing Tree Cover on State Forest Lands in Sindh	Forest Dept. Sindh	On-going	Government of Sindh	4.26	Woody biomass supply will increase as result of increase of forest cover
Sarsabz Sindh Making Sindh Green and Environment Friendly	Forest Dept. Sindh	On-going	Government of Sindh	3.61	Woody biomass supply will increase as result of increase of forest cover
Balochistan					
Ten Billion Tree Tsunami Program (Balochistan Component)	Forest Dept. Balochistan	On-going	Government of Balochistan	20.73	Woody biomass supply will increase as result of increase of forest cover
Total:				**0**	

In terms of biomass resources, the PASBET Project will focus on woody biomass conversion for energy production, including the supply chain such as energy plantations, fuelwood from farmlands and discarded woody waste. However, possibility of using biomass from other sectors e.g., agriculture residues (wheat straw, rice rusk/straw, and cotton sticks) will also be explored. The current biomass conversion practices in Pakistan are direct burning which is both inefficient and polluting in addition CO_2 emissions, whereas modern biomass energy conversion technologies (e.g., direct combustion and dendro-gasification) exists in the market and are already proven technologically viable. In 2020, around 710 GWh[25] is being generated using biomass.

The proposed project will build on these baseline initiatives and technologies for promotion and application of sustainable biomass energy production technologies in Pakistan. The PASBET Project Development Team (PDT) consulted with the stakeholders for the possible resource mobilization from the above list projects. It has been able to receive co-financing commitments on the part of the Ten Billion Tree Tsunami Program where the firewood and woodlot development under the program will be complementing PASBET's supply of the woody biomass fuel component i.e., under fuel wood production component of the project. Further, the Ministry of Energy has already included renewable-energy-based micro-grids as part of ARE Policy 2019. Further, by the time of the PASBET inception necessary regulatory framework for micro-gird is expected to be in place. This will result in complementing the policy and regulatory components of the PASBET project. Due to limited opportunities for direct consultations with the project owners, potential co-financing for the policy and regulatory component will be pursued further by the time of project inception if the project will be approved and implemented.

The successful completion of all the baseline projects and programs listed in **Table 3** will bring Pakistan closer to transitioning towards zero/low-carbon towns and villages and to achieving its NDC energy and environment targets, but still, they will not be enough to fully realize these goals. Improvements in the baseline efforts of the country to achieve zero/low carbon rural town and village development are necessary. Such improvements call for the removal of the identified existing barriers. The removal of such barriers will not only achieve the desired transformation of the current rural development pathway into something that leads to low/zero carbon footprint but also contribute to the realization of the climate change mitigation NDC targets of the country.

[25] https://nepra.org.pk/publications/State%20of%20Industry%20Reports/State%20of%20Industry%20Report%202020.pdf

III. STRATEGY

Applied Project Strategy and Incremental Activities

The PASBET project is expected to bring about an increased utilization of indigenous renewable energy (RE) resources for meeting the electricity and other energy requirements of the country. It will facilitate the realization of an alternative scenario in the country's energy sector characterized by an increased share of RE (excluding hydro) in the power generation mix of the country by 2030 as per the GOP's plan, with woody biomass contributing significantly compared to that at present. Such scenario means Pakistan meeting its growing energy needs, with reduced carbon and environmental footprints. In such scenario, the provinces can enforce energy policy measures and take initiatives for energy production, including linked to power production from renewable resources like biomass.

Using a barrier-removal approach to address the identified barriers, the proposed strategy will involve the implementation of the incremental activities resulting from the gap analysis between what are available from the baseline projects and the existing status of the problems and the desired outcomes that the project aims to attain. Thus, the PASBET will build upon the baseline projects which are being implemented in parallel towards removing all the identified barriers as they are integrated at the national level by the executing agencies. Hence, the program will be sustained by the relevant agencies and stakeholders after the PASBET is completed in five years in pursuit of the desired outcomes. By building on, and complementing, supplementing, or augmenting, PASBET and the baseline projects aim to remove all identified barriers in a combined approach.

Pakistan has experienced a high rate of deforestation over the last three decades. Though the Government of Pakistan has taken some steps to increase forest cover in the country through mass afforestation programs, these measures will not be enough unless a holistic approach is adopted to deliberately to encourage more plantations and at the same time mitigate climate change threats. Substitution of fossil fuel by renewable energy resources such as woody biomass, promoting and facilitating commercial sustainable woody biomass production and supply, and application of energy efficient biomass energy conversion technologies would contribute to the country's efforts to address climate change challenges. The current biomass energy conversion practices in Pakistan are direct burning, which is inefficient and contributing to GHG emissions, whether it is woody biomass used in households or in rural industry thermal operations. Efficient biomass energy conversion technologies have been applied in the country in various energy end use sectors, and have proven successful in reducing, if not eliminating, GHG emissions by virtue of their displacement of fossil fuels that would typically be used in such energy consuming operations and processes. There are indeed major challenges stacked up against the widespread application of energy efficient biomass energy conversion technologies in the country, as well as in sustainable woody biomass production. The GEF resources requested for this project will be targeted to remove these identified major barriers/challenges. The envisioned barrier removal activities that will be carried out under this project are practically the incremental activities that the requested GEF funding will support. Without GEF support, rural communities in Pakistan would continue to rely on traditional use of non-sustainable fuelwood to meet their energy needs. Inefficient wood-fired industrial equipment (e.g., kilns, boilers), coupled with increasing growth of rural energy consuming enterprises, will lead to unsustainable production, and use of biomass resources, thus increasing the chances of further increase in deforestation--loss of carbon stock, hence, increased GHG emissions.

The desired essential changes in the current energy situation in the rural sector in Pakistan will facilitate the widespread application of sustainable biomass energy technologies and such transformations will significantly contribute to the sector's socio-economic development. The widespread application of such energy technologies, particularly those that utilize woody biomass, will also bring about reduction in the greenhouse gas (GHG) emissions from this sector.

These changes are expressed as the following outcomes:

1) effective enforcement of policies and regulations on the sustainable production and use of woody biomass for energy generation and utilization in rural areas of Pakistan.
2) enhanced woody biomass production on forested, non-forested, and on farmlands to cater local biomass energy needs, including for power generation and rural industry operations.
3) increased investments in the application of technologies for the production, and energy efficient utilization of woody biomass energy.
4) improved availability of financial resources available for woody biomass energy technology application projects; and,
5) enhanced local capacity, skills and knowledge in the development, installation, and operation of biomass energy technology systems.

The removal of the current barriers/challenges to the achievement of these changes is therefore key to achieving the objective of **widespread application of sustainable biomass energy technologies for supporting socio-economic development of and reducing greenhouse gas (GHG) emissions from, the rural sector in Pakistan**. By implementing strategic barrier removal activities, as incremental interventions over the ongoing and planned baseline activities, specific outputs will be delivered, which collectively will bring about the desired changes. With the facilitation of the achievement of these major changes, the resulting main global environmental benefit would be reduction of GHG emissions from the rural sector in Pakistan. The strategy is also illustrated in the Theory of Change (ToC) graphical diagram form as presented in **Figure 2** below, which is another version of the Project Results Framework (minus the Indicators, Targets, Means of Verification and Critical Assumptions) in **Section V** of this document, which is also referred usually as the Project Log Frame, as the basis of the design of the project.

***Theory of Change*: Promotion and Application of Sustainable Biomass Energy Technologies in Pakistan (PASBET) – PIMS 6542**

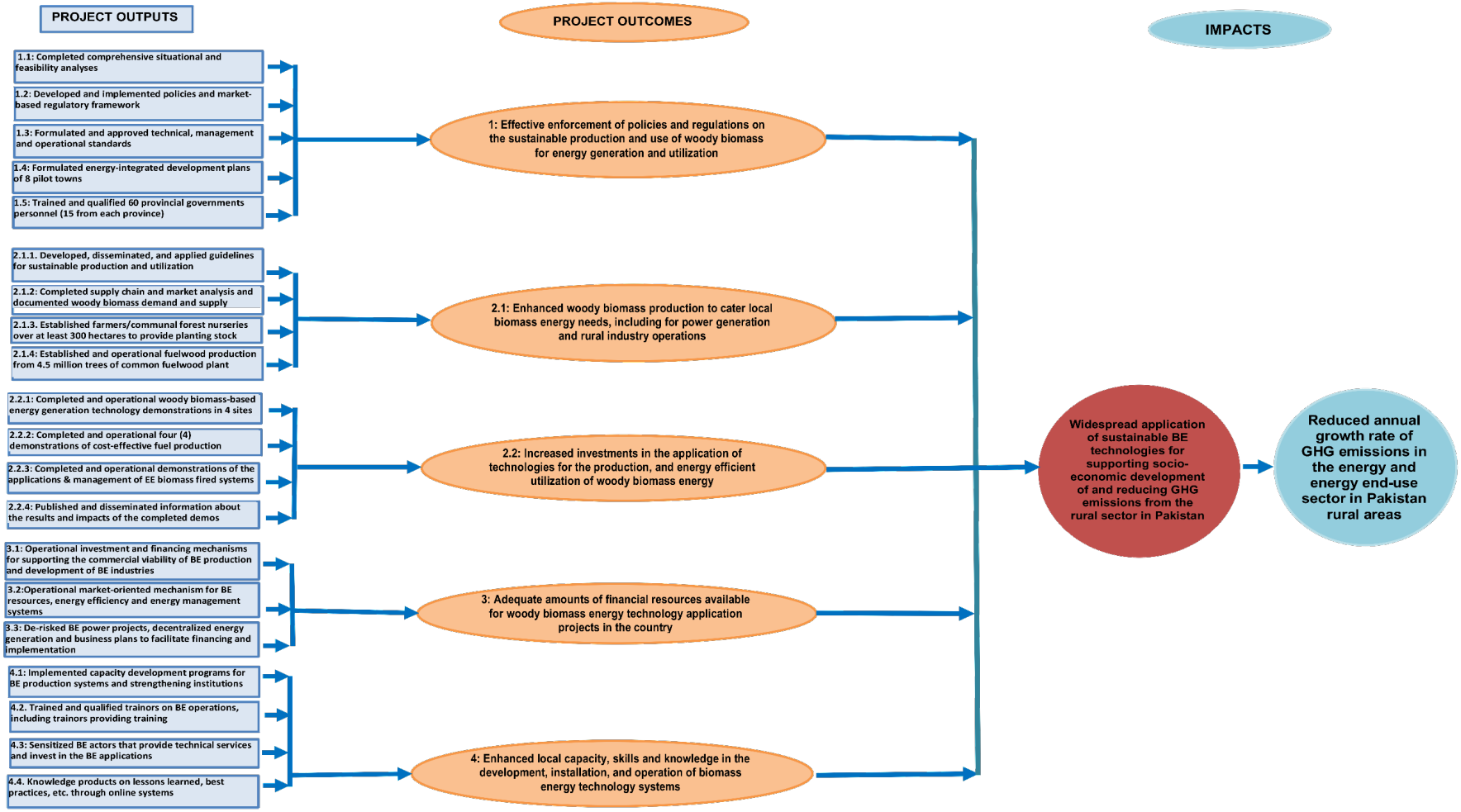


Figure 2. PASBET Theory of Change

The following describe the Components of the project as explained in detail below (illustrated in the TOC Diagram above) and the proposed ways in which the barrier removal strategy will be carried out to realize them:

- **Component 1: Establishment of Policy and Regulatory Framework for Sustainable Woody Biomass Energy Production and Utilization**

Component 1 of the Project addresses the policy, regulatory, institutional barriers in sustainable production and use of woody biomass for promoting efficient biomass energy technologies to reduce GHG emissions from the rural areas. The expected outcome from the delivery of the planned outputs of the activities that will comprise Component 1 is the effective enforcement of policies and regulations through enhancing institutional capacities to support woody biomass production for biomass energy generation and utilizations. The activities involve developing policy guidelines and market-based regulatory framework that support woody biomass production and utilization for energy production both for power and non-power applications. The power applications will focus on decentralized power generation and distribution (through mini/micro-grids) in the unelectrified areas where woody biomass is available either from forest residues or fuel trees that are commercially grown in community-operated woodlots and farmlands. A comprehensive situation analysis will be conducted for mainstreaming woody biomass production into the agriculture and forestry sectors. Based on this situation analysis a national strategy for promotion of sustainable biomass energy production and utilization will be developed with the involvement of key stakeholders. The effective implementation of this strategy would help in establishing community-based energy plantations and promoting fuelwood production on farmlands in potential districts of Punjab, KP, Sindh, and Balochistan. Proper feasibility studies will be conducted, and energy-integrated action plans of 8 pilot towns (2 in each province) will be developed for sustainable woody biomass production and supply for generating biomass energy. A capacity development program will be organized and conducted under Component 4 in support of Component 1 activities and outputs. Technical, management and operational standards will be formulated and approved for biomass energy development and utilization, woody biomass fired equipment and comprehensive biomass energy management.

Component 2: Promotion of Biomass Energy Production and Energy Efficient Utilization Technologies

- **Component 2.1: Woody Biomass Production and Supply**

One of the two sub-components of Component 2 of the Project focuses on understanding the technical aspects of and enhancing woody biomass production on forested, non-forested, and on farmlands to cater the local biomass energy needs, including for power generation and rural industry use in unelectrified or poorly electrified areas of the country. The activities that will be carried out to deliver the outputs that will collectively bring about Outcome 2.1 focus on providing technical support for improved biomass production, including sustainable supply of planting stock of fast-growing native plant species, and managing supply chain of woody biomass with the involvement local entrepreneurs and SMEs. This includes the development and dissemination of guidelines among stakeholders and partners, supply chain and market analysis, related technical and financial support in coordination with the implementation of relevant activities in Component 3.

The expected Outcome 2.1 is enhanced woody biomass production on forested, non-forested, and on farmlands to cater local biomass energy needs, including for power generation and rural industry operations. To enable this, the following guidelines and standards are to be developed:

- Sustainable production and utilization of woody biomass, ESIA and ESMP
- Biomass supply chain and market analysis
- Future commercial operations of nurseries
- Fuel wood plantation, production, and utilization study

Supply of sustainable woody biomass is most critical barrier for this project. Without removing this barrier, the whole project would never achieve its objective. The Project is focusing on generation of sustainable woody biomass in the country starting in the selected 8 districts of the four target provinces. There are already few forested lands in the target districts, but all the existing forests are non-sustainable and if further wood is gathered from them, forest cover will shrink further.

Under the Billion Tree Tsunami Project, Government of Pakistan is planning to grow forests and non-forested lands which could provide sustainable source of woody biomass. Under Component 2.1, a detailed feasibility study (with notations on ESIA results and recommendations, if available from the BTTP), of the available area will be carried out to determine fast growing forest tree species, and best woody biomass harvesting techniques.

- **Component 2.2: Woody Biomass Fuel Production and Utilization**

The second of two sub-components of Component 2 is comprised of activities that will deliver outputs that will collectively bring about increased investments in the application of technologies for the production, and energy efficient utilization of woody biomass energy. The activities that will be carried out will address the technological barriers by promoting and showcasing biomass energy efficient technologies and adoption of innovative practices for woody biomass energy conversion and utilization. Cost-effective applications of EE woody biomass fired technologies and comprehensive energy management systems in selected energy end-use sectors in support of rural socio-economic development will be demonstrated. Among these is the design, engineering, construction, commercial operation, and maintenance of woody biomass fired power generation and distribution systems in the 4 provinces for the supply of electricity to selected unelectrified rural villages. These will showcase the cost-effective application of decentralized woody biomass-based energy generation and distribution (through mini/micro-grids).

There are four (4) sets of demonstration activities under Component 2.2, one set in each province, viz. Punjab, KP, Sindh and Balochistan, all of them are on woody biomass-based energy (power and non-power) generation for rural electrification.

Same technology (biomass gasifier) will be used for each demo plant (including ancillary facilities) to showcase the technology application in each province. However, the size of each demo will vary slightly depending on the detailed biomass assessment study (to be carried out under Component 1). The manner of implementation and business modelling will also differ from province to province since this will depend on the power supply and demand characteristics, rural electrification status and co-financing packaging. Hence, this will consist of four possible typical technology and business cases and diffusion approaches that could find relevant applications as they are replicated in other districts of the province or in some other provinces.

The selection of the biomass gasifier technology is based on the reasoning that it is slightly a mature technology as compared to other available options such as suspension firing, fluidized-bed combustion, etc. Wood gasification is more developed technology in Pakistan and has local expertise. Therefore, it is slightly easier to manage and has better performance confidence based on past studies and technology assessments.

Under each demo design, the following are the major inclusions to be integrated as a system for each of the four provinces in the rural situation:

1) Fuel tree plantation, nursery, and harvesting/gathering system
2) Woody biomass fuel processing, storage, handling, and transportation system.
3) Woody biomass-fed gasifier and gas engine for electricity generation
4) Solar PV power for back up or augmentation to meet the rural electricity demand
5) Mini/micro grid power distribution for rural electrification
6) Industrial burners for industrial thermal (non-power) applications

On the operational aspect, each demo set is an integrated showcasing of the applications of woody-biomass resource development and energy generation that will comprise of the following.

- Distribution of farming/forestry tools and devices for effective and sustained growing and harvesting of woody biomass from fuel wood forests/plantation.

- Design and construction of biomass storage areas. For sustainable biomass supply a multiple storage system will be developed as part of the demo with proper fire safety protocols to minimize the fire hazards. Primary storage system to be planned near the biomass plantation area. While secondary storage facility will be planned near energy generation side. On top of that, multiple biomass collection centers could be developed to collect woody biomass and agriculture residue biomass (if available). Each demo will be sourcing its own woody biomass fuel. The cost of the primary and secondary biomass storage systems is part of the demo's incremental GEF cost budget.

- Development of effective mechanisms for biomass transportation, transportation within the storage area and between the primary and secondary storage area.

- Application of appropriate cost-effective biomass pre-treatment technologies
    - Woodchipper for wood chips to be used in gasifier
    - Biomass pellet machine for wood pellets to be used in energy efficient industrial applications.

- Installation of a woody biomass fired power generation system that will produce and distribute electricity to consumers in the surrounding villages. Biomass gasification technology has been selected for this with an average size of 310 kW for each gasifier. In total there will be 4 gasifiers installed under this project. The reason for the selection of biomass gasifier is because it is slightly mature technology as compared to other available options such as suspension firing, fluidized bed etc. It has local expertise, and it is slightly easier to manage. Several designs are available in the market in Pakistan and other countries. The size may vary for each site depending on the detailed biomass assessment and electrification demand. However, the 310 kW size was determined as

typical for planning and budgeting purposes. The size can be increased with more budget and augmented by co-financing mobilized by the host province. In such cases, the applicable incremental cost of putting up the basic 310 kW gasifier and the woody biomass-fired power generation system is part of the demo's incremental cost budget.

- A mini/micro-grid system will be installed for supplying the electricity to consumers (250 to 300 households in surrounding village). Where possible and feasible, other RE-based power generation units will also be integrated into the system to operate in hybrid mode or as supplementary unit when the main biomass-fired unit is under maintenance. A 250 kW PV Solar system with battery backup could be installed as a back-up facility. Such system will be operated and maintained as a commercial business.

- A key driver of the business model is the collection of fees for electricity sales and the utilization of part of revenues for the operation, repair, and maintenance of the various systems. Each demo is about the commercial operation of a woody biomass-fired energy production and distribution system (power or non-power application). This can be community-owned and operated, or private-sector owned and operated, or government-owned and private sector operated, etc. This further explains the different case studies that can be done for each province typifying the collection system for such energy production and supply business case.

- There will be non-power applications of sustainable, innovative and energy efficient wood-fired industrial equipment as an alternative to coal to reduce GHG emissions from rural industries and commercial uses.

A capacity development program will be organized and conducted under Component 4 in support of the Component 2 activities and outputs for entrepreneurs and individual users in rural areas for woody biomass production and utilization for power and non-power supply.

- **Component 3. Supporting Financial Requirements for Biomass Energy Technologies Initiatives**

Component 3 includes activities that will focus on the removal of financial barriers to woody biomass production and application of energy efficient technologies and adoption of innovative approaches for woody biomass energy conversion and use, including biomass energy production for power and non-power applications. The expected outcome from the delivery of the expected outputs under this component is adequate amounts of financial resources become available for sustainable woody biomass energy technology application projects in the country. The realization of this outcome will involve establishing and operationalizing financial mechanisms for supporting commercial viabilities and operations of woody biomass energy production for both power and non-power applications and development of biomass energy industries at potential locations in targeted districts. The establishment of proper marketing and supply chain would be crucial for the demonstration of the woody biomass energy production technologies. For this purpose, market-oriented mechanism will be established for the development and utilization of biomass energy resources, energy efficiency and comprehensive biomass energy management systems for supporting sustainable rural socio-economic development. These interventions will support creation of "green jobs" and produce benefits for the rural people, especially marginal segments of the society and climate vulnerable communities and help in avoiding potential exposure of these people to COVID-19 or threats from future pandemic. Feasible financial schemes for setting up biomass energy technologies application projects, woody biomass-based energy generation,

and EE wood-fired device/equipment production business will be supported, as well as decentralized biomass-based energy generation initiatives in rural areas. Business plans/models for the district governments and private sector to facilitate financing and implementation of woody biomass energy production projects will be developed.

Component 4: Biomass Energy Technology Capacity Building and Knowledge Management and Gender Mainstreaming

Component 4 is comprised of activities that will primarily address the capacity, awareness and information barriers associated with biomass energy production and utilization technologies. Enhanced local capacity, skills and knowledge in the development, installation, and operation of biomass energy technology systems in rural Pakistan is the expected outcome from the delivery of the planned outputs of this project component. Therefore, building local capacities, creating awareness, generating, and disseminating technical knowledge, and gender mainstreaming steps across the project interventions will be the key elements of this component. This will involve the design (based on a capacity needs assessment) and implementation of gender sensitive capacity development training programs for the design, construction, operation, and maintenance of woody biomass-based energy production systems both for power and non-power applications, as well as for strengthening institutional capacity for supporting sustainable biomass energy production and utilization. Training of trainers will also be carried out to enable capacity development on woodlot operations and management, woody biomass fuel production, and woody biomass-based power generation and distribution, and on the promotion of the widespread applications of energy efficient wood-fired equipment. Outreach activities will also be provided to key value chain actors of biomass energy enterprises to enhance their capacity and motivate them to provide technical services and invest in the applications of cost-effective, climate resilient and energy efficient woody-biomass energy technologies. Knowledge products on lessons learned, best practices will be produced and disseminated through online repository for sharing experiences and replicating and scaling-up use of climate resilient and energy efficient biomass energy technologies.

The other desirable impacts that can be realized with the achievement of the project objective are: (1) improved sustainable socio-economic development; and (2) Improved local environment and minimized unhealthy conditions from the use of inefficient biomass energy technologies and equipment.

With successful project implementation and completion, and achievement of the expected project outcomes and objective, Pakistan's rural areas will be enabled to significantly shift towards a low carbon development path. Such transformation will bring about benefits to the entire rural sector in the following aspects: 1) environmentally, since burning less fossil fuels will allow to reducing the emissions of GHG and particulate matter and with woody-biomass utilization, burning wood to produce electricity emits carbon dioxide into the atmosphere, but planted trees will flourish and capture the emitted carbon dioxide; 2) health benefits for the rural households occupants, since direct combustion of fossil-fuels in households is currently done in a very inefficient and uncontrolled way, with dangerous solid and gaseous emissions in closed spaces and with efficient biomass combustion systems, equivalent power generated will have lesser contribution CO_2 GHG emissions; 3) financially, because PASBET also includes activities devoted to the development and demonstration of the application of proven and biomass-based (particularly woody biomass) energy technologies including their design, engineering, implementation, operation and maintenance and market development; and 4) security, with more indigenous energy generation, the scattered rural villages and towns will be less dependent on imported petroleum and increased supply of reliable and accessible energy transmission and delivery.

The contribution that the PASBET Project will bring in Pakistan's rural sector in the selected provinces, and potentially to other provinces that have good woody biomass potential, is the strategy taken to shift a sizable portion of its energy generation in rural villages and towns towards clean technologies along the low carbon emissions development roadmap. The demos on woody biomass production and utilization have been designed for villages selected in several different regions based on socio-economic and environmental prerequisites. This approach will facilitate replication and scale-up of the demo activities in most, if not all, other rural villages, and towns where woody biomass can be grown systematically and productively.

IV. RESULTS AND PARTNERSHIPS

Goal: Reduced annual growth rate of GHG emissions in the energy and energy end-use sectors in the rural areas of the country

Project Objective: Widespread application of sustainable biomass energy technologies for supporting socio-economic development of and reducing GHG emissions from the rural sector in Pakistan

COMPONENT 1: ESTABLISHMENT OF POLICY AND REGULATORY FRAMEWORK FOR SUSTAINABLE WOODY BIOMASS ENERGY PRODUCTION AND UTILIZATION

This component addresses the policy, regulatory, institutional requirements to address the barriers in sustainable production and use of woody biomass for promoting efficient biomass energy technologies to reduce GHG emissions from the rural areas. The expected outcome is the effective enforcement of policies and regulations through enhancing institutional capacities to support woody biomass production for biomass energy generation and utilizations. This will involve developing policy guidelines and market-based regulatory framework that support woody biomass production and utilization for energy production both for power and non-power applications. The power applications will focus on decentralized power generation and distribution (through mini/micro-grids) in the unelectrified areas where woody biomass is available either from forest residues or fuel trees that are commercially grown in community-operated woodlots and farmlands. A comprehensive and strategic situation analysis will be conducted for mainstreaming woody biomass production into the agriculture and forestry sectors. Based on this situation analysis a national strategy for promotion of sustainable biomass energy production and utilization will be developed with the involvement of key stakeholders. These initial analyses will include the strategic assessment of environmental and social risks/benefits (clarified further in the project's Environmental and Social Management Framework, which clarifies further the methodologies and approaches for ensuring the various strategic assessments proposed also assess S&E impacts.). The effective implementation of this strategy would help in establishing community-based energy plantations and promoting fuelwood production on farmlands in potential districts of Punjab, KP, Sindh, and Balochistan. Proper feasibility studies will be conducted, and energy-integrated action plans of 8 pilot towns (2 in each province) will be developed for sustainable woody biomass production and supply for generating biomass energy. In parallel with the foregoing activities, and based on the initial strategic assessments[26], the Project will formulate a legislative agenda for adoption and implementation by GoP that will guide the preparation and promulgation of a Biomass Energy Regulation consistent with the existing overall National Energy Policy. It will include the needed policies in the technical, management and operational standards in biomass energy development and utilization, woody biomass fired equipment and comprehensive biomass energy management. This will also involve the review of existing and planned legislations, analysis and formulation of preliminary drafts which will be used for consultations with key stakeholders and consultation sessions to present/explain the recommendations and collect feedback from a wider range of stakeholder, resulting in recommended technical, management and operational standards in biomass energy development and utilization, woody biomass fired equipment. The project will build capacity within the various organizations on core principles and practical skills development and the use of strategic measures and tools to enable effective woody biomass production while not compromising forest sustainability, and other environmental and social

[26] This will ensure any upstream policies/strategies/etc. supported by the project are informed by assessment of potential social and environmental impacts.

sustainability factors. A capacity development program will be organized and conducted to train 60 national and local government personnel (15 from each province) that will provide technical support on sustainable production of woody biomass and enforcement of regulatory framework for biomass use for energy production and utilization. Technical, management and operational standards will be formulated and approved for biomass energy development and utilization, woody biomass-fired equipment, and comprehensive biomass energy management.

OUTCOME 1: Effective enforcement of policies and regulations on the sustainable production and use of woody biomass for energy generation and utilization in rural areas of Pakistan

Output 1.1: Completed comprehensive situational and feasibility analyses for mainstreaming woody biomass production in the agriculture and forestry sectors, including assessments of potential areas for sustainable woody biomass production and utilization[27] in Punjab, Khyber Pakhtunkhwa (KP), Sindh and Balochistan provinces[28]

The Project implementation will start with comprehensive situational and feasibility analyses. The results and recommendations will form the essential elements for developing policies and institutionalizing an objective-driven, market-based, and bankable system for the biomass-based production and utilization program. These will be developed in consultation with key stakeholders and regularly updated with inputs from the Forestry Departments of the Provincial Departments and will be housed at the Pakistan Forest Institute. The studies will involve gathering of the pertinent information and recommending the policy and institutional parameters that will be used to benchmark the policy mechanisms and instruments that will guide removal of the main development barriers discussed in the foregoing sections.

Activity 1.1.1: Conduct of Detailed Woody Biomass Resource Assessment. This activity involves the following steps on the fuel supply side: (a) Collection of information regarding availability of woody biomass in Pakistan based on secondary data; (b) Conduct of surveys in selected regions on available biomass resources, land use and land availability for different growing models and establishment of a database on this; (c) Identification and estimation of the pricing of the existing woody biomass vis-à-vis its alternative uses; (d.) Assessment of environmental and social sustainability of potential fuel supply sources; (e) Stakeholder consultation to validate the findings of the assessment, and (f) Development of a GIS based model based on remote sensing for updating energy plantation planning for sustainable woody biomass supply and its update. Based on these data, the assessment will focus on the four (4) selected provinces of Punjab, Khyber Pakhtunkhwa (KP), Sindh and Balochistan[29] and the results and information gathered will be consolidated into a Biomass Resources Report to become the basis for additional resource mobilization and necessary policy guidelines for identifying and planning for additional potential areas for energy plantation.

Activity 1.1.2: Determination of Potential of Woody Biomass-based Power Generation in Pakistan. This activity involves the following steps on the power generation side considering potential market for the biomass-based power: (a) Development of the criteria of drivers for woody biomass power generation in Pakistan, including social and environmental sustainability criteria for the selection of potential wood

[27] This also includes the utilization of available forestry waste such as twigs and branches of standing trees, leftover wood, and branches after commercial harvesting of sustainable forests.

[28] Two districts each in the provinces of Punjab, KP, Sindh, and Balochistan identified as target areas for the planned technology demonstrations, with big potentials for woody biomass production and utilization.

[29] Two districts each in the provinces of Punjab, KP, Sindh, and Balochistan identified as target areas for the planned technology demonstrations, with big potentials for woody biomass production and utilization.

biomass-based power generation plants; (b) Application of the criteria to identify the potential market of woody biomass generation in Pakistan; (c) Identification of the locations of most suitable matching of energy plantation for woody biomass supply and power market for electrification of villages and thermal applications in industrial and commercial establishments; and (d) Stakeholder consultation to validate the findings of the assessment. This involved review and confirmation of the initial site identification and projected power capacities during the PIF stage that will be supported by the Project as biomass technology demonstrations in Component 2, Outcome 2.2, in consultation with the proponents in the selected provinces.

Activity 1.1.3: Conduct of Feasibility Studies for Mainstreaming Woody Biomass Production in the Agriculture and Forestry sectors. This activity involves the following steps to conduct the feasibility studies for the priority districts that have good supply/demand matching among the identified possible sites: (a) Conduct of technical, social, environmental, economic and financial analysis of woody biomass investment projects; (b) Conduct of sensitivity and risk analyses of the various cases determined for a particular site; (c) Conduct of comparative analyses of the base case with BAU and other plausible scenarios to determine the optimal combination for recommending the final Project's Demo sites and financial packaging; (d) Organization and conduct of stakeholder consultation workshop to present and validate the findings and recommendations of the feasibility studies and for endorsement to the Project Board to decide on the procurement and installation plan for the physical facilities, accompanied by the prepared site ESIAs/ESMPs, and in accordance with the project budgets by mid of Year 2; and (e.) Organization and conduct of training/workshop on conducting feasibility studies for the key stakeholders' planning and development staff and the bank representatives (in coordination with the related Component 4 Activities).

GEF support is required for the incremental technical assistance in the conduct of the necessary baseline establishment to promote woody biomass as another viable option for RE-based rural electrification technologies aside from solar and micro-hydro.

Output 1.2: Developed and implemented policies and market-based regulatory framework for supporting woody biomass production and use, including a national strategy for promotion of sustainable biomass energy production and utilization, using community-based woodlots and agroforestry production at the provincial level, and fiscal incentives for private-sector participation in woody biomass energy technology business.

Activity 1.2.1: Development of a tariff structure and policy for woody biomass-based power generation. This activity addresses the absence of a tariff structure and policy as one of the key barriers in the development of woody biomass-based power generation. It involves the following steps: (a) Review of existing tariff structure for biomass i.e., bagasse; (b) Conduct of a comparative study on woody biomass tariffs in the region; (c) Analysis of costs involved in fuel wood supply chain, including recommended fair pricing; (d) Development of a woody biomass pricing model for the tariff; (e) Development of indicative tariff for woody biomass based power generation; (f) Submission of the required documentation with NEPRA for tariff determination; and (g) Facilitating the implementation of the approved tariff policy and structure.

Activity 1.2.2: Policy development at provincial level. Based on the approved national policy and structure on cascading down other elements of sustainable biomass energy to provincial level, this activity shall enable the Provincial government to cascade it at their level through the following steps: (a) Formulation, promotion, approval and adoption of an Industrial Fuel Tree Plantation Rules in Provinces; (b) Formulation

of recommendations on enabling policies and incentive schemes for sustainable fuel wood supply industry (e.g. tax breaks); (c) Establishment and publication of transparent lease rates and lease procedures for fuel wood plantations; (d) Preparation of standard contract templates for lease agreements of land parcels depending on ownership; (e) Formulation of recommendations on enabling policies and incentive schemes for the establishment and operationalization of a fuel wood supply industry (e.g. tax breaks, etc.); and (f) Conduct of local awareness raising and engagement activities to ensure support and ownership of proposed policies/incentive schemes.

Activity 1.2.3: Development of national roadmap to promote sustainable biomass energy systems. Informed by the initial assessments and feasibility studies conducted, a national roadmap, specifically referred to as the National Biomass Energy Road Map (NBERM), to promote and guide the time-bound implementation of sustainable biomass energy systems and its adoption by pertinent government and private sector entities, through the following steps: (a) Review of existing electricity generation planning in Pakistan and role of biomass based distributed/captive systems in meeting the least cost criteria of national planning; (b) Development of targets for biomass generation for captive and distributed generation; (c) Development of Rural Electrification Strategy of Pakistan with specific emphasis on biomass-based power generation; and (d) Stakeholder consultation regarding the inclusion of biomass, in general and woody biomass, in particular, energy production and utilization in the national energy planning process.

GEF support is required for the incremental technical assistance for ensuring that the necessary regulatory and policy environment are created for establishment and promotion of a national woody biomass development and implementation program.

Output 1.3: Formulated and approved technical, management and operational standards in biomass energy development and utilization, woody biomass fired equipment and comprehensive biomass energy management.

Given that the needs of different consumers are often different and site-specific, the design and characteristics of the energy system may vary (or even substantially) from area to area, from customer to customer. At the opportune time, for future replication and scaling up of any technological system, it is preferable that it be standardized to the extent possible so it can be imported or locally manufactured, more cost-effectively. Individual manufacturers could make additional modifications, if they so desire, but the basic design features would remain the same based on the performance evaluation of the pilot demonstrations implemented and the Sustainability & Replication Plan formulated and adopted by GoP in Component 2 in Year 5.

Activity 1.3.1: Formulation and approval of technical, management and operational standards in biomass energy development. This activity focuses on the formulation of standards development, policy adoption and enforcement regarding fuel wood production, preparation and logistical requirements into transportable fuel-grade and consistent-quality forms such chips, pellets, chops, etc. for industrial and commercial scale. This will also be informed by the initial assessments and feasibility studies and will involve the following steps: (a) Review of existing and planned related rules and regulations in other biomass, e.g. bagasse or other solid fuels; (b) Analysis and formulation of policy manuscripts and instruments on biomass-based energy development; (c) Consultations with key stakeholders to present/explain the recommendations and collect feedback from a wider range of stakeholders; (d) Preparation of second drafts of the implementing rules and regulations (IRRs), and recommendations for other relevant policies and plans; (e) Presentation and validation of Final Drafts during a National

Stakeholder Workshop; (f) Submission by MoCC&EC and adoption by the authorized agency of the Final Policy Document and Quality Standards; and (g) Implementation of the standards and regular monitoring and evaluation.

Activity 1.3.2: Formulation and approval of technical, management and operational standards in biomass energy conversion. This Activity focuses on standards development, policy adoption and enforcement on the energy conversion processes using more efficient, safe and environment friendly technologies, techniques, and facilities such as woody biomass gasification, co-generation, multi-fuel kilns and burners, stokers, grates, suspension firing, fluidized-bed combustion, etc. as applied to mini/micro power generation. This will also be informed by the initial assessments and feasibility studies and will involve the following steps: (a) Review of existing rules and regulations, analysis and formulation of preliminary drafts on biomass energy conversion; (b) Analysis and formulation of policy instruments on biomass-based energy development; (c) Consultations with key stakeholders to present/explain the recommendations and collect feedback from a wider range of stakeholders; (d) Preparation of second drafts of the implementing rules and regulations (IRRs), and recommendations for other relevant policies and plans; (e) Presentation and validation of Final Drafts during a National Stakeholder Workshop; (f) Submission by MoCC&EC and adoption by the authorized agency of the Final Policy Document and Quality Standards; and (g) Implementation of the standards and regular monitoring and evaluation.

Activity 1.3.3: Formulation and approval of technical, management and operational standards for comprehensive biomass energy management. This activity involves the development of standards, policy adoption and enforcement with particular focus on the efficient utilization or conservation and energy management of biomass-based energy in various applications, such as conversions of existing oil-fired boilers and kilns into efficient biomass-based energy utilization equipment/systems for industrial and commercial power or thermal (direct heat) uses. This will also be informed by the initial assessments and feasibility studies and will involve the following steps: (a) Review of existing rules and regulations, analysis and formulation of preliminary drafts on biomass energy conversion; (b) Analysis and formulation of policy instruments on biomass-based energy development; (c) Consultations with key stakeholders to present/explain the recommendations and collect feedback from a wider range of stakeholders; (d) Preparation of second drafts of the implementing rules and regulations (IRRs), and recommendations for other relevant policies and plans; (e) Presentation and validation of Final Drafts during a National Stakeholder Workshop; (f) Submission by MoCC&EC and adoption by the authorized agency of the Final Policy Document and Quality Standards; and (g) Implementation of the standards and regular monitoring and evaluation.

GEF support is required for the incremental technical assistance in the standardization of biomass processing, gasification, and utilization technologies so that technological risks regarding introduction of this innovative technology in Pakistan can be minimized.

Output 1.4: Formulated energy-integrated development plans of 8 pilot towns, two in each province of Punjab, KP, Sindh and Balochistan[30]

In parallel with the foregoing activities, the Project will formulate a legislative agenda for adoption and implementation by GoP that will guide the preparation and promulgation of a Biomass Energy Regulation consistent with the existing overall National Energy Policy. It will include the needed policies in the

[30] Based on enhanced provincial data banking systems that include sustainable woody biomass production and utilization, provincial economic and social development parameters and performance indices, energy demand and use trends in each

technical, management and operational standards in biomass energy development and utilization, woody biomass fired equipment and comprehensive biomass energy management. This will also involve the review of existing and planned legislations, analysis and formulation of preliminary drafts which will be used for consultations with key stakeholders and consultation sessions to present/explain the recommendations and collect feedback from a wider range of stakeholder, resulting in recommended technical, management and operational standards in biomass energy development and utilization, woody biomass fired equipment. These activities will need technical input and guidance from Component 2. This will be followed up with submission of Final Policy document and Quality standards. By promoting standardized designs (by a neutral organization, such as AEDB and the National Energy Efficiency and Conservation Authority (NEECA) and other pertinent agencies that use a standardized set of (and as much as possible locally available) parts, better maintenance services can be more easily provided. The policy document mentioned above as the Biomass Energy Policy Guidelines and Road Map (2022) will be reviewed at the Project's Mid-Term and the recommendations will be used to update it into a final National Biomass Energy Policy Guidelines and Road Map (2026) to be the basic document to be annexed to an appropriate legislation to be endorsed to the GoP, as one of the Project's commitment in the Exit Strategy.

Activity 1.4.1: Development and implementation of energy-integrated development plans for 8 pilot towns, two in each province of Punjab, KP, Sindh and Balochistan. This activity involves prior assessment of potential social and environmental risks/benefits and will involve the following steps: (a) Development of selection criteria for the eight (8) pilot towns under the pre-selected districts; (b) Selection of 2 representative towns in each province; (c) Survey to develop existing energy demand profile of the towns and analysis of results to guide the development planning; (d) Development of future demand profile of the town based on past and projected growth; (e) Development of energy supply options for the pilot towns to fulfill current and future energy demands in line with the overall socio, economic and environmental thrusts of the town based on the national development policy and directives; (f) Identification and conduct of feasibility evaluation of the least-cost, energy efficient option in meeting the demand for energy in the medium- to long-term; (g) Development and adoption of the regulatory framework conditions, environmental and social safeguards, financial packaging, institutional and organizational requirements and implementation plan and timetable to integrate woody biomass energy development strategy into the overall energy supply development planning of the town in consultation with the local government, key stakeholders and development partners; (h) Dissemination and coordination of the work with DISCOs in the area in a decision making process to adopt a renewable portfolio standard in general and biomass portfolio standards in particular to achieve electricity access targets as well as improve service delivery quality in line with the energy-integrated development plan as piloted in the towns; and (i) Stakeholder consultation throughout process of development of the plans to standardize the above process in the distributed energy planning at DISCO level.

The lands that will be used for fuelwood plantations are those that are marginal and relatively not so fit for food crops and other food-related products. It will be considered that the community that will be engaged in the project will also be concerned about food security. Therefore, in the planning and allocation of the acreage, this food security and agricultural concern will be taken into consideration in community discussions.

Related areas will be considered in the implementation of energy-integrated development plans for the 8 pilot towns in the selected four provinces in the other aspects of the project, viz. air pollution, biodiversity, land degradation, and job creation co-benefits as part of the integrated approach.

GEF support is required for the incremental technical assistance in development planning to put forth woody biomass-based solution as a technologically and financially viable electrification option for rural off-grid communities of Pakistan.

Output 1.5: Trained and qualified 60 provincial governments personnel (15 from each province) that provide technical support on sustainable production of woody biomass and enforcement of regulatory framework for biomass use for energy production and utilization

On the capacity development needed for Component 1 to be carried out under Component 4, the Project will need to train 60 provincial governments personnel (15 from each province) that provide technical support on sustainable production of woody biomass and enforcement of policy and regulatory framework and institutional requirements for biomass use for energy production and utilization. The project will build capacity within the various organizations on core principles and practical skills development and the use of strategic measures and tools to enable effective woody biomass production while not compromising forest sustainability. Needs assessment will be conducted to develop a comprehensive and targeted training program using individual training modules designed and implemented. The needs' assessment will take into consideration on-going capacity building activities, including at the Pakistan Forest Institute and provincial forestry and wildlife education and training facilities. The training will focus upon enabling stakeholders to apply practical skills in their daily work. The developed modules will form the basis of multi-component training program to be organized and conducted within the Pakistan Forest Institute and/or provincial forestry and wildlife education and training facilities over the life of the project.

Activity 1.5.1: Preparation and planning the implementation of relevant training of provincial governments' personnel. This activity involves coordination with the implementation of the pertinent capacity building activity in the project's Component 4. For the policy and institutional aspects of the tasks, the following steps are to be taken: (a) Preparation and stock-taking and analysis of issues and options on different growing models, including social and environmental impacts and acceptability for (small and large) fuel wood growers; (b) Conduct of a survey to identify gaps in the current management practices and socio economic and environmental benefits that could be derived through different biomass production models promoted in project; and (c) Preparation and dissemination of handbook and training manual on best available approaches for sustainable growing models for different species of fuel wood in Pakistan to be used in the planned training.

Activity 1.5.2: Training of provincial governments' personnel. This activity involves the following steps in the implementation of the training program in coordination with the relevant capacity building activity in the project's Component 2.1: (a) Conduct of training for extension officers on various fuel wood growing models; (b) Organization and implementation of awareness raising events for farmers and plantation owners on benefits of producing fuel wood; (c) Implementation of promotional activities for various growing models; (d) Preparation of outreach materials for the Forest Department to develop fuel wood intercropping model and fuel wood markets; (e) Conduct of outreach activities related to production of fuel wood in cooperation with the Forest Departments of the 4 participating provinces.

Activity 1.5.3: Post-training evaluation of the program for the provincial government personnel. This activity involves an evaluation of the training program in half or one year after training to gauge the quality and impacts of the training program. The evaluation report must be used in improving the design of the follow-up design of the training program (if still necessary).

GEF support is required for the incremental technical assistance in building the capacity of the forest departments and other key staff from other participating departments of the provincial governments so that the sustainable supply of woody biomass for energy purposes can be ensured.

COMPONENT 2: PROMOTION OF BIOMASS ENERGY PRODUCTION AND ENERGY EFFICIENT UTILIZATION TECHNOLOGIES

The main thrust under this component is on ensuring sustainable biomass supply and biomass energy production and utilization through introduction of cost-effective, climate resilient and energy efficient biomass energy technologies. There are 2 expected outcomes under this project component:

- Outcome 2.1: Enhanced woody biomass production on forested, non-forested, and on farmlands to cater local biomass energy needs, including for power generation and rural industry operations

- Outcome 2.2: Increased investments in the application of technologies for production, and energy efficient utilization of woody biomass energy

There are four (4) sets of demonstrations, one set in each of the 4 provinces that will include in each set demonstrations: (a) for showcasing improved woody biomass fuel production technologies for producing wood chips, briquettes, and pellets. (b) for woody biomass-based energy generation technology applications and (c) for the industrial demos for demonstrating the cost-effective applications of energy efficient wood-fired industrial equipment for small and medium enterprises and commercial applications.

The Project is expected to facilitate the installation and operationalization of 5.4 MW of biomass-based power generation capacity during and after the project implementation period. Since the concept of biomass-based energy generation is still at a nascent stage in Pakistan, the Project is designed to introduce the application of the technology by installing and operating 310 kW biomass gasification plants as demonstrations at 4 locations in the selected provinces. These demos are expected to either be replicated or scaled-up during the project implementation period as well as during the project's influence period (i.e., 10 years after project completion).

Lastly, in the light of the current Covid-19 pandemic situation in the country, the investment activities of the project are designed to not only to demonstrate the relevant technologies that the project is promoting but also to support more resilient livelihoods and infrastructure that enables green recovery from COVID-19 impacts and building future resilience. The envisioned demos of the project involve the engagement of the private sector to further bolster the objective of promoting low-carbon, low-pollution, and nature positive solutions to support rural socio-economic development in the country. These actions help in supporting low carbon development through applications of cost-effective renewable energy, low-carbon, and energy efficient technologies.

OUTCOME 2.1: Enhanced woody biomass production on forested, non-forested, and on farmlands to cater local biomass energy needs, including for power generation and rural industry operations -

This outcome features the established plantations of sustainably planted and harvested fast-growing native tree species on farmlands. These plantations also represent improvement and positive impact of the tree cover in the project areas contributing to biodiversity conservation, creation of "green jobs" especially for woman, youth, and marginalized segments of the society in rural communities and recovery

benefits from the Covid-19 pandemic, and future threats of climate change such as flashfloods and prolonged droughts.

Output 2.1.1. Developed, disseminated, and applied guidelines for sustainable production and utilization of woody biomass

Activity 2.1.1.1: Development of general guidelines for commercial production of sustainable biomass in forested, non-forested, and farmlands for energy and non-energy use. This activity focuses on the development of guidelines for producing the biomass resources and involves the following: (a) Inventory of existing and planned biomass production in forested, non-forested, and farmlands in the four (4) selected Provinces and classify them according to energy or non-energy use and publish the results as a Biomass Resource Inventory; (b) Establishment of energy plantations in selected districts of Punjab, KP, Sindh, and Balochistan while considering effective safeguarding measures to be taken to avoid negative effects local ecosystems and biodiversity so that the proposed locations in selected districts will be thoroughly and carefully selected following set criteria and the "Biodiversity Action Plan" prepared for the identification of potential risks and the corresponding appropriate preventive and alleviative risk mitigation measures; (c) Preparation/review of an Environmental Impact Analysis (EIA) of the planned woody biomass plantations and utilization of fuel wood in energy generation and the proposed Environment and Social Management Plan (ESMP) to address those risks (consistent with the project's Environmental and Social Management Framework); (d) Analysis of institutional constraints and coordination mechanisms between government agencies related to sustainable fuel wood production, supply chain and means of utilization of fuel wood into energy; (e) In-depth review of all current policies, rules and regulations regarding fuel wood production, supply and use for power and non-power applications; (f) Development of recommendations on the technical side on the most suitable set-up, responsibilities and operational procedures, social and environmental safeguards and guidelines that will be fed into the policy development process in Component 1 within Year 1; (g) Conduct of consultation workshops among key stakeholders, including local communities and affected people, to develop and endorse guidelines within the purview of existing related policies and guidelines thus amplifying on the gaps and necessary barrier removal measures to reinforce and encourage amended guidelines favoring sustainable production and utilization of woody biomass; (h) Dissemination and conduct seminar workshops to implement the approved guidelines; (i) Establishment of monitoring and evaluation system to track on the progress of implementation not only for this component but for the overall project; and (k.) Development of guidelines for maintaining hazard-free biomass storage facility.

Activity 2.1.1.2: Development of knowledge products and dissemination among project stakeholders for utilization of woody biomass for energy purposes, both for power and non-power applications in rural areas. This activity builds upon the results of the studies, recommendations, and approved guidelines in Activity 2.1.1.1, highlighting the technical requirements and conformance with established standards in biomass production for energy uses. Considering the value of these knowledge products in building the capacities of the target project participants and stakeholders, the information packaging and messaging style with sufficient technical content will be coordinated with the relevant capacity building activities in Component 4 to make them suitable to the capacity development programs that are planned for the Project. This Activity will involve the following: (a) Development of the technical content inputs to the design of training manual, diagrams, illustrations, pictures, materials, and toolkits in coordination with relevant Component 4 capacity building activities; (b) Development of biomass transportation guidelines and standards; and (c) Development of modern technologies for harvesting of forest waste and agri-biomass and provide recommendation for harvesting tools for collection of forest wastes.

GEF support is required for incremental technical assistance in the implementation of the related activities in producing Output 2.1.1.

Output 2.1.2: Completed supply chain and market analysis and documented woody biomass demand and supply in the country.

Activity 2.1.2: Development of a detailed study on supply chain of biomass, including market analysis for determination of biomass consumption, production, and pricing. Based on the assessments and guidelines developed under activity 2.1.1., this assessment targets 8 selected districts and the assessment results are used in the final selection of 4 sites (one in each province). Under the study, the following are the key actions: (a) Review of existing Ten Billion Tree Tsunami Project (TBTTP) and evaluation of the locations of suitable woodlot forest; (b) Identification of the locations of biomass storage facilities; (c) Selection of technology for biomass pre-treatment (for agri-biomass and woody biomass); (d) Determination of Annual Allowable Harvest (AAH) for target forest area; (e) Determination of the cost of collection and transportation of woody biomass (forest wastes) from forest and non-forest sources; (f) Market analysis and future cost forecasting for woody biomass pricing; (g) Identification of industry for installation of industrial sized biomass burner for industrial applications select four industrial partners; and (h) Development of market analysis for determination of required for each area in terms of biomass consumption, production and pricing. The assessment and final selection of the 4 sites will also be informed by assessment of social and environmental risks/benefits, as outlined in the project's ESMF.

GEF support is required for incremental technical assistance in the implementation of activities to produce Output 2.1.2.

Output 2.1.3. Established farmers/communal forest nurseries over at least 300 hectares to provide planting stock for raising woody biomass in rural areas in the provinces of Punjab, KP, Sindh and Balochistan, including energy plantations over 40,000 ha of fast-growing native tree species at non-forested lands in selected districts each of these provinces for woody biomass supply.

Activity 2.1.3: Development of woody biomass supply network. Based on the assessments and guidelines developed under 2.1.1, this activity involves the following steps: (a) Establishment of plantations of fast-growing native tree species at non-forested[31] lands in selected districts under the Project; (b) Establishment of nurseries for sustainable supply for indigenous fuel wood plantation[32]; (c) Identification of nurseries under TBTTP for sustainable supply of native saplings for afforestation of wood lots (energy plantation); and (d) Identification of wood lots (energy plantation) forest and non-forest lands under TBTTP for harvesting of forest waste for supply for woody biomass; and (e.) Implementation and monitoring of environmental and social management plans.

GEF support is partially required for the incremental technical assistance for the implementation of Activity 2.1.3.

Output 2.1.4: Established and operational fuelwood production from 4.5 million trees of common fuelwood plant species on farmlands for supply of woody biomass for production and utilization of biomass energy in rural areas

[31] This will need to ensure current land use and environmental baseline at these sites is understood to ensure no displacement or conversion of habitat.

[32] These are carried out under the Ten Billion Tree Tsunami Project (TBTTP) by each provincial forest department.

Activity 2.1.4: Development and establishment of fuel production processes. This activity involves the development of woody biomass supply system consisting of the following steps (a) Establishment of at least 4.5 million tree of common fuelwood plant species on farmlands for sustainable supply of woody biomass[33]; (b) Development and establishment of sourcing system (nurseries) for supply of fast growing, local species for energy plantation; (c.) Implementation and monitoring of environmental and social management plans, and (d) Dissemination of the information about the developed woody biomass supply system.

The Project design is focused on woody biomass consistent with what has been approved in the PIF. The lands that will be used for fuelwood plantations are those that are allocated by the GoP through the Provincial Forest Departments, including marginal areas and those relatively not so fit for food crops and other products. It will be considered that the community that will be engaged in the project will also be concerned about food security. Therefore, in the planning and allocation of the acreage, this food security and agricultural concern will be taken into consideration in community discussions and decision making.

GEF support is partially required for the incremental technical assistance for Activity 2.1.4 as most of this activity will be covered under the TBTTP.

OUTCOME 2.2: Increased investments in the application of technologies for production, and energy efficient utilization of woody biomass energy

This outcome features the increased volume of investments in the application of technologies for production, and energy efficient utilization of woody biomass energy. The designed activities address the technological barriers by promoting and showcasing biomass energy efficient technologies and adoption of innovative practices for woody biomass energy conversion and utilization. Cost-effective applications of EE woody biomass fired energy generation and utilization technologies and comprehensive energy management systems in selected energy end-use sectors in support of rural socio-economic development are to be demonstrated. Among the activities are the design, engineering, construction, commercial operation, and maintenance of woody biomass fired power generation and distribution systems in the 4 provinces for the supply of electricity to selected unelectrified rural villages. These showcase the cost-effective application of decentralized woody biomass-based energy generation and distribution (through mini/micro-grids). The demonstrations involve the installation of a woody biomass fired power generation system for the production and distribution of electricity to consumers (residential, commercial, and industrial) in the surrounding villages. In addition, a mini/micro-grid system is to be installed for supplying the electricity to consumers. Where possible and feasible, other RE-based power generation units will also be integrated into the system to operate in hybrid mode or as supplementary unit when the main biomass-fired unit is under maintenance. The installed demo woody biomass-fired power generation and distribution system is to be operated and maintained as a commercial business. A key driver of the business model is the collection of fees for electricity sales and using part of revenues for the operation, repair, and maintenance of the system. This involves increased investment both from the public and private sectors for promoting biomass energy efficient technologies and scaling-up innovative practices for biomass energy production and utilization at the rural industry levels. These will be carried out as part of the Component 2.2 developmental activities for supporting the investment for the commercial operation and maintenance of the installed demo system.

[33] Ibid.

Part of the demo is the application of innovative and energy efficient wood-fired industrial equipment to reduce GHG emissions from rural industries. So far, wood briquetting/pelleting technology has not been introduced in the rural areas in Pakistan, though it has been practiced at some localities at a very small scale. Three (3) pilots for showcasing improved woody biomass fuel production technologies for producing wood chips, briquettes and pellets at potential sites are to be implemented. Entrepreneurs in rural areas hardly have access to technical assistance and financial support for installing energy efficient equipment. Technical support and partial financial assistance will be provided to selected local entrepreneurs to produce energy efficient wood-fired equipment. The results and impacts of these investment activities are to be monitored, documented, and published for dissemination.

Output 2.2.1: Completed and operational woody biomass-based energy generation technology demonstrations in 4 selected sites, one in each province of Punjab KP, Sindh and Balochistan showcasing the cost-effective application of decentralized woody biomass-based electricity generation and distribution (through mini/micro-grids)[34]

Activity 2.2.1: Development and implementation of woody biomass-based energy generation technology demonstrations in 4 selected sites. One demonstration is to be implemented in each province of Punjab KP, Sindh and Balochistan showcasing the cost-effective application of decentralized woody biomass-based electricity generation and distribution. This activity includes the following steps: (a) Development of site selection criteria; (b) Selection of four (4) sites for installation of 310 kW biomass gasifier plant in rural village (one in each province); (c) Environmental and social assessment and management plans (consistent with project ESMF); (d) Installation and commissioning of four (4) at 310 kW each woody biomass gasification plant at four different rural areas of Pakistan; (e) Development of micro grid for distribution of electricity to rural households (40-45 households); and (f) Site assessment and Installation of 250 kW PV solar power plant with battery backup, as a backup source of 310 kW gasifier in those times where gasifier will be shut down for maintenance or not working due to unavailability of biomass. No construction activities will be initiated until relevant social and environmental assessments and management plans have been disclosed, consulted, and finalized. Details of the demonstration activity are in **Annex 12A.1**.

GEF support is required for the incremental investment for the needed facilities and operation and maintenance.

Output 2.2.2: Completed and operational four (4) demonstrations of cost-effective production of woody biomass fuels such as wood chips, briquettes, and pellets for use in decentralized woody-biomass based power generation and distribution.

Activity 2.2.2: Implementation of approved demonstrations of cost-effective production of woody biomass fuels in 4 selected sites, one in each province. This activity will also be informed by environmental and social assessments and management plans prepared for the 4 sites for which ESIAs/ESMPs would also need to look at the production impacts. The activity includes the following steps: (a) Preparation of the proposed woody biomass production/pre-treatment facility for each 50 kW bio-gasification plant. Each facility shall be equipped with a wood chipper; (b) Development of biomass treatment facilities near secondary storage area for wood pelleting using wood pelleting machines (one for each site) to produce fuel for operation of energy efficient kilns and boilers to substitute fossil fuel; and (c) Transportation of

[34] This involves the installation of a collective total woody biomass-based power generation capacity of 5.4 MW in the 4 provinces to supply of electricity to supply unelectrified rural villages.

biomass within and/or between the storage areas using 8-ton tractor trolley and two (2) electric fork lifters (with 2 ton weight carrying capacity) for handling woody biomass in storage facility (one lifter each for the primary and secondary storage areas). In total there will be 8 lifters and 8 tractor trollies to be procured for the four sites.

GEF support is required for the incremental investment for the necessary transportation and handling equipment and ancillary requirements.

Output 2.2.3: Completed and operational demonstrations of the cost-effective applications of energy efficient woody biomass fired technologies and comprehensive energy management systems in selected energy end-use sectors in support of rural socio-economic development[35]

Activity 2.2.3: Development and implementation of demonstrations of the applications of energy efficient woody biomass fired technologies and comprehensive energy management systems. This activity will also be informed by the environmental and social assessments and management plans prepared for the 4 sites and involves the following steps: (a) Shortlisting of local industry, one in each province and procurement of four (4) woody biomass based industrial gasifier burners one for each demo site, these burners will be installed at the selected rural industry; (b) Procurement of energy meters for monitoring of energy consumption and billing of electricity usage; energy meters for each site; (c) Procurement of high accuracy energy meter at power plant for monitoring of energy generation to also help to identify any possible leakage in electricity consumption at user side; (d) Installation of four (4) weigh bridges one for each demo site at primary biomass storage area and recording of the total amount of biomass dispatched from the biomass storage area; (e) Installation of four (4) weigh bridges one for each demo site at secondary biomass storage area and recording of the total amount of biomass received at power plant site and notify any potential leakages in it; and (f) Installation 4 weighing scales, one for each demo site for measurement of quantity of chipped wood and pellets prepared by wood-chipper and wood pelletizers.

GEF support is required for the incremental investment for the equipment and facilities to be used in the applications of energy efficient woody biomass fired technologies and comprehensive energy management systems.

Output 2.2.4: Published and disseminated information about the results and impacts of the completed demonstrations as well as developed and adopted a sustainability and replication plan of decentralized woody biomass-based energy production and utilization for power and non-power applications

Activity 2.2.4.1: Published and disseminated information about the results and impacts of the completed demonstrations. This activity involves the following: (a) Development and implementation of a project communication and information dissemination plan to guide the information and knowledge management and products to be disseminated to the government, public and the private sectors for investment promotion and support; (b) Development of annual newsletter for dissemination of project activities, results, outcome for public consumption; and (c) Conduct of annual project stakeholder events for awareness of the results and outcomes of the project. These will be supplemented by information dissemination through the Project website and internet and social media platforms.

[35] This includes local production of energy-efficient wood-fired devices that are cost-effectively utilized in rural industries to improve energy utilization efficiency and reduce GHG emissions.

Activity 2.2.4.2: Developed and adopted a sustainability and replication plan for woody biomass-based production and utilization for power and non-power applications. This activity involves the next generation of sustainable woody biomass-based production and utilization technologies based on the findings of the demonstration projects and correlated with experiences in other woody biomass programs of similarly situated countries. The achievement of PASBET's long-term impact and benefits will only happen when the sustainable and affordable access to energy services from biomass-based technology applications are fully supported and continued by the key stakeholders of the program. A Sustainability and Replication Plan will be developed and adopted by the GoP in the final year of the project through the following: (a) Evaluation of the results of the demonstrations in all the parts of the woody biomass-based value chain in terms of biomass resources, siting location, fuel processing, logistics, user needs, community management models, financing mechanisms and industry support for valuable insights on the underlying conditions that influence success or failure of the local energy interventions; (b) Development of a sustainability and replication strategy framework; (c) Preparation of designs and implementation plans for replication of appropriate, sustainable biomass-based technologies and management systems to improve technical designs and increase the certainty for future investment decisions; (d) Development of technical inputs and standards for the establishment and adoption of the basic designs and operating guidance manuals for the replication program of related biomass-based technologies; (e) Formulation and adoption of the Biomass Sustainability and Replication Plan with targets, resource mobilization and financing strategy; and (f) Establishment of monitoring and evaluation system for the sustainability and replication plan.

GEF support is required for the incremental investment in technical assistance and other requirements for the delivery of Output 2.2.4.

COMPONENT 3: SUPPORTING FINANCIAL REQUIREMENTS FOR BIOMASS ENERGY TECHNOLOGIES INITIATIVES

This component targets the removal of financial barriers to woody biomass production and application of energy efficient technologies and adoption of innovative approaches for woody biomass energy conversion and use, including biomass energy production for power and non-power applications. The expected outcome from the delivery of the expected outputs under this component is adequate amounts of financial resources become available for woody biomass energy technology application projects in the country. The realization of this outcome involves establishing and operationalizing financial mechanisms for supporting commercial viabilities and operations of woody biomass energy production for both power and non-power applications and development of biomass energy industries at potential locations in targeted districts. The establishment of proper marketing and supply chain is crucial for the demonstration of the woody biomass energy production technologies. For this purpose, market-oriented mechanisms are to be established for the development and utilization of biomass energy resources, energy efficiency and comprehensive biomass energy management systems for supporting sustainable rural socio-economic development. These interventions support the creation of "green jobs" and produce benefits for the rural people, especially marginal segments of the society and climate vulnerable communities and help in avoiding potential exposure of these people to COVID-19 or threats from future pandemic. Feasible financial schemes for setting up biomass energy technologies application projects, for woody biomass-based energy generation, and EE wood-fired device/equipment production business are to be supported via provision of access to finance thorough the financing schemes to be developed by the project for rural electrification project and better bankability of community-based projects, as well as decentralized biomass-based energy generation initiatives in rural areas. Business plans/models for the

district governments and private sector to facilitate financing and implementation of woody biomass energy production projects are to be developed.

Outcome 3: Adequate amounts of financial resources available for woody biomass energy technology application projects in the country

Output 3.1: Established and operational investment and financing mechanisms for supporting the commercial viability and operation of woody biomass energy production for power and non-power applications, and the development of biomass energy industries.

Activity 3.1.1: Development of Policy Recommendations State Bank of Pakistan (SBP) RE Refinancing Scheme. This activity involves technical assistance on the development of policy recommendations to be coordinated with the implementation of Activity 1.2.2 in Component 1 for consideration in the policy and regulatory mechanisms development for the biomass-based energy production. It involves the following steps: (a) Review of existing financial schemes, e.g. the State Bank of Pakistan (SBP) Refinancing Scheme for Renewable Energy; (b) Financing schemes gap identification and analysis regarding RE in general and woody biomass in particular; (c) Review of financial institution's (FI's) practices in Pakistan and other countries with successful biomass energy programs in the adoption of financing policies for such programs; (d) Development of policy recommendations and ways forward on appropriate financing for woody biomass-based power, non-power or cogeneration projects based on SBP RE Refinancing Scheme experience. The SBP financing scheme is for SMEs which may also have some relevance to companies or entrepreneurs interested in biomass-based energy generation and woody biomass-based power and non-power (thermal process) or cogeneration applications.

Activity 3.1.2: Inclusion of Woody Biomass Technology Application Projects in the SBP RE Refinancing Scheme or in other Relevant Financial Schemes. This activity focuses on the selection of an appropriate financial institution, such as SBP, based on the results of Activity 3.1.1. It involves the following steps: (a) Estimation of the financial needs of the off-grid sector of Pakistan for biomass-based fuel wood production and processing and mini/microgrid power generation; (b) Development of an appropriate financial and business model applicable to the typical rural power project in the 4 selected provinces; (c) Determination of suitable portfolio that can be accessed by project proponents in power generation, non-power application or cogeneration using biomass in general and woody biomass in particular as energy source; (d) Conduct of consultations with SBP or other FIs on the findings and recommendations of the study; and (e) Development of policy recommendations to and approval by SBP or other qualified FI for inclusion of biomass in general and woody biomass in particular projects into the RE Refinancing Scheme.

Activity 3.1.3: Orientation and Engagement of and Implementation by Selected Banks of the Approved Modified RE Refinancing Scheme. This activity focuses on the implementation of the suitable and approved financing scheme for biomass-based projects based on the results and recommendations of Activity 3.1.2. It involves the following steps: (a) Orientation of SBP and other FIs on the approved financing scheme and engagement of the designated banks through formalization of the financing agreement and fund management plan; (b) Development of toolkits for selected banks' lending schemes and the credit guarantee; and (c) Conduct of training sessions for the 5 major FIs in Pakistan on biomass based electrification projects.

GEF support is required for incremental technical assistance and consultancy services in establishing the necessary conducive framework under which private financing can be made available for woody biomass-based energy projects.

Output 3.2: Established and operational market-oriented mechanism for the enhanced development and utilization of biomass energy resources, energy efficiency and comprehensive biomass energy management systems for supporting sustainable rural socio-economic development.

Activity 3.2.1: Development and Adoption of Business Cases for Biomass-Based Rural Off-Grid Electrification Projects. This activity focuses on establishing bankable, market-oriented business cases for all aspects of biomass-based rural socio-economic development projects. It involves the following steps: (a) Assessment and estimation of the level of financial needs of off-grid electrification sector of Pakistan; (b) Identification and establishment of a portfolio of projects to include qualified biomass-based (particularly woody biomass) projects; (c) Development of a financial and business model for various applications, e.g., micro-grid, CHP, thermal energy etc. ; (d) Preparation of at least 16 business cases of biomass-based rural off-grid electrification projects for different villages (4 for each province); (e) Estimation of the investment levels and value of benefits and advantages of having mini/microgrid biomass-based power systems for rural electrification for each selected village; (f) Evaluation of potential projects for both options consisting of export of surplus electricity to the national grid or establishment of allied SME operations for utilization of excess electricity; and (g) Adoption of feasible business cases for bringing them up for business decision-making and bank support. Social and environmental risks and sustainability aspects will be integral to the development of the business cases to help ensure investment goes towards sustainable biomass energy projects.

Activity 3.2.2: Conduct of Roadshow and Identification of Possible Financing Arrangements for Biomass-based Mini/micro-grids Power Projects for Rural Electrification. This activity focuses on promoting the feasible business cases developed in Activity 3.2.1 bringing about closure of financial packaging and assistance to potential investors. It involves the following steps: (a) Conduct of seminars/workshops for promotion and development of biomass-based mini/microgrid projects for off-grid communities; (b) Conduct of roadshows for FIs, DFIs at national and international levels to inform them of investment opportunities and create traction in motivating financial arrangements for feasible projects; (c) Development of a framework and guidelines for establishing risk-based market-oriented mechanisms for enhanced development of biomass-based energy systems, including consideration of social and environmental risks; and (d) Application for financing arrangements, negotiations and finalization leading to financial closure for at least 8 projects by the communities. This is the measure of success of the demonstrations wherein the proof of concept is well accepted by stakeholders, financing institutions and potential proponent entities and therefore it is important that the demonstrations will become models to attract more investors by conveying this information through the roadshows, workshops, and project info dissemination systems.

GEF support is required for the incremental technical assistance and consultancy services in creating necessary awareness among the relevant stakeholders so that the private capital (both locally and internationally) will be made available for the woody biomass power projects.

Output 3.3: De-risked biomass-based power generation projects, decentralized biomass-based energy generation in rural areas, and business plans for the GoP and private sector to facilitate financing and implementation.

Activity 3.3.1: Development of Risk Portfolio and de-risking framework for biomass-based power generation projects and non-power applications in rural areas. This activity supports the project portfolio developed in Activity 3.2.2 in terms of risk mitigation intended for future biomass-based (including woody

biomass) power generation projects and for non-power applications. The aim is to create a business environment where the technology will be accepted by the intended users, financing agencies and technology development supporters. Cash flow projections of the demo operations will be used to create the business case of financing, which in turn is already a simulation of de-risking mechanisms at the piloting stage towards establishing the proof of concept that interested parties are looking for. The initial technical and economic feasibility studies done during the project design stage show favorable results and to be verified with more detailed consultations during the inception stage of the project.

The de-risking of biomass-based investment projects will assist GoP policymakers to promote and replicate private sector investment. The overall aim is to intervene and catalyze participation of private and public capital cost-effectively. This activity involves the following steps: (a) Identification of financial risks of biomass-based power generation projects, decentralized energy generation in rural areas; (b) Conduct of financial analysis taking into account average baseline (collateral, willingness to pay, financial leverage and liquidity of rural communities) and project investment cost; (c) Conduct of regional comparison to understand best practices to cover the financial risks; (c) Analysis of the risks under the guidance provided by known bank standards, such as Basel III, a global, voluntary regulatory standard on bank capital adequacy, stress testing and market liquidity risk. The UNDP De-risking RE Investment methodology https://www.undp.org/publications/derisking-renewable-energy-investment will be used as the starting point in this regard. The risks mentioned in item (a) above includes risk analysis on technology failure, risk of non-payment from power regulator, risk associated with setting up substations for grid interconnectivity, and additional risk of setting up allied units and exporting power to them as well as related social and environmental risk implications.

Activity 3.3.2: Development and operationalization of a de-risking facility for biomass-based energy development projects. This activity addresses the significant financing gap and barriers that could be experienced in these kinds of projects and catalyze private sector financing. Because of the inherent risks involved when introducing modern technologies, particularly for rural-based technology transfer program, a window operation needs to provide backstopping to financial sector regarding risk evaluation and means of overcoming the financial risks identified of biomass based energy projects. Based on the recommendations on the de-risking framework in Activity 3.3.1, available financial assistance windows of international and multilateral FIs and other appropriate international funding assistance programs will be evaluated. This activity involves the following steps: (a) Discussions with FIs (particularly MFIs) and RE project developers to introduce the de-risking facility approach, to involve them in developing the facility for necessary inputs and feedback on a proposed de-risking facility; (b) Definition (based on the results of the discussions) and approval of the de-risking facility arrangement considering its applicability to all possible and proposed projects and the need to define specific de-risking arrangements for each type of biomass-based project proposal; (c) Development and submission of the proposals from project proponents; (d) Establishment and operationalization of a de-risking facility for biomass-based energy development projects using the approved de-risking facility arrangement[36]; and (e) Monitoring and evaluation of the de-risking facility.

GEF support is required for the incremental technical assistance and consultancy services in the provision of the necessary support for establishing and providing the de-risking facility services for the biomass-

[36] The de-risking facility concept shall be based on the lessons learned and best practices from other RE project de-risking work in bringing in existing global financial mechanisms that could be accessed and be made available to cover the risk of community-managed or industry-owned biomass-based power and non-power project initiatives, as was done with similar work in Pakistan for solar energy with funding support from the USAID.

based technology and rural electrification be minimized and commercial lending can be initiated for woody biomass projects.

COMPONENT 4: BIOMASS ENERGY TECHNOLOGY CAPACITY BUILDING AND KNOWLEDGE MANAGEMENT AND GENDER MAINSTREAMING

This Component addresses the capacity, awareness and information barriers associated with the introduction of biomass energy production and utilization technologies in the selected provinces. Thus, under this Component, it is expected that local capacities are built, awareness created, technical knowledge is generated and disseminated, and gender issues are mainstreamed in steps across the project interventions in mainstreaming gender in the overall biomass-based energy applications in the country. This project component includes the design (based on a capacity needs assessment) and implementation of gender sensitive capacity development training programs for the design, construction, operation, and maintenance of woody biomass-based energy production systems both for power and non-power applications, as well as for strengthening institutional capacity for supporting sustainable biomass energy production and utilization. In general, the training of trainers is to be carried out to enable capacity development on woodlot operations and management, woody biomass fuel production, and woody biomass-based power generation and distribution, and on the promotion of the widespread applications of energy efficient wood-fired equipment. Relevant activities are included for main value chain actors of biomass energy enterprises. The training and workshops are intended to enhance their capacity and motivate them to provide technical services and invest in the applications of cost-effective, climate resilient and energy efficient woody-biomass energy technologies. The extent of women' role and their participation in the implementation of the activities are among the indicators of success in realizing the expected outcome of this project component. Online dissemination of knowledge products on lessons learned and best practices are to be extensively done as well as database development to be used as the repository for sharing experiences and for paving the way for replication and scaling-up use of climate resilient and energy efficient biomass energy technologies.

Outcome 4: Enhanced local capacity, skills and knowledge in the development, installation, and operation of biomass energy technology systems in rural Pakistan

Output 4.1. Developed and implemented gender sensitive capacity development training programs for the design, construction, operation, and maintenance of woody biomass-based energy production systems both for power (e.g., decentralized power generation and distribution) and non-power (e.g., industrial energy efficient wood-fired equipment) applications, as well as for strengthening institutional capacity for supporting sustainable biomass energy production and utilization.

Activity 4.1.1: Village level needs assessment on different gender needs, social and cultural practices to ensure that the training is not affected by/caters to any gender gaps. This activity involves the following steps: (a) Conduct of an assessment to determine and bench mark the baseline situation of the selected project areas, to identify areas of improvement and to enable effective and active monitoring of progress; (b) Assessment on the energy and related needs of both women and men as well as the cultural practices in the region that are being addressed; (c) Gathering of information on the perception of men and women on the planned interventions, and any changes that they would suggest, if any, in the project setup, strategy and technology used as well as the possible roles of both genders in the project activities.

Activity 4.1.2: Development of gender-sensitive training plan. This activity involves the following steps: (a) Conduct of consultations and workshops for determining the elements of a gender-sensitive training plan

that focuses on the design, construction, operation, and maintenance of woody biomass-based energy production systems; (b) Incorporation of the learnings from the village level assessments from Activity 4.1.1 to ensure that the training program is responsive of the specific needs, culture and economy issues of the selected area are reflected in the process; and (c) Adoption and approval of the training plan.

Activity 4.1.3: Development and approval of the implementation of gender sensitive capacity development training plan. This activity involves the following steps: (a) Recruitment of gender expert to support and implement the training plan and sensitize participants; (b) Introduction of the basics of 'why' and 'how to' of mainstreaming gender to project staff and representatives of the various stakeholders; (c) Firming up of the schedule of the training plan, program flow of each of the training modules covering the essential training topics in line with the set objectives; (d) Guidelines in the identification and invitation of speakers, trainers and qualified professionals ; (e) Guidelines in the identification and invitation of participants in consultation with the MoCC&EC, stakeholders and the PMU; (e) Preparation of training evaluation questionnaire and analytical methods to determine best practice, lessons learned and areas for improvement; (f) Guidelines in the implementation of each of the training programs according to the schedule and implementation plan; and (g) Guidelines in the submission of training reports and evaluation summary.

Activity 4.1.4: Implementation of the gender-sensitive capacity development plan. This activity involves carrying out of the plan prepared in Activity 4.1.3 involving the development of appropriate training design, production of the training materials, training of trainers for each province regarding the activities for each demonstration project and organization of a seminar/workshop to specifically discuss and promote the results of the demonstration projects. It is envisaged that the capacity building program for demonstration project personnel will be completed by the second year of implementation for the project with a continuing plan to sustain the availability of the human resources and a back-up plan in case of transfers or resignations and the gender-sensitivities of the program. In coordination with relevant activities of the project, the implementation of capacity development and training program in this Activity 4.1.4 shall include, among others, (a) Livelihood training for women using electricity (e.g., cottage and home industries) and for men in various trades that will make use of the available electricity for productive application, wood-based furniture and other relevant trainings and (b) Basic solar PV power system training and training of trainers for women technicians. The implementation of on-the-job training activities will address all key elements of each demonstration project, including financial arrangements, detailed engineering designs, equipment procurement, construction, installation, commissioning, debugging, operation, and maintenance.

Activity 4.1.5: Monitoring and post evaluation of the impacts of the implemented capacity development plan. This activity aims to determine areas of improvement and sustainability for ensuring human resources availability. The evaluation will be carried out in two stages: (1) through immediate quantitative and qualitative evaluation after each training activity, and (2) impact evaluation of woody biomass energy development, planning and implementation delivered by trainees from the key stakeholders and agencies involved. A seminar/workshop to discuss and promote the results of all demonstration projects as well as progress in the base line projects will be organized as part of the monitoring and evaluation (M&E) activities. In the training seminar/workshop, the hosts of the biomass-based demonstration will report on the accomplishments of the various aspects of the demonstration that was implemented highlighting main experiences, best practices, lessons learned, and their recommendations for development and implementation of more biomass-based energy generation projects in Pakistan. Representatives from other demonstration projects and other stakeholders involved in the baseline projects will also be invited. Further, an overall impact evaluation will be carried out at mid-term in order to determine means to

improve the program and before closing of the project in order to assess the impact of the capacity development program. Recommendations for continuing the capacity development program after the project shall be endorsed to MoCC&EC and a commitment to carry out and improve the follow-up training program shall be part of the post-project arrangements and exit strategy to ensure sustainable biomass-based power and non-power applications program for the replication plans.

GEF support is required for the incremental technical assistance, consultancy services and budgets for workshops, focused group discussions and dialogs in the capacity building activities.

Output 4.2. Trained and qualified people (60 from each province) providing training on woodlot operations and management, woody biomass fuel production, and woody biomass-based power generation and distribution, including trained and qualified community members (600 men and 400 women) providing training on the promotion of the widespread applications of energy efficient wood-fired equipment, including those who will work on the demos.

Activity 4.2.1: Development of training manual, materials, and tool kits. This activity focuses on the production of manuals and training materials, particularly in meeting the capacity building needs of the people to be involved in the implementation of the PASBET's demonstration activities in the 4 selected provinces. This to be carried out in coordination with the relevant Component 2 Activities, i.e., under Output 2.1.3 (in plantations), Output 2.2.1 (fuel processing and utilization), Output 2.2.2 (various aspects in generation, Output 2.2.3 (end-use application) and Output 2.2.4 (power and non-power application), which will be the source of technical information, guidance, and technology. It involves the following steps: (a) Development of training manual, materials and toolkit on various woodlot operations and management, woody biomass fuel production, and woody biomass-based power generation and distribution, relevant environmental and social safeguards and build capacity of trainers for effective implementation; (b) Preparation of training manuals, tool kits and materials which can have separate sections/modules and guidelines that cater to specific aspects of men's and women's development roles, based on their respective needs depending on the province in which the activity is being implemented; and (c) Inclusion of best practices from successful biomass project examples to be incorporated in the training toolkit.

Activity 4.2.2: Selection, invitation and engagement of qualified speakers, trainers, and professionals, especially seeking women professionals. This activity involves the following: (a) Selection of qualified people (60 from each province) as well as community members (600 men, 400 women) to provide training; (b) Identification of government departments and/or organizations from which personnel will be selected; (c) Shortlisting of trainers based on experience, local relevance and necessary skillset; (d) Selection of trainers based on their ability to communicate effectively as well as knowledge of the culture; (e) Selection of qualified community members for training after consultation with village councils or similar respective bodies for every region; and (f) Invitation, engagement and training of the trainers.

Activity 4.2.3: Training delivery and toolkit introduction and use. This activity covers and supports the Project's activities in the design and implementation of biomass-based fuel production and logistics, mini/microgrid power generation installations, and energy efficiency management and involves the following: (a) Conduct of pilot training program on the promotion of the widespread applications of various aspects and areas of biomass-based technologies, equipment, practices to be adopted by the Project, based on training manual and toolkit developed in Activity 4.2.1; and (b) Inclusion of refresher courses at later stages to ensure retaining of knowledge as well as to cater to any changes or enhancements in the training plan and approach. Refresher courses will use the same training materials.

Activity 4.1.5 involves post evaluations whose results can be used in the redesign of training courses and training materials.

GEF support is required for incremental technical assistance, consultancy services and budgets in the development and conduct of training programs.

Output 4.3. Sensitized key value chain actors of biomass energy enterprises that provide technical services and invest in the applications of cost-effective, climate resilient and energy efficient woody-biomass energy technologies

Activity 4.3.1: Selection of energy enterprises. This activity focuses on sensitizing or bringing the key value chain actors to a committed awareness of the Government imperatives and policies on biomass-based technology application in energy. These are the people comprising the biomass energy enterprises that provide technical services and invest in the applications of cost-effective, climate resilient and energy efficient woody-biomass energy technologies. This activity involves the following: (a) Identification and vetting of the potential energy enterprises; (b) Selection of energy enterprises that provide technical services based on their expertise and experience of past projects; and (c) Sharing of experience and success of enterprises in different provinces to be considered, given the differences in culture, and working conditions of all four provinces.

Activity 4.3.2: Development of gender-sensitive training. This activity involves the following: (a) Development of a gender mainstreaming module/training to be used to train staff from different backgrounds; (b) Design of training module to include the basics of mainstreaming gender i.e., 'why' and 'how to' in all aspects of the biomass-based technologies, and includes success stories from the region or areas with similar cultures and issues for women to be used/incorporated in the training module; and (c) Design of training module that includes the gender goals of the project and the sector: economics, social justice, environmental impact.

Activity 4.3.3: Gender sensitization of individuals/teams from the enterprises working on the project. This Activity involves the following: (a) Sensitization of individuals working on the project with respect to gender issues; (b) Project staff to be engaged on gender and environment issues at an early stage including the problems faced by women and cultural practices in the areas and the issues; (c) Briefing of the Project staff on the importance of gender mainstreaming and the potential benefits for women and the population as a whole; and (d) Awareness raising of project personnel on the best possible approaches, good practices and lessons learned from similar projects.

GEF support is required for incremental technical assistance and promotional activities to encourage support from key value chain actors.

Output 4.4. Knowledge products on lessons learned, best practices, etc. that are available through online repository for sharing experiences and replicating and scaling-up use of climate resilient and energy efficient biomass energy technologies

Activity 4.4.1: Development and sharing of knowledge products based on results or on using the monitoring and evaluation information gathered. This activity will further build the capacity of project participants through knowledge exchange and management considering that the four provinces have various levels of capacity to implement projects of this type. Knowledge products will be developed from the experience of implementing the project activities that will include lessons learned, best practices and

other knowledge gathered in the project monitoring and evaluation of the various activities especially the new knowledge in the biomass-based production and power generation and distribution demonstrations. Of particular interest will be the business modeling and fee collection systems that will be implemented to make all the operational aspects sustainable and beneficial. The knowledge gathered and stored in the project repository, and in the "knowledge products" being disseminated in training seminars and information exchange networks will provide crucial decision support system in the replication and scaling-up the biomass-based production and utilization capacities in power supply for rural electrification starting from the selected districts into other districts and provinces as well.

The following is a tentative list of planned knowledge management (KM) products that the project will produce and their respective tentative delivery schedule.

KM Product	Brief Description	Delivery Schedule
Woody Biomass Resources in Pakistan	This is comprised of comprehensive situational and feasibility analyses and supply chain and market analyses on woody biomass production, and sustainable woody biomass production and utilization.	3Q 2024
Woody Biomass Energy Development and Utilization Policies in Pakistan	This is comprised of the results of the project's activities on the development and implementation of policies and market-based regulatory framework for supporting woody biomass production and use, including approved technical, management and operational standards.	2Q 2025
Energy Planning in Rural Areas with Woody Biomass Resources	This presents the formulated energy-integrated development plans that are developed for the pilot towns of the project.	3Q 2024
Woody Biomass Production	This presents the design, establishment, and operational farmers/communal forest nurseries for providing planting stock for raising woody biomass in woodlots and farmlands.	1Q 2026
Woody Biomass Energy Generation Technologies and Applications	This presents the results of the demonstrations of the cost-effective application of decentralized woody biomass-based electricity generation and distribution (through mini/micro-grids).	2Q 2026
Woody Biomass Fuels Production	This is about the results of the demonstration on the establishment and operationalization of cost-effective woody biomass fuels production facilities, and applications of energy efficient woody biomass fired technologies.	3Q 2025
Financing Woody Biomass Energy Production	This is about the investment cost of establishing and commercial operation of woody biomass energy production, as well as potential investment and financing mechanisms for supporting the commercial viability and operation of woody biomass energy production.	4Q 2024
Recommended Woody Biomass Energy Generation Projects in Pakistan	This is about the de-risked biomass-based power generation projects, decentralized biomass-based energy generation in rural areas, and business plans for the GOP and private sector to facilitate financing and implementation.	3Q 2025
Woody Biomass Capacity Development Program	This is about the capacity development training programs for the design, construction, operation, and maintenance of woody biomass-based energy production systems both for power and non-power applications.	1Q 2027

GEF support is required for the incremental technical assistance and budgets to produce the knowledge products.

Global environmental and adaptation benefits

The Project is designed to generate multiple global environmental benefits (GEB). The primary GEB is the reduction in GHG emissions by promoting the cost-effective application of sustainable biomass energy efficient technologies such as biomass-based power generation and innovative practices for woody biomass energy conversion and utilization. Such technologies are intended to directly contribute to reducing GHG emissions in the rural areas and towns, including the introduction and application of improved wood-fired industrial thermal equipment such as efficient wood-fired boilers, kilns, and furnaces and tobacco curing barns.

The GHG emissions that are attributable to the project will come from the woody biomass energy production and woody biomass energy technology application activities that will be carried out under the project. Such applications in the rural sector areas of selected districts of the 4 provinces - Khyber Pakhtunkhwa, Punjab, Sindh and Balochistan) are intended to facilitate reduced GHG emissions from energy activities of the country. Some expected replications of these applications are to be facilitated during the project implementation period and will be in operation towards the end of the project or just after the end of the project. Based on the preliminary line up of demonstration woody biomass-based electricity production as presented in **Annex 12**, the quantity of direct and consequential GHG emission reduction that can potentially be realized from the barrier removal activities of the project is about 3.1 million tons of CO_2 by the end of the project's 10-years influence period. It is estimated that about 64,633 tons of direct GHG emission reduction can be potentially achieved during the five-year implementation period of the PASBET Project. These estimates of GHG emission reductions are based on the identified demos and the potential replications. The actual GHG emission reductions will be determined during the detailed design of the demos under the project's Components 2.1 and 2.2.

Partnerships:

The project will work closely with its key players including MoCC&EC, EAD, MoPDR, Provincial Forestry Departments, AEDB, NEECA, Research and Academic Institutions, Civil Society Organizations/ Community Based Organizations, Farmers' Associations, local communities/governments, and the private sector, which will take active part in the implementation of the PASBET Project activities. They will look after the reliability and availability of wood fuel and other forestry resources in the upstream side of wood fuel production. The four (4) selected participating Provincial Forestry Departments will take the lead and administrative roles in their designated local project plantation sites and provide the link with the other stakeholders in the processing and utilization side or the downstream side of the whole system as they assume either supporting or beneficiary roles which are likewise critical to the achievement of the results of this project. The private sector, e.g., Pakistan Tobacco Company, Independent Power Producers (IPPs), and other private companies and industries that are potential users of fuel wood. which were earlier identified will be encouraged to participate in terms of investment mobilized.

The Government of Pakistan launched a Ten Billion Tree Tsunami Program (TBTT-P), a national program, for the revival of the forestry sector in the country. The PASBET Project builds on the interventions under the forestry component of the TBTT-P. The funds allocated for on-the-ground interventions are placed at the disposal of respective provincial governments, i.e., in Punjab, KP, Sindh and Balochistan, which are likely to be spent in the target districts of the proposed project's pilot demonstrations of the biomass fuel

production and utilization in industries and commercial establishments. These TBTT-P resources are considered investments mobilized from the federal and provincial governments. The provincial governments have shown their interest in promoting farm-forestry and raising woodlots for piloting biomass energy production technology demonstrations. Their allocated resources for these activities are also part and parcel of the proposed project and are considered investments mobilized, including the grants they provide through their public sector development funds. Private sector entities operating in project areas that will benefit from woody biomass-fired energy generation and consuming equipment will be encouraged to co-finance relevant project activities under proper agreements with the project authorities. Similarly, entrepreneurs and Small and Medium Enterprises (SMEs) engaged in woody biomass trading and production activities will be encouraged to make investments by financing at least 25% of activity costs, especially associated with efficient technology transfer for biomass conversion energy.

Co-financing Source	**Co-financing Type**	**Co-financing Amount (USD)**	**Included in project results?**	**If Yes, List the relevant outputs**
Ministry of Climate Change & Environmental Coordination (MoCC&EC), Government of Pakistan	Public Investment (Investment Mobilized)	8,295,414	Yes	Outputs 1.3, 1.5, 2.1, 2.2, 4.1 and 4.2
Ministry of Climate Change & Environmental Coordination (MoCC&EC), Government of Pakistan	In-Kind	2,198,853	Yes	
Government of the Punjab (Forestry, Wildlife & Fisheries Department)	Grant	2,099,547*	Yes	Outputs 1.5, and 2.1
Government of the Punjab (Forestry, Wildlife & Fisheries Department)	In-kind	742,081*	Yes	
Government of Sindh (Forest & Wildlife Department)	Grant	135,747*	Yes	Outputs 1.5, and 2.1
Government of Sindh (Forest & Wildlife Department)	In-kind	135,747*	Yes	
Ten Billion Tree Tsunami Programme, Government of Khyber Pakhtunkhwa	Grant	5,696,832*	Yes	Outputs 1.5, and 2.1
Government of Balochistan (Forests & Wildlife Department)	Grant	63,877*	Yes	Outputs 1.5 and 2.1
Government of Balochistan (Forests & Wildlife Department)	In-kind	1,202,348*	Yes	
UNDP	Grant	80,000	No	UNDP execution support services
	In-kind	500,000	Yes	Administrative support
	Total	**21,150,446**		

**Based on a PKR-USD Exchange Rate of PKR 221 per USD 1. The stated co-financing amount in the co-financing letter is in PKR.*

Risks

Preliminary analysis and screening conducted during the project development phase via UNDP's Social and Environment Screening Procedure (SESP) identified potential social and environmental risks associated with project activities. The screening procedure established that the project falls under Substantial Risk category and that the identified potential social and environmental risks' impacts are manageable through identified mitigation measures.

The SESP template (Annex 5) details the applicable specific environmental and social risks associated with project-supported interventions. The significance of each risk, based on its likelihood of occurrence and extent of impact, has been estimated as being either low or moderate. Based on the significance of these individual risks, the project has been allocated an overall SESP risk categorization rating of Substantial. This was determined by taking a precautionary approach to the overall risk categorisation, due to the identification of a significant number of Moderate risks. It is considered that due to a lack of site-specific information at this stage, a Substantial risk categorization is prudent, as it will enable further assessment and management to confirm the site-specific measures required to uphold SES safeguard standards.

Substantial Risk: is defined by UNDP's SES[37] as "*Projects that include activities with potential adverse social and environmental risks and impacts that are more varied or complex than those of Moderate Risk projects but remain limited in scale and are of lesser magnitude than those of High Risk projects (e.g. reversible, predictable, smaller footprint, less risk of cumulative impacts).*"

The following Moderate Risks have been identified during the SESP:

Risk 1: Given the current security situation in Pakistan, the project could inadvertently expose local communities and/or project workers to adverse security risks/hazards. [P. 5, Standard 3.1, 3.2, 3.3 & 3.8]

Risk 2: The regions in which the project is being proposed have been exposed to increased natural hazard events recently. Physical infrastructure being supported by the project may run the risk of being designed in a maladaptive manner, and could pose a threat to both users and the surrounding community/environment as a whole. [Standard 2.1, 2.3]

Risk 4: Local communities and farmers do not have the capacities or interest to adopt forestry interventions and sustainable production of woody biomass in their agriculture field. As such, there is the potential that unsustainable agricultural practices and forest harvesting could be used to provide fuel for the biomass plants. [Standard 1.1, 1.2]

Risk 5: Introduction of efficient wood-fired equipment in areas having natural forests may create a perverse incentive, which may lead to an increase in the consumption of woody biomass leading further degradation of natural primary forests.. Furthermore, there might be a risk the use of woody biomass becomes more economically attractive in alternative use. [P.2 & P.7; Standard 1.1; 1.3; 1.8]

Risk 6: Woody biomass use could exceed production. Even fast-growing species will take up to five years to mature. Moreover, climate change risks (flash floods and prolonged drought) and rainfall variability could impact production of woody biomass, which could result in the over harvesting of other biomass sources. [Standard 1.6; 2.2; 2.3; 2.4]

Risk 7: Project activities and approaches might not fully incorporate or reflect views of women and girls and ensure equitable opportunities for their involvement and benefit; and there is a risk that a prolonged or recurrent COVID-19 pandemic would exacerbate gender inequality, including risk of GBV. [P.9, P.10, P.11, P.12]

Risk 8: The project proponents may not effectively engage and ensure the participation of all stakeholders during the project design and the implementation phase resulting in human rights violation. [P.2, P.3,P.4, P.6, P.7, P.14, Standard 6.4]

Risk 11: The Project may create adverse biodiversity impacts to habitats and/or ecosystems and ecosystem services, e.g. through habitat loss, conversion or degradation, fragmentation, hydrological changes; and activities within or

[37] UNDP SES, page 57.

adjacent to critical habitats and/or environmentally sensitive areas, including (but not limited to) legally protected areas, areas proposed for protection, or recognized as such by authoritative sources and/or indigenous peoples or local communities. [Standard 1]

Risk 12: Site for project demo and woody biomass wood lot may affect tribal peoples and cultural heritage. [Standard 4.1, 4.2, 6.1, 6.2 6.3]

Risk 13: Working conditions may not meet the requirements of national labor laws and international commitments. [Standard 7.1]

Three Low Risks have been identified during the SESP:

Risk 3: The project could inadvertently increase risk of exposure of communities and/or project workers to Covid19 and other communicable disease through stakeholder consultations and workshops. [P. 7, P. 6; Standard 3.4]

Risk 9: Local communities might face restrictions in access to resources, and anticipated livelihood benefits to local people might not materialize as planned, potentially leading to economic displacement. [Standard 5.2; 5.4]

Risk 10: The project activities could result in multiple sources of ambient pollution and affect the environment in a harmful manner. [Standard 8.1; 8.2; 8.5]

Though all of the risks identified in the SESP are low or moderate, the project's overall risk category is 'substantial'. Therefore, scoped ESIAs are required for relevant sites where project impacts and risks are concentrated. As such, the scoped ESIAs and site-specific ESMPs will further assess and design mitigation/management measures in respect of these activities. It is recommended at this stage that four scoped ESIAs be undertaken at the four demonstration sites as per Output 2.2.1 and 2.2.2. These sites will be used to demonstrate the operationality of EE biomass fired systems under Output 2.2.3. ESIA's will be conducted for all potential demo sites:

- Punjab: Layyah and Bhakkar sites
- Khyber Pakhtunkhwa: Kaghan and Kalam sites
- Sindh: Badin and Sujawal sites
- Balochistan: Jaffarabad/Sohbatpur and Dera Bugti sites

Prior to the commencement of any works, the sites will be reviewed against the exclusion list criteria (below). Sub-projects or activities will be deemed ineligible for the project if they:

- Involve conversion or degradation of natural habitats;
- May cause measurable adverse impacts to critical natural habitats;
- Risk the introduction of invasive alien species;
- May negatively affect endangered species;
- Involve physical or economic displacement of people;
- Do not comply with technical norms and standards;
- Purchase, use or store harmful pesticides or hazardous materials;
- Involve forced labor/ child labour;
- Are likely to create any impact that would be categorized as "High" or "Substantial".

The sites that are not excluded will be individually screened with the UNDP SESP template and then , according to the risk categorization of each site and based on SES requirements, a site-specific environmental and social impact assessment (ESIA) will be conducted in accordance with UNDP's SES policy and the UNDP SES Guidance Note on Assessment and Management. This applies to the following outputs:

- **Output 2.1.3.** Established farmers/communal forest nurseries over at least 300 hectares to provide planting stock for raising woody biomass in rural areas in the provinces of Punjab, KP, Sindh and Balochistan, including energy plantations over 40,000 ha of fast-growing native tree species at non-forested lands in selected districts each of these provinces for woody biomass supply.
- **Output 2.1.4.** Established and operational fuelwood production from 4.5 million trees of common fuelwood plant species on farmlands for supply of woody biomass for production and utilization of biomass energy in rural areas
- **Output 2.2.1.** Completed and operational woody biomass-based energy generation technology demonstration in 4 sites

The ESIA will be used to evaluate the local impacts at the demo sites, specifically for the following risks and specific potential impacts (including but not limited to):

- Risk 5: impacts on biodiversity and natural habitats
- Risk 6: overharvesting of other biomass sources
- Risk 9: potential access restrictions
- Risk 12: ownership and usage rights of lands, territories, and resources

At the Layyah and Bhakkar (Punjab), Sujawal (Sindh) and Dera Bugti (Balochistan) sites, the potential impacts on Indigenous Peoples must be evaluated in detail per the determination made in the *Due diligence Report on the Situation with regard to UNDP SES6 within the context of PASBET* (Annex 5). It was found that impacts on tribal communities surrounding these sites may occur, which has been accounted for in the SESP (Annex 5) and the projects ESMF. Therefore, assessment within the scoped ESIAs for the relevant demo sites must determine whether Indigenous Peoples as defined in UNDP SES6 are present within the project areas surrounding the demo sites. If this is the case, Free Prior and Informed Consent (FPIC) will be required and additional management measures, such as IPPs, may be required to manage potential impacts on Indigenous Peoples.

As such, the ongoing assessment of these activities (Outputs 2.1.3., 2.1.4., and 2.2.1) throughout the conduct of the ESIA process will account for any changes to the likelihood and impact of these risks. Any changes to their significance will result in development of commensurate management/mitigation measures (e.g. ESMPs, IPPs, etc.).

Other than SESP Risks as defined above, the following risks are also there that may hamper the achievement and progress of the project activities:

1) Political instability – Due to frequent changes in Government structure, there is a risk that cooperation level may not be same and also muti-layer approvals may be required.
2) Co-Financing from Government may not be materialized, due to economic instability of the Government.
3) Climatic conditions: Climate induced hazards may hamper the project progress and implementation

Gender equality and Women's Empowerment

The detailed gender analysis and gender action plan is **Annex 9**. The highlights of the gender analysis are as follows:

a. According to the Global Gender Gap Index, Pakistan is currently ranked as the second worst country in the world, scoring very poorly on most of the gender equality parameters. Health is one the primary issues that women face in the country: despite being the sixth most populous country in the world with a remarkably high fertility rate of 3.8 children per woman, there are gaps in the fulfillment of women's reproductive rights and access to quality healthcare for women throughout the country. Pakistan suffers from a lack of access to quality family planning services throughout the country, which has resulted in one of the highest unmet needs in Asia. This prevents women from meaningfully

participating in educational and economic life outside of the household and from making informed choices about their sexual and reproductive lives.

b. Gender-based violence is also prevalent in both the rural and urban areas along with women suffering from domestic abuse, honor killings and sexual violence to institutional discrimination. Barriers also exist in the access to education for women in the country which has resulted in a lower literacy rate for women (47%) compared to men (71%). Enrolment rates in schools are alarmingly low for girls. 32% percent of primary school age girls are out of school, compared with boys**.** This gender gap differential continues into secondary forms of education: only by 13% of young women are still enrolled in middle school. This has long term effect for women in the economic sphere where they constitute a small part of the workforce and their earnings are significantly lower compared to their male counterparts.
c. Moreover, these multidimensional gender inequalities have gotten worse because of covid-19 with regions of high fragility, conflict and poverty suffering the most from this change: the women in these areas are now exposed to greater health risks as well as potential loss of income. With women being confined to their homes for longer period s of time, the burden of unpaid care work on them has increased and they are more vulnerable to gender-based violence.

The key elements of the gender action plan are summarized as follows:

a. Both women and men share various roles and responsibilities in the biomass-based energy production process: both have knowledge about, and varying skills in, use of resources such as fuelwood for energy production use. Women's vital contributions to the process through the collection and use of firewood, however, are not always fully acknowledged and given the importance that it deserves. One of the primary reasons for this is the lack of gender-differentiated perspectives in biomass research and documentation, which itself has not been undertaken at an extensive scale at the provincial or national level. Both genders' perspectives need to be a part of biomass studies to ascertain the efforts required for setting up fuelwood plantations and biomass-based energy production.
b. The gender action plan as elaborated in Annex 9, therefore, is built upon:

    - Comprehensive collection of gender-related information and sex-disaggregated data to get a more holistic picture of the scenario in project areas
    - Incorporation of gender concerns in the project results framework and the monitoring plan that the project is based on which will assist in the formulation of interventions for specific outcomes.

Gender Marker. The Gender Marker Index for PASBET: 2. 'Advancing gender equality is a significant objective but not the principal reason to undertake this project. Gender is reflected in the Conflict Analysis, Implementation/Activities, the Results Framework, and the Budget.'

<u>Innovativeness, Sustainability and Potential for Scaling Up</u>:

Innovation: The project promotes and facilitates commercial production and utilization of sustainable woody biomass in energy deficient remote districts of the country. The application of decentralized power generation and distribution in these areas using available sustainable woody biomass resources is a new and innovative endeavor in Pakistan. To sustainably support the operation of such system, the Project includes the design practical environment friendly fuelwood supply systems that involve the establishment of energy plantations and promoting farm-forestry for enhancing forest cover and woody biomass resources with the involvement of local communities and private sector. This is an innovative model that would generate important lessons and best practices for wider applications across the country. The introduction of energy efficient biomass energy technologies (specifically using sustainable woody

biomass) and the adoption of innovative practices for biomass energy conversion and utilization are expected to contribute to reducing GHG emissions, improving local environment and minimizing unhealthy conditions due to smoke/smog produced from the use of inefficient wood-fired technologies and equipment. Through the project's collaborative approaches, incentive measures and sustainable supply chain development local entrepreneurs' and the private sector's interest in venturing and investing in the production and sustainable use and benefits of woody biomass are expected to be realized.

Sustainability: The sustainability of project interventions is ensured by building on the earlier initiatives of the government and private sector for curbing deforestation and enhancing tree cover in the country. A strong linkage and collaboration with government's on-going initiatives (Clean and Green Pakistan, TBTT-P, REDD+ Readiness project, SFM, SLM projects etc.) helps in creating sustainability. The involvement of relevant federal government agencies, provincial forestry departments, local community leaders, woody biomass traders and entrepreneurs, and private sector into every aspect of project designing and implementation enhances ownership of project interventions and improved/sustainable access and benefit sharing. The project's enabling environment and capacity building efforts for the widespread applications of woody biomass energy generation technologies and energy efficient wood-fired equipment for the industrial sector of the target remote rural areas, as well as the raising of energy plantations for the sustainable supply of woody biomass are expected to help in maintaining health environment promoting local livelihoods through sustainable production and utilization of woody biomass, thus reducing pressure on natural ecosystems and GHG emissions.

Scaling-up: The innovative ideas and practices under the three proposed components of the project can be replicated and scaled-up to other parts of the country, especially in Sindh and Balochistan. Raising energy plantation on barren lands and trees grown on farmlands will provide a major economic incentive to rural communities to expand these activities to other potential sites to earn the hard cash through production and sale of woody biomass to wood-fired thermal equipment used in rural industries and at the same time reducing threats to natural forests and reducing GHG emission from the introduction of efficient technologies. Lessons learned and knowledge generated through on-the-ground project interventions will help in replicating and scaling-up best practices at the national, sub-national and regional levels.

v. PROJECT MANAGEMENT

Cost Efficiency and Effectiveness

Cost efficiency is a critical aspect of any woody biomass project, ensuring its economic viability and long-term sustainability. As it's a key determinant of success for a woody biomass project, it ensures the optimal allocation of resources to achieve the desired outcomes. By leveraging locally available biomass resources, such as forestry residues, wood chips, and sawdust, these projects can effectively harness renewable energy while simultaneously addressing waste management challenges. Implementing cost-efficient practices involves optimizing the entire biomass supply chain, from harvesting and transportation to conversion processes.

Investments in advanced technologies, like efficient biomass boilers and process optimization can significantly improve energy conversion efficiencies, reducing operational costs and maximizing energy output. Moreover, strategic partnerships with local stakeholders and communities can facilitate access to affordable feedstock and foster mutually beneficial collaborations, contributing to the overall success of the woody biomass project.

By carefully assessing the entire project lifecycle, from biomass procurement to energy production and distribution, cost-effective strategies can be implemented to minimize expenses while maximizing benefits. It may also ensure environmental sustainability as well the project will ensure to enhance skills of the relevant government departments as well as the communities so that transition to the use of woody biomass could be ensured from the

indigenous method of fuel wood which is a main reason of deforestation and reason of carbon emissions. Therefore, this project will ensure policy interventions, skill development and resources for its effective roll out in the given locations that may ensure efficiency and effectiveness as a whole.

Project Management

The Implementing Partner for this project is the Ministry of Climate Change & Environmental Coordination (MoCC&EC) with CO support services from UNDP. The project will be implemented based on the proposed implementation modality - UNDP CO Country Office (CO) support to National Implementation Modality (NIM) with the Ministry of Climate Change & Environmental Coordination as implementing partner. UNDP support services to the Government (IP) for CO support to NIM includes the recruitment and recurrent management services of PMU staff, procurement, and engagement of Responsible Parties during the project implementation.

During the 5-year implementation period of the PASBET Project, the MoCC&EC, as the implementing partner will require the cooperation and services of several responsible parties to implement or support the implementation of the project activities. This may include Forest[38] and Energy Departments of respective provinces, Alternative Energy Development Board, National Energy Efficiency and Conservation Authority, Power Distribution Companies and Natural Gas Supply Companies. Under CO Support to NIM, UNDP shall sign a Letter of Agreement (LOA) with the relevant entities among these that will be designated as RPs and engage them to execute the activities in their respective area of responsibility.

From UNDP's side, for coordination and oversight, the Resilience, Environment and Climate Change Unit will be responsible. The staff at UNDP will facilitate work planning, monitoring and programme technical backstopping to the implementing teams, field visits and occasional events. A Project Board will provide oversight to the project through annual review of progress, risks, lessons learnt and alignment to the overall project goal. To ensure best value for money and just use of project financial resources, the UNDP usual annual processes of Spot Check and Annual Audit will be followed as guided by HACT Framework.

[38] Since Forest departments are also Co-Financers, UNDP will only sign the LOAs for GEF funds with respective Forest Departments. The funds contributed by these departments will be managed by their own self.

VI. PROJECT RESULTS FRAMEWORK

This project will contribute to the following Sustainable Development Goal (s): *SDG 7, SDG 13, SDG 8, SDG 11*
This project will contribute to the following country outcome (UNDAF/CPD, RPD, GPD): CPD Outcome 6: Enhanced resilience and socioeconomic development of communities
Applicable Output(s) from the UNDP Strategic Plan: Structural transformation accelerated, particularly green, inclusive, and digital transitions (5.2, 5.2.1)

Strategy	Objectively Verifiable Indicator				Means of Verification	Critical Assumptions
	Description	Baseline	Mid-Term	End-of-Project		
OBJECTIVE: Widespread application of sustainable biomass energy technologies for supporting socio-economic development of and reducing greenhouse gas (GHG) emissions from, the rural sector in Pakistan.	***Indicator 1***: No. of direct project beneficiaries disaggregated by gender (individual people) • Male • Female	• N.A.[39] • N.A.	• 64,000 • 42,800	• 160,380 • 107,000	Project M&E and activity reports. Measurement of increased income of beneficiaries after implementation of activities	No unexpected cultural or security barriers that hinder activities
	Indicator 2: Cumulative GHG emission reduction from the rural sector of Pakistan, tCO_2	**0**	**51,520**	**195,050**	Annual energy supply and consumption reports submitted by relevant national and local governments, etc. Regular periodic operations report from each sustainable biomass energy technology application demo and replication in demo towns/villages. Project M&E and activity reports.	Continuous commitment, support & active participation of the national/local government in low carbon rural development, as well as in meeting the renewable energy targets of the country Realization of committed co-financing from the national/local governments in the implementation of project activities and monitoring systems.

[39] No. of people with access to electricity in rural areas (80.15 mil, 58.7% of rural population) https://trackingsdg7.esmap.org/country/pakistan

Strategy	Objectively Verifiable Indicator				Means of Verification	Critical Assumptions
	Description	Baseline	Mid-Term	End-of-Project		
	Indicator 3: Cumulative Biomass-based electricity in rural areas in Pakistan, GWh	**0**	**1.5**	**98.2**	Annual energy supply and consumption reports in relevant national and local government agencies, etc. Project M&E and activity reports	Continuous commitment, support & active participation of the national/local government in low carbon rural development, as well as in meeting the renewable energy targets of the country.
	Indicator 4: Cumulative number of new jobs created in the application of sustainable biomass energy technologies and techniques in the rural areas. • Male • Female	N.A.[40]	• 1,460 • 1,380	• 3,660 • 3,450	Trade and commerce reports. Project M&E and activity reports. Monitoring report of no. of people employed in all job categories	No unexpected cultural or security barriers that hinder activities
COMPONENT 1: Establishment of Policy and Regulatory Framework for Sustainable Woody Biomass Energy Production and Utilization						
Outcome 1: Effective enforcement of policies and regulations on the sustainable production and use of woody biomass for energy generation and utilization in rural areas of Pakistan.	***Indicator 5***: Cumulative number of sustainable biomass energy projects in rural areas that are facilitated by approved and effectively enforced policies and laws/regulations that promote the commercial production and use of sustainable biomass-based energy.	0	8	32[41]	Project activity (baseline and incremental) reports. Government reports on the implementation of the approved sustainable biomass energy development and utilization policies and regulations. Project M&E reports.	Continuous consistent commitment of the national and local governments in politically supporting RE and EE initiatives in rural towns/villages. No competing value-added uses of woody biomass.

[40] No. of people with jobs in rural areas. Refined activity participation rates in rural areas: 47.1% (Male 68.4%, female 25.6%). Govt of Pakistan Labour Force Survey 2017/18

[41] This is 4 projects x 2 districts x 4 provinces

Strategy	Objectively Verifiable Indicator				Means of Verification	Critical Assumptions
	Description	Baseline	Mid-Term	End-of-Project		
	Indicator 6: Cumulative number of rural villages that develop and implement energy-integrated development plans that include the commercial production and utilization of sustainable woody biomass.	0	**4**	40[42]	Project activity (baseline and incremental) reports. Government reports on the implementation of the approved sustainable biomass energy development and utilization policies and regulations. Project M&E reports.	Continuous consistent commitment of the national and local governments in politically supporting RE and EE initiatives in rural towns/villages. No competing value-added uses of land for wood lots and energy plantations.
	Indicator 7: No. of developed, approved, enforced, and implemented concrete regulations and policy instruments as part of the RE law to include market-based targets of biomass-based energy, particularly woody biomass	**0**	**2**[43]	5[44]	Documentation of the approved and promulgated regulations and policy instruments in the Activity 1.3.1 to 1.3.3 Project activity (baseline and incremental) reports. Project M&E reports	
	Indicator 8: Cumulative no. of standards for equipment to use woody biomass for energy generation and EE utilization established and enforced	**0**	**3**[45]	**4**[46]	Documentation of the approved and promulgated regulations and policy instruments in the Activity 1.3.1 to 1.3.3. Project activity (baseline and incremental) reports. Project M&E reports	

[42] This is 10 villages in each 4 provinces

[44] All five (5) policies that are approved, implemented, and enforced in line with the targets of Biomass based power generation included in the National RE Policy.

[43] This includes: (a) Regulation to promote woody biomass and the corresponding Woody Biomass National Road Map; (b) Tariff application developed and submitted to NEPRA for biomass based electricity; (c) microgrid systems are made part of national grid code; (d) Inclusion of biomass as energy resource in the National Rural electricity policy; and (e) Community-based Natural Resource Management as part of National and Provincial Forestry Policy

[45] Standards developed for: (a) energy plantation; (b) biomass energy conversion; and (c) biomass user technologies

Strategy	Objectively Verifiable Indicator				Means of Verification	Critical Assumptions
	Description	Baseline	Mid-Term	End-of-Project		
COMPONENT 2: Promotion of Biomass Energy Production and Energy Efficient Utilization Technologies						
Outcome 2.1: Enhanced woody biomass production on forested, non-forested, and on farmlands to cater local biomass energy needs, including for power generation and rural industry operations.	***Indicator 9***: No. of established and commercially operated sustainable woody biomass production facilities.	**0**[47]	**1**	**3**[48]	Forest industry reports Trade and commerce reports Project M&E and Activity Reports.	Continuous consistent commitment of the national/local governments and stakeholders in politically supporting sustainable biomass energy technology applications for rural development.
	Indicator 10: Cumulative number of entities that are interested and planned to invest in sustainable woody biomass fuel production. * Local Governments * Private Sector entities	• 0 • 0	• 1 • 1	• 4 • 2	Forest industry reports Trade and commerce reports Project M&E and Activity Reports.	Continuous consistent commitment of the national/local governments and stakeholders in politically supporting sustainable biomass energy technology applications for rural development.

[48] In total there are 3 new sustainable woody biomass production facilities are expected to be completed by the end of the project, one in each province of Punjab, Sindh and Balochistan)

[47] In eight target districts there are few forests from where local community is using fuel wood. However, those forests cannot be characterized as sustainable wood biomass production facilities

[46] Standards opted by relevant authorities and provincial authorities as part of the procurement

Strategy	Objectively Verifiable Indicator				Means of Verification	Critical Assumptions
	Description	Baseline	Mid-Term	End-of-Project		
	Indicator 11: Average annual local production and consumption of sustainable woody biomass fuel, tons/year	0[49]	38,071[50]	**95,178**	Forest industry reports Trade and commerce reports Project M&E and Activity Reports.	Continuous consistent commitment of the national/local governments and stakeholders in politically supporting sustainable biomass energy technology applications for rural development.
Outcome 2.2: Increased investments in the application of technologies for the production, and energy efficient utilization of woody biomass energy.	***Indicator 12***: Cumulative amount of fossil fuel energy displaced by sustainable woody biomass fuels. * Power Applications, toe *Non-Power Applications, toe	 • 0 • 0	 • 1,500 • 150	 • 3,800[51] • 300	Forest industry reports Trade and commerce reports Project M&E and Activity Reports.	Continuous consistent commitment of the national/local governments and stakeholders in politically supporting sustainable biomass energy technology applications for rural development.
	Indicator 13: Cumulative number of private sector entities that are interested and planned to invest in sustainable woody biomass-based energy projects in rural towns/municipalities.	**0**	**3**	**5**	Project activity (baseline and incremental) reports. Documents on each implemented sustainable woody biomass energy	Continuous consistent commitment of the national/local governments and stakeholders in politically supporting

[49] From ESMAP Biomass mapping study it is estimated that 747,250 m^3 fuel is available to be and used for energy generation in the country.
The data for annual average of woody biomass local production and consumption for our eight target districts is not available.
Once all four the provincial forest departments are onboarded to gather the data for existing forests size, and the wood fuel generated from each district. This value cannot be estimated.
It is also important to mention the methodology used by forest department to measure the production & consumption of wood fuel.
Alternatively, we can also put this indicator to 0, as there no existing sustainable woody biomass production and consumption is place.
(Indicator, cannot give any value under this indicator)

[50] To generate 5.4 MW of biomass-based power generation it is estimated that total of 95,178 Tons of woody biomass is required annually. By mid-term at least 40% of this wood should be available from the sourcing.

[51] If 5.4 MW is generated using fossil fuel (e.g., NG based genset) it will require fossil fuel energy equivalent of 3.8 ktoe. Similar energy can be displaced by sustainable woody biomass fuels. By mid-term it is estimated that 40% of the total will be completed.

Strategy	Objectively Verifiable Indicator				Means of Verification	Critical Assumptions
	Description	Baseline	Mid-Term	End-of-Project		
					technology demonstration. Project M&E reports.	sustainable biomass energy technology applications for rural development.
COMPONENT 3: Supporting Financial Requirements for Biomass Energy Technologies Initiatives						
Outcome 3: Adequate amounts of financial resources available for woody biomass energy technology application projects in the country.	***Indicator 14***: Cumulative number of sustainable biomass energy projects whose funding are sourced from other sources of funds apart from national/local government. * Sustainable woody biomass fuel production * Sustainable woody biomass-based power generation and distribution * Sustainable woody biomass-based energy generation and utilization.	• **0** • **0** • **0**	• **4** • **4** • **4**	• **8** • **8** • **8**	Annual Reports on the planned & implemented sustainable woody biomass energy projects that are financed through the adopted financing scheme(s). Project M&E reports Project activity (baseline and incremental) reports. Documents on the financial schemes/ mechanisms from financial institutions.	
	Indicator 15: Cumulative number of financing schemes/mechanisms adopted by financial institutions for supporting sustainable biomass energy initiatives in Pakistan.	**0**	**1**	**2**	Project M&E reports Project activity (baseline and incremental) reports. Documents on the financial schemes/ mechanisms from financial institutions.	Continuous consistent commitment of the national/local governments and stakeholders in politically supporting sustainable biomass energy technology applications for rural development.
COMPONENT 4: Biomass Energy Technology Capacity Building and Knowledge Management and Gender Mainstreaming						
Outcome 4: Enhanced local capacity, skills and knowledge in the	***Indicator 16***: Cumulative number of sustainable biomass woody energy projects designed,	**0**	**4**	**32**	Post-evaluation reports on completed capacity development programs.	Continuous consistent commitment of the rural town/municipal authorities

Strategy	Objectively Verifiable Indicator				Means of Verification	Critical Assumptions
	Description	Baseline	Mid-Term	End-of-Project		
development, installation, and operation of biomass energy technology systems in rural Pakistan.	implemented, and maintained by rural towns/municipalities				Annual Reports on the planned & implemented sustainable biomass energy projects that are designed, operated, and maintained by rural towns/municipalities Project M&E reports Project activity (baseline and incremental) reports.	and stakeholders in supporting sustainable woody biomass energy development and utilization projects.
	Indicator 17: Cumulative number of operational woody biomass-based power generation system installations in rural towns/ municipalities.	**0**	**4**	**16**	Annual Reports on the planned & implemented sustainable biomass energy generation and distribution facilities operated and maintained by rural towns/municipalities Project M&E reports Project activity (baseline and incremental) reports.	Continuous consistent commitment of the rural town/municipal authorities and stakeholders in supporting sustainable woody biomass energy development and utilization projects.
	Indicator 18: Cumulative number of energy consumers in rural areas that are utilizing woody biomass-based energy generating and consuming equipment (persons)	**0**	**1,120**	**2,240**	Project M&E reports Market survey reports and data on reduction in amount of woody biomass used by the beneficiaries on quarterly basis	

VII. MONITORING AND EVALUATION (M&E) PLAN

The project results, corresponding indicators and mid-term and end-of-project targets in the project results framework will be monitored annually and evaluated periodically during project implementation. If baseline data for some of the results indicators is not yet available, it will be collected during the first year of project implementation. The Monitoring Plan included in **Annex 4** details the roles, responsibilities, and frequency of monitoring project results.

Project-level monitoring and evaluation will be undertaken in compliance with UNDP requirements as outlined in the UNDP POPP (including guidance on GEF project revisions) and UNDP Evaluation Policy. The UNDP Country Office is responsible for ensuring full compliance with all UNDP project M&E requirements including project monitoring, UNDP quality assurance requirements, quarterly risk management, and evaluation requirements.

Additional mandatory GEF-specific M&E requirements will be undertaken in accordance with the GEF Monitoring Policy and the GEF Evaluation Policy and other relevant GEF policies[52]. The costed M&E plan included below, and the Monitoring plan in **Annex 4**, will guide the GEF-specific M&E activities to be undertaken by this project.

In addition to these mandatory UNDP and GEF M&E requirements, other M&E activities deemed necessary to support project-level adaptive management will be agreed, including during the Project Inception Workshop, and will be detailed in the Inception Report.

Minimum project monitoring and reporting requirements as required by the GEF:

Inception Workshop and Report: A project inception workshop will be held within 2 months from the First disbursement date, with the aim to:

a. Familiarize key stakeholders with the detailed project strategy and discuss any changes that may have taken place in the overall context since the project idea was initially conceptualized that may influence its strategy and implementation.
b. Discuss the roles and responsibilities of the project team, including reporting lines, stakeholder engagement strategies and conflict resolution mechanisms.
c. Review the results framework and monitoring plan.
d. Discuss reporting, monitoring and evaluation roles and responsibilities and finalize the M&E budget; identify national/regional institutes to be involved in project-level M&E; discuss the role of the GEF OFP and other stakeholders in project-level M&E.
e. Update and review responsibilities for monitoring project strategies, including the risk log; SESP report, Social and Environmental Management Framework and other safeguard requirements; project grievance mechanisms; gender strategy; knowledge management strategy, and other relevant management strategies.
f. Review financial reporting procedures and budget monitoring and other mandatory requirements and agree on the arrangements for the annual audit.

[52] See https://www.thegef.org/gef/policies_guidelines

g. Plan and schedule Project Board meetings and finalize the first-year annual work plan. Finalize the TOR of the Project Board.
h. Formally launch the Project.

GEF Project Implementation Report (PIR):

The annual GEF PIR covering the reporting period July (previous year) to June (current year) will be completed for each year of project implementation. UNDP will undertake quality assurance of the PIR before submission to the GEF. The PIR submitted to the GEF will be shared with the Project Board. UNDP will conduct a quality review of the PIR, and this quality review and feedback will be used to inform the preparation of the subsequent annual PIR.

GEF and/or LDCF/SCCF Core Indicators:

The GEF and/or LDCF/SCCF Core indicators shown in **Annex 14** will be used to monitor global environmental benefits and will be updated for reporting to the GEF prior to MTR and TE. Note that the project team is responsible for updating the indicator status. The updated monitoring data should be shared with MTR/TE consultants prior to required evaluation missions, so these can be used for subsequent ground truthing. The methodologies to be used in data collection have been defined by the GEF and are available on the GEF website.

Independent Mid-term Review (MTR):

An independent mid-term review (MTR) will be completed by the mid-point of the project. The terms of reference, the MTR process and the final MTR report will follow the standard templates and MTR guidance for UNDP-supported GEF-financed projects available on the UNDP Evaluation Resource Center. The MTR must be submitted to the GEF by the mid-point of the project but no later than 48 months after CEO Endorsement. Mid-point is determined by taking the expected ProDoc signature date plus number of months duration and finding the middle date. (For example, if a project has a 5-year duration, the MTR must be submitted 2.5 years after ProDoc signature.) To meet the submission deadline, final MTR reports must be completed and submitted to BPPS NCE team no later than 2 months in advance of the submission deadline to allow sufficient time for internal review/clearance that is required prior to submission.

Provisions must be taken to complete and submit the MTR within the submission deadline. Therefore, the MTR process must start no later than 8 months before the expected date of submission of the MTR.

The MTR will be 'independent, impartial and rigorous'. The evaluator(s) that UNDP will hire to undertake the assignment will be independent from organizations that were involved in designing, executing or advising on the project to be reviewed. Equally, the evaluators should not be in a position where there may be the possibility of future contracts regarding the project under review.

The GEF Operational Focal Point and other stakeholders will be actively involved and consulted during the MTR process. Additional quality assurance support is available from BPPS/NCE.

The final MTR report will be publicly available in English and will be posted on the UNDP ERC by the MTR submission date included on cover page of this project document. A management response to MTR recommendations will be posted in the ERC within six weeks of the MTR report's submission to the GEF.

Terminal Evaluation (TE):

An independent terminal evaluation (TE) will take place upon completion of all major project outputs and activities. The terms of reference, the evaluation process and the final TE report will follow the standard templates and TE guidance for UNDP-supported GEF-financed projects available on the UNDP Evaluation Resource Center. TE must be submitted to the GEF no later than 6 months after the Completion Date. This is a hard deadline that, if not met, can only be extended through a formal extension request. To meet the submission deadline, final TE reports must be completed and submitted to BPPS NCE team no later than 2 months in advance of the deadline to allow sufficient time for internal review/clearance that is required prior to submission.

Provisions must be taken to complete and submit the TE within the submission deadline. Therefore, TE must start no later than 8 months before the expected date of submission of the TE (or 11 months prior to the estimated operational closure date).

The evaluation will be 'independent, impartial and rigorous'. The evaluator(s) that UNDP will hire to undertake the assignment will be independent from organizations that were involved in designing, executing or advising on the project to be evaluated. Equally, the evaluators should not be in a position where there may be the possibility of future contracts regarding the project being evaluated.

The GEF Operational Focal Point and other stakeholders will be actively involved and consulted during the terminal evaluation process. Additional quality assurance support is available from BPPS NCE.

The final TE report will be publicly available in English and posted on the UNDPHY ERC by the TE submission date included on cover page of this project document. A management response to the TE recommendations will be posted to the ERC within six weeks of the TE report submission to the GEF.

Per the GEF Terminal Evaluation requirements, for cancelled full-sized projects, Terminal Evaluations are required if the GEF grant expenditure exceeds more than US$ 2 million.

Final Report:
The project's terminal GEF PIR along with the terminal evaluation (TE) report and corresponding management response will serve as the final project report package. The final project report package shall be discussed with the Project Board during an end-of-project review meeting to discuss lessons learned and opportunities for scaling up.

Agreement on intellectual property rights and use of logo on the project's deliverables and disclosure of information**:** To accord proper acknowledgement to the GEF for providing grant funding, the GEF logo will appear together with the UNDP logo on all promotional materials, other written materials like publications developed by the project, and project hardware. Any citation on publications regarding projects funded by the GEF will also accord proper acknowledgement to the GEF. Information will be disclosed in accordance with relevant policies notably the UNDP Disclosure Policy[53] and the GEF policy on public involvement[54].

[53] See http://www.undp.org/content/undp/en/home/operations/transparency/information_disclosurepolicy/

[54] See https://www.thegef.org/gef/policies_guidelines

Monitoring Plan for the PASBET Project

Monitoring	Indicators	Targets		Description of indicators and targets	Data source & Collection Methods[55]	Frequency	Responsible for data collection	Means of verification	Assumptions
		Mid-term	End of Project						
Project objective: Widespread application of sustainable biomass energy technologies for supporting socio-economic development of and reducing greenhouse gas (GHG) emissions from, the rural sector in Pakistan.	**Indicator 1:** No. of direct project beneficiaries disaggregated by gender (individual people) •Male •Female	•64,000 •42,800	•160,380 •107,000	Beneficiaries attributable to the project activities and outputs	Project M&E and activity reports and interviews of beneficiaries after implementation of activities	Quarterly	PMU/ MoCC&EC	Project M&E and activity reports. Measurement of increased income of beneficiaries after implementation of activities	No unexpected cultural or security barriers that hinder activities
	Indicator 2: Cumulative GHG emission reduction from the rural sector of Pakistan, tCO_2	51,520	195,050	Including biomass energy generation in project demos, replication, and baseline projects attributable to PASBET	Annual energy supply and consumption reports submitted by relevant national and local governments, etc.	Quarterly	PMU/ MoCC&EC	Regular periodic operations report from each sustainable biomass energy technology application demo and replication in demo towns/villages. Project M&E and activity reports.	Continuous commitment, support & active participation of the national/local government in low carbon rural development, as well as in meeting the renewable energy targets of the country Realization of committed co-financing from the national/local governments in the

[55] Data collection methods should outline specific tools used to collect data and additional information as necessary to support monitoring. The PIR cannot be used as a source of verification.

Monitoring	Indicators	Targets		Description of indicators and targets	Data source & Collection Methods	Frequency	Responsible for data collection	Means of verification	Assumptions
		Mid-term	End of Project						
									implementation of project activities and monitoring systems
	Indicator 3: Cumulative Biomass-based Electricity in rural areas in Pakistan, GWh	**1.5**	**98.2**	Relevant generation from biomass attributable to the Project	Project M&E and activity reports	Monthly	PMU, MoCC&EC, DISCOs and rural electrificatio n- agency	Annual energy supply and consumption reports in relevant national and local governments, etc.	Continuous commitment, support & active participation of the national/local government in low carbon rural development
	Indicator 4: Cumulative number of new jobs created in the application of sustainable biomass energy technologies and techniques in the rural areas. •Male •Female	•1,460 •1,380	•3,660 •3,450	Number of jobs related to biomass energy production, power generation and use in various activities	Project M&E and activity reports	Quarterly	PMU, MoCC&EC, DISCOs and rural electrificatio n- agency and local government	Trade and commerce reports. Data gathering on the no of people employed in all job categories	No unexpected cultural or security barriers that hinder activities
Project Outcome 1: Effective enforcement of policies and regulations on the sustainable production and use of woody biomass for	***Indicator 5:*** Cumulative number of sustainable biomass energy projects in rural areas that are	8	32[56]	Biomass-based projects supported from all components	Project M&E and activity reports	Monthly	PMU/ MoCC&EC	Project activity (baseline and incremental) reports. Government reports on the implementation of	Continuous consistent commitment of the national and local governments in politically supporting RE and

[56] This is 4 projects x 2 districts x 4 provinces

Monitoring	Indicators	Targets		Description of indicators and targets	Data source & Collection Methods	Frequency	Responsible for data collection	Means of verification	Assumptions
		Mid-term	End of Project						
energy generation and utilization in rural areas of Pakistan.	facilitated by approved and effectively enforced policies and laws/regulations that promote the commercial production and use of sustainable biomass-based energy.							the approved sustainable biomass energy development and utilization policies and regulations. Project M&E reports	EE initiatives in rural towns/villages. No competing value-added uses of woody biomass.
	Indicator 6: Cumulative number of rural villages that develop and implement energy-integrated development plans that include the commercial production and utilization of sustainable woody biomass.	**4**	40[57]	Villages that have been supported by the project	Project M&E and activity reports	Quarterly	PMU, MoCC&EC, DISCOs and rural electrification- agency and local government	Reports by responsible agency	Continuous consistent commitment of the national and local governments in politically supporting RE and EE initiatives in rural towns/villages

[57] This is 10 villages in each 4 provinces

Monitoring	Indicators	Targets		Description of indicators and targets	Data source & Collection Methods	Frequency	Responsible for data collection	Means of verification	Assumptions
		Mid-term	End of Project						
	Indicator 7: No. of developed, approved, enforced, and implemented and enforced as part of the RE law to include market-based targets of biomass based energy, particularly woody biomass	**2**[58]	**5**[59]	Actual policy instruments passed and implemented	Project M&E and activity reports	Quarterly	PMU/MOC	Documentation of the approved and promulgated regulations and policy instruments in the Output 1.2 activities (baseline and incremental) reports. Project M&E reports.	
	Indicator 8: Cumulative no. of standards for equipment to use woody biomass for energy generation and EE utilization established and enforced	**3**[60]	**4**[61]	Standards that are developed by the project and approved by authorized agencies	Project M&E and activity reports	Quarterly	PMU/ MoCC&EC	Documentation of the approved and promulgated regulations and policy instruments in the Output 1.2 activities (baseline and incremental) reports. Project M&E reports.	

[58] This includes: (a) Regulation to promote woody biomass and the corresponding Woody Biomass National Road Map; (b) Tariff application developed and submitted to NEPRA for biomass based electricity; (c) microgrid systems are made part of national grid code; (d) Inclusion of biomass as energy resource in the National Rural electricity policy; and (e) Community-based Natural Resource Management as part of National and Provincial Forestry Policy

[59] All five (5) policies that are approved and enforced in line with the targets of Biomass based power generation included in the National RE Policy.

[60] Standards developed for: (a) energy plantation; (b) biomass energy conversion; and (c) biomass user technologies

[61] Standards opted by relevant authorities and provincial authorities as part of the procurement

Monitoring	Indicators	Targets		Description of indicators and targets	Data source & Collection Methods	Frequency	Responsible for data collection	Means of verification	Assumptions
		Mid-term	End of Project						
Project Outcome 2.1: Enhanced woody biomass production on forested, non-forested, and on farmlands to cater local biomass energy needs, including for power generation and rural industry operations.	***Indicator 9:*** No. of established and commercially operated sustainable woody biomass production facilities.	**1**	**3**[62]	Woody production facilities supported by the Project	Project M&E and activity reports	Quarterly	PMU/ MoCC&EC	Forest industry reports Trade and commerce reports	Continuous consistent commitment of the national and local governments in politically supporting RE and EE initiatives in rural towns/villages
	Indicator 10: Cumulative number of entities that are interested and planned to invest in sustainable woody biomass fuel production. • Local Gov'ts • Private Sector entities	• • 1 • 1	• • 4 • 2	Number of projects on fuel production supported by the project	Project M&E and activity reports	Quarterly	PMU/ MoCC&EC and forest departments	Forest industry reports Trade and commerce reports	Continuous consistent commitment of the national and local governments in politically supporting RE and EE initiatives in rural towns/villages
	Indicator 11: Average annual local production and consumption of sustainable woody biomass fuel, tons/year	38,071[63]	**95,178**	Including all production and consumption of biomass fuels used of r power and	Project M&E and activity reports	Quarterly	PMU/ MoCC&EC, Department of Energy and forest departments	Forest industry reports Trade and commerce and energy departments reports	Continuous consistent commitment of the national and local governments in politically supporting RE and

[62] In total there are 3 new sustainable woody biomass production facilities are expected to be completed by the end of the project, one in each province of Punjab, Sindh and Balochistan)

[63] To generate 5.4 MW of biomass-based power generation it is estimated that total of 95,178 Tons of woody biomass is required annually. By mid-term at least 40% of this wood should be available from the sourcing.

Monitoring	Indicators	Targets		Description of indicators and targets	Data source & Collection Methods	Frequency	Responsible for data collection	Means of verification	Assumptions
		Mid-term	End of Project						
				non-power uses					EE initiatives in rural towns/villages
Project Outcome 2.2: Increased investments in the application of technologies for the production, and energy efficient utilization of woody biomass energy.	***Indicator 12:*** Cumulative amount of fossil fuel energy displaced by sustainable woody biomass fuels. * Power Applications, ktoe *Non-Power Applications, ktoe	**1,500** **150**	**3,800**[64] **300**	Including all production and consumption of biomass fuels used of r power and non-power uses	Project M&E and activity reports	Monthly	PMU/ MoCC&EC, Department of Energy and forest departments	Forest industry reports Trade and commerce and energy departments reports	Continuous consistent commitment of the national and local governments in politically supporting RE and EE initiatives in rural towns/villages
	Indicator 13: Cumulative number of private sector entities that are interested and plans to invest in sustainable woody biomass-based energy projects in rural towns and municipalities	**3**	**5**	Private companies that have registered and started implementation	Project M&E and activity reports	Monthly	PMU/ MoCC&EC, Department of Energy and forest departments	Forest industry reports Trade and commerce and energy departments reports	Continuous consistent commitment of the national and local governments in politically supporting RE and EE initiatives in rural towns/villages
Outcome 3: Adequate amounts of financial resources available for woody biomass energy technology application projects in	***Indicator 14:*** Cumulative number of sustainable biomass energy projects whose			Projects that are funded by private sector	Project M&E and activity reports	Monthly	PMU/ MoCC&EC, Department of Energy and forest departments	Forest industry reports Trade and commerce and energy departments	Continuous consistent commitment of the national and local governments in politically

[64] If 5.4 MW is generated using fossil fuel (e.g., NG based genset) it will require fossil fuel energy equivalent of 3.8 ktoe. Similar energy can be displaced by sustainable woody biomass fuels. By mid-term it is estimated that 40% of the total will be completed.

Monitoring	Indicators	Targets		Description of indicators and targets	Data source & Collection Methods	Frequency	Responsible for data collection	Means of verification	Assumptions
		Mid-term	End of Project						
the country.	funding are sourced from other sources of funds apart from national/local government. * Sustainable woody biomass fuel production * Sustainable woody biomass-based power generation and distribution * Sustainable woody biomass-based energy generation and utilization.	• **4** • **4** • **4**	• **8** • **8** • **8**					reports	supporting RE and EE initiatives in rural towns/villages
	Indicator 15: Cumulative number of financing schemes/mechanisms adopted by financial institutions for supporting sustainable biomass energy initiatives in Pakistan.	**1**	**2**	Financial mechanisms that are by banks and financial institutions	Project M&E and activity reports	Monthly	PMU/ MoCC&EC, Department of Energy and forest departments	Forest industry reports Trade and commerce and energy department reports	Continuous consistent commitment of the financing institutions, national and local governments in politically supporting RE and EE initiatives in rural towns/villages
Outcome 4: Enhanced	***Indicator 16:***	**4**	**32**	Pertinent	Project M&E	Monthly	PMU/	Forest industry	Continuous

Monitoring	Indicators	Targets		Description of indicators and targets	Data source & Collection Methods	Frequency	Responsible for data collection	Means of verification	Assumptions
		Mid-term	End of Project						
local capacity, skills and knowledge in the development, installation, and operation of biomass energy technology systems in rural Pakistan.	Cumulative number of sustainable biomass woody energy projects designed, implemented, and maintained by rural towns and municipalities			projects supported and attributable to PABET's area of responsibility	and activity reports		MoCC&EC, Department of Energy and forest departments	reports Trade and commerce and energy department reports	consistent commitment of the financing institutions, national and local governments in politically supporting RE and EE initiatives in rural towns/villages
	Indicator 17: Cumulative number of operational woody biomass-based power generation system installations in rural towns and municipalities.	**4**	**16**	Including installations supported by the project in the 4 provinces	Project M&E and activity reports	Monthly	PMU/ MoCC&EC, Department of Energy and forest departments	Forest industry reports Trade and commerce and energy department reports	Continuous consistent commitment of the financing institutions, national and local governments in politically supporting RE and EE initiatives in rural towns and villages
	Indicator 18: Cumulative number of energy consumers in rural areas that are utilizing: woody biomass-based energy generating and consuming equipment (persons)	**1,120**	**2,240**	Number of users of biomass based production and energy generation in the 4 provinces selected	Project M&E and activity reports	Monthly	PMU/ MoCC&EC, Department of Energy and forest departments	Forest industry reports Trade and commerce and energy department reports	Continuous consistent commitment of the financing institutions, national and local governments in politically supporting RE and EE initiatives in rural towns/villages

Table 4: Monitoring and Evaluation Plan and Budget

GEF M&E requirements	Indicative costs (US$)	Time frame
Inception Workshop and report	3,000	Inception Workshop within 2 months of the First Disbursement
M&E of GEF core indicators and project results framework	3,000 per year	Annually and at mid-point and closure.
GEF Project Implementation Report (PIR)	None	Annually typically between June-August
Monitoring all risks	2,000 per year	On-going.
Monitoring of social and environmental safeguard screening	2,000 per year	On-going.
Monitoring of stakeholder engagement plan	2,000 per year	On-going.
Monitoring of gender action plan	2,000 per year	On-going.
Project Board Meetings	2,000 per year	Annually
Reports of Project Board Meetings	None	Annually
Lessons learned and knowledge generation	2,000 per year	Annually
Supervision missions	None	Annually
Oversight missions	None	Troubleshooting as needed
Mid-term GEF and/or LDCF/SCCF Core indicators and METT or other required Tracking Tools	10,000	Before mid-term review
Independent Mid-term Review (MTR)	35,000	3Q 2025
Terminal GEF and/or LDCF/SCCF Core indicators and METT or other required Tracking Tools	10,000	Before terminal evaluation
Independent Terminal Evaluation (TE)	35,000	4Q 2027
Translation of MTR and TE reports into English	None	As needed
TOTAL indicative COST	**168,000**	

VIII. GOVERNANCE AND MANAGEMENT ARRANGEMENTS

Section 1: General roles and responsibilities in the projects' governance mechanism

Implementing Partner: The Implementing Partner for this project is the Ministry of Climate Change & Environmental Coordination (MoCC&EC). The project will be implemented based on the proposed implementation modality - UNDP CO Country Office (CO) support to National Implementation Modality (NIM) with the Ministry of Climate Change & Environmental Coordination as implementing partner. Micro assessments and a review of the resolution of pending assurance activity issues will be conducted to assess whether there have been improvements in the capacities, policies, and internal controls of the IP/MoCC&EC, and review changes to the implementation arrangements and cash transfer modality. UNDP support services to the Government (IP) for CO support to NIM includes the recruitment and recurrent management services of PMU staff, procurement, and engagement of Responsible Parties during the project implementation.

Based on the result of MCA, demonstrating concrete steps taken by MoCC&EC to resolve recurring high risk audit issues relating to the following (but not limited to) internal controls, possible adjustments to the implementation arrangements to a full NIM Implementation modality may be considered along with changes in relevant cash transfer modality in accordance with HACT Framework.

1. Financial management and accounting of UNDP funds where UNDP funds are managed and utilized under Government oversight in accordance with applicable government regulations, rules, and procedures; and
2. compliance with applicable government procurement regulations, rules, and procedures; and
3. compliance with applicable government recruitment and personnel management regulations, policies, and procedures.

However, if the 3rd party assessment results fails to demonstrate the above, UNDP will continue to provide support to MoCC&EC in implementation under CO support to NIM.

The Implementing Partner is the entity to which the UNDP Administrator has entrusted the implementation of UNDP assistance specified in this signed project document along with the assumption of full responsibility and accountability for the effective use of UNDP resources and the delivery of outputs, as set forth in this document. The IP:

- Assumes full responsibility for the effective use of UNDP resources and the delivery of outputs in the signed project document.
- Must report fairly and accurately on project progress against agreed work plans in accordance with the reporting schedule and formats included in the project document; and
- Maintains documentation and evidence of the proper and prudent use of project resources in conformity to the project document and in accordance with applicable regulations and procedures; documentation should be available on request to project monitors and designated auditors.

The Implementing Partner is responsible for executing this project. Specific tasks include:

- Project planning, coordination, management, monitoring, evaluation, and reporting. This includes providing all required information and data necessary for timely, comprehensive, and evidence-based project reporting, including results and financial data, as necessary. The Implementing Partner will strive to ensure project-level M&E is undertaken by national institutes and is aligned with national systems so that the data used and generated by the project supports national systems.
- Risk management as outlined in this Project Document.
- Financial management, including overseeing financial expenditures against project budgets.
- Approving and signing the multiyear workplan.
- Approving and signing the combined delivery report at the end of the year; and,
- Signing the financial report or the funding authorization and certificate of expenditures.

Responsible Parties: During the 5-year implementation period of the PASBET Project, the MoCC&EC, as the implementing partner will require the cooperation and services of several responsible parties to implement or support the implementation of the project activities. This will include Forest and Energy Departments of respective provinces, Alternative Energy Development Board, National Energy Efficiency and Conservation Authority, Power Distribution Companies and Natural Gas Supply Companies. The Project Development Team (PDT) has undertaken necessary consultation with these stakeholders and gathered their initial buy in. Forest departments are partnering with PASBET project as co-financing partners whereas rest will be formally taken on board at the inception stage. Under CO Support to NIM, UNDP shall sign a Letter of Agreement (LOA) with the relevant entities among these that will be designated as RPs and engage them to execute the activities in their respective area of responsibility.

Project Target Groups: The project target groups will be those that will be involved in the project implementation as direct implementers or direct beneficiaries. They will also be involved in the decision-making process in regards the implementation of the project activities. The engagements with these groups will be done though the activities under each project component relevant to policy making, data gathering, participation and procedural documentation. In this regard, the project's governance is designed considering the project management capabilities of the implementing and responsible parties, as well as the target groups.

Project Stakeholders: Please refer to the Stakeholder Engagement Plan in Sec. IV and to **Annex 8**.

The cross-sectoral and participatory nature of the project would require involvement of wide range of stakeholders at different stages of project implementation. Therefore, a comprehensive "stakeholders participation strategy" with a clear road map will be developed during the inception phase of the project. A mechanism for involvement local user groups, traders, and entrepreneurs, including both men and women will be devised. Following are the stakeholders of the Project:

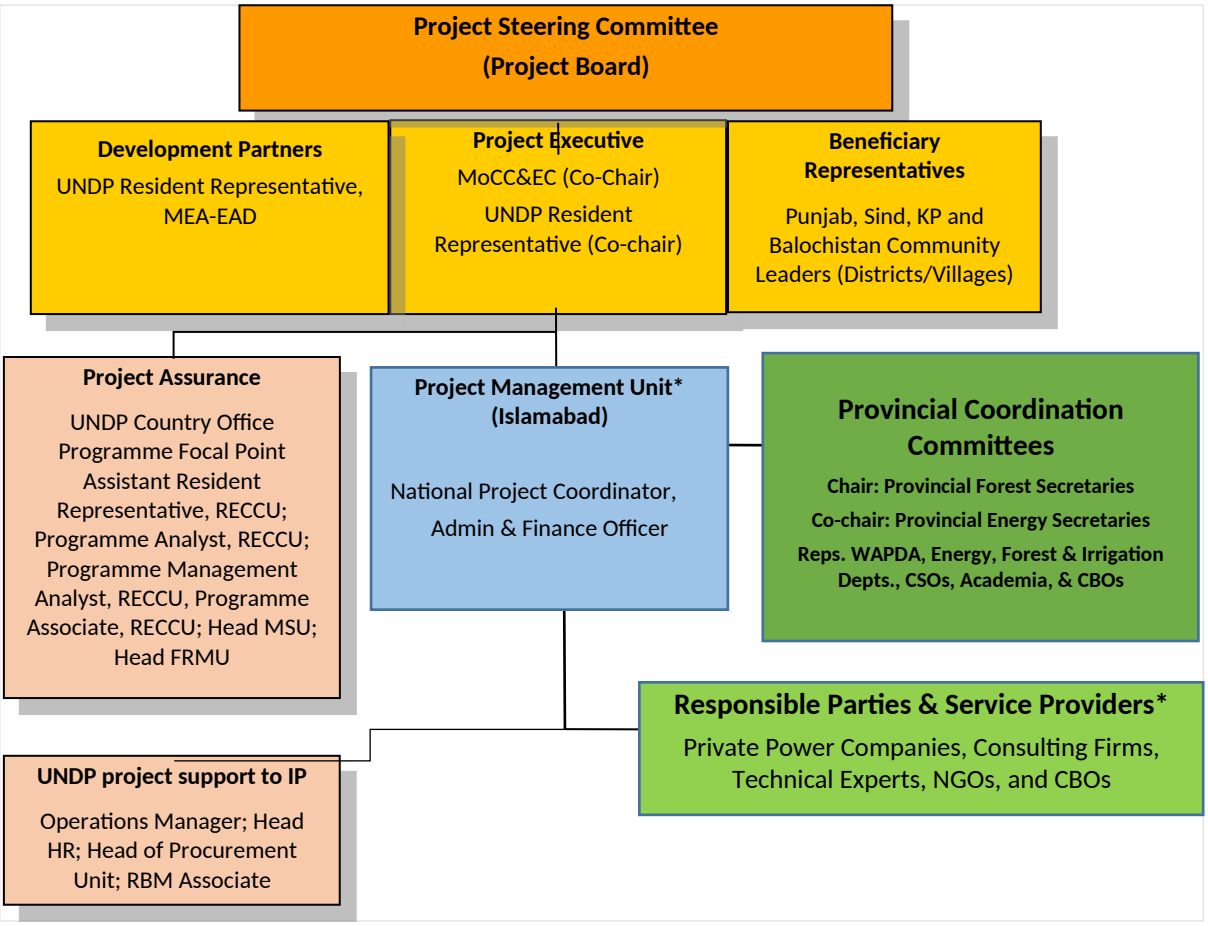
- Ministry of Climate Change & Environmental Coordination (MoCC&EC)
- Provincial Forestry Departments
- Provincial Energy Departments
- Ministry of Energy (Power Division) – Alternative Energy Development Board (AEDB)
- Ministry of Energy (Power Division) - National Energy Efficiency & Conservation Authority (NEECA)
- Civil Society Organizations/ Community Based Organizations/
- Private Sector

- (Electric Power Producers and Rural Industries)
- International NGOs, UN Agencies, and Donors.

UNDP is accountable to the GEF for the implementation of this project. This includes overseeing project execution undertaken by the Implementing Partner to ensure that the project is being carried out in accordance with UNDP and GEF policies and procedures and the standards and provisions outlined in the Delegation of Authority (DOA) letter for this project. **The UNDP GEF Executive Coordinator, in consultation with UNDP Bureaus and the Implementing Partner, retains the right to revoke the project DOA, suspend or cancel this GEF project.** UNDP is responsible for the Project Assurance function in the project governance structure and presents to the Project Steering Committee and attends Project Steering Committee meetings as a non-voting member.

A firewall will be maintained between the delivery of project oversight and quality assurance performed by UNDP and charged to the GEF Fee and any support to project execution performed by UNDP (as requested by and agreed to by both the Implementing Partner and GEF) and may be charged to the GEF project management costs (only if approved by GEF). The segregation of functions and firewall provisions for UNDP in this case is described in the next section.

Section 2: Project governance structure



* The Technical Specialists, M&E Officer and Communication Officer will be hired as consultants. Support staff will be hired as contracted personnel as necessary.
* Responsible Parties & Service Providers shall not serve on the Project board to avoid a conflict of interest.

Figure 3: PASBET Project Organization Structure

First line of defense:

- UNDP oversight of project support to IP cannot be UNDP staff providing project assurance or providing programmatic oversight support to the RR.

Second line of defense:

- Regional Bureau oversees RR and Country Office compliance at portfolio level.
- BPPS NCE RTA oversees technical quality assurance and GEF compliance. BPPS NCE PTA oversees RTA function.
- UNDP GEF Executive Coordinator and Regional Bureau Deputy Director can revoke DOA/cancel/suspend project or provided enhanced oversight.

The UNDP Resident Representative assumes full responsibility and accountability for oversight and quality assurance of this Project and ensures its timely implementation in compliance with the GEF-specific requirements and UNDP's Programme and Operations Policies and Procedures (POPP), its Financial Regulations and Rules and Internal Control Framework. A representative of the UNDP Country Office will assume the assurance role and will present assurance findings to the Project Steering Committee, and therefore attends Project Steering Committee meetings as a non-voting member.

UNDP project support: The Implementing Partner and GEF OFP have requested UNDP to provide support services in the amount of USD 80,000 that will be funded by UNDP resources as specified in the Letter of Agreement (LOA) for the project duration, however this has to be agreed and approved by GEF for UNDP to provide such execution support services. The execution support services – whether financed from the project budget or other sources – will be set out in detail and agreed between UNDP Country Office and the Implementing Partner in a Letter of Agreement (LOA). This LOA is attached to this Project Document (Annex 16).

During the project implementation, a 3rd party micro-capacity assessment (MCA) will be carried out to assess readiness of the proposed IP/MoCC&EC for implementing the project under Full National Implementation Modality (NIM).

Based on the results of the MCA, demonstrating concrete steps taken by MoCC&EC to resolve recurring high risk audit issues relating to the following (but not limited to) internal controls, possible adjustments to the implementation arrangements to a full NIM Implementation modality may be considered along with changes in relevant cash transfer modality in accordance with HACT Framework.

1. Financial management and accounting of UNDP funds where UNDP funds are managed and utilized under Government oversight in accordance with applicable government regulations, rules, and procedures; and
2. compliance with applicable government procurement regulations, rules, and procedures; and
3. compliance with applicable government recruitment and personnel management regulations, policies, and procedures.

However, if the 3rd party results fail to demonstrate above, UNDP will continue to provide support to MoCC&EC in implementation specially carrying out procurements under CO support to NIM. The execution support services – whether financed from the project budget or other sources - have been set out in detail and agreed between UNDP Country Office and the Implementing Partner in the LOA. This LOA is attached to this Project Document in Annex 16.

To ensure the strict independence required by the GEF and in accordance with the UNDP Internal Control Framework, these execution services will be delivered independent from the GEF-specific oversight and quality assurance services.

Section 3: Segregation of duties and firewalls vis-à-vis UNDP representation on the Project Board

As noted in the Minimum Fiduciary Standards for GEF Partner Agencies, in cases where a GEF Partner Agency (i.e. UNDP) carries out both implementation oversight and execution of a project, the GEF Partner Agency (i.e. UNDP) must separate its project implementation oversight and execution duties, and describe in the relevant project document a: 1) Satisfactory institutional arrangement for the separation of implementation oversight and executing functions in different departments of the GEF Partner Agency; and 2) Clear lines of responsibility, reporting and accountability within the GEF Partner Agency between the project implementation oversight and execution functions.

In this case, UNDP's implementation oversight role in the project – as represented in the Project Steering Committee and via the project assurance function – is performed by Assistant Resident Representative, Programme Analyst, Programme Management Analyst, Programme Associate, Resilience Environment and Climate Change Unit (RECCU), Head MSU, Management Support Unit (MSU), Finance Manager, Finance & Resource Management Unit (FRMU). UNDP's execution role in the project (as requested by the implementing partner and approved by the GEF) is performed by the operations staff who are accountable to Operations Manager and these staff will have no oversight roles. They are Head of Human Resources Unit, Travel Associate, HR Officer, Head of Procurement Unit, Procurement Officer, and RBM Associate, Management Support Unit.

Roles and Responsibilities of the Project Organization Structure

a. **Project Board**: All UNDP projects must be governed by a multi-stakeholder board or committee established to review performance based on monitoring and evaluation, and implementation issues to ensure quality delivery of results. The Project Board (also called the Project Steering Committee in some projects) is the most senior, dedicated oversight body for a project. The members of the PB/PSC will be finalized after the CEO endorsement of the project by GEF and in accordance with UNDP Programme and Operation Policies and Procedures (POPP).

The two main (mandatory) roles of the Project Board are as follows:

1) **High-level oversight of the execution of the project by the Implementing Partner** (as explained in the "Provide Oversight" section of the POPP). This is the primary function of the Project Board and includes annual (and as-needed) assessments of any major risks to the project, and decisions/agreements on any management actions or remedial measures to address them effectively. The Project Board reviews evidence of project performance based on monitoring, evaluation, and reporting, including progress reports, evaluations, risk logs and the combined delivery report. The Project Board is responsible for taking corrective action as needed to ensure

the project achieves the desired results. The project quality assurance report and vertical fund reports should be discussed with the PSC, along with management actions to improve quality. PSC decisions are made in accordance with standards to ensure management for development results: best value for money, fairness, integrity, transparency, and effective international competition.

2) **Approval of strategic project execution decisions of the Implementing Partner** with a view to assess and manage risks, monitor and ensure the overall achievement of projected results and impacts and ensure long term sustainability of project execution decisions of the Implementing Partner (as explained in the "Manage Change" section of the POPP).

Requirements to serve on the Project Board:

- ✓ Agree to the Terms of Reference of the Board and the rules on protocols, quorum, and minuting.
- ✓ Meet annually; at least once.
- ✓ Disclose any conflict of interest in performing the functions of a Project Board member and take all measures to avoid any real or perceived conflicts of interest. This disclosure must be documented and kept on record by UNDP.
- ✓ Discharge the functions of the Project Board in accordance with UNDP policies and procedures.
- ✓ Ensure highest levels of transparency and ensure Project Board meeting minutes are recorded and shared with project stakeholders.

Responsibilities of the Project Board

- ✓ Consensus decision making:
  - The Project Board provides overall guidance and direction to the project, ensuring it remains within any specified constraints, and providing overall oversight of the project implementation.
  - Review project performance based on monitoring, evaluation, and reporting, including progress reports, risk logs and the combined delivery report.
  - The Project Board is responsible for making management decisions by consensus.
  - To ensure UNDP's ultimate accountability, Project Board decisions should be made in accordance with standards that shall ensure management for development results, best value money, fairness, integrity, transparency, and effective international competition.
  - In case consensus cannot be reached within the Board, the UNDP Resident Representative on the board will mediate to find consensus and, if this cannot be found, will take the final decision to ensure project implementation is not unduly delayed.

- ✓ Oversee project execution:
  - Agree on National Project Coordinator (NPC)'s tolerances as required, within the parameters outlined in the PASBET project document, and provide direction and advice for exceptional situations when the NPC's tolerances are exceeded.
  - Appraise annual work plans (AWPs) prepared by the Implementing Partner for the Project; review Combined Delivery Reports (CDRs) prior to certification by the implementing partner.
  - Address any high-level project issues as raised by the National Project Coordinator and project assurance.
  - Advise on major and minor amendments to the project within the parameters set by UNDP and the donor and refer such proposed major and minor amendments to the UNDP

BPPS Nature, Climate and Energy Executive Coordinator (and the GEF, as required by GEF policies).
- Provide high-level direction and recommendations to the project management unit to ensure that the agreed deliverables are produced satisfactorily and according to plans.
- Track and monitor co-financed activities and realization of co-financing amounts of this project.
- Approve the Inception Report, GEF annual project implementation reports, mid-term review and terminal evaluation reports.
- Ensure commitment of human resources to support project implementation, arbitrating any issues within the project.

✓ Risk Management:
- Provide guidance on evolving or materialized project risks and agree on mitigation and management actions to address specific risks.
- Review and update the project risk register and associated management plans based on the information prepared by the Implementing Partner. This includes risks related that can be directly managed by this project, as well as contextual risks that may affect project delivery or continued UNDP compliance and reputation but are outside of the control of the project. For example, social and environmental risks associated with co-financed activities or activities taking place in the project's area of influence that have implications for the project.
- Address project-level grievances.

✓ Coordination:
- Ensure coordination between various donor and government-funded projects and programmes.
- Ensure coordination with various government agencies and their participation in project activities.

The composition of the Project Board must include the following roles:

1. **Project Executive:** This is an individual who represents ownership of the project and chairs the Project Board. The Executive is normally the senior national counterpart for nationally implemented projects. The Project Executive is designated after the project is CEO endorsed by GEF in accordance with UNDP Programme and Operation Policies and Procedures (POPP).

   MoCC&EC, as the implementing partner, and UNDP RR represents the Project Executive, and chairs the Project Board. The Executive is responsible for the project, supported by the Beneficiary Representatives and the Development Partners. The Executive's role is to ensure that the project is focused throughout its life cycle on achieving its objectives and delivering outputs that will contribute to higher level outcomes. The Executive must ensure that the project gives value for money, ensuring cost-conscious approach to the project, balancing the demands of Beneficiary Representatives and Development Partners.

   Specific Responsibilities: (as part of the above responsibilities for the Project Board)

   - Ensure that there is a coherent project organization structure and logical set of plans.

- Set tolerances in the AWP and other plans as required for the National Project Coordinator.
- Monitor and control the progress of the project at a strategic level.
- Ensure that risks are being tracked and mitigated as effectively as possible.
- Brief relevant stakeholders about project progress.
- Organize and chair Project Board meetings.

2. **Beneficiary Representatives**: Individuals or groups representing the interests of those groups of stakeholders who will benefit from the project. These will not be any entity who has been engaged as Responsible Partner during project implementation. Their primary function within the board is to ensure the realization of project results from the perspective of project beneficiaries. Often representatives from civil society, industry associations, or other government entities benefiting from the project can fulfil this role. There can be multiple beneficiary representatives on a Project Board. The Beneficiary representatives are:

    - Provincial Government, Punjab
    - Provincial Government, Sind
    - Provincial Government, KP
    - Provincial Government, Balochistan

    The Beneficiary Representatives are responsible for validating the needs and for monitoring that the solution will meet those needs within the constraints of the project. The Beneficiary Representative role monitors progress against targets and quality criteria. This role may require more than one person to cover all the beneficiary interests. For the sake of effectiveness, the role should not be split between too many people.

    Specific Responsibilities (as part of the above responsibilities for the Project Board)

    - Prioritize and contribute beneficiaries' opinions on Project Board decisions on whether to implement recommendations on proposed changes.
    - Specification of the Beneficiary's needs is accurate, complete, and unambiguous.
    - Monitoring of the implementation of activities at all stages is monitored to ensure that they will meet the beneficiary's needs and are progressing towards that target.
    - Evaluation of the impacts of potential changes is evaluated from the beneficiary point of view.
    - Risks to the beneficiaries are frequently monitored.

3. **Development Partners:** Individuals or groups representing the interests of the parties concerned that provide funding, strategic guidance and/or technical expertise to the project. The Development Partners is United Nations Development Programme (UNDP).

    The Development Partners' primary function within the Board is to provide guidance regarding the technical feasibility of the project.

    Specific Responsibilities (as part of the above responsibilities for the Project Board)

    - Make sure that progress towards the outputs remains consistent from the Development Partners perspective.

- Promote and maintain focus on the expected project output(s) from the point of view of project development management.
- Ensure that the technical and financial resources required for the project are made available.
- Contribute technical and financial opinions on Project Board decisions on whether to implement recommendations on proposed changes.
- Arbitrate on, and ensure resolution of, any technical and financial priority or resource conflicts.

b. **Project Assurance:** Project assurance is the responsibility of each project board member; however, UNDP has a distinct assurance role for all UNDP projects in carrying out objective and independent project oversight and monitoring functions. UNDP performs quality assurance and supports the Project Board (and Project Management Unit) by carrying out objective and independent project oversight and monitoring functions, including compliance with the risk management and social and environmental standards of UNDP. The Project Board cannot delegate any of its quality assurance responsibilities to the National Project Coordinator. Project assurance is independent of project execution.

c. **Project Management – Execution of the Project**: The National Project Coordinator (NPC) has 2 sets of tasks to carry out for the PASBET Project. These are: (a) coordination of the management and implementation of the project activities (20%); and (b) provision of technical coordination of, and technical support to, the promotion, design, implementation of the project's demonstration activities (80%). The National Project Coordinator (NPC) is the senior most representative of the PMU and is responsible for coordinating the day-to-day management of the project on behalf of the Implementing Partner, including the mobilization of all project inputs, supervision over project staff, responsible parties, consultants, and sub-contractors. The NPC shall present key deliverables and documents to the board for their review and approval, including progress reports, annual work plans, adjustments to tolerance levels and risk registers.

   The NPC will also provide technical and coordination roles in collaboration with MoCC&EC, and other project partners (e.g., for demonstrations), provincial officers and provincial government institutions, and UNDP[65]. UNDP shall recruit PMU staff to execute their respective duties under their area of responsibility. The recruitment will be carried out by UNDP under the LoA signed with the IP/MoCC&EC for CO support to NIM. UNDP will also provide induction and regular training to PMU staff on UNDP procedures and policies so that they may effectively execute their roles and responsibilities.

[65] Per the agreement among the project proponent and key stakeholders, the National Project Coordinator (NPC) will be coordinating the management and implementation of the PASBET Project activities on behalf of the Implementing Partner (MoCC). Selected MoCC personnel will be assigned to assist the NPC in the management of the project activities. Moreover, the PMU will also be supported by the UNDP in the execution of the project. While also involved in project activities coordination, bulk of the tasks of the NPC involves the provision of technical coordination and technical assistance work in the design, implementation, and evaluation of the project's demo activities in the partner provinces. Specifically, the technical tasks involve the: (1) provision of advice and guidance in the establishment of the technical specifications, standards to be followed; (2) provision of technical inputs in TOR/Contract preparation, RFQ and bidding documentation, selection and engagement of the biomass energy technology service providers and contractors for the demo activities; (3) technical evaluation of bid proposals for the supply, installation, operation and systems from contractors and individual proponents; (4) provision of technical inputs and support to the PMU in the performance tracking system and contract performance monitoring of contractors and consultants to deliver expected outputs and deliverables, including technical review of reports prepared and submitted by contractors, as basis for contract payments; and, (5) Supervision and provision of technical advice and inputs in the testing, initial operation, and performance evaluation of the demo units under Component 2.2.

A designated representative of the PMU is expected to attend all board meetings and support committee processes as a non-voting representative. The primary PMU representative attending board meetings is the NPC.

IX. FINANCIAL PLANNING AND MANAGEMENT

The total cost of the project is *USD 24,589,487*. This is financed through a GEF grant of *USD* 3,439,041 and USD 80,000 UNDP resource to be administered by UNDP and USD 21,070,446 co-financing (USD 16,291,417 Investment mobilized, and USD 4,779,029 in-kind). UNDP, as the GEF Implementing Agency, is responsible for the oversight of the GEF resources and the cash co-financing transferred to UNDP bank account only.

Co-financing: The actual realization of project co-financing will be monitored by the UNDP Country Office and the PMU on an annual basis in the GEF PIF and will be reported to the GEF during the *mid-term review* and terminal evaluation process as follows:

Table 5: List of PASBET Project Co-Financing

Co-financing Source	Co-financing type	Amount, US$	Co-financed Activity	Risks	Risk Mitigation Actions
Ministry of Climate Change & Environmental Coordination (MoCC&EC), Government of Pakistan	Public Investment	8,295,414	Activities 1.3.1, 1.3.2, 1.3.3, 1.5.1, 1.5.2, 2.1.1, 2.1.2, 2.2.3, 4.1.3, 4.2.3	Change of Political Government resulting in lack of funding for TBBT Lack of interest on part of Ministry in Distributed Rural Electrification Change in GoP policy regarding support for RE/EE Projects Lack of interest due to social and technological barriers	Having Formal Co-Financing Commitment, Due feasibility, identify target areas not depending on TBBT interventions alone Using community mobilization to create political ownership of rural electrification projects Adopt sustainable business models for micro grids and energy efficient industrial combustion systems, not depending on GoP subsidies Raising the capacity and interest through community engagement and ownership as part of Component 4 of the
	In-kind	2,198,853			

Co-financing Source	Co-financing type	Amount, US$	Co-financed Activity	Risks	Risk Mitigation Actions
					project
Government of the Punjab (Forestry, Wildlife & Fisheries Department)	In-kind	742,081	Activities 1.5.1, 1.5.2, 2.1.3, 2.1.4	Change of Political Government resulting in lack of funding for TBBT	Having formal co-financing commitment, due feasibility, identify target areas not depending on TBBT interventions alone
	Grant	2,099,547			
Ten Billion Tree Tsunami Programme, Government of Khyber Pakhtunkhwa	Grant	5,696,832	Activities 1.5.1, 1.5.2, 2.1.3, 2.1.4	Change of Political Government resulting in lack of funding for TBBT	Having formal co-financing commitment, due feasibility, identify target areas not depending on TBBT interventions alone
Government of Sindh (Forest & Wildlife Department)	In-kind	135,747	Activities 1.5.1, 1.5.2, 2.1.3, 2.1.4	Change of Political Government resulting in lack of funding for TBBT	Having formal co-financing commitment, due feasibility, identify target areas not depending on TBBT interventions alone
	Grant	135,747			
Government of Balochistan (Forests & Wildlife Department)	In-kind	1,202,348	Activity 1.5.1, 1.5.2, 2.1.3, 2.1.4	Change of Political Government resulting in lack of funding for TBBT	Having formal co-financing commitment, due feasibility, identify target areas not depending on TBBT interventions alone
	Grant	63,877			
UNDP	In-kind	500,000	Administrative support to the whole project and its activities	Not applicable as project will be building on the work of sustainable forestation already done by UNDP under Sustainable Forest Project	
	Grant	80,000	UNDP execution support services to IP	N/A	N/A
Total Co-Financing		**21,150,446**			

Budget Revision and Tolerance: As per UNDP POPP, the project board may agree with the project manager on a tolerance level for each detailed plan under the overall multi-year workplan. The agreed tolerance should be written in the project document or approved project board meeting minutes. It should normally not exceed 10 percent of the agreed annual budget at the activity level, but within the overall approved multi-year workplan at the activity level. Within the agreed tolerances, the project manager can operate without intervention from the project board. Restrictions apply as follows:

Should the following deviations occur, the Project Manager/IP through UNDP Country Office will seek the approval of the BPPS/NCE-VF team to ensure accurate reporting to the GEF. It is **strongly encouraged** to maintain the expenditures within the approved budget at the budgetary account and at the component level:

a) Budget reallocations must prove that the suggested changes in the budget will not lead to material changes in the results to be achieved by the project. A strong justification is required and will be approved on an exceptional basis. Budget re-allocations among the components (including PMC) of the approved Total Budget and Work Plans (TBWP) that represent a value greater than 10% of the total GEF grant.
b) Introduction of new outputs/activities (i.e. budget items) that were not part of the agreed project document and TBWP that represent a value greater than 5% of the total GEF grant. The new budget items must be eligible as per the GEF and UNDP policies.
c) Project management cost (PMC): budget under PMC component is capped and cannot be increased.

UNDP is not in a position to increase the total budget above the amount approved by the donor, therefore any over-expenditure would have to be absorbed from non-GEF resources by the Implementing Partner (GEF Executing Entity)

Project extensions: The UNDP-BPPS-NCE team Executive Coordinator must approve all requests for extension of the Project Completion Date and for other milestone extensions with hard deadlines. All extensions impose additional time and cost burdens at all levels and the GEF project budget cannot be increased beyond its originally approved amount. A single extension may be granted on an exceptional basis and subject to the conditions and maximum durations set out in the UNDP POPP. The project management costs during the extension period must remain within the originally approved amount, and any increase in PMC costs shall be covered by non-GEF resources; the additional UNDP oversight costs during the extension period must be covered by non-GEF resources, in accordance with UNDP's policy as set out in UNDP POPP.

For any extension request, UNDP CO and IP will consult and jointly present a clear plan indicating how and from which specific sources the additional oversight costs that will be incurred by UNDP will be covered during the extended period. The BPPS-NCE Executive Coordinator will consult the Regional Bureaux (RBX) and may reject the extension request if no (external co-financing by the IP or internal UNDP CO resources) can be identified.

All extension requests, along with all supporting documentation, shall be submitted by the IP to the UNDP CO in line with the requirements and within the deadlines set out in the UNDP SOPs and policies in UNDP POPP.

Audit: The project will be audited as per UNDP Financial Regulations and Rules and applicable audit policies. Audit cycle and process must be discussed during the Inception workshop. If the Implementing Partner is an UN Agency, the project will be audited according to that Agencies applicable audit policies.

Transfer or disposal of assets: In consultation with the Implementing Partner and other parties of the project, UNDP is responsible for deciding on the transfer or other disposal of assets. Transfer or disposal of assets is recommended to be reviewed and endorsed by the project board following UNDP rules and regulations. Assets may be transferred to the government for project activities managed by a national institution at any time during the life of a project, however **must be done before the operational closure date**. In all cases of transfer, a transfer document must be prepared and kept on file[66]. The transfer should be done before Project Management Unit complete their assignments.

Completion Date: The project completion date is the date of Project Document Signature plus project duration. This date can only be extended through a formal extension request. Prior to completion date, all UNDP-financed inputs must be provided and related activities for the Project completed. No activities, except for the final clearance of the

[66] See https://popp.undp.org/_layouts/15/WopiFrame.aspx?sourcedoc=/UNDP_POPP_DOCUMENT_LIBRARY/Public/PPM_Project%20Management_Closing.docx&action=default.

Terminal Evaluation Report and the corresponding management response and the end-of-project review Project Board Meeting should take place after the Completion Date.

Project Closure: Project closure will be conducted as per UNDP requirements outlined in the UNDP POPP. All costs incurred to close the project must be included in the project closure budget and reported as final project commitments presented to the Project Board during the final project review. The only costs a project may incur following the final project review are those included in the project closure budget.

- **Operational Closure**: **Operational closure must happen within 9 months from project completion date.** Prior to operational closure, the Terminal Evaluation must have been submitted and the corresponding TE management response and the end-of-project review Project Board meeting must have been completed. The Implementing Partner through a Project Board decision will notify the UNDP Country Office when operational closure has been completed. Before Operational Closure, the project must have completed the transfer or disposal of any equipment that is still the property of UNDP.
- **Financial Closure: Financial closure must happen within 6 months of operational closure or after the date of cancellation.** The project will be financially closed when the following conditions have been met: a) the project is operationally completed or has been cancelled; b) the Implementing Partner has reported all financial transactions to UNDP; c) UNDP has closed the accounts for the project; d) UNDP and the Implementing Partner have certified a final Combined Delivery Report (which serves as final budget revision).
  Between operational and financial closure, the implementing partner will identify and settle all financial obligations and prepare a final expenditure report. The UNDP Country Office will send the final signed closure documents including confirmation of final cumulative expenditure and unspent balance to BPPS/NCE for confirmation before the project will be financially closed in Quantum by the UNDP Country Office.

Cancellation and Suspension: All projects considering going through cancellation or suspension must follow UNDP and GEF requirements. Guidance can be found in the UNDP POPP (SOPs for management actions of Vertical Fund projects escalated to the Executive Coordinator and Guidance for GEF project revisions).

Refund to GEF: Should a refund of unspent funds to the GEF be necessary, this will be managed directly by the BPPS/NCE team Directorate in New York. No action is required by the UNDP Country Office on the actual refund from UNDP project to the GEF. Unspent project balance is not permitted to be transferred to any other projects.

X. TOTAL BUDGET AND WORK PLAN

	Total Budget and Work Plan		
Quantum Business Unit	UNDP-PAK		
Quantum Project ID:	00123744	Quantum Project Title:	CCM and PASBET (GEF-7)-UNDP-PAK-00123744
Quantum Award ID:	00123744.1	Quantum Award Title:	PAK_ASL_23744
UNDP-GEF PIMS No.	6542		
Implementing Partner	Ministry of Climate Change & Environmental Coordination and Environmental Coordination, Government of Pakistan with full CO support to NIM by UNDP		

Quantum Outcome (GEF Component)	Quantum Output (GEF Outcome)	Quantum Activity (GEF Output)	Quantum Responsible Party (UNDP, IP, or Responsible Party)	Quantum Fund ID	Quantum Donor ID	Quantum Budgetary Account Code	Quantum Budget Account Description	Amount Year {2023} (USD) (1)	Amount Year {2024} (USD) (2)	Amount Year {2025} (USD) (3)	Amount Year {2026} (USD) (4)	Amount Year {2027} (USD) (5)	Total (USD)	See Budget Note: (reference as number, not as letter)
Component 1: Establishment of Policy and Regulatory Framework for Sustainable Woody Biomass Energy Production and Utilization	1.1. Effective enforcement of policies and regulations on the sustainable production and use of woody biomass for energy generation and utilization in rural areas of Pakistan.	1.1.1. Completed comprehensive situational and feasibility analyses for mainstreaming woody biomass production in the agriculture and forestry sectors, including assessments of potential areas for sustainable woody biomass production and utilization in Punjab, Khyber Pakhtunkhwa (KP), Sindh and Balochistan provinces	UNDP	62000	010003	71200	International Consultants			25,000			25,000	1
				62000		71300	National Consultants		25,000				25,000	2
				62000		71600	Travel	3,000	2,000				5,000	3
				62000		72100	Contractual Services - Companies	50,000					50,000	4
				62000		75700	Training, Workshops, and Conferences	5,000					5,000	6
			Total UNDP					58,000	27,000	25,000	-	-	110,000	
			Total Output 1.1.1					58,	27,	25,			110,	

								000	000	000	-	-	000	
		1.1.2. Developed and implemented policies and market-based regulatory framework for supporting woody biomass production and use, including national strategy for promotion of sustainable biomass energy production and utilization, using community-based woodlots and agroforestry production at the provincial level, and fiscal incentives for private-sector participation in woody biomass energy technology business.	UNDP	62000	010003	71200	International Consultants			20,000	5,000		25,000	1
				62000		71600	Travel		3,000				3,000	3
				62000		72100	Contractual Services - Companies	15,000	15,000				30,000	4
			Total UNDP					15,000	18,000	20,000	5,000	-	58,000	
		Total Output 1.1.2						15,000	18,000	20,000	5,000	-	58,000	
		1.1.3. Formulated and approved technical, management and operational standards in biomass energy development and utilization, woody biomass fired equipment and comprehensive biomass energy management.	UNDP	62000	010003	71200	International Consultants				25,000		25,000	1
				62000		71300	National Consultants			25,000			25,000	2
				62000		71600	Travel			5,000	7,000		12,000	3
				62000		72100	Contractual Services - Companies	15000	15,000				30,000	4
			Total UNDP					15,000	15,000	30,000	32,000	-	92,000	

		Total Output 1.1.3						15,000	15,000	30,000	32,000	-	92,000		
		1.1.4. Formulated energy-integrated development plans of 8 pilot towns two in each province of Punjab, KP, Sindh and Balochistan	UNDP	62000		71300	National Consultants				25,000		25,000	2	
				62000		71600	Travel	5,500					5,500	3	
				62000		72100	Contractual Services - Companies	45,000					45,000	4	
			Total UNDP					50,500	-	-	25,000	-	75,500		
		Total Output 1.1.4						50,500	-	-	25,000	-	75,500		
		1.1.5. Trained and qualified 60 provincial governments personnel (15 from each province) that provide technical support on sustainable production of woody biomass and enforcement of regulatory framework for biomass use for energy production and utilization	UNDP	62000	010003	71200	International Consultants				40,000		40,000	1	
				62000		71300	National Consultants			25,000			25,000	2	
				62000		71600	Travel			2,000	45,000		47,000	3	
				62000		72100	Contractual Services - Companies			4,000	4,000		8,000	4	
				62000		75700	Training, Workshops, and Conferences	2,000	5,000	10,000	10,000		27,000	6	
			Total UNDP					2,000	5,000	41,000	99,000	-	147,000		
		Total Output 1.1.5						2,000	5,000	41,000	99,000	-	147,000		
	Total Outcome 1.1							140,500	65,000	116,000	161,000	-	482,500		
Total Component 1								140,500	65,000	116,000	161,000	-	482,500		
Component 2: Promotion of Biomass Energy Production and Energy Efficient	2.1. Enhanced woody biomass production on forested, non-forested,	2.1.1. Developed, disseminated, and applied guidelines for sustainable production and utilization of	UNDP	62000	010003	71200	International Consultants	30,000					30,000	7	
				62000		71600	Travel	2,000	2,000				4,000	9	
				62000		72100	Contractual Services -		20,000				20,000	10	

							Companies							
		woody biomass		62000		75700	Training, Workshops, and Conferences	3,000					3,000	13
			Total UNDP					35,000	22,000	-	-	-	57,000	
		Total Output 2.1.1						35,000	22,000	-	-	-	57,000	
		2.1.2. Completed supply chain and market analysis and documented woody biomass demand and supply in the country.	UNDP	62000		71300	National Consultants	20,000					20,000	8
				62000		71600	Travel	2,000					2,000	9
				62000		74200	Audio Vis Visual & Print Prod Costs	3,000	1,500				4,500	12
				62000		75700	Training, Workshops, and Conferences	3,000					3,000	13
			Total UNDP					28,000	1,500	-	-	-	29,500	
Utilization Technologies	and on farmlands to cater local biomass energy needs, including for power generation and rural industry operations.	Total Output 2.1.2						28,000	1,500	-	-	-	29,500	
		2.1.3. Established farmers/communal forest nurseries over at least 300 hectares to provide planting stock for raising woody biomass in rural areas in the provinces of Punjab, KP, Sindh and Balochistan, including energy plantations over 40,000 ha of fast-growing native tree species at non-forested lands in selected districts each of these provinces for	UNDP	62000		71300	National Consultants	-	20,000				20,000	8
				62000		75700	Training, Workshops, and Conferences		3,000				3,000	13
			Total UNDP					-	23,000	-	-	-	23,000	

		woody biomass supply												
		Total Output 2.1.3						-	23,000	-	-	-	23,000	
		2.1.4. Established and operational fuelwood production from 4.5 million trees of common fuelwood plant species on farmlands for supply of woody biomass for production and utilization of biomass energy in rural areas.	UNDP	62000		71300	National Consultants	20,000					20,000	8
				62000		71600	Travel	2,000	2,000	-	-	-	4,000	9
				62000		72500	Supplies		5,000				5,000	11
				62000		75700	Training, Workshops, and Conferences	3,000					3,000	13
			Total UNDP					25,000	7,000	-	-	-	32,000	
		Total Output 2.1.4						25,000	7,000	-	-	-	32,000	
Total Outcome 2.1								88,000	53,500	-	-	-	141,500	
	2.2. Increased investments in the application of technologies for the production, and energy efficient utilization of woody biomass energy.	2.2.1. Completed and operational woody biomass-based energy generation technology demonstrations in 4 selected sites, one in each province of Punjab KP, Sindh and Balochistan showcasing the cost-effective application of decentralized woody biomass-based electricity generation and distribution (through mini/micro-grids)	UNDP	62000	010003	71200	International Consultants	13,343	15,000	-			28,343	14
				62000		71300	National Consultants		5,000	5,000	5,000	2,500	17,500	15
				62000		72100	Contractual Services - Companies		4,000	4,000	4,000	4,000	16,000	18
			Total UNDP					13,343	24,000	9,000	9,000	6,500	61,843	

		Total Output 2.2.1						13,343	24,000	9,000	9,000	6,500	61,843	
		2.2.2. Completed and operational four (4) demonstrations of cost-effective production of woody biomass fuels such as wood chips, briquettes, and pellets for use in decentralized woody-biomass based power generation and distribution	UNDP	62000	010003	71200	International Consultants		10,000	15,000			25,000	14
				62000		71300	National Consultants		5,000	5,000	5,000	2,500	17,500	15
				62000		72200	Equipment and Furniture		383,000	429,900	491,900	370,968	1,675,768	19
				62000		71400	Contractual Services - Individuals	23,619	25,981	28,579	31,437	34,581	144,197	17
			Total UNDP					23,619	423,981	478,479	528,337	408,049	1,862,465	
		Total Output 2.2.2						23,619	423,981	478,479	528,337	408,049	1,862,465	
		2.2.3. Completed and operational demonstrations of the cost-effective applications of energy efficient woody biomass fired technologies and comprehensive energy management systems in selected energy end-use sectors in support of rural socio-economic development	UNDP	62000	010003	71200	International Consultants			10,000	-	25,000	35,000	14
				62000		71300	National Consultants		5,000	5,000	5,000	2,500	17,500	15
				62000		71600	Travel		1,600	1,700	1,700	1,600	6,600	16
			Total UNDP					-	6,600	16,700	6,700	29,100	59,100	
		Total Output 2.2.3						-	6,600	16,700	6,700	29,100	59,100	
		2.2.4. Published and disseminated information about the results and impacts of the	UNDP	62000	010003	71200	International Consultants				25,000	-	25,000	14
				62000		71300	National Consultants		5,000	5,000	5,000	2,500	17,500	15
				62000		75700	Training,			2,	3,	3,	8	20

							Workshops, and Conferences			800	000	000	,800	
		completed demonstrations as well as developed and adopted a sustainability and replication plan of decentralized	Total UNDP					-	5,000	7,800	33,000	5,500	51,300	
		Total Output 2.2.4						-	5,000	7,800	33,000	5,500	51,300	
	Total Outcome 2.2							36,962	459,581	511,979	577,037	449,149	2,034,708	
Total Component 2								124,962	513,081	511,979	577,037	449,149	2,176,208	
3. Supporting Financial Requirements for Biomass Energy Technologies Initiatives	3.1 Adequate amounts of financial resources available for woody biomass energy technology application projects in the country	3.1.1 Established and operational investment and financing mechanisms for supporting the commercial viability and operation of woody biomass energy production for power and non-power applications, and the development of biomass energy industries.	UNDP	62000	010003	71300	National Consultants			10,000			10,000	21
				62000		72100	Contractual Services - Companies			30,000			30,000	22
			Total UNDP					-	-	40,000	-	-	40,000	
		Total Output 3.1.1						-	-	40,000	-	-	40,000	
		3.2.2 Established and operational market-oriented mechanism for the enhanced	UNDP	62000	010003	71300	National Consultants				7,000		7,000	21
				62000		72100	Contractual Services - Companies				50,000		50,000	22

				62000		75700	Training, Workshops, and Conferences				90,000		90,000	23
		development and utilization of biomass energy resources, energy efficiency and comprehensive biomass energy	Total UNDP					-	-	-	147,000	-	147,000	
			Total Output 3.1.2					-	-	-	147,000	-	147,000	
		3.3.3 De-risked biomass-based power generation projects, decentralized biomass-based energy generation in rural areas, and business plans for the GoP and private sector to facilitate financing and implementation.	UNDP	62000	010003	71300	National Consultants					7,000	7,000	21
				62000		72100	Contractual Services - Companies					20,000	20,000	22
			Total UNDP					-	-	-	-	27,000	27,000	
			Total Output 3.1.3					-	-	-	-	27,000	27,000	
	Total Outcome 3.1							-	-	40,000	147,000	27,000	214,000	
Total Component 3								-	-	40,000	147,000	27,000	214,000	
4: Biomass Energy Technology Capacity Building and Knowledge Management and Gender Mainstreaming	4.1. Enhanced local capacity, skills and knowledge in the development, installation,	4.1.1. Developed and implemented gender sensitive capacity development training programs for the design, construction, operation, and	UNDP	62000	010003	71300	National Consultants	10,000	18,000	11,000	19,000		58,000	25
				62000		71600	Travel	6,500	4,500	5,000	6,500		22,500	26
				62000		75700	Training, workshops, and conference	5,500	5,500	6,000	6,000		23,000	31
			Total UNDP					22,	28,	22,	31,		103,	

							000	000	000	500	-	500	
and operation of biomass energy technology systems in rural Pakistan	maintenance of woody biomass-based energy production systems both for power (e.g., decentralized power generation and distribution) and non-power (e.g., industrial energy efficient wood fired												
	Total Output 4.1.1						22,000	28,000	22,000	31,500	-	103,500	
	4.1.2. Trained and qualified people (60 from each province) providing training on woodlot operations and management, woody biomass fuel production, and woody biomass-based power generation and distribution, including trained and qualified community members (600 men and 400 women) providing training	UNDP	62000	010003	71200	International Consultants	12,000	6,000				18,000	24
			62000		71600	Travel	6,500	4,500	5,000	6,500		22,500	26
			62000		72500	Supplies	1,000	1,000	600	600		3,200	29
			62000		74200	Audio Vis Visual & Print Prod Costs	10,000	5,000	5,000	2,369		22,369	30
			62000		75700	Training, workshops, and conference	5,500	5,500	6,000	6,000	4,000	27,000	31
		Total UNDP					35,000	22,000	16,600	15,469	4,000	93,069	

	on the promotion of the widespread applications of energy efficient													
	Total Output 4.1.2						35,000	22,000	16,600	15,469	4,000	93,069		
	4.1.3. Sensitized key value chain actors of biomass energy enterprises that provide technical services and invest in the applications of cost-effective, climate resilient and energy efficient woody-biomass energy technologies.	UNDP	62000	010003	72100	Contractual Services - Companies	5,000	5,000	1,000			11,000	27	
		Total UNDP					5,000	5,000	1,000	-	-	11,000		
	Total Output 4.1.3						5,000	5,000	1,000	-	-	11,000		
	4.1.4. Knowledge products on lessons learned, best practices, etc. that are available through online repository for sharing experiences and replicating and scaling-up use of climate resilient and energy efficient biomass energy technologies.	UNDP	62000	010003	72200	Equipment and Furniture	15,000	6,000	6,000			27,000	28	
		Total UNDP					15,000	6,000	6,000	-	-	27,000		
	Total Output 4.1.4						15,000	6,000	6,000	-	-	27,000		
Total Outcome 4.1							77,000	61,000	45,600	46,969	4,000	234,569		

Total Component 4								77,000	61,000	45,600	46,969	4,000	234,569	
Monitoring and Evaluation (M&E)	M&E	Monitoring and Evaluation (M&E)	UNDP	62000	010003	71200	International Consultants	-	30,000	-	-	30,000	60,000	32
				62000		71300	National Consultants	8,000	28,000	8,000	8,000	28,000	80,000	33
				62000		71600	Travel	-	10,000	-	-	10,000	20,000	34
				62000		75700	Training, Workshops, and Conferences	-	4,000	-	-	4,000	8,000	35
			Total UNDP					8,000	72,000	8,000	8,000	72,000	168,000	
Total M&E								8,000	72,000	8,000	8,000	72,000	168,000	
Project Management Cost (PMC)	PMC	PMC	UNDP	62000	010003	71400	Contractual Services - Individuals	24,518	24,141	25,630	27,710	29,712	131,711	36
				62000		74100	Professional Services	4,820	4,820	4,820	4,820	4,820	24,100	39
				62000		72500	Supplies	1,000	1,000	1,000	1,000	1,000	5,000	37
				62000		75700	Training, Workshops, and Conferences	480	480	480	680	833	2,953	38
			Total UNDP					31,363	31,177	33,548	32,903	34,773	163,764	
Total Project Management Cost								31,363	31,177	33,548	32,903	34,773	163,764	
Total Project Grant								381,825	742,258	755,127	972,909	586,922	3,439,041	

** Remark: UNDP will provide full CO support services to the IP/MoCC&EC.*

Atlas Activity (GEF Component)	Outcome	Atlas Implementing Agent	Donor Name	Atlas Fund ID	Atlas Budgetary Account Code	ATLAS Budget Description	Amount Year 1 (USD)	Amount Year 2 (USD)	Amount Year 3 (USD)	Amount Year 4 (USD)	Amount Year 5 (USD)	Total (USD)	See Budget Note:
Project Management Cost	PMC	UNDP	UNDP	04000	74596	Services to projects - GOE	21,450	17,586	15,426	13,501	12, 037	80,000	40

Summary of Funds: [67]

	Year 1	Year 2	Year 3	Year 4	Year 5	Total
GEF grant administered by UNDP	**381,825**	**742,258**	**755,127**	**972,909**	**586,922**	**3,439,041**
UNDP Co-financing administered by UNDP (cash and in-kind)	**121,450**	**117,586**	**115,426**	**113,501**	**112,037**	**580,000**
Governments Co-financing	**2,220,124**	**4,126,150**	**4,203,310**	**5,494,122**	**4,526,740**	**20,570,446**
TOTAL	**2,723,399**	**4,985,994**	**5,073,863**	**6,580,532**	**5,225,699**	**24,589,487**

[67] *Summary table should include all financing of all kinds: GEF financing, co-financing, cash, in-kind, etc…*

Budget Notes	Description of cost items
1	**71200. International Consultants** Activity 1.3.1: Technical and operational standards for biomass-based power generation (Year 3) (US$ 800/day, total US$ 25,000) Activity 1.3.2: Technical and operational standards for energy conservation (Year 3 & 4) (US$ 800/day, total US$ 25,000) Activity 1.3.3: Technical and operational standards for energy management system (Year 4) (US$ 800/day, total US$ 25,000) Activity 1.5.2: International trainer (Year 4) (US$ 800/day for 50 person-days, total US$ 40,000)
2	**71300 National Consultants** Activity 1.2.1: Biomass tariff expert (Year 2) (US$ 400/day, total US$ 25,000) Activity 1.2.2: Provincial policy for off-grid sustainable biomass power generation (Year 3) (US$ 400/day, total US$ 25,000) Activity 1.2.3: National roadmap for sustainable biomass generation (Year 4) (US$ 400/day, total US$ 25,000) Activity 1.5.1: Consultant for conducting trainings for government staff at PFI (US$ 400/day, total US$ 25,000)
3	**71600 Travel** Activity 1.1.1: Woody biomass assessment (Year 1) (US$ 3,000) Activity 1.1.2: Technical and financial feasibility (Year 2) (US$ 3,000) Activity 1.4.1: Site selection for 8 sites (Year 1) (US$ 5,500) Activity 1.2.1: Biomass tariff expert (Year 2) (US$ 2,000) Activity 1.2.2: Provincial policy for off-grid sustainable biomass power generation (Year 3) (US$ 2,000) Activity 1.3.1: Technical and operational standards for biomass-based power generation (Year 3) (US$ 3,500) Activity 1.3.2: Technical and operational standards for energy conservation (Year 3&4) (US$ 3,500) Activity 1.5.2: International study tour (Year 4) (US$ 45,000) Activity 1.2.3: National roadmap for sustainable biomass generation (Year 4) (US$ 2,000) Activity 1.3.3: Technical and operational standards for energy management system (Year 4) (US$ 3,000)
4	**72100 Contractual Services – Companies** Technical assistance by consultancies, companies, or other service providers; or though appropriate agreement between the MoCC&EC and provincial line departments. If the IP requests UNDP to provide support service in engaging Responsible Parties (RPs) through the Letter of Agreement, UNDP will engage RPs on behalf of the IP and follow UNDP's rules and regulation. Activity 1.1.1: Conduct woody biomass resource assessment as contracted services (Year 1) (Lump sum total US$ 50,000) Activity 1.4.1, Site selection for 8 sites (Year 1) (Lump sum total US$ 45,000) Activity 1.1.2: Conduct technical feasibility of power generation potential at selected sites (Year 1&2) (Lump sum total US$ 30,000) Activity 1.1.3: Conduct financial feasibility and risk analysis between different technologies (Year 1&2) (Lump sum total US$ 30,000)
5	**74200 Audio Visual & Print Prod Costs** Activity 1.5.1: Printing of project reports and promotional materials with assistance by the national consultant (Year 3) (US$ 4,000) Activity 1.5.2: Printing of project reports and promotional materials with assistance by the international consultant (Year 4) (US$ 4,000)
6	**75700 Training, workshops, and conference** Activity 1.1.1: Stakeholder consultation (Year 1) (US$ 5,000) Activity 1.4.1: Training and development workshop (Year 1) (US$ 2,000) Activity 1.1.2: Stakeholder consultation & training session (Year 2) (US$ 2,500) Activity 1.1.3: Stakeholder consultation & training session (Year 2) (US$ 2,500) Activity 1.5.1: National training and development workshop for the preparation and planning of the capacity building program for provincial governments personnel (Year 3) (US$ 10,000)

Budget Notes	Description of cost items
	Activity 1.5.2: Implementation of the approved national training and development workshops in coordination with the capacity building program in Component 2.1 (Year 4) (US$ 10,000)
7	**71200. International Consultants** Activity 2.1.1: For development of guidelines of sustainable production and utilization of woody biomass, with Biodiversity Action Plan drawn from the international experience and development standards that is expected to be brought in by the IC. The IC will develop the action plan and recommend strategies in close coordination with the NC, as well as and leading the whole activity so that the desired Outputs will be realized within the timeline and quality standards. (Year 1) (US$ 800$/day, total US$ 30,000)
8	**71300 National Consultants** Activity 2.1.2: Assistance to the IC in gathering data, coordination with local authorities and stakeholders and documentation for development of guidelines of sustainable production and utilization of woody biomass, with Biodiversity Action Plan (year 1) (US$ 400/day, total US$ 20,000) Activity 2.1.4: Established and operational fuelwood production (Year 1) (US$ 400/day, total US$ 20,000) Activity 2.1.3: Biomass supply chain and market analysis (Year 2) (US$ 400/day, total US$ 20,000)
9	**71600 Travel** Activity 2.1.1: National travel for development of guidelines of sustainable production and biodiversity action plan (Year 1) (US$ 2,000) Activity 2.1.1: National travel for EIA and ESMP (Year 2) (US$ 2,000) Activity 2.1.2: National travel to support the facilitation and coordination of the development of a detailed study on supply chain of biomass, including market analysis for determination of biomass consumption, production, and pricing by the international and national consultants (Year 1) (US$ 2,000) Activity 2.1.4: Established and operational fuelwood production (Year 1) (US$ 2,000) Activity 2.1.4: National travel to facilitate and coordinate the development and establishment of sourcing system (nurseries) for supply of fast growing, local species for energy plantation; implementation and monitoring of environmental and social management plans; and dissemination of the information about the developed woody biomass supply system(Year 2) (US$ 2,000)
10	**72100 Contractual Services – Companies** Technical assistance by consultancies, companies, or other service providers; or though appropriate agreement between the MoCC&EC and provincial line departments. If the IP requests UNDP to provide support service in engaging Responsible Parties (RPs) through the Letter of Agreement, UNDP will engage RPs on behalf of the IP and follow UNDP's rules and regulation. Activity 2.1.1: To review the ESSP screening checklist to update the risks and mitigation measures as part of the EIA process of the government before the actual installation of the demos and develop the ESMP as may be required in the updated assessment (Year 2) (Lump sum total US$ 20,000)
11	**72500 Supplies** Activity 2.1.4: Budget for fuel wood plantation in target districts (Year 2) (US$ 5,000)
12	**74200 Audio Visual & Print Prod Costs** Activity 2.1.2: Printing of project reports and promotional materials with assistance by the international and national consultant (Year 1 & 2) (US$ 4,500)
13	**75700 Training, workshops, and conference** Activity 2.1.1 & 2.1.2 & 2.1.4: Awareness raising events each year for dissemination of results and stakeholder consultation sessions (Year 1) (US$ 9,000) Activity 2.1.3: Awareness raising events each year for dissemination of results and stakeholder consultation sessions (Year 2) (US$ 3,000)
14	**71200 International Consultant** Cost of International Technical Consultant Services prorated over the year (Year 1 - 5) to provide technical advice, review of other consultants' and EPC contractors' deliverables and support to PMU on project demos implementation and in related activities in the Components. The IC is tasked to design and produce the information packages from the information as gathered by the project and recommend a communication and

Budget Notes	Description of cost items
	dissemination plan. The following details on the activities are part of the International Technical Consultant inputs that needs to be budgeted as deliverables. Activity 2.2.1: Development and implementation of woody biomass-based energy generation technology demonstrations in 4 selected sites (Year 1 &2) (US$ 800/day, total US$ 28,343) Activity 2.2.2: Implementation of approved demonstrations of cost-effective production of woody biomass fuels in 4 selected sites, one in each province (Year 2 & 3) (US$ 800/day, total US$ 25,000) Activity 2.2.3: Development and implementation of demonstrations of the applications of energy efficient woody biomass fired technologies and comprehensive energy management systems. (Year 3-5) (US$ 800/day, total US$ 35,000) Activity 2.2.4.1: Produce the information to be published and facilitate the disseminated information about the results and impacts of the completed demonstrations (Year 4 &5) (US$ 800/day, total US$ 13,000) Activity 2.2.4.2: Developed and adopted a sustainability and replication plan for woody biomass-based production and utilization for power and non-power applications (Year 4) (US$ 800/day, total US$ 12,000)
15	**71300 National Consultants** Salary of Gender-Communications Consultant to provide technical services in aspect of gender mainstreaming in the project demos in the selected demo areas and its population in support for Activity 2.2.1, 2.2.2 and 2.2.3 (Year 2-4) (US$ 400/day, total US$ 30,000) Salary of M&E – Safeguards Consultant to provide technical services in aspect of monitoring and evaluation (M&E) as well as implementing the risk-based management (RBM) action plans and developing social and environmental management framework, grievance redress mechanism; conducting feasibility studies; monitoring of activities including risk log and SESP report. in support for Activity 2.2.1, 2.2.2 and 2.2.3 (Year 2-5) (US$ 400/day, total US$ 40,000)
16	**71600 Travel** Travelling for project staff for M&E of activities under component 2.2 (Year 2 - 5) (US$ 6,600)
17	**71400 Contractual Services – Individual** This is 80% of the total budget for the National Project Coordinator (NPC), representing the cost for NPC's task of providing technical coordination and technical assistance work in the design, implementation, and evaluation of the project's demo activities in the partner provinces. Specifically, the technical tasks involve the: (1) provision of advice and guidance in the establishment of the technical specifications, standards to be followed; (2) provision of technical inputs in TOR/Contract preparation, RFQ and bidding documentation, selection and engagement of the biomass energy technology service providers and contractors for the demo activities; (3) technical evaluation of bid proposals for the supply, installation, operation and systems from contractors and individual proponents; (4) provision of technical inputs and support to the PMU in the performance tracking system and contract performance monitoring of contractors and consultants to deliver expected outputs and deliverables, including technical review of reports prepared and submitted by contractors, as basis for contract payments; and, (5) Supervision and provision of technical advice and inputs in the testing, initial operation, and performance evaluation of the demo units under Component 2.2. The NPC is also responsible for assisting the PMU in the overall coordination of the management and implementation of project activities (20%). Please Refer to Budget Note # 36. This amount (USD 144,197) plus the budget allocation (20%) for the project management tasks of the NPC is based on the standard NPSA scale Step 10 starting with US$ 31,142 for combined technical and management functions, with provisions for increases starting Year 2).
18	**72100 Contractual Services – Companies** Technical assistance by consultancies, companies, or other service providers; or though appropriate agreement between the MoCC&EC and provincial line departments. If the IP requests UNDP to provide support service in engaging Responsible Parties (RPs) through the Letter of Agreement, UNDP will engage RPs on behalf of the IP and follow UNDP's rules and regulation. Other contractual services including daily manpower services facility operation, maintenance, housekeeping, and other ancillary support services during commissioning of demos (Year 2-5) (US$ 4,000 $ every year)
19	**72200 Equipment and Furniture**

Budget Notes	Description of cost items
	Purchase and installation of equipment for demo projects, including the critical spare parts and O&M cost of demo plants (Year 2-5) $ 1,675,768 Under each demo, the following are the major items that will collectively make up an integrated system for woody biomass-based power generation and utilization in the selected rural sites in each of the four provinces: • Biomass Collection, Storage, Transportation, Pre-treatment (US$ $ 193,632) • Woody biomass-fed gasifier plant and gas engine for electricity generation (US$ $ 119,392) • Solar PV power for back up or augmentation to meet the village electricity demand (US$ $ 30,913) • Mini/micro grid power distribution for rural electrification, 40kW for 40-50 households (US$ $ 53,999) • Woody biomass combustion systems for industrial thermal (non-power) applications (US$ $ 4,162) • Measuring equipment (US$ $ 16,844) Total for each provincial site (US$ $ 418,942) Total for the 4 sites (US$ $ 1,675,768)
20	**75700 Training, workshops, and conference** Awareness raising events each year for dissemination of results
21	**71300 National Consultants** Activity 3.1 Financial Expert (Year 3) (10,000) Activity 3.2 Financial Expert (Year 4) (US$ 7,000) Activity 3.3 Financial Expert (Year 5) (US$ 7,000)
22	**72100 Contractual Services – Companies** Technical assistance by consultancies, companies, or other service providers; or though appropriate agreement between the MoCC&EC and provincial line departments. If the IP requests UNDP to provide support service in engaging Responsible Parties (RPs) through the Letter of Agreement, UNDP will engage RPs on behalf of the IP and follow UNDP's rules and regulation. Activity 3.1: Financial Advisory Firm to advise on strategies and action plans in ensuring adequate amounts of financial resources available for woody biomass energy technology application projects in the country (Year 3) (Lump sum total US$ 30,000) Activity 3.2: Business Case Development for Biomass Based Off Grid Electrification (Year 4) (Lump sum total US$ 50,000) Activity 3.3: Consultancy Firm for development of risk portfolio and de-risking framework and its implementation (Year 5) (Lump sum total US$ 20,000)
23	**75700 Training, workshops, and conference** Awareness raising events for dissemination of results and stakeholder consultation sessions
24	**71200 International Consultant** International consultant for capacity building (US$ 800/day, total US$ 18,000)
25	**71300 National Consultant** National consultants for capacity building (US$ 400/day, total US$ 58,000)
26	**71600 Travel** Local travel and related expenses for capacity building, awareness, knowledge management, partnerships, co-financing consultations, workshops, and demonstration sites for Consultants and PMU staff.
27	**72100 Contractual Services – Companies** Technical assistance by consultancies, companies, or other service providers; or though appropriate agreement between the MoCC&EC and provincial line departments. If the IP requests UNDP to provide support service in engaging Responsible Parties (RPs) through the Letter of Agreement, UNDP will engage RPs on behalf of the IP and follow UNDP's rules and regulation. Contracted services for local companies providing support and ancillary services on how to operate and maintain biomass technologies for demonstration sites not covered by the EPC contractors.
28	**72200 Equipment and Furniture**

Budget Notes	Description of cost items
	Equipment including IT devices and services needed for the Projects various activities and knowledge management storage of projects data/information
29	**72500 Supplies** For various supplies needed in the capacity building activities
30	**74200 Audio Visual & Print Prod Costs** Printing & production including print of awareness materials etc. for capacity building activities
31	**75700 Training, workshops, and conference** Training, workshops, and conferences for capacity building activities, the specific breakdown will be determined based on the results of the needs analysis and recommendations.
32	**71200 International Consultant** International Consultant and Team Head for Mid-Term Review (Year 2), including travel costs (US$ 30,000 $) International Consultant and Team Head for Terminal Evaluation (Year 5), including travel cost (US$ 30,000)
33	**71300 National Consultants** National consultants for Mid-Term Review and Terminal Evaluation, including travel costs
34	**71600 Travel** Travel for Mid Term Review (Year 2), for monitoring related M&E activities regarding GEF and/or LDCF/SCCF Core indicators and METT or other required Tracking Tools (US$ 10,000) Travel for Terminal Review (Year 5), for monitoring related M&E activities regarding GEF and/or LDCF/SCCF Core indicators and METT or other required Tracking Tools (US$ 10,000)
35	**75700 Training, workshops, and conference** M&E related training, workshops, and conferences (Year 2) (US$ 4,000) M&E related training, workshops, and conferences (Year 5) (US$ 4,000)
36	**71400 Contractual Services – Individual** (a) For salary of the Admin/Financial Officer (AFO) at USD 16,995 in Year 1 & 2 increasing at around 10% annually from Year 3 up to Year 5, or 5-year total of USD 95,658 (b) This is 20% of the total budget for the National Project Coordinator (NPC), representing the cost for the NPC's project management related tasks. The 80% represents the cost for the NPC's technical tasks (mainly for the technical coordination and technical assistance tasks on the project demonstration activities (See Budget Note #17). The total NPC cost corresponds to the standard NPSA Step10 rate that includes technical and management functions with provisions for annual increases (USD 36,053).
37	**72500 Supplies** General office stationaries (Year 1-5) (US$ 1,000 per year)
38	**75700 Training, workshops, and conference** Annual board meeting and workshops for 5 years
39	**74100 Audit** Project audit Year 2 to 5 (US$ 4,820 per year)
40	**74596 Services to projects – GOE** UNDP support services to the Government (IP) for CO support to NIM includes the recruitment and recurrent management services, procurement, and engagement of Responsible Parties during the project implementation. The support services will be provided based on the IP request and are calculated on the basis of estimated actual or transaction-based costs. See Agreement in Annex 16 for details. Total cost estimated for the project duration is US$ 80,000 to be funded by non-GEF funding sources.

XI. LEGAL CONTEXT

The project document shall be the instrument envisaged and defined in the Supplemental Provisions to the Project Document, attached hereto and forming an integral part hereof, as "the Project Document".

This project will be implemented by the Ministry of Climate Change & Environmental Coordination ("Implementing Partner") in accordance with its financial regulations, rules, practices and procedures only to the extent that they do not contravene the principles of the Financial Regulations and Rules of UNDP. Where the financial governance of an Implementing Partner does not provide the required guidance to ensure best value for money, fairness, integrity, transparency, and effective international competition, the financial governance of UNDP shall apply.

XII. RISK MANAGEMENT

1. Consistent with the Article III of the SBAA *[or the Supplemental Provisions to the Project Document]*, the responsibility for the safety and security of the Implementing Partner and its personnel and property, and of UNDP's property in the Implementing Partner's custody, rests with the Implementing Partner. To this end, the Implementing Partner shall:
   a) put in place an appropriate security plan and maintain the security plan, taking into account the security situation in the country where the project is being carried;
   b) assume all risks and liabilities related to the Implementing Partner's security, and the full implementation of the security plan.

2. UNDP reserves the right to verify whether such a plan is in place, and to suggest modifications to the plan when necessary. Failure to maintain and implement an appropriate security plan as required hereunder shall be deemed a breach of the Implementing Partner's obligations under this Project Document.

3. The Implementing Partner agrees to undertake all reasonable efforts to ensure that no UNDP funds received pursuant to the Project Document are used to provide support to individuals or entities associated with terrorism and that the recipients of any amounts provided by UNDP hereunder do not appear on the United Nations Security Council Consolidated Sanctions List, and that no UNDP funds received pursuant to the Project Document are used for money laundering activities. The United Nations Security Council Consolidated Sanctions List can be accessed via https://www.un.org/securitycouncil/content/un-sc-consolidated-list.

4. The Implementing Partner acknowledges and agrees that UNDP will not tolerate sexual harassment and sexual exploitation and abuse of anyone by the Implementing Partner, and each of its responsible parties, their respective sub-recipients and other entities involved in Project implementation, either as contractors or subcontractors and their personnel, and any individuals performing services for them under the Project Document.

   (a) In the implementation of the activities under this Project Document, the Implementing Partner, and each of its sub-parties referred to above, shall comply with the standards of conduct set forth in the Secretary General's Bulletin ST/SGB/2003/13 of 9 October 2003, concerning "Special measures for protection from sexual exploitation and sexual abuse" ("SEA").

   (b) Moreover, and without limitation to the application of other regulations, rules, policies and procedures bearing upon the performance of the activities under this Project Document, in the implementation of activities, the Implementing Partner, and each of its sub-parties referred to above, shall not engage in any form of sexual harassment ("SH"). SH is defined as any unwelcome conduct of a sexual nature that might reasonably be expected or be perceived to cause offense or humiliation, when such conduct interferes with work, is made a condition of employment or creates an intimidating, hostile or offensive work environment. SH may occur in the workplace or in connection with work. While typically involving a pattern of conduct, SH may take the form of a single incident. In assessing the reasonableness of expectations or perceptions, the perspective of the person who is the target of the conduct shall be considered.

5. a) In the performance of the activities under this Project Document, the Implementing Partner shall (with respect to its own activities), and shall require from its sub-parties referred to in paragraph 4 (with respect to their activities) that they, have minimum standards and procedures in place, or a plan to develop and/or improve such standards and procedures in order to be able to take effective preventive and investigative action. These should include: policies on sexual harassment and sexual exploitation and abuse; policies on whistleblowing/protection against retaliation; and complaints, disciplinary and investigative mechanisms. In line

with this, the Implementing Partner will and will require that such sub-parties will take all appropriate measures to:

i. Prevent its employees, agents or any other persons engaged to perform any services under this Project Document, from engaging in SH or SEA;

ii. Offer employees and associated personnel training on prevention and response to SH and SEA, where the Implementing Partner and its sub-parties referred to in paragraph 4 have not put in place its own training regarding the prevention of SH and SEA, the Implementing Partner and its sub-parties may use the training material available at UNDP;

iii. Report and monitor allegations of SH and SEA of which the Implementing Partner and its sub-parties referred to in paragraph 4 have been informed or have otherwise become aware, and status thereof;

iv. Refer victims/survivors of SH and SEA to safe and confidential victim assistance; and

v. Promptly and confidentially, record and investigate any allegations credible enough to warrant an investigation of SH or SEA. The Implementing Partner shall advise UNDP of any such allegations received and investigations being conducted by itself or any of its sub-parties referred to in paragraph 4 with respect to their activities under the Project Document, and shall keep UNDP informed during the investigation by it or any of such sub-parties, to the extent that such notification (i) does not jeopardize the conduct of the investigation, including but not limited to the safety or security of persons, and/or (ii) is not in contravention of any laws applicable to it. Following the investigation, the Implementing Partner shall advise UNDP of any actions taken by it or any of the other entities further to the investigation.

b) The Implementing Partner shall establish that it has complied with the foregoing, to the satisfaction of UNDP, when requested by UNDP or any party acting on its behalf to provide such confirmation. Failure of the Implementing Partner, and each of its sub-parties referred to in paragraph 4, to comply of the foregoing, as determined by UNDP, shall be considered grounds for suspension or termination of the Project.

6. Social and environmental sustainability will be enhanced through application of the UNDP Social and Environmental Standards (http://www.undp.org/ses) and related Accountability Mechanism (http://www.undp.org/secu-srm).

7. The Implementing Partner shall: (a) conduct project and programme-related activities in a manner consistent with the UNDP Social and Environmental Standards, (b) implement any management or mitigation plan prepared for the project or programme to comply with such standards, and (c) engage in a constructive and timely manner to address any concerns and complaints raised through the Accountability Mechanism. UNDP will seek to ensure that communities and other project stakeholders are informed of and have access to the Accountability Mechanism.

8. All signatories to the Project Document shall cooperate in good faith with any exercise to evaluate any programme or project-related commitments or compliance with the UNDP Social and Environmental Standards. This includes providing access to project sites, relevant personnel, information, and documentation.

9. The Implementing Partner will take appropriate steps to prevent misuse of funds, fraud or corruption, by its officials, consultants, responsible parties, subcontractors and sub-recipients in implementing the project or using UNDP funds.

10. In the implementation of the activities under this Project Document, UNDP places reasonable reliance upon the Implementing Partner for it to apply its laws, regulations and processes, and applicable international laws regarding anti money laundering and countering the financing of terrorism, to ensure consistency with the principles of then in force the UNDP Anti-Money Laundering and Countering the Financing of Terrorism Policy.

11. The Implementing Partner will ensure that its financial management, anti-corruption, anti-fraud and anti-money laundering and countering the financing of terrorism policies are in place and enforced for all funding received from or through UNDP.

12. The requirements of the following documents, then in force at the time of signature of the Project Document, apply to the Implementing Partner: (a) UNDP Policy on Fraud and other Corrupt Practices and (b) UNDP Office of Audit and Investigations Investigation Guidelines. The Implementing Partner agrees to the requirements of the above documents, which are an integral part of this Project Document and are available online at www.undp.org.

13. In the event that an investigation is required, UNDP has the obligation to conduct investigations relating to any aspect of UNDP projects and programmes in accordance with UNDP's regulations, rules, policies and procedures. The Implementing Partner shall provide its full cooperation, including making available personnel, relevant documentation, and granting access to the Implementing Partner's (and its consultants', responsible parties', subcontractors' and sub-recipients') premises, for such purposes at reasonable times and on reasonable conditions as may be required for the purpose of an investigation. Should there be a limitation in meeting this obligation, UNDP shall consult with the Implementing Partner to find a solution.

14. The signatories to this Project Document will promptly inform one another in case of any incidence of inappropriate use of funds, or credible allegation of fraud or corruption with due confidentiality.

    Where the Implementing Partner becomes aware that a UNDP project or activity, in whole or in part, is the focus of investigation for alleged fraud/corruption, the Implementing Partner will inform the UNDP Resident Representative/Head of Office, who will promptly inform UNDP's Office of Audit and Investigations (OAI). The Implementing Partner shall provide regular updates to the head of UNDP in the country and OAI of the status of, and actions relating to, such investigation.

15. UNDP shall be entitled to a refund from the Implementing Partner of any funds provided that have been used inappropriately, including through fraud, corruption or other financial irregularity, or otherwise paid other than in accordance with the terms and conditions of the Project Document. Such amount may be deducted by UNDP from any payment due to the Implementing Partner under this or any other agreement. Recovery of such amount by UNDP shall not diminish or curtail the Implementing Partner's obligations under this Project Document.

    Where such funds have not been refunded to UNDP, the Implementing Partner agrees that donors to UNDP (including the Government) whose funding is the source, in whole or in part, of the funds for the activities under this Project Document, may seek recourse to the Implementing Partner for the recovery of any funds determined by UNDP to have been used inappropriately, including through fraud, corruption or other financial irregularity, or otherwise paid other than in accordance with the terms and conditions of the Project Document.

    Note: The term "Project Document" as used in this clause shall be deemed to include any relevant subsidiary agreement further to the Project Document, including those with responsible parties, subcontractors and sub-recipients.

16. Each contract issued by the Implementing Partner in connection with this Project Document shall include a provision representing that no fees, gratuities, rebates, gifts, commissions or other payments, other than those shown in the proposal, have been given, received, or promised in connection with the selection process or in

contract execution, and that the recipient of funds from the Implementing Partner shall cooperate with any and all investigations and post-payment audits.

17. Should UNDP refer to the relevant national authorities for appropriate legal action any alleged wrongdoing relating to the project, the Government will ensure that the relevant national authorities shall actively investigate the same and take appropriate legal action against all individuals found to have participated in the wrongdoing, recover and return any recovered funds to UNDP.

18. The Implementing Partner shall ensure that all of its obligations set forth under this section entitled “Risk Management” are passed on to each responsible party, subcontractor and sub-recipient and that all the clauses under this section entitled “Risk Management Standard Clauses” are included, mutatis mutandis, in all sub-contracts or sub-agreements entered into further to this Project Document.

XIII. ANNEXES

Mandatory Annexes

Annex 1: GEF Budget Template
Annex 2: Project Map and geospatial coordinates of the project area
Annex 3: Multiyear Workplan
Annex 4: Monitoring Plan
Annex 5: Social and Environmental Screening Procedure (SESP)
Annex 6: UNDP Risk Register
    Annex 6A: PASBET Climate Risks Assessment
    Annex 6B: PASBET Project and Covid-19
Annex 7: Overview of technical consultancies/subcontracts
Annex 8: Stakeholder Engagement Plan

Supplementary Annexes

Annex 9: Gender Analysis and Gender Action Plan
Annex 10: Project Knowledge Management
Annex 11: Procurement Plan – for first year of implementation
Annex 12: GEF focal area specific annexes
    Annex 12A - PASBET Project Demonstration Profiles
    Annex 12B: Estimation of Potential GHG Emission Reduction from the PASBET Project
Annex 13: Additional agreements: Co-Financing Letters
Annex 14: GEF and/or LDCF/SCCF Core indicators
Annex 15: GEF Taxonomy
Annex 16: Letter of Agreement for CO Support to NIM
Annex 17: GEF CEO Endorsement
Annex 18: Results of the Partners Capacity Assessment Tool (PCAT) and HACT Micro Assessment
Annex 19: UNDP Project Quality Assurance Report (to be completed in UNDP online corporate planning system)

ANNEX 1: GEF BUDGET

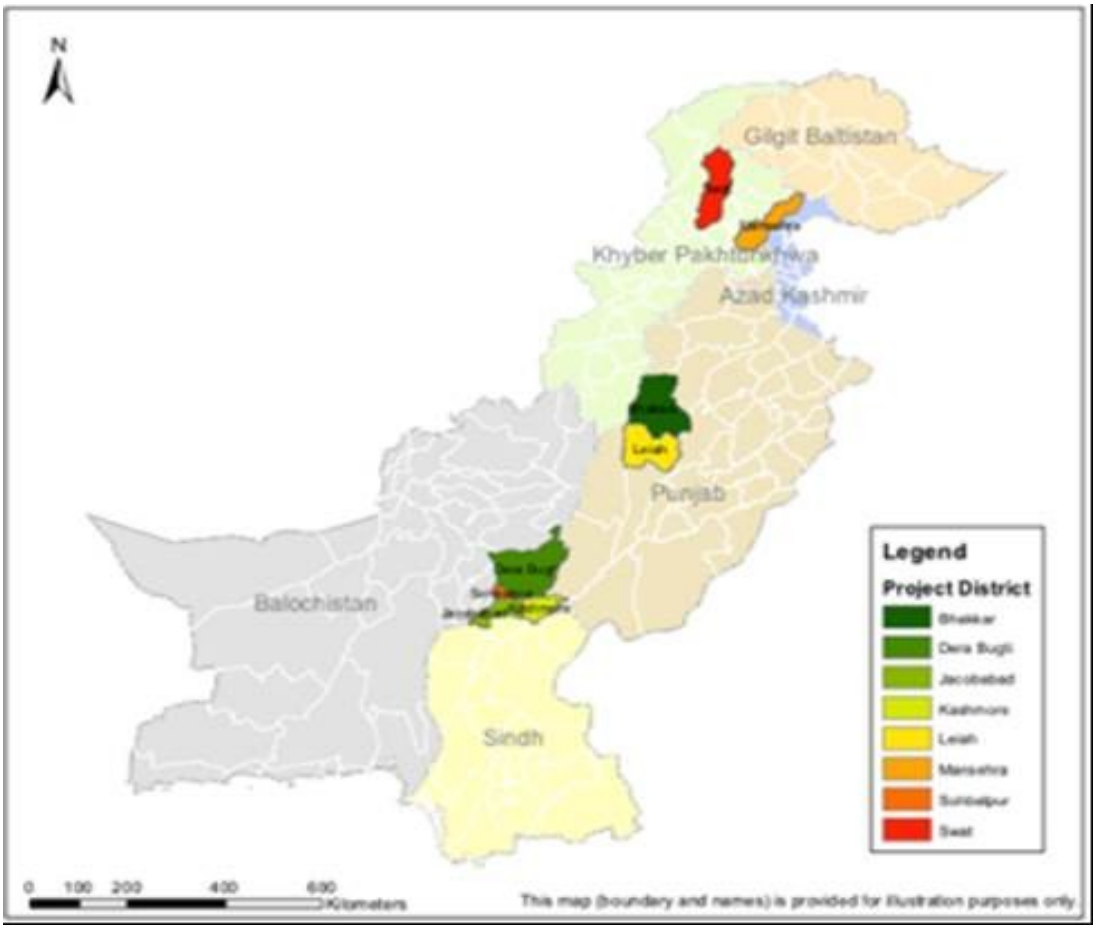
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ANNEX 2: PROJECT MAP AND GEOSPATIAL COORDINATES OF PROJECT SITES

Geo-reference/Coordinates Proposed PASBET Districts – For updating on the districts and coordinates

Province/District	Coordinates
Punjab	
Layyah	31.0998° N, 71.0022° E
Bhakkar	31.8621° N, 71.3824° E
Khyber Pakhtunkhwa	
Kaghan	34°50′N 73°31′E
Kalam	35.4801°N 72.5874°E
Sindh	
Badin	24°39′26″N 68°50′26″E
Sujawal	24°36'23" N and 68°4'19"E
Balochistan	
Jaffarabad/Sohbatpur	28.3009° N, 68.1908° E/28.4871° N, 68.6440° E
Dera Bugti	29.0278° N, 69.0970° E

Statement on Any Territorial Dispute: There is no territorial dispute with other countries near or within the vicinity or border of the



provinces where the project sites are located.

ANNEX 3: MULTI YEAR WORK PLAN

Outcomes	Outputs	Activities (Titles in Sec. IV)	Year 1 (Nov 23-Oct 24)				Year 2 (Nov 24-Oct 25)				Year 3 (Nov 25-Oct 26)				Year 4 (Nov 26-Oct 27)				Year 5 (Nov 27-Oct 28)			
			Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4
COMPONENT 1: ESTABLISHMENT OF POLICY AND REGULATORY FRAMEWORK FOR SUSTAINABLE WOODY BIOMASS ENERGY PRODUCTION AND UTILIZATION																						
Outcome 1. Effective enforcement of policies and regulations on the sustainable production and use of woody biomass for energy generation and utilization in rural areas of Pakistan.	Output 1.1. Completed comprehensive situational and feasibility analyses for mainstreaming woody biomass production in the agriculture and forestry sectors, including assessments of potential areas for sustainable woody biomass production and utilization[68] in Punjab, Khyber Pakhtunkhwa (KP), Sindh and Balochistan provinces[69]	**Activity 1.1.1:**																				
		Activity 1.1.2:																				
		Activity 1.1.3																				
	Output 1.2. Developed and implemented policies and market-based regulatory framework for supporting woody biomass production and use, including national strategy for promotion of sustainable biomass energy production and utilization, using community-based woodlots and agroforestry production at the provincial level, and fiscal incentives for private-sector participation in woody biomass energy technology business.	**Activity 1.2.1:**																				
		Activity 1.2.2:																				
		Activity 1.2.3																				
	Output 1.3. Formulated and approved technical, management and operational standards in biomass energy development and utilization, woody biomass fired equipment and comprehensive biomass energy management	**Activity 1.3.1:**																				
		Activity 1.3.2:																				
		Activity 1.3.3:																				
	Output 1.4. Formulated energy-integrated development plans of 8 pilot towns two in	**Activity 1.4.1:**																				

[68] This also includes the utilization of available forestry waste such as twigs and branches of standing trees, leftover wood, and branches after commercial harvesting of sustainable forests.

[69] Two districts each in the provinces of Punjab, KP, Sindh, and Balochistan identified as target areas for the planned technology demonstrations, with big potentials for woody biomass production and utilization.

Outcomes	Outputs	Activities (Titles in Sec. IV)	Year 1 (Nov 23-Oct 24)				Year 2 (Nov 24-Oct 25)				Year 3 (Nov 25-Oct 26)				Year 4 (Nov 26-Oct 27)				Year 5 (Nov 27-Oct 28)			
			Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4
	each province of Punjab, KP, Sindh and Balochistan[70].																					
	Output 1.5. Trained and qualified 60 provincial governments personnel (15 from each province) that provide technical support on sustainable production of woody biomass and enforcement of regulatory framework for biomass use for energy production and utilization.	**Activity 1.5.1:**																				
		Activity 1.5.2:																				
		Activity 1.5.3																				
COMPONENT 2: PROMOTION OF BIOMASS ENERGY PRODUCTION AND ENERGY EFFICIENT UTILIZATION TECHNOLOGIES																						
Outcome 2.1. Enhanced woody biomass production on forested, non-forested, and on farmlands to cater local biomass energy needs, including for power generation and rural industry operations	Output 2.1.1. Developed, disseminated, and applied guidelines for sustainable production and utilization of woody biomass	**Activity 2.1.1.1:**																				
		Activity 2.1.1.2:																				
	Output 2.1.2. Completed supply chain and market analysis and documented woody biomass demand and supply in the country	**Activity 2.1.2:**																				
	Output 2.1.3. Established farmers/communal forest nurseries over at least 300 hectares to provide planting stock for raising woody biomass in rural areas in the provinces of Punjab, KP, Sindh and Balochistan, including energy plantations over 40,000 ha of fast-growing native tree species at non-forested lands in selected districts each of these provinces for woody biomass supply	**Activity 2.1.3:**																				
	Output 2.1.4. Established and operational fuelwood production from 4.5 million trees of common fuelwood plant species on farmlands for supply of woody biomass for production and utilization of biomass energy in rural areas.	**Activity 2.1.4:**																				
Outcome 2.2. Increased investments	Output 2.2.1. Completed and operational woody biomass-based energy generation technology demonstrations in 4 selected	**Activity 2.2.1:**																				

[70] Based on enhanced provincial data banking systems that include sustainable woody biomass production and utilization, provincial economic and social development parameters and performance indices, energy demand and use trends in each province.

Outcomes	Outputs	Activities (Titles in Sec. IV)	Year 1 (Nov 23-Oct 24)				Year 2 (Nov 24-Oct 25)				Year 3 (Nov 25-Oct 26)				Year 4 (Nov 26-Oct 27)				Year 5 (Nov 27-Oct 28)			
			Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4
in the application of technologies for the production, and energy efficient utilization of woody biomass energy.	sites, one in each province of Punjab KP, Sindh and Balochistan showcasing the cost-effective application of decentralized woody biomass-based electricity generation and distribution (through mini/micro-grids)[71]																					
	Output 2.2.2. Completed and operational four (4) demonstrations of cost-effective production of woody biomass fuels such as wood chips, briquettes, and pellets for use in decentralized woody-biomass based power generation and distribution	**Activity 2.2.2:**																				
	Output 2.2.3. Completed and operational demonstrations of the cost-effective applications of energy efficient woody biomass fired technologies and comprehensive energy management systems in selected energy end-use sectors in support of rural socio-economic development[72]	**Activity 2.2.3:**																				
	Output 2.2.4. Published and disseminated information about the results and impacts of the completed demonstrations as well as developed and adopted a sustainability and replication plan of decentralized woody biomass-based energy production and utilization for power and non-power applications	**Activity 2.2.4**																				
COMPONENT 3: SUPPORTING FINANCIAL REQUIREMENTS FOR BIOMASS ENERGY TECHNOLOGIES INITIATIVES																						
Outcome 3. Adequate amounts of financial resources available for	**Output** 3.1. Established and operational investment and financing mechanisms for supporting the commercial viability and operation of woody biomass energy production for power and non-power applications, and the development of	**Activity 3.1.1:**																				
		Activity 3.1.2:																				
		Activity 3.1.3:																				

[71] This involves the installation of a collective total woody biomass-based power generation capacity of 5.4 MW in the 4 provinces to supply of electricity to supply unelectrified rural villages.

[72] This includes local production of energy-efficient wood-fired devices that are cost-effectively utilized in rural industries to improve energy utilization efficiency and reduce GHG emissions.

Outcomes	Outputs	Activities (Titles in Sec. IV)	Year 1 (Nov 23-Oct 24)				Year 2 (Nov 24-Oct 25)				Year 3 (Nov 25-Oct 26)				Year 4 (Nov 26-Oct 27)				Year 5 (Nov 27-Oct 28)			
			Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4
woody biomass energy technology application projects in the country	biomass energy industries																					
	Output 3.2. Established and operational market-oriented mechanism for the enhanced development and utilization of biomass energy resources, energy efficiency and comprehensive biomass energy management systems for supporting sustainable rural socio-economic development	**Activity 3.2.1:**																				
		Activity 3.2.2:																				
	Output 3.3. De-risked biomass-based power generation projects, decentralized biomass-based energy generation in rural areas, and business plans for the GoP and private sector to facilitate financing and implementation	**Activity 3.3.1:**																				
		Activity 3.3.2:																				
COMPONENT 4: BIOMASS ENERGY TECHNOLOGY CAPACITY BUILDING AND KNOWLEDGE MANAGEMENT AND GENDER MAINSTREAMING																						
Outcome 4: Enhanced local capacity, skills and knowledge in the development, installation, and operation of biomass	Output 4.1. Developed and implemented gender sensitive capacity development training programs for the design, construction, operation, and maintenance of woody biomass-based energy production systems both for power (e.g., decentralized power generation and distribution) and non-power (e.g., industrial energy efficient wood-fired equipment) applications, as well as for strengthening institutional capacity for supporting sustainable biomass energy production and utilization	**Activity 4.1.1:**																				
		Activity 4.1.2:																				
		Activity 4.1.3:																				
		Activity 4.1.4																				
		Activity 4.1.5																				

Outcomes	Outputs	Activities (Titles in Sec. IV)	Year 1 (Nov 23-Oct 24)				Year 2 (Nov 24-Oct 25)				Year 3 (Nov 25-Oct 26)				Year 4 (Nov 26-Oct 27)				Year 5 (Nov 27-Oct 28)			
			Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4
energy technology systems in rural Pakistan	Output 4.2. Trained and qualified people (60 from each province) providing training on woodlot operations and management, woody biomass fuel production, and woody biomass-based power generation and distribution, including trained and qualified community members (600 men and 400 women) providing training on the promotion of the widespread applications of energy efficient wood-fired equipment	**Activity 4.2.1:**																				
		Activity 4.2.2:																				
		Activity 4.2.3:																				
	Output 4.3. Sensitized key value chain actors of biomass energy enterprises that provide technical services and invest in the applications of cost-effective, climate resilient and energy efficient woody-biomass energy technologies	**Activity 4.3.1:**																				
		Activity 4.3.2:																				
		Activity 4.3.3:																				
	Output 4.4. Knowledge products on lessons learned, best practices, etc. that are available through online repository for sharing experiences and replicating and scaling-up use of climate resilient and energy efficient biomass energy technologies	**Activity 4.4.1:**																				

ANNEX 4: MONITORING PLAN

This Monitoring Plan and the M&E Plan and Budget in Section VI of this project document will both guide monitoring and evaluation at the project level for the duration of project implementation. Please NOTE that there are indicators for M&E of the Gender Action Plan in **Annex 9** that should be taken into consideration when implementing this project M&E Plan.

Monitoring	Indicators	Targets		Description of indicators and targets	Data source & Collection Methods[73]	Frequency	Responsible for data collection	Means of verification	Assumptions
		Mid-term	End of Project						
Project objective: Widespread application of sustainable biomass energy technologies for supporting socio-economic development of and reducing greenhouse gas (GHG) emissions from, the rural sector in Pakistan.	**Indicator 1:** No. of direct project beneficiaries disaggregated by gender (individual people) •Male •Female	 •64,000 •42,800	 •160,380 •107,000	Beneficiaries attributable to the project activities and outputs	Project M&E and activity reports and interviews of beneficiaries after implementation of activities	Quarterly	PMU/ MoCC&EC	Project M&E and activity reports. Measurement of increased income of beneficiaries after implementation of activities	No unexpected cultural or security barriers that hinder activities
	Indicator 2: Cumulative GHG emission reduction from the rural sector of Pakistan, tCO_2	51,520	195,050	Including biomass energy generation in project demos, replication, and baseline projects attributable to PASBET	Annual energy supply and consumption reports submitted by relevant national and local governments, etc.	Quarterly	PMU/ MoCC&EC	Regular periodic operations report from each sustainable biomass energy technology application demo and replication in demo towns/villages. Project M&E and activity reports.	Continuous commitment, support & active participation of the national/local government in low carbon rural development, as well as in meeting the renewable energy targets of the country Realization of committed co-financing from the national/local governments in the implementation of

[73] Data collection methods should outline specific tools used to collect data and additional information as necessary to support monitoring. The PIR cannot be used as a source of verification.

Monitoring	Indicators	Targets		Description of indicators and targets	Data source & Collection Methods	Frequency	Responsible for data collection	Means of verification	Assumptions
		Mid-term	End of Project						
									project activities and monitoring systems
	Indicator 3: Cumulative Biomass-based Electricity in rural areas in Pakistan, GWh	**1.5**	**98.2**	Relevant generation from biomass attributable to the Project	Project M&E and activity reports	Monthly	PMU, MoCC&EC, DISCOs and rural electrification - agency	Annual energy supply and consumption reports in relevant national and local governments, etc.	Continuous commitment, support & active participation of the national/local government in low carbon rural development
	Indicator 4: Cumulative number of new jobs created in the application of sustainable biomass energy technologies and techniques in the rural areas. •Male •Female	 •1,460 •1,380	 •3,660 •3,450	Number of jobs related to biomass energy production, power generation and use in various activities	Project M&E and activity reports	Quarterly	PMU, MoCC&EC, DISCOs and rural electrification - agency and local government	Trade and commerce reports. Data gathering on the no of people employed in all job categories	No unexpected cultural or security barriers that hinder activities
Project Outcome 1: Effective enforcement of policies and regulations on the sustainable production and use of woody biomass for energy generation and utilization in rural areas of Pakistan.	***Indicator 5:*** Cumulative number of sustainable biomass energy projects in rural areas that are facilitated by approved and effectively enforced policies and laws/regulations that promote the commercial production and use	8	32[74]	Biomass-based projects supported from all components	Project M&E and activity reports	Monthly	PMU/ MoCC&EC	Project activity (baseline and incremental) reports. Government reports on the implementation of the approved sustainable biomass energy development and utilization policies and regulations. Project M&E reports	Continuous consistent commitment of the national and local governments in politically supporting RE and EE initiatives in rural towns/villages. No competing value-added uses of woody biomass.

[74] This is 4 projects x 2 districts x 4 provinces

Monitoring	Indicators	Targets		Description of indicators and targets	Data source & Collection Methods	Frequency	Responsible for data collection	Means of verification	Assumptions
		Mid-term	End of Project						
	of sustainable biomass-based energy.								
	Indicator 6: Cumulative number of rural villages that develop and implement energy-integrated development plans that include the commercial production and utilization of sustainable woody biomass.	**4**	40[75]	Villages that have been supported by the project	Project M&E and activity reports	Quarterly	PMU, MoCC&EC, DISCOs and rural electrification - agency and local government	Reports by responsible agency	Continuous consistent commitment of the national and local governments in politically supporting RE and EE initiatives in rural towns/villages
	Indicator 7: No. of developed, approved, enforced, and implemented and enforced as part of the RE law to include market-based targets of biomass based energy, particularly woody biomass	**2**[76]	**5**[77]	Actual policy instruments passed and implemented	Project M&E and activity reports	Quarterly	PMU/MOC	Documentation of the approved and promulgated regulations and policy instruments in the Output 1.2 activities (baseline and incremental) reports. Project M&E reports.	

[75] This is 10 villages in each 4 provinces

[76] This includes: (a) Regulation to promote woody biomass and the corresponding Woody Biomass National Road Map; (b) Tariff application developed and submitted to NEPRA for biomass based electricity; (c) microgrid systems are made part of national grid code; (d) Inclusion of biomass as energy resource in the National Rural electricity policy; and (e) Community-based Natural Resource Management as part of National and Provincial Forestry Policy

[77] All five (5) policies that are approved and enforced in line with the targets of Biomass based power generation included in the National RE Policy.

Monitoring	Indicators	Targets		Description of indicators and targets	Data source & Collection Methods	Frequency	Responsible for data collection	Means of verification	Assumptions
		Mid-term	End of Project						
	Indicator 8: Cumulative no. of standards for equipment to use woody biomass for energy generation and EE utilization established and enforced	**3**[78]	**4**[79]	Standards that are developed by the project and approved by authorized agencies	Project M&E and activity reports	Quarterly	PMU/ MoCC&EC	Documentation of the approved and promulgated regulations and policy instruments in the Output 1.2 activities (baseline and incremental) reports. Project M&E reports.	
Project Outcome 2.1: Enhanced woody biomass production on forested, non-forested, and on farmlands to cater local biomass energy needs, including for power generation and rural industry operations.	***Indicator 9:*** No. of established and commercially operated sustainable woody biomass production facilities.	**1**	**3**[80]	Woody production facilities supported by the Project	Project M&E and activity reports	Quarterly	PMU/ MoCC&EC	Forest industry reports Trade and commerce reports	Continuous consistent commitment of the national and local governments in politically supporting RE and EE initiatives in rural towns/villages
	Indicator 10: Cumulative number of entities that are interested and planned to invest in sustainable woody biomass fuel production. • Local Gov'ts • Private Sector entities	 • 1 • 1	 • 4 • 2	Number of projects on fuel production supported by the project	Project M&E and activity reports	Quarterly	PMU/ MoCC&EC and forest departments	Forest industry reports Trade and commerce reports	Continuous consistent commitment of the national and local governments in politically supporting RE and EE initiatives in rural towns/villages

[78] Standards developed for: (a) energy plantation; (b) biomass energy conversion; and (c) biomass user technologies

[79] Standards opted by relevant authorities and provincial authorities as part of the procurement

[80] In total there are 3 new sustainable woody biomass production facilities are expected to be completed by the end of the project, one in each province of Punjab, Sindh and Balochistan)

Monitoring	Indicators	Targets		Description of indicators and targets	Data source & Collection Methods	Frequency	Responsible for data collection	Means of verification	Assumptions
		Mid-term	End of Project						
	Indicator 11: Average annual local production and consumption of sustainable woody biomass fuel, tons/year	38,071[81]	**95,178**	Including all production and consumption of biomass fuels used of r power and non-power uses	Project M&E and activity reports	Quarterly	PMU/ MoCC&EC, Department of Energy and forest departments	Forest industry reports Trade and commerce and energy departments reports	Continuous consistent commitment of the national and local governments in politically supporting RE and EE initiatives in rural towns/villages
Project Outcome 2.2: Increased investments in the application of technologies for the production, and energy efficient utilization of woody biomass energy.	***Indicator 12:*** Cumulative amount of fossil fuel energy displaced by sustainable woody biomass fuels. * Power Applications, ktoe *Non-Power Applications, ktoe	**1,500** **150**	**3,800**[82] **300**	Including all production and consumption of biomass fuels used of r power and non-power uses	Project M&E and activity reports	Monthly	PMU/ MoCC&EC, Department of Energy and forest departments	Forest industry reports Trade and commerce and energy departments reports	Continuous consistent commitment of the national and local governments in politically supporting RE and EE initiatives in rural towns/villages
	Indicator 13: Cumulative number of private sector entities that are interested and plans to invest in sustainable woody biomass-based energy projects in rural towns and municipalities	**3**	**5**	Private companies that have registered and started implementation	Project M&E and activity reports	Monthly	PMU/ MoCC&EC, Department of Energy and forest departments	Forest industry reports Trade and commerce and energy departments reports	Continuous consistent commitment of the national and local governments in politically supporting RE and EE initiatives in rural towns/villages

[81] To generate 5.4 MW of biomass-based power generation it is estimated that total of 95,178 Tons of woody biomass is required annually. By mid-term at least 40% of this wood should be available from the sourcing.

[82] If 5.4 MW is generated using fossil fuel (e.g., NG based genset) it will require fossil fuel energy equivalent of 3.8 ktoe. Similar energy can be displaced by sustainable woody biomass fuels. By mid-term it is estimated that 40% of the total will be completed.

Monitoring	Indicators	Targets		Description of indicators and targets	Data source & Collection Methods	Frequency	Responsible for data collection	Means of verification	Assumptions
		Mid-term	End of Project						
Outcome 3: Adequate amounts of financial resources available for woody biomass energy technology application projects in the country.	***Indicator 14:*** Cumulative number of sustainable biomass energy projects whose funding are sourced from other sources of funds apart from national/local government. * Sustainable woody biomass fuel production * Sustainable woody biomass-based power generation and distribution * Sustainable woody biomass-based energy generation and utilization.	• **4** • **4** • **4**	• **8** • **8** • **8**	Projects that are funded by private sector	Project M&E and activity reports	Monthly	PMU/ MoCC&EC, Department of Energy and forest departments	Forest industry reports Trade and commerce and energy departments reports	Continuous consistent commitment of the national and local governments in politically supporting RE and EE initiatives in rural towns/villages
	Indicator 15: Cumulative number of financing schemes/mechanisms adopted by financial institutions for supporting sustainable biomass energy initiatives in Pakistan.	**1**	**2**	Financial mechanisms that are by banks and financial institutions	Project M&E and activity reports	Monthly	PMU/ MoCC&EC, Department of Energy and forest departments	Forest industry reports Trade and commerce and energy department reports	Continuous consistent commitment of the financing institutions, national and local governments in politically supporting RE and EE initiatives in rural towns/villages
Outcome 4: Enhanced local	***Indicator 16:*** Cumulative number	**4**	**32**	Pertinent projects	Project M&E and activity	Monthly	PMU/ MoCC&EC,	Forest industry reports	Continuous consistent

Monitoring	Indicators	Targets		Description of indicators and targets	Data source & Collection Methods	Frequency	Responsible for data collection	Means of verification	Assumptions
		Mid-term	End of Project						
capacity, skills and knowledge in the development, installation, and operation of biomass energy technology systems in rural Pakistan.	of sustainable biomass woody energy projects designed, implemented, and maintained by rural towns and municipalities			supported and attributable to PABET's area of responsibility	reports		Department of Energy and forest departments	Trade and commerce and energy department reports	commitment of the financing institutions, national and local governments in politically supporting RE and EE initiatives in rural towns/villages
	Indicator 17: Cumulative number of operational woody biomass-based power generation system installations in rural towns and municipalities.	**4**	**16**	Including installations supported by the project in the 4 provinces	Project M&E and activity reports	Monthly	PMU/ MoCC&EC, Department of Energy and forest departments	Forest industry reports Trade and commerce and energy department reports	Continuous consistent commitment of the financing institutions, national and local governments in politically supporting RE and EE initiatives in rural towns and villages
	Indicator 18: Cumulative number of energy consumers in rural areas that are utilizing: woody biomass-based energy generating and consuming equipment (persons)	**1,120**	**2,240**	Number of users of biomass based production and energy generation in the 4 provinces selected	Project M&E and activity reports	Monthly	PMU/ MoCC&EC, Department of Energy and forest departments	Forest industry reports Trade and commerce and energy department reports	Continuous consistent commitment of the financing institutions, national and local governments in politically supporting RE and EE initiatives in rural towns/villages

ANNEX 5: UNDP SOCIAL AND ENVIRONMENTAL SCREENING PROCEDURE

Project Information	
1. Project Title	Project Title: Combating Climate Change through the Promotion and Application of Sustainable Biomass Energy Technologies in Pakistan (PASBET)
2. Project Number (i.e. Atlas project ID, PIMS+)	PIMS 6542
3. Location (Global/Region/Country)	Pakistan
4. Project stage (Design or Implementation)	Design
5. Date	3 July 2023

Part A. Integrating Programming Principles to Strengthen Social and Environmental Sustainability

QUESTION 1: How Does the Project Integrate the Programming Principles in Order to Strengthen Social and Environmental Sustainability?
Briefly describe in the space below how the project mainstreams the human rights-based approach
The primary objective of the project i.e., widespread application of sustainable biomass energy technologies for supporting socio-economic development will ensure that the project's approach is in in line with the human rights-based approach. The key elements of the project including the building of local capacities, creating awareness, disseminating technical knowledge and gender mainstreaming will help improve local livelihoods and promote co-management of resources in the all the target areas. The project is being implemented in the four provinces of Punjab, Sindh, KP and Balochistan which are home to population groups that differ greatly in terms of their culture, social structure, gender roles, economic conditions and use of natural resources including firewood. These differences elevate the importance of a human rights-based approach in the project strategy in terms of identifying and respecting distinct cultures and the implementing policies to accommodate them and the surrounding environment. Human rights principles have been incorporated in the project design by ensuring that the targeted population, women and children in particular, benefit equally from the planned interventions and opportunities. A participatory approach has been used/deployed at every stage of the project to involve the target population keeping in mind the cultural context in the areas for example through consultations with community members in specific areas where demonstrations will showcase the application of RE and EE technologies for woody biomass energy production. The participatory approach in the project design is to ensure that there is a two-way process incorporating the views of community members in the project: and ensuring that the requirements of the beneficiaries are considered to ensure that their interests are taken care of and the project leads to demand oriented solutions. Information exchange and learning for all members of the society is an integral part of the knowledge management and information dissemination activities. Effective stakeholder engagement approaches on these lines assisted in devising and adapting strategies for the implementation of RE-based energy generation activities. Capacity building activities include the training of qualified people (60 from each province, with a target to achieve 30% gender ratio) on woodlot operations and management, woody biomass fuel production, and woody biomass-based power generation and distribution who in turn, provide this training to selected community members (600 men and 400 women) on these essential aspects. The capacity building plans include equal access to information sharing sessions for both men and women. Moreover, gender disaggregated data is aimed to be collected during trainings, workshops, discussions, interviews and focus group interactions to assess the impact of all activities on women in the long run. The principles of equality and non-discrimination will also be a part of the gender action plan. The project features the mainstreaming of women into decision making mechanisms to ensure that any environmental and health-based threats that they face due to biomass-based energy production are reduced.
Briefly describe in the space below how the project is likely to improve gender equality and women's empowerment
GEF recognizes gender as a cross-cutting priority that is integral to the GEF's ability to achieve global environmental benefits. The GEF policy on gender equality seeks to ensure equal

opportunities for women and men to participate in, contribute and benefit from GEF financed activities. This project has been deigned on the basis of these principles to ensure that gender and women's empowerment are incorporated at every stage.

At the project preparation grant stage (PPG), a Gender, Social and Environmental Specialist was assigned to ensure integration of the gender and gender mainstreaming actions in the GEF project planning processes and various stages of the project cycle. Based on the findings of research conducted on the issue, gender disparity exists within the activities related to biomass energy technology extension, education, and more importantly, the collection and use of fuel wood. These issues arise primarily due to the prevailing cultural, and religious practices in rural in Pakistan. Gender considerations, therefore, are an integral part of the project strategy i.e., component four of the project focuses on 'Biomass Energy Technology Capacity Building and Knowledge Management and Gender Mainstreaming'. The interventions that will be guided by this component will improve gender equality and promote women's empowerment.

Gender mainstreaming in the project is based on a gender action plan mainstreams and crosscuts gender across all planned activities. Barriers that hinder the provision of equal rights to women have been identified in all provinces and recommendations on gender sensitive policy design to remove these barriers to ensure that the application of biomass energy production and utilization technologies are gender responsive are incorporated. Gender equity within the project design is a key criterion for the provision of goods and services to the communities in the project areas.

The number of women that will benefit from the project is approximately 107,000, with the majority of them being part of the technology demo projects: women will constitute 40% of the beneficiaries in this activity which is a significant number given the cultural constraints and barriers that they face. The project will also ensure capacity building of 400 qualified women from the project communities on the promotion of the widespread applications of energy efficient wood-fired equipment and devices. Women's representation in the energy plantation activity will also result in 250 beneficiaries (50%).

Consultations with women will continue to be an integral feature at all stages of the project, from design to implementation: qualified and capable women have been involved in the strategic process design and capacity development interventions by giving them decision making powers in culturally relevant ways in all provinces. This step is particularly essential given that women's health and safety are the biggest issues in the biomass energy process due to their roles in firewood collection and cooking which is harmful in multiple ways. The gender mainstreaming process will also ensure that women have equal access to information through the knowledge management and information dissemination activities of the project.

Gender has also been incorporated in the monitoring and evaluation aspect of the project: gender disaggregated data will be collected in the capacity development activities i.e., trainings, workshops, discussions, interviews as well as project indicators and targets. Moreover, both male and female beneficiaries will be consulted on an equal level during baseline data assessments, mid-term evaluations and terminal evaluations in the project sites. The results at different stages i.e., knowledge products will be disseminated, tailored to the literacy and cultural circumstances of the local communities, to ensure equitable gender and social inclusion.

The establishment of women organization has been incorporated in the project design where possible to support the smooth integration of these processes in all provinces. Existing local governance structures in the agriculture and forest sectors that promote equality and leadership for women at all levels will be strengthened to support implementation. Moreover, the project staff is to be sensitized on gender issues at all stages: this could be achieved through the appointment of an experienced gender specialist to support the implementation, and monitoring of gender mainstreaming strategy at both the internal and external level. Lastly, both men and women will have equal opportunities in the development and promotion of service provision jobs in the relevant downstream activities of the application of woody biomass energy technologies in the rural energy sector and rural industries.

With regards to employment opportunities for women, the project will result in approximately 3,448 jobs for women with a vast majority of them being related to biomass collection. Other sectors/activities that will create jobs for women include the setting up of biomass-based power plants (432) as well as their 'Operation and Management' activities (162). Jobs

created in these categories as well as Project Execution Staff (4) will comprise almost half of the overall jobs created in the project.

Briefly describe in the space below how the project mainstreams sustainability and resilience

The project is designed to mainstream sustainability and resilience by focusing on local communities and environment. The project's objectives revolve around capacity building and economic uplifting of the targeted population. This goal is coupled with ecological and conservation efforts through promoting the use of a sustainable resource for power in the form of woody biomass. In terms of supporting the community, the project will strengthen relation relations between multiple stakeholders and empower community-based organizations/groups. This approach will ensure that the benefits to the community and environment will continue beyond the lifetime of the project and add to the resilience and sustainability of the landscape

The project is based on the sustainable production of woody biomass including the establishment, operation and maintenance of dedicated fuelwood lots and the processing of wood wastes into woody biomass resources. The application of these renewable resources by targeted communities will help reduce the use of fossil fuels currently under consumption. When processed in efficient forms such as chips, pellets and briquets, woody biomass results in effective combustion and reduces energy use as well as emissions. Moreover, the use of efficient technologies also leads to better energy utilization of this sustainable resource, thus helping to attain environmental and socio-economic benefits when put to use by community members. This process will be complemented by the plantation of trees that are part of the government's flagship program' Ten Billion Tree Tsunami'.

The promotion of renewable energy to communities in the rural areas through the project directly contributes to the achievement of GEF-7 core indictors 1,4,6,11as well as the following Sustainable Development Goals (SDGs) of the United Nations:

- Goal 7: "Ensure access to affordable, reliable, sustainable and modern energy for all"
- Goal 8: "Promote sustained, inclusive and sustainable economic growth, full and productive employment and decent work for all",
- Goal 11: "Make cities and human settlements inclusive, safe, resilient and sustainable",
- Goal 13: "Take urgent action to combat climate change and its impacts by regulating emissions and promoting developments in renewable energy",

The increased use of sustainable energy resource in the form of woody biomass will contribute to the government's goal of increasing the share of renewable resources to 30% by the year 2030. Moreover, the project will generate climate co-benefits through the reduction in the emissions of greenhouse gases by switching to efficient energy production methods and reduction in fossil fuels. The emission reductions are also in line with Pakistan's Intended Nationally Determined Contribution (PAK-INDC) as part of the Paris Agreement within the UNFCCC. The project also encompasses the strengthening of policy frameworks on natural forest use, enhancing capacities for implementing sustainable natural resource management practices and improving the ecosystem services in these areas. The lessons learned and best practices observed through monitoring of the project could be shared with other communities and used to implement similar projects.

The project is expected to provide financial support to farmers and community organizations establishing farmers/communal nurseries over around 300 ha to provide planting stock for raising woody biomass on forested, non-forested and farmlands. The project will also establish energy plantations (over 40,000) to support woody biomass production that will take care of the biomass energy needs for power generation. Farmers in the targeted population will be encouraged to support sustainable fuelwood production through the planting of locally adaptive tree species. Measures will also be taken to safeguard the environment and ensure that there are no unintended harmful effects on the ecosystems and local biodiversity. For this purpose, feasibility studies will be carried out in the selected sites by a biodiversity and safeguards specialist during the project design and development (PPG) stage and a Biodiversity Action Plan will also be prepared during to alleviative any risks.

Briefly describe in the space below how the project strengthens accountability to stakeholders

The project aims to strengthen the accountability mechanisms by supporting the development or strengthening of policies related to sustainable use of renewable resources. Component 1 of the project focuses on 'technical assistance activities addressing the policy/regulatory and institutional barriers, and the expected outcome is the effective enforcement of policies and regulations on the sustainable production and use of woody biomass for energy generation and utilization in rural areas of Pakistan.' The project will aim to ensure that these policies constitute a mechanism of protecting the interests of all stakeholders involved, particularly the target communities. By providing this policy and statutory umbrella to ecosystem conservation and promotion of renewable resources, as well as engagement of the community, the project will directly result in strengthened accountability of the project activities and interventions.

The scale and cross-cutting design of the project means that it constitutes multiple governance and landscape approaches. The project strategy is therefore reliant on engagement, among different administrative levels (e.g., national, provincial, district), across different sectors (e.g., energy, forestry, agriculture, disaster management, community development, etc.), and involves a multitude of stakeholders including governmental, civil society, private sector, and research-academic institutions.

Stakeholder participation is an integral part of the project with a particular focus on women, children and any marginalized groups in all of the project sites. Stakeholder consultation activities carried out under the/as part of the Government's Ten Billion Tree Tsunami Program were focused on the technical and financial support that will be provided to local farmers and community organizations to help them establish farmers/communal nurseries to provide planting stock for raising woody biomass.

Under the project agreement, implementing partners of the project are responsible for

- Conducting project and programme-related activities in a line with the UNDP Social and Environmental Standards
- Implement the project Environmental and Social Management Framework, including any management or mitigation plan(s) prepared for the project to comply with such standards
- Engage in a productive manner to address any concerns and complaints raised, including through the project level grievance redress mechanism (clarified in ESMF) and through UNDP's Accountability Mechanism (www.undp.org/secu-srm) . Ensure that communities and other project stakeholders are aware of and have access to the relevant grievance mechanisms.

Part B. Identifying and Managing Social and Environmental Risks

QUESTION 2: What are the Potential Social and Environmental Risks? *Note: Complete SESP Attachment 1 before responding to Question 2.*	**QUESTION 3: What is the level of significance of the potential social and environmental risks?** *Note: Respond to Questions 4 and 5below before proceeding to Question 5*			**QUESTION 6: Describe the assessment and management measures for each risk rated Moderate, Substantial or High**
Risk Description (broken down by event, cause, impact)	***Impact and Likelihood (1-5)***	***Significance (Low, Moderate Substantial, High)***	***Comments (optional)***	***Description of assessment and management measures for risks rated as Moderate, Substantial or High***
Risk 1: Given the current security situation in Pakistan, the project could inadvertently expose local communities and/or project	I = 3 L = 2	**Moderate**	Those at risk to be affected by any security issues includes both direct and indirect workers as well as local communities.	Though the overall security situation of the country is considered to be safe, there are differences within some of the provinces. Project sites in Punjab, Sindh and Balochistan do not face any current or foreseeable security threat. However, some parts of KP can be deemed to be vulnerable because of the regional geo-political situation. Apart from affecting project staff and

workers to adverse security risks/hazards. **[P. 5, Standard 3.1, 3.2, 3.3 & 3.8]**				community members, particularly women, any acts of violence could also affect the landscape and result in potential harmful effects on the environment. As part of the plan to avoid any possible security threats, target areas will be selected with the aim of avoiding any insecure/politically sensitize areas. Effective coordination with provincial and district administration (and security agencies if needed) will help to reduce any possible threats in the region. Any threats to the communities inhabiting the region, project personnel and women will be assessed periodically, with preventative measures being established when/where required through the assistance of relevant government personnel. The project will be adopting a participatory approach and comprehensively involving all local stakeholders. This will assist in countering any social instability and maintain the flow of activities: strong partnerships based on mutual trust with local businesses and traders working with biomass will ensure that any minor or moderate security threat will not affect business and the livelihoods of community members associated with it. This will be guided by the Environmental and Social Management Plan (ESMP) to be developed for each demo site that will require installation, construction and operations following the Environmental and social Management Framework (ESMF) as part of the project design to ensure this risk is further assessed and relevant management measures in place.
Risk 2: The regions in which the project is being proposed have been exposed to increased natural hazard events recently. Physical infrastructure being supported by the project may run the risk of being designed in a maladaptive manner, and could pose a threat to both users and the surrounding community/environment as a whole. **[Standard 2.1, 2.3]**	I = 3 L = 3	**Moderate**	Pakistan is ranked very high on the list of countries vulnerable to climate change along with being one of the least prepared to face any disaster hazards. Some of the ecosystems in the project are vulnerable to the impacts of climate change including glacial lake outburst floods, storms, flooding, landslides, extreme heat, earthquakes, change in rainfall patterns and water scarcity.	Given the country's vulnerable climate change events and destruction caused by heavy floods in Punjab and Sindh over the last decade or so, disaster hazards present a moderate threat to the project's activities. Moreover, high mountain ecosystems that are part of the project landscape such as those in KP are also vulnerable to climate change: unpredictable events such as higher than usual rainfall, storms etc. could impact the lives of project beneficiaries as well as the environment including trees and newly planted saplings used for woody biomass. This increases the importance of the need to integrate climate uncertainties in the management models. **Annex 6A**: PASBET Climate Risks Assessment (PASBET ProDoc) presents the climatic risk situation in each project district of the 4 provinces that are involved in the project. The ESMF also prescribes that further assessment will be carried out during the project implementation in coordination with the relevant national and regional government agencies (e.g., MoCC&EC, Pakistan Metrological Department, District Disaster

				Management Authorities). The project design has been centered around the resilience of forest plantations as well as the communities in the areas. In the case of natural disasters such as floods, improved woody biomass management, including raising wood lots and growing fuelwood trees on farmlands will help minimize the damage to the environment and communities. Similarly, the restoration and agroforestry activities will revolve around local species that can adapt quickly to conditions and more resilient in nature. Demo units in these areas will be compliant with widely accepted internationally standards and state-of-the-art infrastructure and engineering designs. Moreover, management measures such as early warning systems and district and village level emergency response teams, etc. will be incorporated in the projects planning. This will require strong communication with the respective governments in the area and an awareness of the strength of their disaster management facilities. Similarly, with Risk 1, ESMPs will be prepared for each demo site. **Annex 6A**: PASBET Climate Risks Assessment (PASBET ProDoc) provides the initial information about the potential climate change-related events that may happen in Pakistan, particularly for the districts selected. This information will be further augmented during project inception, and with the development of the demo site ESMPs.
Risk 3: The project could inadvertently increase risk of exposure of communities and/or project workers to Covid19 and other communicable disease through stakeholder consultations and workshops. . **[P7.6; Standard 3.4]**	I = 2 L = 2	**Low**	Though Pakistan has managed to contain the threat of COVID-19, the fourth wave is still on-going in the country. Coupled with the relative lack of precautions taken in rural areas, it presents a moderate threat to the activities in project areas in case of a breakout.	While the SES state that no assessment and management measures are needed for low risk activities, Annex 6B: PASBET Project and Covid-19 (PASBET ProDoc) presents the initial assessment and plan of action to address Covid-19 during the implementation of the project. A more definitive **Covid19 Analysis and Action Plan** will be prepared in coordination with the preparation of the project's **ESMF** to ensure that the potential community health impacts of these eventualities are addressed.
Risk 4: Local communities and farmers do not have the capacities or interest to adopt forestry interventions and sustainable production of woody biomass in their agriculture field. As	I = 3 L =2	**Moderate**	Local communities and farmers could have limited capacities for or be reluctant to adopt new technologies given the possible financial risks associated with the	To better understand as well as alleviate the fears of targeted community members, stakeholder engagement (with the Stakeholder Engagement Plan initially prepared as part of the PPG and to be updated during the inception period) along with public awareness campaigns will be implemented to engage and inform farmers of new technologies and their potential benefits, as well as the possible advantages over the current ones.

such, there is the potential that unsustainable agricultural practices and forest harvesting could be used to provide fuel for the biomass plants. **[Standard 1.1, 1.2]**			changes. Given that the project will be targeting communities falling under low-income areas, farmers/community members could be reluctant to adhere to newly introduced practices and systems. Most of the targeted population relies on a single source of resource/activity for their livelihoods and adapting to a new technology could be seen as a move which could have serious effects on their finances if it doesn't not work as planned.	Moreover, strong market linkages created as part of the project will encourage farmers to switch fuelwood production on less productive lands, given the security of income associated with it. It is expected that the participatory nature of the project activities couples with incentives linked to investing in energy plantations as well as the use of efficient technologies will encourage communities/farmers adhere to the proposed interventions. The project will also implement measures to monitor compliance to the new technologies used by farmers to minimize the risk of environmental damage. Similarly, with Risk 1, ESMP will be prepared for each demo site considering this risk at the inception phase.
Risk 5: Introduction of efficient wood-fired equipment in areas having natural forests may create a perverse incentive, which may lead to an increase in the consumption of woody biomass leading further degradation of natural primary forests. Furthermore there might be a risk the use of woody biomass becomes more economically attractive in alternative use. **[P.2 & P.7; Standard 1.1; 1.3; 1.8]**	I = 3 L = 3	**Moderate**	Successful implementation of activities and adaption by farmers/community members could see excessive use of the forests for fuelwood i.e., increased energy generation at the cost of more resources being used than replaced in standing natural forests. This could lead to a greater use of inefficient methods and a long-term degradation of the forest if left unchecked	This threat will be countered by activity-design, ensuring that the power generation and EE equipment introduced in applicable rural industries have appropriate licenses and following regulatory guidelines. These will be monitored through relevant bodies applicable to provinces i.e., provincial forest departments and local authorities. This could be implemented through a certification scheme for the use of proper equipment. This will be complemented through strict checks and balances to limit the use of woody biomass and ensure its sustainability. Another possible step is the avoidance of areas close to natural forests as a choice of projects sites. Finally, alternate biomass e.g., crop residuals could also be looked at as an alternate to avoid an imbalance in the environment through the use of forest wood at an unsustainable rate. Similarly with Risk 1, ESMP will be prepared for each demo site considering this risk at the inception phase. Risks to biodiversity and natural habitats will be assessed as part of the demo site ESIAs that are to be undertaken before the implementation of related outputs.
Risk 6: Woody biomass use could exceed production. Even fast-growing	I = 3 L = 3	**Moderate**	This displacement of biomass resources at an unsustainable rate will result in the increase	Success in the initial stages of the project could result in the excessive use of natural resources by the community without timely replacements i.e., community members/farmers could start depleting resources of firewood

species will take up to five years to mature. Moreover, climate change risks (flash floods and prolonged drought) and rainfall variability could impact production of woody biomass, which could result in the over harvesting of other biomass sources. **[Standard 1.6; 2.2; 2.3; 2.4]**			in overall emissions in the region due to a depletion of wood resources	without replacing it at the same/sustainable rate. Not only do species used for firewood take years to mature, the use of alien or unsuitable species to speed up the process can have harmful effects on the environment and hamper the sustainability process in the long run. The project will have measures in place to avoid excessive and unsustainable use of firewood in the designated areas. This will be achieved by developing strict, specific criteria to monitor areas that provide woody biomass to rural industries. Checks and balances on the following will allow effective monitoring of the situation: • Availability of woody biomass, • Access to market supplies, and past production experience Similarly, rural industries in the regions will be closely monitored in terms of their use of firewood and will also be regulated with the assistance of the forest department and any other relevant body. Moreover, the PPG phase will be used to develop more criteria to keep a check on the demand and supply numbers of firewood. Similar with Risk 1, ESIAs and ESMP will be prepared for each demo site that will consider this risk.
Risk 7: Project activities and approaches might not fully incorporate or reflect views of women and girls and ensure equitable opportunities for their involvement and benefit; and there is a risk that a prolonged or recurrent COVID-19 pandemic would exacerbate gender inequality, including risk of GBV. **[P.9, P.10, P.11, P.12]**	I = 3 L = 3	**Moderate**	Gender mainstreaming should be an essential part of all the activities in the project design phase	Women play an essential role in the biomass energy cycle in terms of their share of work i.e., they are responsible for collection of firewood in most of the project areas which leaves them vulnerable to sexual violence or gender-based violence (GBV) as a serious violation of human rights and a life-threatening health and protection issue. This has been flagged as a Yes in the risk checklist below. Moreover, women are primarily responsible for the use of firewood when cooking in their homes which exposes them to harmful emissions from the use of firewood and has several long-term implications on their health and well-being. In a number of areas, often due to prevalent cultural practices, women do not get the opportunity to take decisions regarding the use of fuelwood-based energy generation even though they suffer the most in terms of the risks associated with the process Gender mainstreaming is an integral part of the process and will be incorporated at all levels of implementation. The use of environmentally friendly technology will play a role in reducing the harmful effects of fuelwood use for women. More importantly, women will be involved be involved in participatory discussions at all stages of the projects keeping in

				mind the cultural and social norms of the region. Women will also be given decision making powers at all levels to ensure that their input makes a difference and helps to improve their lives. A gender expert will also be hired to sensitize project and government staff on gender issues so that any steps and polices that are formulated inculcate the concerns of women in the region. A **Gender Action Plan** has been prepared during the PPG phase and will be implemented for the project. This plan, includes and addresses GBV issues and considers the issues faced by women in all project aspects and steps that can be taken to involve women in the decision-making process, empower them in different positions, reduce any violence and threats that they face including those related to the environment. In addition, the project **ESMF** will define specific environmental and social safeguard procedures for the project, including consideration of gender related risks. The **Stakeholder Engagement Plan** also ensures participation and engagement of women and girls to ensure that all relevant issues are tackled, and appropriate management measures are put in place.
Risk 8: The project proponents may not effectively engage and ensure the participation of all stakeholders during the project design and the implementation phase resulting in human rights violation. **[P.2, P.3,P.4, P.6, P.7, P.14]** Standard 6.4	I = 3 L = 4	**Moderate**	Participation of the local community and integration of their views in the project plans is a vital cog in the successful implementation of the project	The participation of the targeted community during the project planning stage is critical activity that needs to be planned effectively. Project staff could make the mistake of using a 'one size fits all' model with proved effectiveness in some other part of the world, without consulting essential issues with the locals whose lives will be affected by these decisions. A participatory approach will be in place at all stages of the project to maximize the involvement of the community members keeping in mind the cultural context in the areas: consultations will be held with community members in all project areas where demonstrations will showcase the application of RE and EE technologies for woody biomass energy production. The discussions will be a two-way process to ensure that the input of communities is also integrated in the project: the views and requirements of the beneficiaries will be considered to ensure that their interests are catered to. Information exchange and learning for all members of the society will be an integral part of the knowledge management and information dissemination activities. Effective stakeholder engagement approaches on these lines will ensure that local communities do not face restrictions in access to resources, and anticipated livelihood benefits materialize.

				A **Stakeholder Engagement Plan** is prepared for the project to ensure effective and meaningful stakeholder engagement throughout. In addition, the **ESMF** will ensure stakeholders are engaged in the environmental and social assessment and management processes and a **grievance mechanism** is in place to address concerns.
Risk 9: Local communities might face restrictions in access to resources, and anticipated livelihood benefits to local people might not materialize as planned, potentially leading to economic displacement. **[Standard 5.2; 5.4]**	I = 3 L = 2	**Moderate**		Sites that would result in economic or physical displacement will be avoided. However, this risk could be caused by not considering the local conditions and requirements of the project in different areas and/or incorporating the locals/beneficiaries in the decision-making process, leading to inadvertent economic displacement. This could also be caused by implementation issues and eventually result in the beneficiaries losing their livelihood resources. Given that this might/could be the primary source of income for the beneficiaries, issues with implementation could possibly lead to economic displacement for some of them. The project will seek to avoid such impacts. The potential for physical displacement will also be confirmed and avoided. Effective stakeholder engagement approaches will ensure that local communities do not face restrictions in access to resources, and anticipated livelihood benefits materialize i.e., beneficiaries will be involved at all stages of the project to ensure that the final product incorporates their requirements and their livelihood is not affected in a perverse way. A **Stakeholder Engagement Plan** is prepared for the project to ensure effective and meaningful stakeholder engagement throughout. In order that the risk will be addressed fully, the ESMF, ESIAs and subsequent ESMPs will assess this risk further to confirm that initial assumptions that the project will not result in economic or physical displacement are correct. Any physical displacement will be avoided. If economic displacement risks are identified and can not be avoided, appropriate and realistically implemented and sustainable Livelihood Action Plans will be prepared and promoted and the project SESP also updated accordingly to reflect the relevance of SES Standard 5.
Risk 10: The project activities could result in multiple sources of ambient pollution and affect the environment in a harmful manner. **[Standard 8.1; 8.2; 8.5]**	I = 2 L = 2	**Low**	Activities including the plantation of trees as well as waste generation from biomass production could pollute the environment. The project could be deemed	No measures needed as the risk is Low.

			detrimental to the environment through processes associated with planting of new trees; as well as generation of waste from the production of biomass and the release of pollutants to the environment. The planting of new trees can involve the use of pesticides or other materials that can have a negative effect on the environment and human health. There is also the possibility of the consumption of other raw materials and water for the plantation to be effective. The biomass production setups could be the source of hazardous or non-hazardous waste being released. The biomass-based power production technology being used in the project will ensure the reduction of pollution compared to the current use of biomass prevalent in the project regions. This will be particularly beneficial for women in the long run who are exposed to harmful emittances from biomass use in traditional ways. The new technologies used for power generation are likely to reduce any potential hazardous or non-hazardous waste and	

			ensure that the environment is not compromised in any manner. The formation of nurseries and plantation of trees in selected sites is in line with similar initiatives of the government. The use of pesticides in such projects is managed based on regulations defined by the government: pesticides that are banned/being phased out are not used given their harmful effects on the soil.	
Risk 11: The Project may create adverse biodiversity impacts to habitats and/or ecosystems and ecosystem services, e.g. through habitat loss, conversion or degradation, fragmentation, hydrological changes; and activities within or adjacent to critical habitats and/or environmentally sensitive areas, including (but not limited to) legally protected areas, areas proposed for protection, or recognized as such by authoritative sources and/or indigenous peoples or local communities. **[Standard 1]**	I = 3 L = 2	**Moderate**	This will be included in the overall ESMF and subsequent ESMPs with due consultation and participation with appropriate agencies dealing with animal habitats as well as, if any administrators of legally protected areas. As indicated as having a Yes in the checklist below, the project may result in adverse impacts to habitats could be deemed detrimental to the environment through processes associated with planting of new trees; as well as generation of waste from the production. This risk will be considered in the project ESMF as well as subsequent ESMPs completed during the early part of Year 1. A participatory approach	No measures needed as the risk is Low.

			will be in place at all stages of the project to maximize the involvement of the appropriate agencies dealing with animal habitats and ecosystems, community members keeping in mind the legally protected areas as well as the cultural context in the areas.	
Risk 12: Site for project demo and woody biomass wood lot might affect tribal peoples and cultural heritage. **[Standard 4.1, 4.2]** **Standard 6.1, 6.2 6.3**	I=4 L=2	**Moderate**		Site specific impacts on cultural heritage may occur, further assessment of this will take place, as part of the ESIA process, once the exact project demo sites have been determined. The presence of "indigenous peoples", as per UNDP SES6, was assessed during project design, with an accompanying due diligence report being produced included in the ESMF. The findings of this exercise indicated that SES6 should be triggered, and that groups falling within the UNDP SES6 criteria could be present at sites in the following provinces: Punjab, Sindh, and Balochistan. The DD report determined that outputs 2.1.3,2.1.4, 2.2.1 may have impacts on the livelihoods, rights, and resources of these potentially affected indigenous peoples. As such, the examination of ownership and usage rights to lands, territories and resources of IPs will be included within the scope of the ESIAs at each of the demo sites. Potential impacts on cultural heritage will be considered in the ESMF and chance find procedures as well as measures to avoid and mininize impacts on intangible cultural heritage put in place where needed, including in the elaboration of site-specific ESMPs. If adverse impacts on IPs are identified, IPPs must be developed and FPIC pursued prior to initiation of any activities that may impact indigenous peoples. This will tentatively be required for Outputs 2.1.3, 2.1.4, and 2.2.1 and in the areas surrounding the demo sites, where the establishment of forest nurseries is foreseen, and in the areas where fuelwood will initially be sourced. The affected sites and surrounding areas are likely to be around the following demo sites: ▪ Sindh – Sujawal ▪ Punjab – Bhakkar and Layyah ▪ Balochistan – Dera Bughti

Risk 13: Working conditions may not meet the requirements of national labor laws and international commitments. **[Standard 7.1]**	I=3 L=3	**Moderate**		As the project will involve labor for construction, on plantations and potentially in other setting, there is a risk of national labor laws and international commitments not being met. Hence, monitoring of labor and working conditions must be integrated into the ESIA/ESMP(s) that will be prepared for each demo site. Particularly, health and safety issues should be prevented by implementing appropriate training for workers on site. To manage the risks Labor Management Procedures should be used to provide an overview of Labor Use in the Project, further assess the key potential labor risks, assign responsible staff, define policies and procedures and outline an appropriate actions plan.

QUESTION 4: What is the overall project risk categorization?

Low Risk	☐	
Moderate Risk	☐	
Substantial Risk	**X**	While all risks are identified as "Moderate" or below, since there are multiple risks across many of the standards, the overall risk-rating for the project is "Substantial".
High Risk	☐	

QUESTION 5: Based on the identified risks and risk categorization, what requirements of the SES are triggered? (check all that apply)

Question only required for Moderate, Substantial and High Risk projects

Is assessment required? (check if "yes")	**X**			***Status? (completed, planned)***
if yes, indicate overall type and status		**X**	Targeted assessment(s)	
		X	- Gender Analysis and Gender Action Plan	Completed
		X	- Landscape profiles	to be prepared under project as activity
		X	- Climate and Disaster Risk Screening	Completed
		X	- COVID-19 Analysis and Action Framework	Completed
		X	ESIA (Environmental and Social Impact Assessment)	Planned during project implementation. As per the ESMF these will cover the demo

					sites. ESIAs at the demo sites in Sindh, Punjab and Balochistan will include targeted assessment of SES6 issues.
	Are management plans required? (check if "yes)	**X**			
	If yes, indicate overall type		**X**	Targeted management plans	
			X	- Gender Analysis and Gender Action Plan	Completed during project preparation
			X	- Stakeholder Engagement Plan	Livelihood Action Plans to be adopted in conjunction with the Stakeholder Engagement and ESMP
			X	- COVID-19 Analysis and Action Framework	Completed during project preparation
				- Indigenous Peoples Plan (IPP)	Applicability to undertake an IPP will be determined during the conduct of demo site ESIAs, as prescribed in the ESMF.
			X	ESMP (Environmental and Social Management Plan which may include range of targeted plans)	Planned during project implementation per the ESMF. The ESMF has prescribed one ESMP per demo site.
			X	ESMF (Environmental and Social Management Framework)	To be prepared prior to project approval.
	Based on identified <u>risks</u>, which Principles/Project-level Standards triggered?		**Comments**		
	Overarching Principle: Leave No One Behind				
	Human Rights	**X**			

	Gender Equality and Women's Empowerment	**X**	
	Accountability	**X**	
	1. Biodiversity Conservation and Sustainable Natural Resource Management	**X**	
	2. Climate Change and Disaster Risks	**X**	
	3. Community Health, Safety and Security	**X**	
	4. Cultural Heritage	**X**	
	5. Displacement and Resettlement	☐	**Low.** (To be further confirmed during ESMF and ESMPs and ongoing stakeholder engagement. If risk is identified then SESP will be updated and relevant safeguards measures put in place per the SES.) .
	6. Indigenous Peoples	☐	This will be confirmed further through ongoing stakeholder engagement and site-specific ESMPs.
	7. Labor and Working Conditions	**X**	
	8. Pollution Prevention and Resource Efficiency	☐	**Low.** (To be further confirmed during ESMF and ESMPs and ongoing stakeholder engagement. If risk is identified then SESP will be updated and relevant safeguards measures put in place per the SES.)

Final Sign Off

Final Screening at the design-stage is not complete until the following signatures are included

Signature	***Date***	***Description***
QA Assessor		UNDP staff member responsible for the project, typically a UNDP Programme Officer. Final signature confirms they have "checked" to ensure that the SESP is adequately conducted.
QA Approver		UNDP senior manager, typically the UNDP Deputy Country Director (DCD), Country Director (CD), Deputy Resident Representative (DRR), or Resident Representative (RR). The QA Approver cannot also be the QA Assessor. Final signature confirms they have "cleared" the SESP prior to submittal to the PAC.
PAC Chair		UNDP chair of the PAC. In some cases PAC Chair may also be the QA Approver. Final signature confirms that the SESP was considered as part of the project appraisal and considered in recommendations of the PAC.

SESP Attachment 1. Social and Environmental Risk Screening Checklist

Checklist Potential Social and Environmental Risks		
INSTRUCTIONS: The risk screening checklist will assist in answering Questions 2-6 of the Screening Template. Answers to the checklist questions help to (1) identify potential risks, (2) determine the overall risk categorization of the project, and (3) determine required level of assessment and management measures. Refer to the SES toolkit for further guidance on addressing screening questions.		
Overarching Principle: Leave No One Behind **Human Rights**		**Answer (Yes/No)**
P.1	Have local communities or individuals raised human rights concerns regarding the project (e.g. during the stakeholder engagement process, grievance processes, public statements)?	Yes
P.2	Is there a risk that duty-bearers (e.g. government agencies) do not have the capacity to meet their obligations in the project?	Yes
P.3	Is there a risk that rights-holders (e.g. project-affected persons) do not have the capacity to claim their rights?	Yes
Would the project potentially involve or lead to:		
P.4	adverse impacts on enjoyment of the human rights (civil, political, economic, social or cultural) of the affected population and particularly of marginalized groups?	Yes
P.5	inequitable or discriminatory impacts on affected populations, particularly people living in poverty or marginalized or excluded individuals or groups, including persons with disabilities? [83]	No
P.6	restrictions in availability, quality of and/or access to resources or basic services, in particular to marginalized individuals or groups, including persons with disabilities?	Yes
P.7	exacerbation of conflicts among and/or the risk of violence to project-affected communities and individuals?	Yes
Gender Equality and Women's Empowerment		
P.8	Have women's groups/leaders raised gender equality concerns regarding the project, (e.g. during the stakeholder engagement process, grievance processes, public statements)?	No
Would the project potentially involve or lead to:		
P.9	adverse impacts on gender equality and/or the situation of women and girls?	Yes
P.10	reproducing discriminations against women based on gender, especially regarding participation in design and implementation or access to opportunities and benefits?	Yes
P.11	limitations on women's ability to use, develop and protect natural resources, taking into account different roles and positions of women and men in accessing environmental goods and services? *For example, activities that could lead to natural resources degradation or depletion in communities who depend on these resources for their livelihoods and well being*	Yes
P.12	exacerbation of risks of gender-based violence? *For example, through the influx of workers to a community, changes in community and household power dynamics, increased exposure to unsafe public places and/or transport, etc.*	Yes
Sustainability and Resilience: Screening questions regarding risks associated with sustainability and resilience are encompassed by the Standard-specific questions below		
Accountability		
Would the project potentially involve or lead to:		
P.13	exclusion of any potentially affected stakeholders, in particular marginalized groups and excluded individuals (including persons with disabilities), from fully participating in decisions that may affect them?	Yes

[83] Prohibited grounds of discrimination include race, ethnicity, sex, age, language, disability, sexual orientation, gender identity, religion, political or other opinion, national or social or geographical origin, property, birth or other status including as an indigenous person or as a member of a minority. References to "women and men" or similar is understood to include women and men, boys and girls, and other groups discriminated against based on their gender identities, such as transgender and transsexual people.

P.14	grievances or objections from potentially affected stakeholders?	Yes
P.15	risks of retaliation or reprisals against stakeholders who express concerns or grievances, or who seek to participate in or to obtain information on the project?	No
Project-Level Standards		
Standard 1: Biodiversity Conservation and Sustainable Natural Resource Management		
Would the project potentially involve or lead to:		
1.1	adverse impacts to habitats (e.g. modified, natural, and critical habitats) and/or ecosystems and ecosystem services? *For example, through habitat loss, conversion or degradation, fragmentation, hydrological changes*	Yes
1.2	activities within or adjacent to critical habitats and/or environmentally sensitive areas, including (but not limited to) legally protected areas (e.g. nature reserve, national park), areas proposed for protection, or recognized as such by authoritative sources and/or indigenous peoples or local communities?	Yes
1.3	changes to the use of lands and resources that may have adverse impacts on habitats, ecosystems, and/or livelihoods? (Note: if restrictions and/or limitations of access to lands would apply, refer to Standard 5)	Yes
1.4	risks to endangered species (e.g. reduction, encroachment on habitat)?	No
1.5	exacerbation of illegal wildlife trade?	No
1.6	introduction of invasive alien species?	Yes
1.7	adverse impacts on soils?	No
1.8	harvesting of natural forests, plantation development, or reforestation?	Yes
1.9	significant agricultural production?	No
1.10	animal husbandry or harvesting of fish populations or other aquatic species?	No
1.11	significant extraction, diversion or containment of surface or ground water? *For example, construction of dams, reservoirs, river basin developments, groundwater extraction*	No
1.12	handling or utilization of genetically modified organisms/living modified organisms?[84]	No
1.13	utilization of genetic resources? (e.g. collection and/or harvesting, commercial development)[85]	No
1.14	adverse transboundary or global environmental concerns?	No
Standard 2: Climate Change and Disaster Risks		
Would the project potentially involve or lead to:		
2.1	areas subject to hazards such as earthquakes, floods, landslides, severe winds, storm surges, tsunami or volcanic eruptions?	Yes
2.2	outputs and outcomes sensitive or vulnerable to potential impacts of climate change or disasters? *For example, through increased precipitation, drought, temperature, salinity, extreme events, earthquakes*	Yes
2.3	increases in vulnerability to climate change impacts or disaster risks now or in the future (also known as maladaptive or negative coping practices)? *For example, changes to land use planning may encourage further development of floodplains, potentially increasing the population's vulnerability to climate change, specifically flooding*	Yes
2.4	increases of greenhouse gas emissions, black carbon emissions or other drivers of climate change?	Yes
Standard 3: Community Health, Safety and Security		
Would the project potentially involve or lead to:		
3.1	construction and/or infrastructure development (e.g. roads, buildings, dams)? (Note: the GEF does not finance projects that would involve the construction or rehabilitation of large or	Yes

[84] See the Convention on Biological Diversity and its Cartagena Protocol on Biosafety.

[85] See the Convention on Biological Diversity and its Nagoya Protocol on access and benefit sharing from use of genetic resources.

	complex dams)	
3.2	air pollution, noise, vibration, traffic, injuries, physical hazards, poor surface water quality due to runoff, erosion, sanitation?	Yes
3.3	harm or losses due to failure of structural elements of the project (e.g. collapse of buildings or infrastructure)?	Yes
3.4	risks of water-borne or other vector-borne diseases (e.g. temporary breeding habitats), communicable and noncommunicable diseases, nutritional disorders, mental health?	Yes
3.5	transport, storage, and use and/or disposal of hazardous or dangerous materials (e.g. explosives, fuel and other chemicals during construction and operation)?	No
3.6	adverse impacts on ecosystems and ecosystem services relevant to communities' health (e.g. food, surface water purification, natural buffers from flooding)?	No
3.7	influx of project workers to project areas?	No
3.8	engagement of security personnel to protect facilities and property or to support project activities?	Yes
Standard 4: Cultural Heritage		
Would the project potentially involve or lead to:		
4.1	activities adjacent to or within a Cultural Heritage site?	Yes
4.2	significant excavations, demolitions, movement of earth, flooding or other environmental changes?	No
4.3	adverse impacts to sites, structures, or objects with historical, cultural, artistic, traditional or religious values or intangible forms of culture (e.g. knowledge, innovations, practices)? (Note: projects intended to protect, and conserve Cultural Heritage may also have inadvertent adverse impacts)	No
4.4	alterations to landscapes and natural features with cultural significance?	No
4.5	utilization of tangible and/or intangible forms (e.g. practices, traditional knowledge) of Cultural Heritage for commercial or other purposes?	No
Standard 5: Displacement and Resettlement		
Would the project potentially involve or lead to:		
5.1	temporary or permanent and full or partial physical displacement (including people without legally recognizable claims to land)?	No
5.2	economic displacement (e.g. loss of assets or access to resources due to land acquisition or access restrictions – even in the absence of physical relocation)?	Yes
5.3	risk of forced evictions?[86]	No
5.4	impacts on or changes to land tenure arrangements and/or community based property rights/customary rights to land, territories and/or resources?	Yes
Standard 6: Indigenous Peoples		
Would the project potentially involve or lead to:		
6.1	areas where indigenous peoples are present (including project area of influence)?	Yes
6.2	activities located on lands and territories claimed by indigenous peoples?	Yes
6.3	impacts (positive or negative) to the human rights, lands, natural resources, territories, and traditional livelihoods of indigenous peoples (regardless of whether indigenous peoples possess the legal titles to such areas, whether the project is located within or outside of the lands and territories inhabited by the affected peoples, or whether the indigenous peoples are recognized as indigenous peoples by the country in question)? *If the answer to screening question 6.3 is "yes", then the potential risk impacts are considered significant and the project would be categorized as either Substantial Risk or High Risk*	Yes
6.4	the absence of culturally appropriate consultations carried out with the objective of achieving FPIC on matters that may affect the rights and interests, lands, resources, territories and	Yes

[86] Forced eviction is defined here as the permanent or temporary removal against their will of individuals, families or communities from the homes and/or land which they occupy, without the provision of, and access to, appropriate forms of legal or other protection. Forced evictions constitute gross violations of a range of internationally recognized human rights.

	traditional livelihoods of the indigenous peoples concerned?	
6.5	the utilization and/or commercial development of natural resources on lands and territories claimed by indigenous peoples?	No
6.6	forced eviction or the whole or partial physical or economic displacement of indigenous peoples, including through access restrictions to lands, territories, and resources? *Consider, and where appropriate ensure, consistency with the answers under Standard 5 above*	No
6.7	adverse impacts on the development priorities of indigenous peoples as defined by them?	No
6.8	risks to the physical and cultural survival of indigenous peoples?	No
6.9	impacts on the Cultural Heritage of indigenous peoples, including through the commercialization or use of their traditional knowledge and practices? *Consider, and where appropriate ensure, consistency with the answers under Standard 4 above.*	Yes
Standard 7: Labor and Working Conditions		
Would the project potentially involve or lead to: (note: applies to project and contractor workers)		
7.1	working conditions that do not meet national labor laws and international commitments?	Yes
7.2	working conditions that may deny freedom of association and collective bargaining?	No
7.3	use of child labor?	No
7.4	use of forced labor?	No
7.5	discriminatory working conditions and/or lack of equal opportunity?	No
7.6	occupational health and safety risks due to physical, chemical, biological and psychosocial hazards (including violence and harassment) throughout the project life-cycle?	Yes
Standard 8: Pollution Prevention and Resource Efficiency		
Would the project potentially involve or lead to:		
8.1	the release of pollutants to the environment due to routine or non-routine circumstances with the potential for adverse local, regional, and/or transboundary impacts?	Yes
8.2	the generation of waste (both hazardous and non-hazardous)?	Yes
8.3	the manufacture, trade, release, and/or use of hazardous materials and/or chemicals?	No
8.4	the use of chemicals or materials subject to international bans or phase-outs? *For example, DDT, PCBs and other chemicals listed in international conventions such as the Montreal Protocol, Minamata Convention, Basel Convention, Rotterdam Convention, Stockholm Convention*	No
8.5	the application of pesticides that may have a negative effect on the environment or human health?	Yes
8.6	significant consumption of raw materials, energy, and/or water?	No

ANNEX 6: UNDP RISK REGISTER

Risks:

As per standard UNDP requirements, the National Project Coordinator will monitor risks quarterly and report on the status of s to the UNDP Country Office. The UNDP Country Office will record progress in the UNDP ATLAS risk log. Risks will be reported as critical when the impact and probability are high (i.e., when impact is rated as 5, and when impact is rated as 4 and probability is rated at 3 or higher). Management responses to critical risks will also be reported to the GEF in the annual Project Implementation Review (PIR).

The risks that might prevent the project objectives from being achieved, during the project implementation, are listed as follows:

#	Event	Cause	Impact(s)	Risk Category and Sub-category *(including Risk Appetite)*	Impact, Likelihood & Risk Level *(see Annex 3 Risk Matrix)*	Risk Valid From/To	Risk Owner *(individual accountable for managing the risk)*	Risk Treatment and Treatment Owner
1	There is a risk that Unexpected cultural or security barrier/situation (e.g., law and order)	As a result Of may delay project implementation in some project districts or hinder activities	Which will impact in meeting the objectives of the project	8. SAFETY AND SECURITY (8.2. Political instability) - UNDP Risk Appetite: CAUTIOUS	Likelihood: **3 - Moderately likely** Impact: **1 - Negligible** Risk level: **LOW (equates to a risk appetite of CAUTIOUS)**	From: 01-Oct-23 To: 31-Dec-25	Provincial departments PMU MoCC&EC	Risk Treatment 1.1: Awareness campaigns and information drive on the value and benefits of the project for all stakeholders and local leaders that they may be motivated to continuously support the successful implementation of the project. Insecure project sites were avoided in the selection of target demonstration and logistics areas. By adopting a participatory approach and involving all local stakeholders, risks related to social instability is reduced. In addition, the project's partnerships with local entrepreneurs and biomass traders will be established through regular consultation and feedback to ensure that project interventions continue even under moderate security threats.

#	Event	Cause	Impact(s)	Risk Category and Sub-category *(including Risk Appetite)*	Impact, Likelihood & Risk Level *(see Annex 3 Risk Matrix)*	Risk Valid From/To	Risk Owner *(individual accountable for managing the risk)*	Risk Treatment and Treatment Owner
								Risk Treatment Owner: …
								Risk Treatment 1.2 Proper coordination with the provincial line department, security agencies and district administration during field activities reduce/eliminate security threats or cultural barriers during the field operations. Consultations with the local leaders and stakeholders on the nature of the concern or barrier and involve them in the solution of the problem and means to avoid recurrence with the realization of the overall benefit of the project interventions to the communities and to realize sense of ownership. Risk Treatment Owner: …

#	Event	Cause	Impact(s)	Risk Category and Sub-category *(including Risk Appetite)*	Impact, Likelihood & Risk Level *(see Annex 3 Risk Matrix)*	Risk Valid From/To	Risk Owner *(individual accountable for managing the risk)*	Risk Treatment and Treatment Owner
2	There is a risk that there might be a resistance or low level participation of individuals, entrepreneurs and SMEs	As a result of this project activities may not be implemented in full spirit	Which will impact in achieving overall project objectives	1. SOCIAL AND ENVIRONMENTAL (1.4. Biodiversity conservation and sustainable natural resource management) - UNDP Risk Appetite: CAUTIOUS	Likelihood: **1 - Not likely** Impact: **3 - Intermediate** Risk level: **LOW (equates to a risk appetite of MINIMAL)**	From: 01-Oct-23 To: 31-Mar-24	PMU MoCC&EC	Risk Treatment 2.1: The project will be implemented with the involvement of local communities, leaders, biomass traders and rural industries to enhance appreciation of the benefits. Wherever possible, their legitimate interests and socio-economic needs will be safe guarded. Risk Treatment Owner: …
								Risk Treatment 2.2: The project will explore the possibilities of providing incentives (social, economic, and technological) to local farmers and entrepreneurs to encourage their participation in project designing and implementation. Risk Treatment Owner: …

#	Event	Cause	Impact(s)	Risk Category and Sub-category *(including Risk Appetite)*	Impact, Likelihood & Risk Level *(see Annex 3 Risk Matrix)*	Risk Valid From/To	Risk Owner *(individual accountable for managing the risk)*	Risk Treatment and Treatment Owner
								Risk Treatment 2.3: … Risk Treatment Owner: …
3	There is a risk that Project activities might not be designed in a way to respond the gender inclusion and and approaches might not fully incorporate or reflect views of women and girls and ensure equitable opportunities for their involvement and benefit; and there is a risk that a prolonged or recurrent COVID-19 pandemic would exacerbate gender inequality, including risk of GBV. And potential occurrence of discriminations against women based on gender, especially	As a result of not appropriately planning the activities, approaches might not fully incorporate or reflect views of women and girls and ensure equitable opportunities for their involvement and benefit	Which will impact in missing out the gender based targets in the project lifecycle	1. SOCIAL AND ENVIRONMENTAL (1.2. Gender equality and women's empowerment) - UNDP Risk Appetite: CAUTIOUS	Likelihood: **3 - Moderately likely** Impact: **1 - Negligible** Risk level: **LOW (equates to a risk appetite of MINIMAL)**	From: 01-Oct-23 To: 31-Dec-24	PMU MoCC&EC	Risk Treatment 3.1: *Preventive:* The project will implement gender mainstreaming as an integral part of the process and will be incorporated at all levels of implementation. The use of environmentally friendly technology will play a role in reducing the harmful effects of fuelwood use for women. More importantly, women will be involved in participatory discussions at all stages of the projects keeping in mind the cultural and social norms of the region. Women will also be given decision making powers at all levels to ensure that their input makes a difference and helps to improve their lives. A gender expert will also be hired to sensitize project and government staff on gender issues so that any steps and polices that are formulated inculcate the concerns of women in the region. A **Gender Action Plan** has been prepared

#	Event	Cause	Impact(s)	Risk Category and Sub-category *(including Risk Appetite)*	Impact, Likelihood & Risk Level *(see Annex 3 Risk Matrix)*	Risk Valid From/To	Risk Owner *(individual accountable for managing the risk)*	Risk Treatment and Treatment Owner
	regarding participation in design and implementation or access to opportunities and benefits.							and will be implemented for the project. This plan, includes and addresses GBV issues and considers the issues faced by women in all project aspects and steps that can be taken to involve women in the decision-making process, empower them in different positions, reduce any violence and threats that they face including those related to the environment. In addition, the project **ESMF** will define specific environmental and social safeguard procedures for the project, including consideration of gender related risks. The **Stakeholder Engagement Plan** and M&E Plan also ensure participation and engagement of women and girls so that all relevant issues are tackled, and appropriate management measures are put in place. *Alleviative:* The project's M&E system has included a more detailed set of indicators (at the level of project Activities supportive of the Outcome level indicators) that will be ensure that the actual

#	Event	Cause	Impact(s)	Risk Category and Sub-category *(including Risk Appetite)*	Impact, Likelihood & Risk Level *(see Annex 3 Risk Matrix)*	Risk Valid From/To	Risk Owner *(individual accountable for managing the risk)*	Risk Treatment and Treatment Owner
								achievement of the gender mainstreaming in all project activities is carried out. Risk Treatment Owner: …
								Risk Treatment 3.2: … Risk Treatment Owner: …
								Risk Treatment 3.3: … Risk Treatment Owner: …

#	Event	Cause	Impact(s)	Risk Category and Sub-category *(including Risk Appetite)*	Impact, Likelihood & Risk Level *(see Annex 3 Risk Matrix)*	Risk Valid From/To	Risk Owner *(individual accountable for managing the risk)*	Risk Treatment and Treatment Owner
4	There is a risk that natural disasters may occur	As a result of this, the project activities may slow down	Which will impact in achieving timely results from the project	1. SOCIAL AND ENVIRONMENTAL (1.5. Climate change and disaster risks) - UNDP Risk Appetite: CAUTIOUS	Likelihood: **3 - Moderately likely** Impact: **3 - Intermediate** Risk level: **MODERATE (equates to a risk appetite of EXPLORATORY)**	From: 01-Oct-23 To: 31-Dec-25	PMU MoCC&EC	Risk Treatment 4.1: : The project is designed for increasing resilience of forest plantations and among communities prone to natural disasters. Selection of demo sites has considered likelihood of these possibilities. In the event of a natural disaster (e.g., floods, drought) improved woody biomass management, including raising wood lots and growing fuelwood trees on farmlands will reduce the damages and impacts of climate change. Considering the potential climate change events (e.g., monsoon flooding, high ambient temperatures, drought), the site selection of project demo units and logistics facilities have been considered in the stakeholder consultation in the PPG stage. During the

#	Event	Cause	Impact(s)	Risk Category and Sub-category *(including Risk Appetite)*	Impact, Likelihood & Risk Level *(see Annex 3 Risk Matrix)*	Risk Valid From/To	Risk Owner *(individual accountable for managing the risk)*	Risk Treatment and Treatment Owner
								implementation, the facilities will be designed in such a way that these will be resilient and comply with internationally accepted infrastructure design and engineering standards Risk Treatment Owner: …
								Risk Treatment 4.2: Should there be any occurrence during the implementation, the project management will inform immediately the local disaster management units to activate pre-arranged response plans in case of emergencies to protect lives and properties. Risk Treatment Owner: …
								Risk Treatment 4.3: … Risk Treatment Owner: …

#	Event	Cause	Impact(s)	Risk Category and Sub-category *(including Risk Appetite)*	Impact, Likelihood & Risk Level *(see Annex 3 Risk Matrix)*	Risk Valid From/To	Risk Owner *(individual accountable for managing the risk)*	Risk Treatment and Treatment Owner
5	There is a risk that COVID 19 may re-occur	As a result of this mobility and communication among stakeholders, especially travel to field project sites and stakeholders' consultations during the project designing and implementation phases would be limited	Which will impact in hindrance in monitoring of project objectives	3. OPERATIONAL (3.7. Occupational safety, health and well-being) - UNDP Risk Appetite: EXPLORATORY TO OPEN	Likelihood: **3 - Moderately likely** Impact: **2 - Minor** Risk level: **LOW (equates to a risk appetite of MINIMAL)**	From: 01-Oct-23 To: 31-Dec-24	PMU MoCC&EC	Risk Treatment 5.1: in the situation of prolonged effects of COVID-19, the MoCC&EC as Implementing Partner, in coordination with the Provincial Departments and the PMU will continue to make use of the strategies applied in the GOP's and the UNDP's (and other UN Agencies' and Donors') successes in curtailing the spread of COVID-19 and suppressing social and economic impacts of this pandemic in the implementation of the design and operational requirements of the installations of the project. These include safety-net for the poor, localized, micro level smart-lockdowns, and use of electronic media for communicating with the provincial governments, development agencies, donors, and local community leaders. The lessons learned during

#	Event	Cause	Impact(s)	Risk Category and Sub-category *(including Risk Appetite)*	Impact, Likelihood & Risk Level *(see Annex 3 Risk Matrix)*	Risk Valid From/To	Risk Owner *(individual accountable for managing the risk)*	Risk Treatment and Treatment Owner
								the pandemic will be fully considered in maintaining effective communication with stakeholders and partners. Risk Treatment Owner: …
								Risk Treatment 5.2: Moreover, during the implementation, should there be inevitable situations due to surges of cases affecting the communities that would necessitate remedial and stop gap measures, the Provincial Departments and local hosts of the demo facilities and fuel logistics systems shall implement corrective activities for decentralized actions, adjustments and geographic clustering that would be limiting the extent of travelling to and from and within the project sites in accordance with established local protocols while

#	Event	Cause	Impact(s)	Risk Category and Sub-category *(including Risk Appetite)*	Impact, Likelihood & Risk Level *(see Annex 3 Risk Matrix)*	Risk Valid From/To	Risk Owner *(individual accountable for managing the risk)*	Risk Treatment and Treatment Owner
								minimizing the effects to the project implementation and to attendant costs, e.g. cost escalations, quarantine expenses, etc. and disruption of project timetable. In any case, the related project emergency response plans should be established using online and telecom to facilitate planning, operations, reporting and monitoring systems. Risk Treatment Owner: … Risk Treatment 5.3: … Risk Treatment Owner: …
6	There is a risk that Local communities and farmers do not adhere to forestry interventions and sustainable production of woody biomass in their agriculture fields.	As a result of this project activities may not be implemented as desired	Which will impact in getting the desired results	1. SOCIAL AND ENVIRONMENTAL (1.5. Climate change and disaster risks) - UNDP Risk Appetite: CAUTIOUS	Likelihood: **3 - Moderately likely** Impact: **3 - Intermediate** Risk level: **MODERATE (equates to a risk appetite of EXPLORATORY)**	From: 01-Oct-23 To: 31-Dec-23	PMU MoCC&EC	Risk Treatment 6.1: Participatory nature and incentives measures associated with raising energy plantations and use of efficient technologies are built-in design to encourage communities/farmers to adhere to the proposed interventions. Furthermore, compliance measures will be ensured with

#	Event	Cause	Impact(s)	Risk Category and Sub-category *(including Risk Appetite)*	Impact, Likelihood & Risk Level *(see Annex 3 Risk Matrix)*	Risk Valid From/To	Risk Owner *(individual accountable for managing the risk)*	Risk Treatment and Treatment Owner
								sufficient guidelines and local organizational requirements that involve proper checks and balances during the implementation of the project will help in mitigating this risk. Public awareness on benefits and program requirements shall reinforce the targeted campaigns, as well as market linkages and financing windows will be implemented to encourage farmers to fuelwood production to maximize fuelwood production on less productive lands. Risk Treatment Owner: …
								Risk Treatment 6.2: In case of actual deviations from agreed plans and commitments, attention of responsible parties and local leadership should be warranted, and appropriate decisions and corrective actions should be initiated promptly.

#	Event	Cause	Impact(s)	Risk Category and Sub-category *(including Risk Appetite)*	Impact, Likelihood & Risk Level *(see Annex 3 Risk Matrix)*	Risk Valid From/To	Risk Owner *(individual accountable for managing the risk)*	Risk Treatment and Treatment Owner
								Effective monitoring of corrective actions and adherence to project objectives shall be encouraged and if possible, incentives in whatever form shall be mobilized from co-financed resources until commercial mode of operation follow from successful demonstrations. Risk Treatment Owner: … Risk Treatment 6.3: … Risk Treatment Owner: …
7	There is a risk that may create a perverse incentive that leads to more production than replacement of inefficient methods	As a result of Introduction of improved wood-fired equipment in areas having standing natural forests	Which will impact in leading further degradation of natural forest.	1. SOCIAL AND ENVIRONMENTAL (1.4. Biodiversity conservation and sustainable natural resource management) - UNDP Risk Appetite: CAUTIOUS	Likelihood: **1 - Not likely** Impact: **3 - Intermediate** Risk level: **PLEASE SELECT**	From: 01-Oct-23 To: 31-Dec-24	PMU MoCC&EC	Risk Treatment 7.1: The project will ensure that all power generation and EE equipment introduced in applicable rural industries have appropriate licenses and following regulatory standards and guidelines to be monitored by the provincial forest departments and local authorities. For this purpose, a proper certification scheme is part of project design.

#	Event	Cause	Impact(s)	Risk Category and Sub-category *(including Risk Appetite)*	Impact, Likelihood & Risk Level *(see Annex 3 Risk Matrix)*	Risk Valid From/To	Risk Owner *(individual accountable for managing the risk)*	Risk Treatment and Treatment Owner
								Moreover, areas with closer natural forests will be avoided for project interventions. The project shall implement the awareness and capacity building activities that are designed to encourage efficient fuel wood burning equipment and forest conservation practices. Risk Treatment Owner: …
								Risk Treatment 7.2: Proper check and balance will be maintained to ensure that use of woody biomass is sustainable and does not exceed the production limits. The possibility of using alternate biomass e.g., crop residuals will also be explored Risk Treatment Owner: …
								Risk Treatment 7.3: … Risk Treatment Owner: …

#	Event	Cause	Impact(s)	Risk Category and Sub-category *(including Risk Appetite)*	Impact, Likelihood & Risk Level *(see Annex 3 Risk Matrix)*	Risk Valid From/To	Risk Owner *(individual accountable for managing the risk)*	Risk Treatment and Treatment Owner
8	There is a risk that Woody biomass use could exceed production	As a result of Even fast-growing species will take up to five years to mature. Moreover, climate change risks (flash floods and prolonged drought) and rainfall variability could impact production of woody biomass	Which will impact in reduction of wood/forest	1. SOCIAL AND ENVIRONMENTAL (1.4. Biodiversity conservation and sustainable natural resource management) - UNDP Risk Appetite: CAUTIOUS	Likelihood: **1 - Not likely** Impact: **4 - Extensive** Risk level: **LOW (equates to a risk appetite of CAUTIOUS)**	From: 01-Oct-23 To: 31-Dec-24	PMU MoCC&EC	Risk Treatment 7.1: The project will develop specific criteria during the implementation to validate the areas that support operations of rural industries that use wood as fuel that will include evidence of (i) availability of woody biomass, (ii) access to market supplies, and past production experience. The project will implement the standards and applicable criteria in fuel plantations' location planning and permitting in coordination with the Provincial Departments. Risk Treatment Owner: … Risk Treatment 7.2: In case of supply/demand imbalance during the project implementation, a monitoring system will be established and sustained. The use of wood biomass by the relevant rural industries will be carefully monitored and

#	Event	Cause	Impact(s)	Risk Category and Sub-category *(including Risk Appetite)*	Impact, Likelihood & Risk Level *(see Annex 3 Risk Matrix)*	Risk Valid From/To	Risk Owner *(individual accountable for managing the risk)*	Risk Treatment and Treatment Owner
								regulated with active involvement of provincial forestry departments and the local community leadership. Risk Treatment Owner: …
								Risk Treatment 7.3: … Risk Treatment Owner: …
9	There is a risk that woodlots were not well-established and managed.	As a result of Improved and efficient wood-fired equipment to be introduced may not be suitable under the local conditions and plantation-based woody biomass production	Which will impact in overall supply side of the project	1. SOCIAL AND ENVIRONMENTAL (1.4. Biodiversity conservation and sustainable natural resource management) - UNDP Risk Appetite: CAUTIOUS	Likelihood: **1 - Not likely** Impact: **2 - Minor** Risk level: **LOW (equates to a risk appetite of MINIMAL)**	From: 01-Oct-23 To: 31-Dec-23	PMU MoCC&EC	Risk Treatment 7.1: The project will encourage the use of modern fuelwood-fed gasifiers, energy-efficient kilns and boilers that will be demonstrated by the project which have already been used, technically proven and their performance accepted. The findings and performance results provided by recent relevant project studies have been useful in selecting practical and economically viable fuel wood technologies. These technologies will be adopted for reducing GHG emissions that are indeed the appropriate

#	Event	Cause	Impact(s)	Risk Category and Sub-category *(including Risk Appetite)*	Impact, Likelihood & Risk Level *(see Annex 3 Risk Matrix)*	Risk Valid From/To	Risk Owner *(individual accountable for managing the risk)*	Risk Treatment and Treatment Owner
								technology to be introduced for wood-fired commercial/residential applications and industrial equipment operations. Moreover, technology transfer must be adaptable to local communities' preferences, affordable, and easily accessible. The project seeks to mitigate this risk by recommending appropriate technologies based on the proper feasibility studies. Risk Treatment Owner: …
								Risk Treatment 7.2: A mechanism for certification appropriate/efficient technologies/equipment will be introduced and enforced with the involvement of concerned departments and agencies and potential users. In addition, disincentives (fines, penalties etc.) will

#	Event	Cause	Impact(s)	Risk Category and Sub-category *(including Risk Appetite)*	Impact, Likelihood & Risk Level *(see Annex 3 Risk Matrix)*	Risk Valid From/To	Risk Owner *(individual accountable for managing the risk)*	Risk Treatment and Treatment Owner
								be introduced to reduce the severity of this risk. Risk Treatment Owner: … Risk Treatment 7.3: … Risk Treatment Owner: …
10	There is a risk that adverse biodiversity impacts may occure	As a result of impacts to habitats, ecosystems, and ecosystem services; and activities in the communities.	Which will impact in natural climate habitat strcuture	1. SOCIAL AND ENVIRONMENTAL (1.4. Biodiversity conservation and sustainable natural resource management) - UNDP Risk Appetite: CAUTIOUS	Likelihood: **2 - Low likelihood** Impact: **2 - Minor** Risk level: **LOW (equates to a risk appetite of MINIMAL)**	From: 01-Oct-23 To: 31-Dec-23	PMU MoCC&EC	*The risk assessment will be validated during the early part of Year 1 as part of the ESMF and ESMP development and adoption. A participatory approach will be in place at all stages of the project to maximize the involvement of the appropriate agencies dealing with animal habitats and ecosystems, community members keeping in mind the legally protected areas as well as the cultural context in the areas. A mechanism for a biodiversity action plan and implementation will be adopted and enforced with the involvement of concerned departments*

#	Event	Cause	Impact(s)	Risk Category and Sub-category *(including Risk Appetite)*	Impact, Likelihood & Risk Level *(see Annex 3 Risk Matrix)*	Risk Valid From/To	Risk Owner *(individual accountable for managing the risk)*	Risk Treatment and Treatment Owner
								and agencies and potential users. In addition, disincentives (fines, penalties etc.) will be introduced to reduce the severity of this risk.
								In case of reported incidences of adverse biodiversity impacts to habitats, ecosystems, and ecosystem services; and activities within or adjacent to critical habitats and/or environmentally sensitive areas, including legally protected areas, areas proposed, proper investigation, consultation and remedial measures will be adopted and enforced to prevent recurrence. Due penalty for responsible acts will be bestowed based on approved sanctions and disincentives.

#	Event	Cause	Impact(s)	Risk Category and Sub-category *(including Risk Appetite)*	Impact, Likelihood & Risk Level *(see Annex 3 Risk Matrix)*	Risk Valid From/To	Risk Owner *(individual accountable for managing the risk)*	Risk Treatment and Treatment Owner
11	There is a risk that project assets may suffer a loss	As a result of lack of insurance policies at the Implementing Partner	Which will impact in achieving project objectives	2. FINANCIAL (2.2. Value for money) - UNDP Risk Appetite: MINIMAL TO CAUTIOUS	Likelihood: **3 - Moderately likely** Impact: **3 - Intermediate** Risk level: **MODERATE (equates to a risk appetite of EXPLORATORY)**	From: 01-Oct-23 To: 31-Dec-24	PMU MoCC&EC	*Since MoCC&EC has no insurance cover for any of the assets, the MoCC&EC will obtain insurance cover at least for all its significant assets to avoid risk of any major losses in case of manmade or natural disaster* *In case of reported incidences of major losses due to manmade or natural disaster, the insurance coverage will be applied following the terms and conditions of the insurance policy and the appropriate adjustments and instruments to cushion the impact of the unfavorable event.*

#	Event	Cause	Impact(s)	Risk Category and Sub-category *(including Risk Appetite)*	Impact, Likelihood & Risk Level *(see Annex 3 Risk Matrix)*	Risk Valid From/To	Risk Owner *(individual accountable for managing the risk)*	Risk Treatment and Treatment Owner
12	There is a risk that loss of data during project implementation	As a result of lack of proper financial system in place for project implementation	Which will impact in loss of data and inaccurate reporting or audit issues	2. FINANCIAL (2.3. Corruption and fraud) - UNDP Risk Appetite: MINIMAL TO CAUTIOUS	Likelihood: **3 - Moderately likely** Impact: **3 - Intermediate** Risk level: **MODERATE (equates to a risk appetite of EXPLORATORY)**	From: 01-Oct-23 To: 31-Dec-24	PMU MoCC&EC	*Following the recommendation made in the October 2021 micro assessment, the MoCC&EC will implement an automated financial system to increase the integrity of data and to optimize the benefit of a financial and management reporting system* *Should there be breach of the integrity and security of the automated financial system, due investigation and validation will be immediately done and when there is proven breach and security intrusion, corresponding penalty and sanction will be levied to the responsible individuals. Corrective actions in the system should be developed and adopted immediately.*

#	Event	Cause	Impact(s)	Risk Category and Sub-category *(including Risk Appetite)*	Impact, Likelihood & Risk Level *(see Annex 3 Risk Matrix)*	Risk Valid From/To	Risk Owner *(individual accountable for managing the risk)*	Risk Treatment and Treatment Owner
13	There is a risk that value for money may not be obtained	As a result of lack of proper procurement system in place	Which will impact in loss of value for money from project resources	2. FINANCIAL (2.2. Value for money) - UNDP Risk Appetite: MINIMAL TO CAUTIOUS	Likelihood: **3 - Moderately likely** Impact: **3 - Intermediate** Risk level: **MODERATE (equates to a risk appetite of EXPLORATORY)**	From: 01-Oct-23 To: 31-Dec-24	PMU MoCC&EC	*Similarly, with recommendations of the micro assessment of MoCC&EC conducted in October 2021, MoCC&EC will develop and implement an automated procurement system; prepare procurement reports; formalize a process of vendor prequalification; and develop a system of maintaining supplier database ad performance evaluation.* *Should there be breach of the integrity and security of the automated procurement and contract administration system, due investigation and validation will be immediately done and when there is proven breach and security intrusion, corresponding penalty and sanction will be levied to the*

#	Event	Cause	Impact(s)	Risk Category and Sub-category *(including Risk Appetite)*	Impact, Likelihood & Risk Level *(see Annex 3 Risk Matrix)*	Risk Valid From/To	Risk Owner *(individual accountable for managing the risk)*	Risk Treatment and Treatment Owner
								responsible individuals. Corrective actions in the system should be developed and adopted immediately.

Annex 6A: PASBET Climate Risks Assessment

a) Climate Change Projections/Scenarios for Pakistan

Though Pakistan's total annual GHG emission is significantly lower than many developed countries, it has been a victim of numerous of climate change stresses and climate induced natural disasters. Pakistan now ranks as the 5th most vulnerable country to the impacts of climate change on the long-term Global Climate Risk Index (German Watch 2020). The country's Ministry of Climate Change & Environmental Coordination (MoCC&EC) has estimated in 2013 that Pakistan needs to spend around than 7 to 14 billion US$ annually to adapt to climate change (MoCC&EC 2013).

More importantly, the future scenarios for climate change in Pakistan are painting a grim picture. Despite limitations in downscaled climate data and indigenous climate modeling tools, the estimates are suggesting an overall increase in temperatures of 1.4 ^{0}C – 3.7 ^{0}C and +4 mm/day increase in mean annual precipitation by 2050 (USAID 2017)[87]. After the release of the IPCC 5th assessment report in 2014, Pakistan Metrological Department (PMD) released a daily gridded downscaled precipitation and temperature time series from 2010 to 2099 climate change scenarios. These downscaled results show a 3℃–5℃ temperature rise in mean temperature under the Representative Concentration Pathways (RCP) and 4.5 emission scenario[88]. The mean trend of temperature under RCP8.5 indicates a 4℃–6℃ rise by the end of the century with a sharp increase after 2050. The rainfall is highly variable in both spatial and temporal domains, where average rainfall in Pakistan shows a large inter-annual variability and sharp rising peaks give an indication of extreme weather events, while negative peaks indicate frequent droughts. On the other hand, snow-covered mountain peaks in northern Pakistan show a larger increase in mean temperature compared to central and southern regions under both RCP scenarios. However, RCP8.5 shows a more abrupt increase in temperature in the region after 2060 and up to 10℃–12 ^{o}C, especially in northern Pakistan, whereas RCP4.5 shows a similar increasing trend, but with less intensity.

Other studies also suggest that by 2020, summer rainfall peaks will shift toward August, while those of winter will shift toward March. These shifts in the rainfall patterns will continue even up to the end of the 21st century. RCP4.5 scenario shows an increase of 4 mm/day in annual mean precipitation with a shift in maxima toward the northeastern part of the country until 2050. After 2050, the precipitation pattern shifts toward northwest until the end of the 21st century with the same magnitude and wet situation in the southern region of the country. A similar pattern is seen under the RCP8.5 scenario, but with less magnitude of up to 2–3 mm/day and more spatial spread (ADB 2017)[89]

b) Key Potential Hazards that may Impact on the PASBET Project

The climate scenarios described above suggest, the increase in temperature is set to accelerate the melting of glaciers in northern parts of the country. Precipitation will not only increase by 4 mm per annum, but significant spatio-temporal shifts are expected in the onset of monsoon and winter rainfall

[87] USAID (2017). Climate Change Risk Profile-Pakistan (Fact Sheet): 5 pp.

[88] The IPCC Fifth Assessment Report defined a set of four new scenarios, denoted RCPs, RCP2.6, RCP4.5, and RCP6; and RCP8.5 where the numbers (2 to 8.5) denote approximate total radiative forcing in year 2100 relative to 1750, ranging from 2.6 watts per square meter (W/m2) to 8.5 W/m2. RCP4.5 is a stabilization scenario where GHG emissions stabilize by 2100. RCP8.5 is another scenario with very high GHG emissions where radiative forcing does not peak by 2100. Each RCP provides spatially resolved data sets of land use change and sector-based emissions of air pollutants, and specify annual GHG concentrations and anthropogenic emissions up to 2100

[89] Climate Change Profile of Pakistan (2017), Asian Development Bank: 130 pp.

patterns (PMD 2015). This variability in precipitation patterns coupled with increased runoff in the rivers due to glacial melt is expected to increase the frequency and intensity of riverine floods. Events of cloud outbursts caused by erratic precipitation patterns are going to increase the frequency of flash flooding across the country. Implications of such climate scenarios on the proposed project sites would depend on how the departure from normal rainfall, mean annual, minimum, and maximum temperatures will be at the district scale. Unfortunately, lack of availability of high-resolution downscaled climate datasets at the level of district scale are yet to be developed. This affects accuracy of predictions about climate change at the district level. Nonetheless, qualitative correlations can be drawn from sub-national ensemble outputs and observed climate trends in each district to estimate the developing climate change scenarios by 2050.

Sindh (Districts Badin and Sujawal))

The project districts in Sindh Province, namely Badin and Sujawal have a history of climate induced disasters associated with high temperatures, flash floods, and riverine floods.

Badin has a hot desert climate (Köppen climate classification BWh). The climate of the district is on average moderate and is tempered by the sea breeze which blows for eight months of the year from March to October, making the hot weather somewhat cooler than for other parts of Pakistan. During the monsoon period, the sky is cloudy, but there is usually no rainfall. The climate in summer is moist and humid. The cold weather in Badin starts from the beginning of November when a sudden change from the moist sea breeze to the dry and cold north-east wind brings about, as a natural consequence, an immediate fall in temperature. Cyclones and floods are hit because of sea.

Sujawal's climate is moderate, winter is cold, and summer is bit hot. As well humidity in this area too. Sujawal is near the sea. During winter seasonal birds migrate from Siberia and Russia in this area lakes.

Khyber Pakhtunkhwa (Kaghan and Kalam)

Kalam is in the northern upper section of Swat valley along the banks of the Swat River in the Khyber Pakhtunkhwa province of Pakistan. Due to its geographical location (adjacent to Swat river), the district is highly susceptible to riverine and flash flooding during monsoon season. The climate change scenarios have further aggravated the situation, making parts of the district, along the riverbanks, more vulnerable to flash-flooding. The floods have been experienced in 2010, 2015 and 2022, resulting in economic loss, destroyed houses and hoteling industry infrastructure. The weather station at Saidu Sharif was established in 1973 and has recorded increase of 12.91% in rainfall during the last decade.

Kaghan is an alpine valley located in the Mansehra District of Khyber Pakhtunkhwa, Pakistan with climate pattern like that of Kalam. Landslides triggered by the devastating 2005 Kashmir earthquake destroyed many passes leading into the valley, though roads have since been rebuilt. River Kunar flows through the district. As is that case with Kalam, the district is highly susceptible to riverine and flash flooding during monsoon season. Weather station at Balakot, established in 1957, has recorded decrease of 39.3% in average rainfall during last decade. The climate change scenarios have further aggravated the situation, making parts of the district, along the riverbanks, more vulnerable to flash-flooding. The same has been experienced in year 2022 as well when the record monsoon rains and floods swept away houses, buildings, roads, bridges, and crops.

Punjab (Layyah and Bhakkar)

Districts of Bhakkar and Layyah fall under the Doaba (i.e., land between the Indus and Jehlum Rivers) and Thal Desert, with the Indus River on the west boundary of both districts. These districts are susceptible to riverine flooding from the Indus River, because these are part of the river's old alluvial plain. Moreover, the eastern parts of these districts experience frequent droughts. Mean annual maximum temperatures in summer months exceed 46°C creating the likelihoods of heat waves, whereas rainfall is scanty and fluctuates between 100 – 300mm in the monsoon and 50-100 mm during winter, respectively. More recently the districts have been suffering from the menace of locust attacks, which are attributed to changing climate, presence of vegetation and availability of adequate breeding grounds in eastern parts of these districts. Both districts fall in the Punjab Province's Barani and Northern Irrigated Plains zones. The changes in mean temperatures and rainfall for the past century in both the zones are a total of 0.22°C increase in mean temperatures and 6.2 mm increase in average rainfall, over the past century (1909 – 2016) and most of these changes have occurred during in the past 2 decades (1990 – 2016)[90]. If these trends continued, it is expected that these districts will experience more volatile weather pattern till 2050, as the region is vulnerable to prolonged drought, sandstorms, and insects' infestation.

Balochistan (Dera Bugti and Sohbatpur/Jaffarabd)

The project districts in Balochistan Province are Jaffarabad (including Sohbatpur) and Dera Bugtiy. These districts have a history of climate induced disasters associated with high temperatures, flash floods, and prolonged drought, especially in Dera Bugti. Both districts are adjacent to proposed project districts of Sindh, Jacobabad and Kashmore. The climate in these districts is hot and becomes extremely hot during summer, where dust storms are common during the summer. Jaffarabad/Sohbatpur District falls in flood plains with elevation around 1500 above sea level having inadequate rainfall. According to 2012 total climate change vulnerability assessment Dera Bugti falls in 20 districts with high to medium vulnerability rankings, which corresponds to low levels of economic activities. Moreover, these districts have history of natural disasters, like severe drought (Dera Bugti) and flash-flooding causing heavy damage to crops and infrastructure. It is expected that changing climate scenarios in the next three decades will further exacerbate the occurrence of these hazards by 2050. Dera Bugti is also highly vulnerable to soil erosion triggered by flash-flooding.

c) Potential hazards for the project related to the climate change scenarios

The following are the expected climate changes that could impact the implementation of the PASBET Project, and the continuous and sustainable operation of the project sites during the project lifetime and beyond:

1) Increased temperature of 1.4°-3.7°C by the 2060 in most of the project districts, with projected increases higher during the winter.
2) Increased frequency of heavy rainfall and cloudburst from January-June and decreased from July to September, with increased frequency and intensity of extreme climate events.
3) Increased glacier melt, and decreased glacier volume in the north.
4) Increased frequency and duration of droughts.

Potential Risks:

[90] Sathar Z.A. and K. Khan (eds). 2019. Climate, Population, and Vulnerability in Pakistan: Exploring Evidence of Linkages for Adaptation. Islamabad: Population Council. (Data Source; The World Bank Climate Knowledge Portal)

Woodlots/Tree Farms/Forestry: Loss of productive woodlots and farmlands for growing fuel trees; reduced fuel tree production; reduced water availability; reduced tree survival; damage to energy plantations (woodlots); and reduced woody biomass production.
Water Supply: Increased inundation of water storage by floodwaters; increased soil-erosion; reduced surface and ground water availability; reduced water storage capacity; decreased water availability and supply (drought); increased water logging and salinity, and increased water related conflicts.
Power and Energy: Reduced operation of woody biomass fired power generation facilities; increased demand for dry fuelwood supply; increased electricity demand (e.g., for air conditioning); increased power outages and resulting economic losses, and increased demand of renewable energy sources. Regarding the latter, it is already happening - more people and small and medium sized industrial operations are opting for alternate/renewable energy.
Human Health: Increased heatwave related mortalities, increased water-borne diseases, increased disease outbreaks, loss of livelihoods, malnutrition, human displacement (climate migrant).

d) Climate Risk Assessment and Climate Risk Mitigation Measures Identification during PASBET Project Design.

Because of limited travel and meeting opportunities during the PPG stage, detailed site-specific Climate Risk Assessments cannot be conducted in the identified project sites where the project interventions will be carried out. Because of the importance of this climate risk assessments, the project implementation team during the inception phase will collaborate with the concerned government agencies and development partners engaged in analysis of downscaled subnational datasets (e.g., PRECIS, etc.) for the updated list of districts using the best suited model, (e.g., CMIP5, MIROC, etc.). The findings of these assessments will be calibrated with observed data for the district to predict changes in temperature and rainfall by the year 2050. To further ground truth the data a climate change perception survey based on the PRA methodology and participatory assessments will be carried out in each project district with the support of District Disaster Management Authorities (DDMAs) during the inception phase. Based on the secondary data and findings of climate modelling conducted by relevant government agencies, data will be triangulated to present a clear picture for the changing climate under various RCPs. Finally, the risk mitigation measures included in this Project Document and CEO Endorsement will be validated during the inception phase consistent with the findings of the risk vulnerability assessments overall at the project, district, and local levels.

Annex 6B: PASBET Project and Covid-19

a. How the pandemic will be addressing the Covid-related issues during project implementation, including associated impacts, risks, and opportunities.

The PASBET project is designed and developed during challenging times wherein the COVID-19 has gripped the globe, which means there is still the possibility of addressing this pandemic by the time the project is ready for implementation. The prevalent conditions of COVID-19 have shaped the world posing multiple challenges to businesses and development across countries. The possible impacts that PASBET Project may face by the time it is ready for implementation includes nationwide fragile economy, interrupted trade, investment and travel, pre-occupied government institutions coping with the pandemic. Hence changed priorities. The project personnel, local communities and government officials may face health issues and may not achieve the desired results being planned at that stage—the condition that may arise due to changed priorities. On the other hand, the project, local and global stakeholders may benefit from opportunities such as accelerating digital transformation, shifting working patterns, rapid innovation, and access to skills. In such circumstances, the PASBET project will continue to identify measures at the early implementation stage. The indicative actions are covered under (b) and (c) below.

b. Risk analysis: How risks from COVID-19 will be analyzed and mitigation strategies incorporated into the design of the project implementation

A comprehensive COVID-19 Risk Analysis that was not conducted during the project design and development (PPG) stage, will have to be carried out in the inception stage by involving expert professionals to carefully analyze COVID-19 related risks and to suggest mitigation measures during project implementation in the form of a 'COVID-19 Risk Management Plan (CRMP)'. The project staff will be trained to ensure compliance with the CRMP during its implementation. With the lessons learned and experiences in addressing the Covid-19 issues in the country, the Project consultant for the purpose, together with the UNDP and experts, will prepare the CRMP. While it is acknowledged at this stage of project conceptualization (i.e., PIF development) that all that can be said is general risk statement that COVID-19 will negatively impact the social-economic development of the planned project areas, during the detailed PPG stage, the Covid-19 risk analysis will be better informed by the project development team's knowledge of the nature and impacts of this pandemic. Where necessary, local public health officials in the project areas will also be consulted to get a wide perspective view of the risk issues and produce the appropriate preventive and alleviative mitigation measures. Moreover, COVID-19 will be made part and parcel of the project risk log which will be regularly analyzed, carefully monitored, and periodically updated. Particular attention will be paid to ensuring the safety, health and welfare of the project implementation team, and the project partners that they work with during the project. Moreover, the safety, health, and welfare of PASBET project staff, local communities and concerned government/non-government personnel involved in project implementation till completion will accordingly be facilitated.

c. Opportunity analysis in the midst of COVID-19 to deliver GEBs and Outputs/or climate adaptation and resilience benefits and contribute toward green recovery and building back better.

Both COVID-19 Risk, and Opportunity analyses go side-by-side. The Risk Analysis conducted at the project design stage indicated potential opportunities in addressing the identified risks and to deliver global environmental benefits. The lessons learned drawn from other development projects in Pakistan in the wake of the pandemic will also help in delivering benefits pertinent to climate change adaptation as well

mitigation. The results of the assessments of potential Covid-19 risks, and the lessons learned and experience of these entities during the Covid-19 pandemic can be used also to determine opportunities for mitigating the impacts of Covid-19. For example, Covid-19 could result in the heightened concern of people about infectious diseases from flooding events that are often experienced also in the identified project districts presents an opportunity for the local government and the partner rural industries and woodlot/fuelwood plantations to work together to reduce the impacts of flooding which during a pandemic can raises anxieties and concerns about water-borne diseases, as well as destroy physical assets. This presents opportunities, which can also be facilitated by the PASBET Project, to incorporate technologies/techniques that will make installations of woody biomass-fired equipment not only robust but also climate resilient, as well as opportunities for employing measures that will lessen the inundation of floodwaters in woodlots/fuelwood plantations. In the context of this proposed project, the lessons learned, experiences and knowledge gain during the Covid-19 pandemic provides opportunities for the project proponents and their partners to formulate effective mitigating actions to prevent and/or alleviate any negative impacts of a persistent pandemic on the project interventions that are intended directly to bring about GHG emissions, as well as actions that will sustain/maintain the achievements that would be realized. Other indirect opportunities include the accelerated digital transformation that helped avoiding air travel, consuming less energy and use of plastics (during workshops, meetings, and seminars) thus reducing GHG emissions and abating the effects of climate change. During the implementation stage, more of such opportunities will be identified to do better and avoid harm to nature enhancing resilience of communities and ecosystems.

ANNEX 7: OVERVIEW OF PROJECT STAFF AND TECHNICAL CONSULTANCIES

Consultant	Time Input	Tasks, Inputs and Outputs
Local / National contracting		
National Project Coordinator	60 months / over 5 years	The National Project Coordinator (NPC) has 2 sets of tasks to carry out for the PASBET Project. These are: (a) coordination of the management and implementation of the project activities (20%); and (b) provision of technical coordination of, and technical support to, the promotion, design, implementation of the project's demonstration activities (80%). The NPC will be mainly responsible for the provision of technical assistance and coordination work to PMU in implementing the project's demonstration activities. He will also be involved in the overall coordination of the management and implementation of the project activities with the assistance of the project's Administrative and Financial Officer, in coordination with UNDP and MoCC&EC as the IP. Duties and Responsibilities *Project Management (20%) – 12 months (over 5 years)* • *The NPC will carry out the following tasks in coordination with the MoCC&EC and UNDP:* • *Coordination of the overall management and implementation of the project activities with the contractors and consultants engaged by the project in a timely and results-oriented manner with the assistance of the project's Administrative/Financial Officer and in coordination with UNDP and MoCC&EC as the IP.* • *Coordination of the planning of the implementation of the activities of the project and monitor progress against the approved workplan.* • *Monitoring of events as determined in the project monitoring plan, and update the plan as required.* • *Performance of regular progress reporting to the project board as agreed with the board and to UNDP for the GEF reports, including measures to address challenges and opportunities.* • *Development and preparation of revisions to the multi-year workplan, as needed, as well as annual and quarterly plans if required.* • *Preparation of the inception report no later than one month after the inception workshop.* • *Ensuring that the indicators included in the project results framework are monitored annually in connection with the GEF Project Implementation Review (PIR) preparation.* • *Supporting the conduct of the Mid-term review and Terminal Evaluation process.* *Technical (80%) – 48 months (over 5 years)* *This coordination work will focus on the facilitation of the demonstration activities in the partner provinces, and*

Consultant	Time Input	Tasks, Inputs and Outputs
		includes the mobilization of all committed inputs to, and implementation arrangements for, all planned demonstrations (socialization/promotion, design, engineering, coordination with the owners/implementers of the subsumed baseline projects that will be modified and implemented as demonstration projects of the PASBET Project. Apart from the coordination with the project partners, the task also includes supervision over technical personnel and sub-contractors that will carry out the incremental activities of the demonstrations on behalf of the IP (MoCC&EC) with particular attention in the delivery of the planned outputs under Component 2.2. The NPC will carry out the following tasks in coordination with the project partners involved in the demonstration activities (e.g., Forestry Departments of the partner provincial governments), MoCC&EC and UNDP. • *Provision of organizational and coordination work and technical advice in the planning (e.g., technical aspects of site clearing, surveying, plan design and investigation, etc.) and implementation (e.g., technical requirements for permitting process, quality assurance and quality control, etc.) of the project's demonstration activities in the partner provinces.* • *Provision of technical advice and support in the design of the planned demonstrations in coordination with the baseline project owners/implementers and the designated project technical specialist for the demonstrations* • *Provision of advice and guidance in the establishment of the technical specifications, standards to be followed, providing technical inputs in TOR/Contract preparation, RFQ and bidding documentation, selection and engagement of the biomass energy technology service providers and contractors for the demo activities.* • *Assist in the technical evaluation of bid proposals for the supply, installation, operation and systems from contractors and individual proponents* • *Provide technical inputs and support to the PMU in the performance tracking system and contract performance monitoring of contractors and consultants to deliver expected outputs and deliverables, including technical review of reports prepared and submitted by contractors, as basis for contract payments* • *Supervision and provision of technical advice and inputs in the testing, initial operation, and performance evaluation of the demo units.*
Admin/Finance Officer	60 months / over 5 years	Duties and Responsibilities • Keep records of project funds and expenditures and ensure all project-related financial documentation are well maintained and readily available when required by the National Project Coordinator. • Review project expenditures and ensure that project funds are used in compliance with the Project Document and GoP financial rules and procedures. • Validate and certify FACE forms before submission to UNDP. • Provide necessary financial information as and when required for project management decisions. • Provide necessary financial information during project audit(s).

Consultant	Time Input	Tasks, Inputs and Outputs
		• Review annual budgets, project expenditure reports, and notify the National Project Coordinator if there are any discrepancies or issues. • Consolidate financial progress reports submitted by the responsible parties for implementation of project activities and prepare the required reports by UNDP and MoCC&EC as the IP. • Liaise and follow up with the responsible parties for implementation of project activities in matters related to project funds and financial progress reports.
For Technical Assistance		
International / Regional and global contracting		
International Technical Consultants for woody biomass resource assessment, utilization in biomass-based power technology applications, capacity building and related evaluations	280 days / over 5 years	<u>Duties and Responsibilities</u> • Provide technical assistance in the project components and integrating the project-level technical requirements and inputs in coordination with the specialized international and local consultants. • Assist the National Project Coordinator and PMU in the specific issues and requirements in project implementation that will be requested to achieve project objectives. • Assist in the development of relevant Terms of References for all consultants and contractors in coordination with the PMU. • Advise on scheduling, procurement, and monitoring issues. • Provide written advice and comments on policy, regulatory, institutional, financial, technical, and awareness issues. • Advise on establishing and managing monitoring and evaluation programs for all aspects of the project following the Log Frame of the project and on the preparation and finalization of the Annual Project Report/Project Implementation Review as required by the UNDP/GEF M&E system. • Advise the UNDP, PM and PMO and independent evaluators in the conduct of the Mid-term Review (MTR) and Terminal Evaluation (TE), exit strategy and the project closure process and other UNDP-GEF requirements. • Provide support and input to the PM for the preparation of the project final report (including the best practices, lessons learnt etc.). • Any other relevant tasks as assigned by the PMU.
For the Outcomes		
Local / National contracting		
• Regulatory Expert • GIS Expert	*as indicated in the TBWP and*	*<u>Main Duties and Responsibilities</u>* • Regulatory Expert o Assist in all policy and regulatory needs of the project and provide policy advice to the Project Board

Consultant	Time Input	Tasks, Inputs and Outputs
• Woody biomass Expert • M&E-Safeguards Consultant for Risk-based Management (RBM), EIA and ESMP • Gender-Communications Consultant • Financial Expert	*validated during the AWP and project review process*	and to the PMU under the Director ○ Develop a legislative and policy agenda for the project's regulatory requirements in Component 1 ○ Review and recommend for position or strategies to the proposed regulations or legislations paying particular attention to current relevant biomass-based resource production and energy generation legislation pertinent to the PASBET objectives ○ Develop, establish and if possible, initiate implementing a monitoring and evaluation system to track the impacts of enforcement on policy, pricing and regulatory measures involving biomass-based activities and project • GIS Expert ○ Conceptualize, design, implement, operate, and monitor all GIS related activities of PASBET in coordination with MoCC&EC including GIS-database and geo-processing models. ○ Identify relevant repository of GIS-databases among international organizations, government departments and ministries, and collaborate with them to obtain these databases. ○ Undertake GIS mapping, analysis, data capture, data management and geo-processing appropriate for forest and energy plantation resources, ○ Develop and perform research methods and analytics for synthesis of GIS applications to produce maps, figures, and tables to support all PASBET activities ○ Build and expand the in-house GIS capabilities of PASBET in coordination with MoCC&EC needs • Woody biomass Expert ○ Consolidate current best practices in the utilization of sustainable biomass and application of biomass technologies both nationally and internationally ○ Develop appropriate performance norms or standards for the manufacture, installation, operation, and performance testing of these systems based on international best practice standards and local considerations ○ Specify a testing regime for biomass-based selected equipment to measure and compare energy performance, maintenance requirements and standards development ○ Identify the issues and needs for the four provinces on demonstration of selected technologies and applicable industry segments and technology systems for power and non-power systems where standard setting would enhance market opportunities ○ Assess existing Pakistan and international systems and components, both electricity and thermal

Consultant	Time Input	Tasks, Inputs and Outputs
		production equipment, for operating performance and manufacturing/ installation practices and capabilities. • Gender-Communications Consultant o Provide support to the PMU in developing PASBET's communications products, contents writing and editing, to promote PASBET's work and its visibility and deliver a range of communication and information products and services as indicated in the Gender Action Plan and mainstreaming activities o Lead the development and implementation of a multimedia communication campaign on gender-based violence (GBV) and domestic violence relevant to biomass production, utilization, and logistics for woody-biomass applications for energy. o Develop and implement PASBET's communication plan o Provide technical assistance and support in the submission of Annual Work Plan(s) which outlines how the communication activities and outputs will be implemented and produced over the specified duration of the contract. o Support and evaluate the results of the communication activities/ campaigns with the PMU • M&E-Safeguards Consultant for RBM, ESIA and ESMP o Update, prepare and validate an M&E plan that follows a Risk-Based Management (RBM) approach to ensure that the identified project implementation barriers, risks, and challenges in the risk log are addressed and mitigated o Prepare an Environmental and Social Assessment (ESIA) and its respective Environmental and Social Management Plan (ESMP) for the design, installation, and operation of the project demos to ensure the socio-environmental sustainability of its different components. o Analyze, evaluate, and propose measures to prevent, control, mitigate, restore and/or compensate the potential environmental and social impacts of the project so that the project complies with UNDP's SES and national legislation o Recommend management plans and other instruments detailing environmental and social requirements, in particular to guide the final design of the demo projects and their components, o Develop a consultation/stakeholder engagement plan, including an analysis of interested and affected parties, detailing documentation requirements, and dissemination of information about the project.

Consultant	Time Input	Tasks, Inputs and Outputs
		○ Support the implementing partner (and responsible party) in carrying out meaningful consultations. • Financial Expert ○ Install and implement the project's financial systems and procedures, in accordance with UNDP's existing guidelines on national execution ○ Design and specification of the financial schemes and lending programs needed in line with PASBET demos, replication, and sustainability ○ Develop institutional framework for financing mechanisms ○ Recommend appropriate financial assistance mechanisms, tariffs, etc. considering program sustainability, industrial/individual incentives for commitment ○ Design financial monitoring/evaluation and determine financial institution's requirements and compliance ○ Develop financial scheme guidelines and management arrangements ○ Determine criteria for funding eligibility ○ Selection process for funding project proposals to build the replication plan ○ Identify capacity building requirements in the financial aspect and assist in the development of requisite training materials NOTE: All TORs must specify that the project consultants should have technical experience, familiarity, or specialization as maybe necessary to the tasks involved in pertinent aspects and practices in biomass-based resource fuel production, processing, logistics and distribution and utilization for energy purposes.
International contracting		
• Project Technical Specialist for Demos		*Duties and Responsibilities* • Provide advice to the PMU in all aspects of the PASBET Demonstrations and providing technical support to users and communities • Evaluate proposals and TORs for the supply, installation, operation and systems from contractors and individual proponents • Design a performance tracking system and contract performance monitoring of contractors and consultants to deliver expected outputs and deliverables as basis for contract payments • Maintaining system functionality and attend coordination meetings as assigned by PMU

ANNEX 8: PROJECT STAKEHOLDER ENGAGEMENT PLAN

Based on the stakeholder analysis, the project's key players include MoCC&EC, EAD, MoPDR, Provincial Forestry Departments, AEDB, NEECA, Research and Academic Institutions, Civil Society Organizations/ Community Based Organizations/Farmers' Associations MOF, NDRC, MEE, local communities/governments, and the private sector. They will take active part in the implementation of the PASBET project activities and cooperate in the monitoring and reporting as well as knowledge management process in preserving and disseminating knowledge on lessons learned and practices to aid in decision making while the others will assume either supporting or beneficiary roles.

The cross-sectoral and participatory nature of the project requires involvement of wide range of stakeholders at different stages of project implementation. Below is the framework for the stakeholder areas of responsibilities. Since there was limited opportunity to conduct a meeting with intended representatives of the project stakeholders, the list and roles of stakeholders will be validated and a comprehensive "stakeholders participation strategy" with a clear road map will be developed from the following stakeholders' roles and duties during the inception phase of the project involving also local user groups, traders, and entrepreneurs, including both men and women will be devised.

The key stakeholders that were identified through the project development process are listed in table below.

Table 7: Stakeholders Roles and Duties during the Project Implementation

Stakeholders	Roles and Duties during Project Implementation
Ministry of Climate Change & Environmental Coordination (MoCC&EC)	MoCC&EC oversees inter-provincial coordination of forestry-related matters and is responsible for overseeing the implementation of the Project with the provincial forest departments, chairs the Project Board in monitoring and decision-making in policy, strategy, and action plans for the Project to ensure sustainable management of woody biomass in the country. Being the Implementing Partner, it is also responsible for meeting international obligations under various Multilateral Environmental Agreements (MEAs). The Ministry will also be the executing agency to facilitate the inter-provincial coordination during implementation phase of the project and ensure inter-sectoral coordination with the relevant ministries (particularly for energy, forestry, environment sectors for planning, implementing, monitoring and taking actions on implementation issues and barrier removal for sustainable biomass production, decentralized power generation and introducing efficient woody biomass energy conversion technologies.
Economic Affairs Division (EAD) and Ministry of Planning Development and Reforms (MoPDR)	The EAD and MoPDR will provide guidance in project planning and implementation phase, including coordination with Finance, and Development Planning ministries to realize the co-financing commitments of the project. The Centre for Rural Economy (CRE) operating under the MoPDR will support and recommend ways and means in filling gaps in woody biomass based rural economy and to study its value chain and recommend ways and means to improve

Stakeholders	Roles and Duties during Project Implementation
	supply chain for meeting needs producers and consumers, as well small farmers.
Provincial Forestry Departments	The provincial Forest Departments of Punjab, KP, Sindh, and Balochistan will actively plan and implement the project activities, procurement and technology demonstrations that are committed by the Departments in terms of co-financing resources, site selection, planning, technical support, implementation, and stakeholder engagements. They will also oversee the management, operation and maintenance of the demonstration facilities, plantations, fuel processing and logistics, and the storage facilities for in the, as well as provide technical support for raising energy plantation and ensure realization of the provincial co-financing inputs of the project.
Ministry of Energy (Power Division) – Alternative Energy Development Board (AEDB)	AEDB, being mandated by the Federal Government to facilitate, promote, and encourage development of Renewable Energy in Pakistan and with a mission to introduce Alternative and Renewable Energies (AREs) at an accelerated rate, it will provide technical support in the implementation of the project activities on the formulation of policies, programs and initiatives through private sector that are supportive of woody biomass energy generation, in line with achieving sustainable economic growth. It will provide advice on the in the planning and implementation of transfer of technology and development of local technical services on woody biomass resources.
Ministry of Energy (Power Division) - National Energy Efficiency & Conservation Authority (NEECA)	NEECA, being the federal focal agency for initiating, catalyzing, and coordinating all energy efficiency and conservation activities in different sectors of economy, will provide support in developing mechanisms and procedures for conservation and efficient use of energy. NEECA has been charged with a wide range of responsibilities. They will support the development and enforcement of policies and regulatory frameworks during the project implementation. NEECA will also take lead role for technology transfer, capacity development and organization of trainings for capacity building of individual, SMEs, and private sector in designing and using energy efficient wood-fired industrial equipment.
Research and academic institutions	National and provincial research institutions including Pakistan Forest Institute and National University Science and Technology (NUST) will be engaged in the development and implementation of capacity building programs for farmers, woody biomass producers, traders, and other actors involved in biomass supply chain. Moreover, the Global Change Impact Studies Centre (GCISC) of MoCC&EC will lead in the relevant research and development activities of the project and support establishing online Knowledge Management (KM) repository.
Civil Society	CSOs/local CBOs including Women Organizations being important

Stakeholders	Roles and Duties during Project Implementation
Organizations/ Community Based Organizations/ Farmers' Associations[91]	stakeholders will be informed and be engaged through consultative meeting and stakeholders' workshops to seek their input and identify their roles during implementation of project activities. Farmers' association will be established and involved in raising woodlots and promoting fuelwood production on farmlands.
Private Sector (Electric Power Producers and Rural Industries)	Private sector such electric power producers, woody biomass traders, rural industry companies will be involved in the project planning and implementation by continuing to hold special consultative sessions with biomass energy consumers and individual meetings as well as ensuring the active participation, leading to investments by the private sector during implementation of the project. This will include controlling deforestation, raising energy plantations, and introducing energy efficient technologies. They will also be encouraged to provide co-financing the project activities and derive GHG emissions reduction benefits and making investments for raising woodlots and technology transfer.
International NGOs, UN Agencies, and Donors.	International NGOs like International Renewable Energy Agency (IRENA), ICIMOD, and ACARDA which have been involved in renewable energy (including biomass energy) studies, assessments, and mapping, will share information resources in relevant areas necessary in project implementation. Similarly, UN Agencies—UNIDO, FAO and UNDP will continue to be involved in special studies and bio-energy resource mapping. Multi-lateral donors such World Bank and Asian Development Bank will continue assisting Pakistan for assessment of and tapping renewable energy potential in the country.

There is a big potential for the private sector to participate in the Project implementation and become a major key stakeholder and player with regards to woody biomass production and utilization for energy generation in power and non-power application. At present, there is yet to happen more significant private sector engagement. To remove one of the barriers identified and is herewith being addressed by the Project is the development and facilitation of the adoption of policies, rules, and regulations on the private sector engagements in biomass-based development activities at the provincial level. The country has adopted a decentralized approach in the execution and administration of the government programs

[91] The discussions with the local communities were mainly part of the stakeholder consultation activities that they have done under the Government's Ten Billion Tree Tsunami Program. The discussions under that program were done through workshops and focused consultations and were in line with the objective to pave the path for job creation for common people through green initiatives. The government is focused to increase the number of private nurseries for raising plants to generate the economic opportunities. Hence, the discussions with the local communities in the planned project districts was, among others, on how the proposed project will build on the planned technical and financial support that will be provided to local farmers and community organizations under the program for establishing farmers/communal nurseries to provide planting stock for raising woody biomass on forested, non-forested and farmlands by involving individual farmers and community organizations. Among the discussions that were carried out with stakeholders were with the technical personnel of the Provincial Forestry Departments, who themselves are regularly in touch with local communities including those communities in the planned project districts. The Forestry personnel informed the project proponents about the suggestions of the local communities in comprehending the key challenges to tackling deforestation and in coming up with concrete measures to tackle the issue with other innovative measures such as the planned interventions of the proposed project.

at the provincial level. There are some cases of privately owned energy plantations which are referred to as woodlots on private lands. The situation is varied across the four provinces selected under the Project. The Ten Billion Tree Tsunami (TBTT) Project will provide a big boost and resources on public and private farmlands. The provincial governments through the Forest Departments are providing various forms of subsidies to the tree farmers through different development projects to increase tree cover on private farmlands using varied modes of participation in achieving the goals in tree plantation coverage. On the fuel wood utilization for energy generation, the Government has launched many related projects and established platforms where private sector participation will be encouraged. The Project has included feasibility studies of sustainable woody biomass fuel production facilities which are envisaged to attract private sector investments, particularly in Component 2.2, where commercial models will be demonstrated during the Project and planned for replication and upscaling in the coming years.

The main private actors will be woody biomass-based small and medium enterprises (SMEs), and businesses engaged in woody biomass fuel production, their role/involvement in the rural electrification program of the country, in promoting energy efficient biomass energy technologies in rural industries, raising energy plantation, and woody biomass energy production and supply. These entities will be encouraged to participate in: (a) woody biomass energy generation and supply; (b) woody biomass supply chain analysis and identification of market needs; (c) raising energy plantations and wood lots; (d) development of business-based energy models for woody biomass based SMEs in collaboration with local communities; (e) development of partnership with the project for transfer of technology and introduction of best practices for reducing GHG emissions from the energy consuming operations in rural industries; and, (f) capacity building and knowledge management interventions targeting biomass production and promoting biomass energy efficient technologies in power and non-power applications, and public awareness raising campaigns. The Project will develop and facilitate the adoption of a platform for synchronizing its policy and regulatory development activities with significant policies that are found relevant to the development of biomass energy for rural electrification with potential private sector involvement such as in: National Electricity Policy 2021, Energy Regulations, NEPRA licensing, public-private partnership in power generation of the Government that shifted to the provincial administration where there are different approaches, legal frameworks and policies for each province in a decentralized authority and execution as overseen by the Public Private Partnership Authority (PPPA) as amended in 2021. The Project has therefore included in its activities the policy development and adoption for widening the public private participation in energy generation.

Follow-up Stakeholder Consultations

During the project development stage, the project development team (PDT) including the concerned UNDP-Pakistan personnel conducted consultation meetings with the key stakeholders such as the Ministry of Climate Change & Environmental Coordination, Ministry of Planning Development and Reforms, provincial Forest Departments, and provincial Energy Departments. However, there was limited opportunity for a broader and detailed stakeholder consultation, particularly with local communities and other local government departments due to COVID-19 restrictions. The public health restrictions during the pandemic led to closure of government offices and private businesses and strict protocols for keeping social distancing. Hence, the limited interaction with stakeholders in the communities where many of the planned project activities (e.g., demonstrations) will be implemented.

During the inception phase of the project, the project team and the technical personnel that will be onboarded will come up with a concrete plan for carrying out more detailed consultations with the project stakeholders particularly those that were not consulted during project design stage, such as other

provincial government departments, community leaders and district/local government institutions in the project areas. Follow-up consultations will also be done, when and if necessary, with the key stakeholders that were consulted during the design phase of the project. The following is a summary of stakeholder engagement activities that will be carried out in preparation for, and during the project inception phase:

- Telephone calls to stakeholders to organize meetings, follow-up with appointments and provide further information for stakeholders;
- Email exchange with stakeholders to provide further information on project scope, demonstrations, and value-adding initiatives for the project;
- Attendance in specific meetings with the PDT staff and the identified potential co-financers, and implementers of identified baseline project to learn about potential synergies from such projects and share project information;
- Organized stakeholder consultations;
- Field visits and focus group discussions on project plans, benefits, risks, impacts and community interest and engagement; and,
- Focus group discussion with women on gender roles related to the planned project activities, benefits, risks, impacts and interest and engagement.

Subsequently, the information that will be gathered during these consultations will be used in preparing a more detailed Stakeholder Engagement Plan. Such plan will also specify, among others, the roles, and responsibilities of various project participants/stakeholders such as: (a) Linkages and coordination between participants and activities; (b) Role of each participant in the delivery of project outputs; and (c) Strengthening of links to project stakeholders. The plan will be presented during the project's inception meeting for discussions and endorsement of the stakeholders.

ANNEX 9: GENDER ANALYSIS AND GENDER ACTION PLAN

1. Project Overview

This document presents the gender analysis and action plan for the UNDP-GEF project "Combating Climate Change through the Promotion and Application of Sustainable Biomass Energy Technologies in Pakistan (PASBET)."

The objective of the PASBET Project is the widespread application of sustainable biomass energy technologies for supporting socio-economic development of and reducing greenhouse gas (GHG) emissions from, the rural sector in Pakistan. The project's activities are expected to increase the use of indigenous renewable energy (RE) resources with woody biomass being the primary contributor. The project's goals are also in line with the government's plan of raising the share of renewables in the power generation mix by 2030.

The transformation in the rural energy situation through the use of sustainable biomass energy technologies at a larger scale will massively contribute to the socio-economic development in the region along with the following changes:

- Effective enforcement of policies and regulations on the sustainable production and use of woody biomass for energy generation and utilization
- Enhanced woody biomass production on forested, non-forested, and on farmlands for woody biomass energy supply
- Increased investments in the application of technologies for the production, and energy efficient utilization of woody biomass energy
- Improved availability of financial resources available for woody biomass energy technology application projects
- Enhanced local capacity, skills and knowledge in the development, installation, and operation of biomass energy technology systems

In order to achieve these outcomes, the project will attempt to address policy, planning, financial, and knowledge barriers inhibiting the production of sustainable energy through biomass. The project has four main components with an overarching strategy of removing current/existing barriers to help achieve project objective.

Component 1: Establishment of Policy and Regulatory Framework for Sustainable Woody Biomass Energy Production and Utilization
Component 2: Promotion of Woody Biomass Energy Production and Energy Efficient Utilization Technologies
Component 3: Supporting Financial Requirements for Biomass Energy Technologies Initiatives
Component 4: Biomass Energy Technology Capacity Building and, Knowledge Management and Gender Mainstreaming

2. Gender Mainstreaming Strategy for the Project

Gender mainstreaming is an integral part of the project, given the role and involvement of women in the biomass energy cycle; and this will be reflected in the project activities throughout its duration. Women's representation and participation in the decision-making process will be a high priority along with access to woody biomass and energy efficient technology/equipment used in houses and rural cottage industries. Women play a critical role in the collection, use and management of natural resources used for biomass-based energy production. Moreover, women's health and safety are also severely affected in the process and a majority of rural area residents are not fully aware of the extent of damages caused to women during these processes. Overall, the project will contribute towards improved health and better socio-economic conditions of women in rural areas.

Based on the research conducted on the biomass energy scenario and the role and status of women involved, a gender mainstreaming strategy was formed in the Gender Action Plan and has been incorporated in the overall project strategy. The strategy is designed to guarantee women's participation in teams and capacity building to help them fulfil their roles. The project targets and indicators have been disaggregated by gender which will help to define the role of women beneficiaries: gender considerations have been included at all project levels. Overall, the project will attempt to provide equal opportunities for women to participate in the project and benefit through the following set of activities:

- Discussions with stakeholders/women in project areas on the status and needs of women in the area and the potential barriers that the project will face in achieving gender objectives
- Strategies to tackle the challenges and issues/barriers that women face based on cultural constraints in the regions and their socio-economic status
- Inclusion of gender-responsive measures into the project logical framework, monitoring, evaluation, and reporting process such as gender-sensitive indicators, sex-disaggregated targets, and reporting requirements.
- Activity planning to ensure equal participation of women at all stages of the project
- Measures to ensure that women are part of all teams including the project implementation staff

The gender mainstreaming strategy is consistent with Sustainable Development Goal (SDG) 5. This project is also in line with Government's priorities to reduce gender inequality by advancing women's social, economic, and political empowerment.[92]

3. Gender Analysis

3.1 Gender Situation in Pakistan

According to the Global Gender Gap Index, Pakistan is currently ranked as the second worst country in the world, scoring very poorly on most of the gender equality parameters. Health is one the primary issues that women face in the country: despite being the sixth most populous country in the world with a remarkably high fertility rate of 3.8 children per woman, there are gaps in the fulfillment of women's reproductive rights and access to quality healthcare for women throughout the country. Pakistan suffers from a lack of access to quality family planning services throughout the country, which has resulted in one of the highest unmet needs in Asia. This prevents women from meaningfully participating in educational

[92] United Nations, Country program document for Pakistan (2018-2022)

and economic life outside of the household and from making informed choices about their sexual and reproductive lives.[93]

Gender-based violence is also prevalent in both the rural and urban areas along with women suffering from domestic abuse, honor killings and sexual violence to institutional discrimination. Barriers also exist in the access to education for women in the country which has resulted in a lower literacy rate for women (47%) compared to men (71%). Enrolment rates in schools are alarmingly low for girls. 32% percent of primary school age girls are out of school, compared with boys. This gender gap differential continues into secondary forms of education: only by 13% of young women are still enrolled in middle school.[94] This has long term effect for women in the economic sphere where they constitute a small part of the workforce, and their earnings are significantly lower compared to their male counterparts.

Moreover, these multidimensional gender inequalities have gotten worse because of covid-19 with regions of high fragility, conflict and poverty suffering the most from this change: the women in these areas are now exposed to greater health risks as well as potential loss of income. With women being confined to their homes for longer period s of time, the burden of unpaid care work on them has increased and they are more vulnerable to gender-based violence.[95]

3.2 Status of Rural Women

Women face greater inequalities in terms of rights and access to facilities in rural areas partly because of the pervasive conservatism. Rural areas are home to almost two-thirds of Pakistan's population with agriculture being the main labor activity of rural women: approximately 75 percent of women and girls are employed in the agriculture sector. Women involved in the sector spend over 34 hours per week on agriculture in addition to their reproductive and care work at home which can equally time consuming. Given the average family size in rural areas is 6.8 persons, the reproductive work burden along with other tasks coupled with poor health services often takes a toll on women's bodies and results in high malnutrition rates. Women have extremely low access to quality health facilities in rural areas which results in chronic health issues throughout the country. Moreover, lack of proper nutrition in their diets due to low household incomes means that they are vulnerable to multiple diseases and weak in general, which in turn also affects the health of their children. This is further compounded by an increased burden of on-farm and off-farm activities, lack of clean drinking water and appropriate sanitation facilities.

Moreover, women's work remains unrecognized, unpaid, or underpaid and is based on need and often without choice in most cases. Moreover, women farmers perform most of the activities manually and increases their workload which affecting women's productivity and wage rate.[96] This is so the case with women involved in the collection and use of firewood for the production of biomass-based energy in the project areas.

Gender inequalities also exist in the form of land ownership: male household members hold land and women are denied access to land even when they have rights mostly due to cultural settings. This, in turn, also affects women's ability to access credit to set up any kind of business initiative. The literacy rate

[93] https://pakistan.unfpa.org/en/topics/gender-equality-14

[94] https://edufinance.org/latest/blog/2021/empowering-girls-through-education-in-pakistan#:~:text=In%20Pakistan%2C%20an%20estimated%2022.5,compared%20with%2021%25%20of%20boys.

[95] https://www.theigc.org/blog/has-covid-19-exacerbated-gender-inequalities-in-pakistan/

[96] UN Women (2018)

among rural women is extremely low because of a lack of access to education facilities, conservative culture, and early marriages: Rural women ages 15-64 have aliteracy rate of 35% compared to 47% for women urban areas. Though social relations and cultures vary in provinces and within districts, the discriminatory social gender norms persist and affect all rural women.

Pakistan is part of key international commitments to gender equality and women's human rights – the Universal Declaration of Human Rights, Beijing Platform for Action, the Convention on the Elimination of all forms of Discrimination against Women, and the Sustainable Development Goals. Moreover, national commitments in place include a National Policy for Development and Empowerment of Women, Protection against Harassment of Women at Workplace Act, Criminal Law (Amendment) (Offences in the name or pretext of Honor) Act, Criminal Law (Amendment) (Offences Relating to Rape) and a National Plan of Action on Human Rights. Local commitments adopted include Gender Equality Policy Frameworks and Women's Empowerment Packages and Initiatives. However, work needs to be done to ensure that the status of the women in the country is improved in the long run through the proper implementation if these acts. "The empowerment of rural women and girls and the realization of their human rights and gender equality are essential to the achievement of the Beijing Declaration and Platform for Action, the Sustainable Development Goals (SDGs) as well as the Government of Pakistan's Vision 2025"[97]
However, it is important to recognize that the main the majority of issues that women are facing including constraints on mobility, health problems, and access to and control over farm assets and incomes are the result of pervasive socio-cultural norms, more than the legal or regulatory systems.

This does not undermine the importance of regulatory reforms and agreements in place in strengthening the position of women. This is not to underplay the importance of legal and regulatory reforms to improve the position of women, but to point out that such reforms by themselves will not be remarkably effective unless the importance of socio-cultural norms are considered in formulating effective strategies to improve the position of women. Certainly, the sociocultural norms themselves are changing everywhere in Pakistan, including in the rural areas, but they do and cannot be changed quickly or easily through changes in policies and regulatory reforms alone. Practical policies to improve the position of rural women, including policy and institutional reforms in horticultural markets, must consider this sociocultural reality.

3.3 GEF Gender Implementation Strategy

The GEF Gender Implementation Strategy identifies **three gender gaps** most relevant to GEF projects and programs in the GEF-7 programming directions: **Unequal Access to and Control of Natural Resources, Unbalanced Participation** and Decision Making in Environmental Planning and Governance at All, **Uneven Access to Socio-Economic Benefits** and Services. The gender analysis for this project found some similar challenges and have been discussed above.

3.4 Biomass and Women's Health

Bioenergy is the primary source of energy production in rural areas of the country: fuelwood and other forms of biomass are used by over 90% of rural families to fulfill their energy needs[98] Burning biomass results in the emission of carbon dioxide along with a number of other harmful chemicals which are

[97] https://asiapacific.unwomen.org/-/media/field%20office%20eseasia/docs/publications/2018/08/status-of-the-rural-women-in-pakistan-report.pdf?la=en&vs=3739

[98] https://www.mdpi.com/2077-0472/11/7/596

categorized under Indoor air Pollution (IAP). Exposure to the fumes resulting from burning biomass increases the risk of a variety of diseases including pneumonia, chronic respiratory diseases, and lung cancer.[99] Women and children are particularly vulnerable to these risks because of the prevalent gender roles and responsibilities: women are responsible for cooking and spending time indoors along with children. Coupled with the size of the kitchens which are often small and not properly ventilated, the high levels of pollutants put Pakistan high on the list of countries affected by IAP. According to estimates, solid fuelwood was responsible for almost 70,700 every year while the percentage of national burden of disease attributed to solid fuel use is 4.6%, as compared to less than 1% seen in the developed world.[100] Moreover, exposure to the smoke from inefficiently burnt biomass fuels increases the chances of stillbirths among women as well as low birth weight (LBW) deliveries. The introduction of energy efficient biomass burning technologies, therefore, can have a major impact on the lives and health of women in the region.

3.5 Importance of Gender Mainstreaming in the project

Both women and men share different roles and responsibilities in the biomass-based energy production process: both have knowledge about, and varying skills in, use of resources such as fuelwood for energy production use. Women's vital contributions to the process through the collection and use of firewood, however, are not always fully acknowledged and given the importance that it deserves. One of the primary reasons for this is the lack of gender-differentiated perspectives in biomass research and documentation, which itself has not been undertaken at an extensive scale at the provincial or national level. Both genders' perspectives need to be a part of biomass studies to ascertain the efforts required for setting up fuelwood plantations and biomass-based energy production. Women's voices on the subject matter need to be amplified and their needs and requirements understood to ensure the success of projects in rural areas where women are involved.

The gender action plan, therefore, is built upon:

- Comprehensive collection of gender-related information and sex-disaggregated data to get a more holistic picture of the scenario in project areas
- Incorporation of gender concerns in the theory of change that the project is based on which will assist in the formulation of interventions for specific outcomes

4. Gender Action Plan

4.1 Indicative Gender-Responsive Activities

Indicative gender-responsive project activities are listed below in **Table 2**:

Component 1: Establishment of Policy and Regulatory Framework for Sustainable Woody Biomass Energy Production and Utilization				
Outcome 1: Effective enforcement of policies and regulations on the sustainable production and use of woody biomass for energy generation and utilization in rural areas of Pakistan				
Project Output	**Intended Gender Equality Results**	**Gender Responsive Actions**	**Indicators to be included in M&E**	**Timeline**

[99] https://pubmed.ncbi.nlm.nih.gov/20717550/

[100] https://pubmed.ncbi.nlm.nih.gov/20717550/

Output 1.1: Completed comprehensive situational and feasibility analyses for mainstreaming woody biomass production in the agriculture and forestry sectors, including assessments of potential areas for sustainable woody biomass production and utilization[101] in Punjab, Khyber Pakhtunkhwa (KP), Sindh and Balochistan provinces	Situational and feasibility analyses to be gender sensitive and inclusive of women's knowledge and opinions	• Incorporation of gender equality and women empowerment as essential elements as part of feasibility analysis on mainstreaming woody biomass production. This could be done including gender-environment experts and local women who are involved in the process in discussions, conducting background studies; and building gender-specific capacity among stakeholders at all levels. • The data on the collection, management, control, and usage of resources should be gender-disaggregated: women and men's knowledge on these issues varies and it is essential that the data on different needs and requirements of both genders should be documented • Women-based organizations who have worked on similar projects involving gender disparities in areas should be involved in the research process to ensure that the best practices that they have implemented are effectively used.	Number of gender experts involved Proportion/ Percentage of women involved in the discussions for feasibility analysis	Year 1
Output 1.2: Developed and implemented policies and market-based regulatory framework for supporting woody biomass production and use, including national strategy for promotion of sustainable biomass energy production and utilization, using community-based woodlots and agroforestry production at the provincial level, and fiscal incentives for private-sector participation in woody biomass energy technology business	Gender mainstreaming in regulatory frameworks	• Policies and market-based regulatory frameworks that will be developed should include a gender mainstreaming dimension. Officials from government agencies, as well as non-governmental organizations that are part of the development process should have specific/reasonable proportion of women involved in the strategy and implementation process. • Legislations developed by the government that have successfully incorporated gender mainstreaming should be studied and used as an example during the development process. One such legislation which has included gender sensitization in a comprehensive manner and can provide lessons and best practices is the Punjab Protected Areas Act, 2020. Though the culture and social settings in rural areas among the provinces differs to some degree, the approach taken/used by Punjab and can be molded to implement	Level of participation of women in development and implementation of policy frameworks	Year 1

[101] This also includes the utilization of available forestry waste such as twigs and branches of standing trees, leftover wood, and branches after commercial harvesting of sustainable forests.

		policies in KP, Balochistan and Sindh.		
Output 1.3: Formulated and approved technical, management and operational standards in biomass energy development and utilization, woody biomass fired equipment and comprehensive biomass energy management	Women's voices to be included in the formulation process	• Ensure the inclusion of women in the committees/groups responsible for the formation of technical, management and operational standards. The women involved in the process should have equal decision-making powers so that their participation can influence the final outcomes.	Level of participation of women in formation of standards to ensure gender considerations are included	Year 1
Output 1.4: Formulated energy-integrated development plans of 8 pilot towns two in each province of Punjab, KP, Sindh and Balochistan	Women's needs to be incorporated through their participation in the decision-making process	• Development plans to incorporate cultural values as well as gender relation and roles in different provinces: implementation processes to involve women in various capacities based on the prevalent scenario in each province	Level of participation of women in community consultations leading to the development plans	Year 1 & 2
Output 1.5: Trained and qualified 60 provincial governments personnel (15 from each province) that provide technical support on sustainable production of woody biomass and enforcement of regulatory framework for biomass use for energy production and utilization	Equal representation of women as trainers	• Equal selection of women in the government staff from each province that will provide technical support. • Capacity of selected government staff on gender issues to be improved through gender sensitization training. The training material can be developed with the help of women-based organization working in the country (or province specific if required). • Gender Focal Points need to be activated in the forest departments of all four provinces. These will constitute existing staff members and their primary role will be the promotion of gender issues on the project and assistance in monitoring progress towards achievement of gender mainstreaming objectives.	Number /proportion of female trainings and the training sessions provided to them Number of gender focal points in relevant Government departments	Year 1 & 2
Component 2: Promotion of Biomass Energy Production and Energy Efficient Utilization Technologies				
Outcome 2.1: Enhanced woody biomass production on forested, non-forested, and on farmlands to cater local biomass energy needs, including for power generation and rural industry operations				
Project Output	**Intended Gender Equality Results**	**Gender Responsive Actions**	**Indicators to be included in M&E**	**Timeline**
Output 2.1.1: Developed, disseminated, and applied guidelines for sustainable	Representation of women in the guideline development and disseminating	• Women's participation to be ensured in the guideline development process at all levels: people have complementary knowledge about biomass	Number of women participants in the guideline development	Year 1

production and utilization of woody biomass	process	resources and utilization and incorporating this knowledge is essential in ensuring that the guideline document is comprehensive and useful for the community. • Ensure the presence of women when disseminating guidelines to users of woody biomass. Women are the primary users of woody biomass in some of the regions and it is essential that they are part of this learning process. Cultural differences in provinces and rural areas will have to be kept in mind in this process: women trainers/facilitators will have to be used/hired for the capacity building of women in some of the areas	process Number/ proportion of women who receive guidelines for sustainable production and utilization of woody biomass	
Output 2.1.2: Completed supply chain and market analysis and documented woody biomass demand and supply in the country	Representation of women in the market analysis process	• Women's views and opinions to be comprehensively represented in research during supply chain and market analysis. This can be achieved by ensuring that all genders have equal participation in the surveys carried out for this analysis. • Women to be involved in survey and report preparation: qualified women to be selected through involving women-based organizations working in the regions	No of women involved in the research / market analysis process	Year 1
Output 2.1.3: Established farmers/communal forest nurseries over at least 300 hectares to provide planting stock for raising woody biomass in rural areas in the provinces of Punjab, KP, Sindh and Balochistan, including energy plantations over 40,000 ha of fast-growing native tree species at non-forested lands in selected districts each of these provinces for woody biomass supply	Equitable distribution of nurseries/ woodlots for all genders	• Involvement through agroforestry and reforestation across all targeted districts could be ensured for women. Nurseries can be established at the community or household level where women can be trained on the growth and management of fast-growing native tree species at non-forested lands for raising woody biomass stock • Exclusively female-led nurseries could also be established (with a separate budget) particularly in regions where cultural practices do not allow men and women to work together. Given that this employment for women will benefit the household as a whole while incorporating traditions and values, it will be easier to get male household members on board to support this incentive. This will act as additional opportunity for women's employment that will not threaten the livelihoods of men in the area.	No of women who receive ownership of nurseries / proportion of nurseries with female owners	Year 1-3

		• The benefits of involving women in nurseries and forest management will have long term benefits other than the economic growth that it will offer at the household level. At a community level, these changes will create an impact on education, health, economic uplift of women, voice, and agency for women, and contribute to the socio-economic empowerment of the women in the communities.		
Output 2.1.4. Established and operational fuelwood production from 4.5 million trees of common fuelwood plant species on farmlands for supply of woody biomass for production and utilization of biomass energy in rural areas.	Involvement / representation of women/females in the production of fuelwood	• Equal/high number of women to be trained in the operation of fuelwood production in all regions • Given that women are currently involved in the process of fuelwood production, training and employing women will be smooth and easier to implement, and will contribute to enhancing their skills • Along with empowering women in terms of work, involvement of women will also give them job security and a reliable source of income in the long run based on targets set for fuelwood production	No of women involved in the fuelwood production process	Year 1-3
Outcome 2.2: Increased investments in the application of technologies for the production, and energy efficient utilization of woody biomass energy				
Project Output	**Intended Gender Equality Results**	**Gender Responsive Actions**	**Indicators to be included in M&E**	**Timeline**
Output 2.2.1: Completed and operational woody biomass-based energy generation technology demonstrations in 4 selected sites, one in each province of Punjab KP, Sindh and Balochistan showcasing the cost-effective application of decentralized woody biomass-based electricity generation and distribution (through mini/micro-grids)	Selection of women as technical trainers	• Technical training for selected women in the area who will be responsible for/tasked with technology demonstrations along with men is one of the which women can be involved in this process. Women in general and those in rural areas in particular are not usually employed in technical jobs in the country. However, this project represents a od opportunity to involve women given their association with the collection and use of fuelwood for biomass energy at their homes.	Number/ Proportion of women selected as technical trainers	Year 1
Output 2.2.2: Completed and operational three (3) demonstrations of cost-effective production of woody biomass fuels such as	Participation of women in demonstrations	• Being the primary source of collection and use of woody biomass at homes, involving women in the demonstrations and training them is essential in implementing these processes/activities in the long run. The project should aim to	Number and percentage of women who are part of the demonstrations	Year 1

wood chips, briquettes, and pellets for use in decentralized woody-biomass based power generation and distribution.		target an equitable representation of women (50%) in this activity.		
Output 2.2.3: Completed and operational demonstrations of the cost-effective applications of energy efficient woody biomass fired technologies and comprehensive energy management systems in selected energy end-use sectors in support of rural socio-economic development	Participation of women in demonstrations	• Given that the project will be implemented at a large scale and the number of women required to operate biomass plants will continue with the expansion of the project, women can be trained and employed along with men where cultural conditions allow it. This can be initiated with the training of a batch of selected women from the communities with their proportion to be increased with the passage of time.	Number of women trained from the communities to operate biomass plants	Year 1
Output 2.2.4: Published and disseminated information about the results and impacts of the completed demonstrations as well as developed and implemented sustainability and replication plan of decentralized woody biomass-based energy production and utilization for power and non-power applications	Involvement of women in all stages of the documentation and dissemination process through equitable representation	• Women to be involved in the documentation, publication and dissemination of the results and impacts of the demonstrations. Constructing a gender-balanced team that works on the results will also help in documenting information that will be collected from women in some of the more conservative regions. • Success stories for women farmers/workers to be an integral part of the published information: the case studies will depict the changes in economic conditions of these women and the improvements that families will see at a household level • Partnerships with students form universities (and female students in particular) could also help with for mapping the impact of the technologies through various economic and social indicators.	Number of success stories of women who have benefitted from the completed demonstrations	Year 1
Component 3: Supporting Financial Requirements for Biomass Energy Technologies Initiatives				
Outcome 3: Adequate amounts of financial resources available for woody biomass energy technology application projects in the country				
Project Output	**Intended Gender Equality Results**	**Gender Responsive Actions**	**Indicators to be included in M&E**	**Timeline**
Output 3.1: Established and operational investment and financing	Improved access to finance for women	• Create and strengthen existing partnerships between business enterprises and local communities through integrating/the use of social associations (community	Number of women with improved access to finance/credit for setting up	Year 1 & 2

mechanisms for supporting the commercial viability and operation of woody biomass energy production for power and non-power applications, and the development of biomass energy industries		groups, village committees, women's groups, etc.), and raising awareness of financial opportunities available particularly for women. Direct market linkages with biomass-based industries for the communities collecting firewood with a focus on women • Improved access to credit/finance for communities and women in particular interested in setting up biomass-based production. Women face a number of barriers when accessing credit from banks or other institutions and the process is particularly difficult for those residing in rural areas. Institutions with a good track record of strengthening women's access to finance can be identified to setup a supportive system for the project	biomass-based systems	
Output 3.2: Established and operational market-oriented mechanism for the enhanced development and utilization of biomass energy resources, energy efficiency and comprehensive biomass energy management systems for supporting sustainable rural socio-economic development	Improved access to finance and new technologies for women	• Improved access for women to new and improved technology along with the establishment of credit lines to assist them in setting up biomass-based projects. Women in rural areas are not usually involved in technical aspects of the industry despite playing a major role in the collection and use of firewood. Giving them access to improved technology would allow them to make use of their existing differential knowledge and will assist them in improving their social and economic conditions through the adaptation of new technologies. • The promotion of energy efficient technologies (particularly for women) will be an improvement on their current work status and will act in the creation of new green jobs for them. Along with the financial and social benefits, this will also improve women's health in the long term given that hazards thy face from the burning of fuelwood at home.	Number of women with improved access to finance/credit for setting up biomass-based energy systems Number of women benefitting from the introduction of new technologies through sustainable and energy efficient biomass plants	Year 1 & 2
Output 3.3: De-risked biomass-based power generation projects, decentralized biomass-based energy generation in rural areas, and business plans for the GoP and private sector to facilitate				

financing and implementation				
Component 4: Biomass Energy Technology Capacity Building and Knowledge Management and Gender Mainstreaming				
Outcome 4: Enhanced local capacity, skills and knowledge in the development, installation, and operation of biomass energy technology systems in rural Pakistan				
Project Output	**Intended Gender Equality Results**	**Gender Responsive Actions**	**Indicators to be included in M&E**	**Timeline**
Output 4.1.1: Developed and implemented gender sensitive capacity development training programs for the design, construction, operation, and maintenance of woody biomass-based energy production systems both for power (e.g., decentralized power generation and distribution) and non-power (e.g., industrial energy efficient wood-fired equipment) applications, as well as for strengthening institutional capacity for supporting sustainable biomass energy production and utilization	Successful implementation of gender sensitive capacity development training Gender sensitization of community field staff	• Development of training toolkit which incorporates gender issues in the regions: best practices from existing training plans can be used as well as the services of a gender specialist to design the training • Training, extension services and other initiatives especially designed for rural women based on their needs and requirements • Gender sensitization of community field staff through the training program to familiarize them with issues that women face in the region and steps required to ensure that women benefit from the project goals as planned. A critical aspect of this activity is to training female staff to effectively mobilize women who are project beneficiaries • The following approaches for design of agriculture and energy-based projects have proven to work well with regards to gender sensitization and should be incorporated in the training content: • Promote collective action among women • Do not exclude men • Protect women's rights and control over economic gains • Take necessary measures to • ensure women's voice and • representation • Recruit and train women as service providers • Involve rural women in the design of innovative products and services • Strengthen investment in research and sound impact assessment.	Number of people trained in gender sensitive capacity development Number of community staff trained on gender issues and gender sensitization	Year 1 & 2
Output 4.1.2: Trained and qualified people (60 from each province) providing training on woodlot	Equal representation of women as master trainers and highlighting of	• Women to be given equal opportunities as master trainer (60 from each province) given their role in wood collection and management. This will also be	Number of women master trainers Number of	Year 1 & 2

operations and management, woody biomass fuel production, and woody biomass-based power generation and distribution, including trained and qualified community members (600 men and 400 women) providing training on the promotion of the widespread applications of energy efficient wood-fired equipment	women's health issues	essential in some of the regions where women are more comfortable being trained by their own gender primarily due to cultural values. • Awareness raising and advocacy campaigns focused on specific themes and aimed at defined target groups, e.g., farmers, women's groups, to complement the training process with females playing an important part in the process • Women's health issues to be highlighted in the training process given their roles as collectors and users of firewood in rural areas, they are most vulnerable to emissions resulting from the process: discussion on the advantages to their health and related social benefits from using energy efficient wood-fired devices will make the training more effective	female trainees with improved knowledge on women's health issues	
Output 4.1.3: Sensitized key value chain actors of biomass energy enterprises that provide technical services and invest in the applications of cost-effective, climate resilient and energy efficient woody-biomass energy technologies	Gender sensitization of key value chain actors	• Impart gender sensitization training to build capacity of all key value chain actors on the issue. • Overview of key gender issues in the country and in the specific energy sub-sector to be addressed so that all project related personnel have a good understanding of the needs of gender mainstreaming in the project: • Context (national policy/organizational policy) that the project is operating within • Key country and province level gender and energy issues that the project must be aware of, and could deal with • Existing experiences and opportunities for gender mainstreaming in the energy sub-sector • Aims of the project from a gender standpoint	Number of key value chain actors who have received training on gender mainstreaming in all processes	Year 1
Output 4.1.4 Knowledge products on lessons learned, best practices, etc. that are available through online repository for sharing experiences and replicating and scaling-up use of climate resilient and	Timely updating of the project's Gender Analysis and Gender Action Plan	• Regularly review and update the project's Gender Analysis and Gender Action Plan, encouraging adaptive management during the implementation of the project	Bi-annual / Annual update of the project's Gender Analysis and Gender Action Plan Development of an Impact Report at the end of the project to	Year 1-5

energy efficient biomass energy technologies			culminate best practices, lessons learned and create an online guidebook for scaling up energy efficient & climate resilient biomass energy technologies	

4.3 Gender Mainstreaming Framework

The project will support women's participation and directly benefit women in all regions in the following ways:

- Incorporating the knowledge and views of women through consultation at all stages of the project including the design of the livelihood improvement and incentive mechanisms aspect i.e., establishment of communal nurseries. Making use of women's perspectives and knowledge will ensure that the resulting interventions cater to the of the needs and requirements of both genders. In regions where cultural barriers pose difficulties in including women in consultations, separate sessions facilitated by females will be organized to make sure that they are aware of the activities being implemented and can voice their opinions accordingly. More importantly, men will be sensitized on gender issues within their areas to help them understand the requirements of women which will shape their views on the project activities.
- Equal consideration of women in surveys monitoring baseline knowledge, attitudes, and practices: project areas where women play key roles such as collectors and users of firewood for biomass production will be identified and women's knowledge will be recognized to ensure that their traditional experience and capabilities can be used and further developed. This will result in empowering them and highlighting their importance to the biomass-based energy generation process.
- The introduction of energy efficient and environmentally friendly designs of biomass-based power generation in rural areas will directly improve women's health in these areas. Women are exposed to emissions from the burning of fuelwood in traditional ways which has been proven to be detrimental for their health in the long run. The use of new and improved technologies will significantly improve this aspect of their lives as well as the children wo are also exposed to these emissions.
- Targeted training and technical assistance to women beneficiaries of the project to ensure that least 50% of total direct beneficiaries are women. Moreover, training will be deigned in a manner that guarantees to empower women and assist them in improving their livelihoods. This will also result in the improvement of women's health and their contribution to their families' well-being because of their critical role in society.
- The project also includes the training of government officials on technical support on sustainable production of woody biomass and enforcement of regulatory frameworks in all areas. Women officials will be given an equal opportunity and trained to help in the implementation the associated activities. Though the selection of women will have challenges given the relatively low number of women employed in government offices, selecting, and training them as part of the project will empower these individuals and have long term implications on the status and importance of women in these areas.

- The project will also ensure equal opportunity recruitment of women for positions within the project management office and associated positions that will be available once the implementation activities are initiated. This also includes master trainers (60 from each province) who will provide training on woodlot operations and management, woody biomass fuel production, and woody biomass-based power generation and distribution; half of which will be qualified women.
- Women will be trained on the growth and management of forests as part of the project's activity of establishing of communal forest nurseries at the community and household level to provide planting stock for raising woody biomass. Exclusively female-led nurseries could also be established (with a separate budget) particularly in regions where cultural practices do not allow both men and women to work together. This will provide women with a new and green/environmentally friendly source of employment and contribute towards an improved role and status in the society.
- The project will support the establishment of biomass-based business ventures are run by women or ensure a majority participation by women, as well as facilitating opportunities for women and women groups to participate in project activities aimed at introducing improved livelihoods activities.
- Specific principles and associated actions are described below in **Table 3**.

Table 3- Framework for Gender Outcomes

Strategic principle	Actions
1. Facilitating women empowerment	Equal representation of women in all teams including the government officials to be trained, head trainers for the use of biomass energy technologies and the project staff.
2. Enhancing gender equality	Ensure that women receive equal proportion of benefits (50%) through the use of new technologies as well as provision of nurseries to raise woodlots
3. Ensuring gender integration	Gender consideration to be mainstreamed in polices ta the national and provincial level
4. Promoting gender awareness	Gender awareness to be promoted throughout the life of the project. UNDP's Principles of Gender Sensitive Communications[102] to be used for communicating gender-sensitive awareness to all relevant stakeholders which will lead to empowerment of women in the region Guidance and support on how to detect, intercept, respond to, and prevent (or refer cases) of sexual harassment, gender-based violence or related problems that could arise will also be a part of gender awareness training

Gender Marker

Gender Marker Index: 2

'Advancing gender equality is a significant objective but not the principal reason to undertake this project. Gender is reflected in the Conflict Analysis, Implementation/Activities, the Results Framework, and the Budget.'

- Gender equality and women's empowerment is a significant objective of the project as highlighted in the Gender Action Plan

[102] http://www.jm.undp.org/content/dam/jamaica/docs/gender/JM-AUG-29-UNDP%20Gender%20Seal-Principles%20of%20gender-sensitive%20communications.pdf

- The distinct requirements, skills, and capabilities of all genders have been incorporated in the project description and the activities related to outcomes
- Men, women, boys, and girls are targeted by the project and their distinct needs and capacities are reflected in the project description and multiple activities address barriers to gender equality and women's empowerment.
- Component 4 of the project focuses on gender equality and women's empowerment: 'Biomass Energy Technology Capacity Building and Knowledge Management and Gender Mainstreaming'. The outputs falling under this component will directly contribute to gender mainstreaming and women's empowerment at all levels of the project.
- Multiple outputs from all four components in the Gender Action Plan directly contribute to gender equality
- Moreover, all data that will be collected for various activities will be disaggregated by gender and age wherever possible to better reflect women's situation in the region.

ANNEX 10 - PROJECT KNOWLEDGE MANAGEMENT

The achievement of PASBET's development outcomes at the broader national level, and most specially at the provincial level, particularly for it as one case with provincial scope of activities and outputs, anchors in the knowledge management strategy and plan for the project. This system encourages an environment where people's and institution's experience, learning and wisdom are valued; and where internal processes are structured to support people or its users in creating, sharing, and using the pertinent knowledge gained.

KM is very crucial for the long-term success and sustainability of results of the PASBET Project, during and after the implementation period for the following reasons:

a) The temporary nature of the project (for five years of implementation) requires that knowledge and information related to the project need to be captured, created, shared, and reused within the project structure and the stakeholders and be carried over and institutionalized before the end of the project.
b) Establishment and enhancement of a KM system will enable faster response time particularly on knowledge and information relating to biomass-based resource production and energy generation.
c) KM enables a decision support system and empowers decision making capability for the project and transfer the practice to the institution which will sustain the program.
d) KM improves knowledge sharing on biomass-based program implementing experiences in Pakistan and can be shared among countries that are similarly situated between and among UNDP, relevant ministries in Pakistan and other stakeholders.
e) The application of KM in the PASBET will improve the quality of climate mitigation and biomass-based development work that will strive to reduce greenhouse emissions in the energy sector and end-use sectors including communities electrified, etc.
f) KM avoids repeating the same mistakes through sharing lessons learned in the past or ongoing similar biomass-based projects.
g) KM provides significant level of confidence and influences policy optimization, strategic thinking, and broader perspectives through rooting from validated experience and results, especially in the context of the barrier-removal approach of PASBET.
h) KM improves the efficiency, relevancy, and effectiveness of the country's development work at the national and local contexts.
i) KM preserves notable best practices that can be sustained and replicated in other sites, districts, and applications, thereby maximizing the level of the intended energy saving and GHG reduction impact.
j) KM assists in the development of strong networks of people and organizations inspired by common objectives.

KM within the Project Structure and Stakeholder Network – The main KM participants, sources, users and disseminators of data/information that were generated by the Project are: (a.) the project proponent (MoCC&EC), which is also the designated implementing partner of UNDP for this project; (b.) the Forestry Departments of the 4 provinces (Punjab, K.P., Sindh and Balochistan) that will be involved in the project implementation particularly in the demonstration activities and results alongside the other project components; (c,) two (2) units under the Ministry of Energy (Power Division), which are the Alternative Energy Development Board (AEDB) and National Energy Efficiency & Conservation Authority (NEECA); and (d.) the other stakeholders listed in the stakeholder engagement plan. The information that these entities include, among others, the baseline data and information on climate change scenarios for Pakistan (2020-2050); data on biomass (and woody biomass) production, forest industry data; and annual power

generation and electricity consumption (by sector) data from each province. These entities also provide information about the results of previous and ongoing projects/programs that are related and relevant to the development of the biomass-based program.

KM Plans by involving relevant projects, programs, initiatives & evaluations in Pakistan - The project will involve active participation, providing inputs and deriving usage in the KM system within the project structure by major stakeholders and other project development partners working on related biomass projects in the country. These are by UN agencies (e.g., UNIDO, FAO), other relevant institutions and organizations such as the World Bank, Asian Development Bank, bilateral country cooperation programs, government such as the National Energy Policy and Plans, National Electrification Policy, public-private partnership programs in electricity generation, independent power production, etc. as well as related provincial initiatives. The knowledge contents are regarding lessons learned on biomass technologies applications, best practices, and capacity development from the projects that the project participants have been supporting or implemented in Pakistan. These will also consist of climate change mitigation and adaptation, and related socio-economic development projects particularly in the 4 project provinces.

KM processes to capture, assess and document info, lessons, best practice & expertise generated during implementation - Actions to be taken for information gathering (e.g., reporting arrangements for baseline and incremental activities, field missions, etc.) on the results of implemented project activities will be organized into a computer-aided, internet-based system built-in the project's monitoring and implementation plans. A system for gathering, processing, and compiling this information will be developed to make it useful, easily available, and accessible. This will be integrated into an online repository for knowledge management products for project partners and general public. The repository will ensure that all information, pertinent data, lessons learned, best practice & expertise generated during project implementation are widely disseminated to concerned stakeholders and emerging experts not only across the country but the region. Information gathered and stored in the repository will provide the information that is to be reviewed and if required edited to develop practical, user friendly media publications and information packages that highlight production, energy use and environment issues and how these are being addressed in the energy sector, particularly relevant to biomass-based production and energy generation. The Stakeholder Engagement Plan in Annex that will be developed and implemented during the project implementation will be on interaction with the various rural industries in the demo districts to define and articulate methods of information exchange that will be an important attribute to the successful delivery of this initiative.

KM tools and methods for knowledge exchange, learning & collaboration – The information repository that will be developed and maintained for data banking purposes of the project is among the tools and processes that will be used for the knowledge management. It will also serve as a user friendly information sharing system. Such system will include all accumulated information about biomass energy resource production and biomass (including woody biomass) based energy generation for power and non-power applications that are applicable in the context of Pakistan and can be shared with other countries that have the same topographical and climate conditions. It will also include the results and impacts of the project activities, particularly the demonstrations. The local and international repository and information sharing system are essential sources of data/information that continuously support the knowledge management objective of the project.

KM outputs to be produced and shared with stakeholders – The project will deliver various outputs that will be documented and included in the planned repository for sharing experiences in making the demonstrations successful and in replicating and scaling them up further after the project is completed

using the demonstrated climate resilient and energy efficient biomass energy technologies. These include all pertinent information about the following: (1) Comprehensive situational and feasibility analyses and supply chain and market analyses on woody biomass production, and sustainable woody biomass production and utilization; (2) Developed and implemented policies and market-based regulatory framework for supporting woody biomass production and use, including approved technical, management and operational standards; (3) Formulated energy-integrated development plans of pilot towns; (4) Established and operational farmers/communal forest nurseries for providing planting stock for raising woody biomass in woodlots and farmlands; (5) Installed and operational woody biomass-based energy generation technology demonstrations showcasing the cost-effective application of decentralized woody biomass-based electricity generation and distribution (through mini/micro-grids); (7) Established and operational cost-effective demo woody biomass fuels production facilities, and applications of energy efficient woody biomass fired technologies; (8) Established and operational investment and financing mechanisms for supporting the commercial viability and operation of woody biomass energy production; (9) De-risked biomass-based power generation projects, decentralized biomass-based energy generation in rural areas, and business plans for the GOP and private sector to facilitate financing and implementation; and, (10) Capacity development training programs for the design, construction, operation, and maintenance of woody biomass-based energy production systems both for power and non-power applications.

The following is a tentative list of planned knowledge management (KM) products that the project will produce and their respective tentative delivery schedule.

KM Product	Brief Description	Delivery Schedule
Woody Biomass Resources in Pakistan	This is comprised of comprehensive situational and feasibility analyses and supply chain and market analyses on woody biomass production, and sustainable woody biomass production and utilization.	3Q 2024
Woody Biomass Energy Development and Utilization Policies in Pakistan	This is comprised of the results of the project's activities on the development and implementation of policies and market-based regulatory framework for supporting woody biomass production and use, including approved technical, management and operational standards.	2Q 2025
Energy Planning in Rural Areas with Woody Biomass Resources	This presents the formulated energy-integrated development plans that are developed for the pilot towns of the project.	3Q 2024
Woody Biomass Production	This presents the design, establishment, and operational farmers/communal forest nurseries for providing planting stock for raising woody biomass in woodlots and farmlands.	1Q 2026
Woody Biomass Energy Generation Technologies and Applications	This presents the results of the demonstrations of the cost-effective application of decentralized woody biomass-based electricity generation and distribution (through mini/micro-grids).	2Q 2026
Woody Biomass Fuels Production	This is about the results of the demonstration on the establishment and operationalization of cost-effective woody biomass fuels production facilities, and applications of energy efficient woody biomass fired technologies.	3Q 2025
Financing Woody Biomass Energy Production	This is about the investment cost of establishing and commercial operation of woody biomass energy production, as well as potential investment and financing mechanisms for	4Q 2024

	supporting the commercial viability and operation of woody biomass energy production.	
Recommended Woody Biomass Energy Generation Projects in Pakistan	This is about the de-risked biomass-based power generation projects, decentralized biomass-based energy generation in rural areas, and business plans for the GOP and private sector to facilitate financing and implementation.	3Q 2025
Woody Biomass Capacity Development Program	This is about the capacity development training programs for the design, construction, operation, and maintenance of woody biomass-based energy production systems both for power and non-power applications.	1Q 2027

Contribution of KM and learning to overall project impact and sustainability – The PASBET project will continue to develop and implement a knowledge management strategy and plan during project implementation based on this KM framework to achieve the improved awareness and capacity development of the pertinent GOP ministries and agencies and all stakeholders that will be involved in the project at the national and provincial levels. These will be about cost-effective applications of woody biomass-based energy generation for decentralized power generation and distribution, and also for industrial productive use applications. The essential components of the plan e.g., online repository, and information sharing system, will be useful not only for the woody biomass-fired power generation companies, and rural industries, but also GOP agencies in sustainable rural development planning. It will also be useful to international donors, implementing partners and consultants when designing and implementing similar biomass-based energy generation projects in Pakistan. The successful implementation of the KM strategy and plan will improve awareness, communications, and education and ensure that the experiences and results of the proposed project are properly captured and disseminated to the project stakeholders and partners, including UNDP and GEF. The proper management of knowledge will require transparent and timely sharing of data, and other information through proper communication, whether printed or in digital forms.

Plans for dissemination and strategic communications – Disseminating the results of the various activities of the project is essential in achieving the widespread application of woody biomass-based energy generation in the country starting from the provincial level achievements, and in other countries of the same circumstances as Pakistan. Hence, among the barrier removal strategies that will be employed in the PASBET project is knowledge management and dissemination and decision support systems. Such strategy requires the: (a) generation of knowledge that the pertinent sectors of the country, e.g., energy, forestry, agriculture, etc. need to progress towards the next level in the application of best available technologies; and (b) dissemination of that knowledge to a broad range of entities within these sectors in an accessible and affordable manner. In this regard, the activities of the project will be designed in such a way that these are included in the project's knowledge management plan. As part of the information sharing and dissemination aspects of the project, communicating the results of feasibility analyses and engineering design of all technology application demonstrations will be used for decision making in the operation, maintenance, and sustainability of the demo units, and provide information to help convince potential users to replicate and upscale the demos into more commercial applications using the business and financial models developed.

Knowledge management is therefore a primary function of the Project Management Unit in close coordination with the pertinent MoCC&EC unit's regular activity as the Implementing partner. This will enable the latter to sustain the KM system and activities after the project is completed and making sure that ample budget is allocated yearly to maintain its operational requirements. A knowledge management strategy and plan are therefore required to be developed starting the inception phase based on this KM

framework and adopted and operationalized during the entire project implementation. It is budgeted within the activities during implementation. As the project is completed, the sustainability of the KM system will form part of the exit strategy that aims that the knowledge base be handed over to MoCC&EC before the Project ends for it to support the implementation of the replication plan and post-project sustainability to achieve the desired project's long term outcomes in GHG emission reduction and economic and social benefits.

ANNEX 11: PROCUREMENT PLAN – FOR FIRST YEAR OF IMPLEMENTATION

Description	Project Component	Estimated no. of Contracts	Estimated Commencement	Estimated Value Year 1 (USD)	Estimated Value Cumulative (USD)	Recruitment Method	Comments
71200							
International consultant for development of guidelines of sustainable production and utilization of woody biomass, with Biodiversity Action Plan	C2.1	1	Q3/2023	30,000	30,000	International advertisement at UNDP portal, request for proposal	Procurement of services
International consultant for capacity building	C4	1	Q4/2023	12,000	12,000	International advertisement at UNDP portal, request for proposal	Procurement of services
Technical Consultants	C2.2	1	Q1/2023	13,343	113,343	National recruitment through UNDP portal	Procurement of services
71300							
Development of guidelines of sustainable production and utilization of woody biomass with biodiversity action plan	C2.1	1	Q3/2023	20,000	20,000	National advertisement at UNDP portal, request for proposal	Procurement of services
Established and operational fuelwood production guidelines	C2.1	1	Q3/2023	20,000	20,000	National advertisement at UNDP portal, request for proposal	Procurement of services
National consultants for capacity	C4	1	Q3/2023	10,000	10,000	National	Procurement of

Description	Project Component	Estimated no. of Contracts	Estimated Commencement	Estimated Value Year 1 (USD)	Estimated Value Cumulative (USD)	Recruitment Method	Comments
building						advertisement at UNDP portal, request for proposal	services
71400/71800							
National Project Coordinator	C2.2	1	Q1/2023	31,142	180,250	National recruitment through UNDP portal	Hiring of personnel
Admin/Finance Officer	PMC	1	Q1/2023	6,285	38,369	National recruitment through UNDP portal	Hiring of personnel
72100							
Woody Biomass Survey Firm	C1	1	Q4/2023	50,000	50,000	National advertisement at UNDP portal, request for proposal	Procurement of services
Site selection for biomass gasification plant	C1	1	Q4/2023	45,000	45,000	National advertisement at UNDP portal, request for proposal	Procurement of services
Technical feasibility of power generation potential at selected sites	C1	1	Q4/2023	15,000	30,000	National advertisement at UNDP portal, request for proposal	Procurement of services
Financial feasibility of power	C1	1	Q4/2023	15,000	30,000	National	Procurement of

Description	Project Component	Estimated no. of Contracts	Estimated Commencement	Estimated Value Year 1 (USD)	Estimated Value Cumulative (USD)	Recruitment Method	Comments
generation potential at selected sites						advertisement at UNDP portal, request for proposal	services
Local companies providing trainings on how to operate and maintain biomass technologies for demonstration sites	C4	1	Q3/2023	5,000	11,000	National advertisement at UNDP portal, request for proposal	Procurement of services
Other							
Equipment including IT devices and services needed for the Projects various activities and knowledge management storage of projects data/information	C4	1	Q1/2023	15,000	21,000	National advertisement at UNDP portal, request for proposal	Procurement of services

ANNEX 12: GEF FOCAL AREA SPECIFIC ANNEXES

Annex 12A - PASBET Project Demonstration Profiles

The PASBET Project Demonstrations profiles that are included in this Annex are in accordance with the following Outputs in Component 2, Outcome 2.2 as follows:

Output 2.2.1. Completed and operational woody biomass-based energy generation technology demonstrations in 4 selected sites, one in each province of Punjab KP, Sindh and Balochistan showcasing the cost-effective application of decentralized woody biomass-based electricity generation and distribution (through mini/micro-grids)[103] (**Annex 12A.1**).

Output 2.2.2. Completed and operational three (3) demonstrations of cost-effective production of woody biomass fuels such as wood chips, briquettes, and pellets for use in decentralized woody-biomass based power generation and distribution (**Annex 12A.2**).

Output 2.2.3. Completed and operational demonstrations of the cost-effective applications of energy efficient woody biomass fired technologies and comprehensive energy management systems in selected energy end-use sectors in support of rural socio-economic development[104] (**Annex 12A.3)**

PASBET Project Demonstrations in Four (4) Provinces

The second of two sub-components of Component 2 is comprised of activities that will deliver outputs which will collectively bring about increased investments in the application of technologies for the production, and energy efficient utilization of woody biomass energy. The activities that will be carried out will address the technological barriers by promoting and showcasing biomass energy efficient technologies and adoption of innovative practices for woody biomass energy conversion and utilization. Cost-effective applications of EE woody biomass fired technologies and comprehensive energy management systems in selected energy end-use sectors in support of rural socio-economic development will be demonstrated. Among these is the design, engineering, construction, commercial operation, and maintenance of woody biomass fired power generation and distribution systems in the 4 provinces for the supply of electricity to selected unelectrified rural villages. These will showcase the cost-effective application of decentralized woody biomass-based energy generation and distribution (through mini/micro-grids).

There are four (4) sets of demonstrations, one set for each province, and all of them are on an integrated system of woody biomass-based energy (power and non-power) generation and utilization to support rural electrification and rural development. The same technology (biomass gasifier) will be used for each demo in each province. However, the size of each demo may vary slightly depending on the detailed biomass assessment study (planned under Component 1). The manner of implementation and business modelling may also differ from province to province since this will depend on the power supply and demand characteristics, rural electrification status and co-financing packaging. Hence, this will consist of

[103] This involves the installation of a collective total woody biomass-based power generation capacity of 5.4 MW in the 4 provinces to supply of electricity to supply unelectrified rural villages.

[104] This includes local production of energy-efficient wood-fired devices that are cost-effectively utilized in rural industries to improve energy utilization efficiency and reduce GHG emissions.

four possible typical technology and business cases and diffusion approaches that could find relevant applications as they are replicated in other districts of the province or in some other provinces.

The suitable potential demo villages have been identified during the project design stage. These are villages that are located near sites (and potential sites) of sustainable woody biomass production for energy use. The specific demo villages will be confirmed during the project implementation. These will be in the partner provinces and are within the general vicinity of the geographical coordinates indicated in Annex 2.

For the woody biomass energy conversion technology that will be applied in the demos, the biomass gasifier technology was selected. This selection is based on the reasoning that such technology is a mature and proven one compared to other available options such as suspension firing, fluidized-bed combustion, etc. Wood gasification is a known technology in Pakistan and technical expertise locally and in other countries is available[105]. Therefore, it is slightly easier to manage and has better performance confidence based on past studies and technology assessments.

Under each demo, the following are the major items that will collectively make up an integrated system for woody biomass-based power generation and utilization in the selected rural sites in each of the four provinces:

- Demo Component 1
  - Woody biomass-fed gasifier and gas engine for electricity generation
  - Solar PV power for back up or augmentation to meet the village electricity demand
  - Mini/micro grid power distribution for rural electrification
- Demo Component 2
  - Fuel tree plantation, nursery, and harvesting/gathering system
  - Woody biomass fuel processing, storage, handling, and transportation system.
- Demo Component 3
  - Woody biomass combustion systems for industrial thermal (non-power) applications

[105] A gasifier can be procured from "Gasifier and Equipment Manufacturing Corporation (GEMCOR), Philippines" which is the world's largest gasification manufacturing facility. They produce around 3,000 units/yr. with a size range of 10 to 250 kW. Apart from imported technology there are few local manufacturers of woody biomass gasifiers such as Royal Bio Energy Company (Lahore, Pakistan).

Each demo set is an integrated showcasing of the applications of woody-biomass resource development and energy generation, distribution and utilization comprising of the following.

- Distribution of farming/forestry tools and devices for effective and sustained growing and harvesting of woody biomass from fuel wood forests/plantation.

- Design and construction of biomass storage areas. For sustainable biomass supply a multiple storage system will be developed as part of the demo with proper fire safety protocols to minimize the fire hazards. Primary storage system to be planned near the biomass generation area (plantation area). While secondary storage facility will be planned near energy generation side. On top of that multiple biomass collection centers could be developed to collect woody biomass and agriculture residue biomass (if available).

- Development of effective mechanisms for biomass transportation, transportation within the storage area and between the primary and secondary storage area.

- Application of appropriate cost-effective biomass pre-treatment technologies
    - Woodchipper for wood chips to be used in gasifier
    - Biomass pellet machine for wood pellets to be used in energy efficient industrial combustions applications.

- Installation of a woody biomass fired power generation system that will produce and distribute electricity to consumers in the surrounding villages. Biomass gasification technology has been selected. The initial size of the bio-gasifier based on discussions with provincial representatives is 50kW. Several bio-gasifier designs are available in the market in Pakistan and other countries. The size may vary for each site depending on the detailed biomass assessment and electrification demand. However, the 50kW size was determined as typical for planning and budgeting purposes. The size can be increased with more budget and augmented by co-financing mobilized by the host province.

- A mini/micro-grid system will be installed for supplying the electricity to consumers (40 to 50 households in the demo village). Where possible and feasible, other RE-based power generation units will also be integrated into the system to operate in hybrid mode or as supplementary unit when the main biomass-fired unit is under maintenance. A 40kW PV Solar system with battery backup will be installed to serve as a back-up facility.

- The demo will be operated and maintained as a commercial business. Therefore, a key driver of the business model is the collection of fees for the electricity services and using part of the revenues for the operation, repair, and maintenance of the various systems. Each demo is about the commercial operation of a woody biomass-fired energy production and distribution system (power or non-power application). This can be community-owned and operated, or private-sector owned and operated, or government-owned and private sector operated, etc. This further explains the different case studies that can be done for each province typifying the collection system for such energy production and supply business case.

- There will be non-power applications of innovative and energy efficient wood-fired industrial equipment as an alternative to coal to reduce GHG emissions from rural industries[106]..

A capacity development program will be organized and conducted under Component 4 in support of the Component 2 activities and outputs for entrepreneurs in rural areas for woody biomass production and utilization for power and non-power supply.

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Annex 12A.1: Demo Component 1
Mini Grid based Woody Biomass Gasification Power Generation Plant

This component of the demo is showcasing the cost-effective application of decentralized woody biomass-based electricity generation and distribution through a mini-grid. The system to be designed, engineered, installed, operated, and maintained is an integral component of the demonstration of an integrated system of woody biomass-based energy (power and non-power) generation and utilization to support rural electrification and rural development.

Building on the provincial plans for the electrification of the unelectrified villages in the 4 provinces (Punjab KP, Sindh and Balochistan) the electrification of a selected village will be facilitated with the installation of a woody biomass gasification power generation plant and the distribution of the generated electricity to the village households through a mini grid that will also be installed. While this system is part of a demonstration, it will be operated and maintained as a commercial business.

Demo Component Description: Based on the initial information about the typical energy needs of the potential demo villages, a woody biomass gasifier will be installed to generate synthetic gas (syngas) that will be used to operate a 310 kW gas engine. The gasifier will be fed with processed woody biomass (e.g., wood chips).

Technology Application: Based on the electricity demand of the host village, which would typically include for powering of tube wells and household consumption; and woody biomass availability, a woody biomass gasifier and 310 kW gas engine which operates on syngas produced by gasification of the processed woody biomass (e.g., wood chips) will be installed. Based on the typical demand of the prospective demo villages, a downward draft gasifier will be appropriate and most suitable for small capacity gasification power generation projects. Wood chips will be used in the bio-gasifier, and the syngas that will be produced will be used for electricity generation in a gas engine. The electricity produced (net of power plant use) will be distributed through a mini grid that will be installed. The demo will also feature the installation and use of back up 250 W solar PV system (with battery energy storage). Also, to be used are energy monitoring and control systems such as the main electricity meter at the power plant site; and the electricity meters that will be installed in each house that will be connected to the mini grid. The main electricity meter will be extremely sensitive and will indicate and record the following parameters (Voltage, Current, Power Factor, kWh, kW). Accuracy of the meter shall not be less than 99.9%. The electricity meters (normal kWh meter) are for billing purposes, will have data logging capacity and with accuracy of not be less than 99.5%.

The operation of the woody biomass gasification power plant will generate 54,303 kWh/year of electricity that will be supplied to, and utilized in, the demo village. This includes the electricity generation from the back-up solar PV system, based on the anticipated downtime of the biomass-gasifier. This translates to a 145 tons of CO_2 emission avoidance (based on a grid emission factor of 0.586 tCO_2/MWh).

Annex 12A.2: Demo Component 2
Cost-effective Production of Woody Biomass Fuel

This component of the demo is on the showcasing of a cost-effective system for harvesting, storage, transportation, and processing of woody biomass fuel. The system to be designed, engineered, installed, operated, and maintained is an integral component of the demonstration of an integrated system of woody biomass-based energy (power and non-power) generation and utilization to support rural electrification and rural development.

In the four districts in the 4 partner provinces ((Punjab KP, Sindh and Balochistan) where the prospective demo villages are located, there are already few riverine or Bela Forests that were developed under the Ten Billion Tree Tsunami Project (TBTTP), which is Government of Pakistan's flagship afforestation program to improve existing forest land and grow new forests in all parts of the country. Currently, there is no community managed sustainable forests in the country, and no infrastructure for treatment, storage and transportation of woody biomass is available in the country. Woody biomass is presently utilized for domestic usage such as heating and cooking. Apart from being used domestically woody biomass is also used in few industrial applications such as brick kilns, and in steam generation in the sugar and textile industries. The use of biomass (e.g., agri-waste) for power generation have been done in the country, but not specifically woody biomass. However, woody biomass-based power generation is a new concept for rural communities in Pakistan as no such initiative has never been taken by either government or private sector in the country previously. Building on the forestry development plans and rural electrification plans of the 4 provinces the planned woody biomass-based power generation and distribution system (i.e., Demo Component 1) will be supplied with processed biomass (e.g., wood chips) from a demo woody biomass production system. The system will build on suitable plans of the forestry departments for tree nurseries. While this system is part of a demonstration, it will be operated and maintained as a commercial business.

Demo Component Description: Considering the forestry development plans, tree nurseries will be established for the sustainable supply of samplings of fast-growing native tree species[107]. At least one nursery is established near the demo site to reduce the transportation cost. Looking at the climate of the selected districts, there will be no need for heating or cooling for nurseries (greenhouses). Under the Ten Billion Tree Tsunami Project (TBTTP) forest departments of each province will set up nurseries. Woody biomass will be grown and harvested in designated forests and plantations based on the agreements with the forestry departments[108]. There will be two storage facilities required to store harvested woody biomass that will be supplied for the sustainable operations of the demo. One site will be near the forest side which will be the primary storage facility. The second storage facility will be constructed near the gasification plant side[109]. It is recommended to have a woody biomass fuel storage of at least 3 months. In rural areas, woody biomass is transported using bullock/ camel carts for short distances. The woody biomass has to be transported from the primary storage to secondary storage facility, and the

[107] There should be contractual agreement with the state-owned forests or private forests for the nurseries and plantations where the fuel trees will be grown. The plant species of Shisham (Dalbergia sissoo), Kikar (Accacia nilotica), Tun (Cedrella toona), Arjan (Terminalia arjuna), Toot (Morus alba), Acacia Albida etc. will be grown as per soil conditions at the selected sites. Eucalyptus species may also be planted at problematic areas where soil conditions are water-logged or saline.

[108] Few of the government owned forests already have contracts with private entities to collect forestry residue and wood waste.

[109] Outdoor storage facilities are recommended, as establishment of stack lets (tall storage buildings with no walls) will increase the woody biomass fuel price, and as the past experience with outdoor storage facilities shows that the quality of woody biomass is not reduced if it is covered with plastic sheets, and the stack is placed on a drained apron. The storage facility should comply with the international fire safety compliance standards and people operating this should be trained accordingly.

recommended transportation solution is the use of agricultural tractors with trolleys. The woody biofuel from the secondary storage will be processed for use in the bio-gasifier, wood-fired cooking stoves and wood-fired industrial equipment. These can be in the form of wood chips, wood briquets or wood pellets.

Technology Application: This demo component will feature the application of equipment for woody biomass fuel production (e.g., high-capacity, and heavy duty harvesting tools, woodchippers and wood briquetting and pelletizing machines). It will also feature equipment and systems for the sustainable handling, storage, and transportation of woody biomass fuel. The use of appropriate production management control systems will also be featured such as robust and accurate weigh bridges at the primary and secondary storage facilities, as well as weighing scale for pellets and wood chips at the secondary storage facility, as well as data recording and calibration systems for these instruments.

The processed woody biomass fuels produced from the facility is primarily for the demos. These are for use in biogasifiers (wood chips); and industrial equipment (wood pellets). There are also other expected markets for the sale of the woody biomass fuels outside of the project's demo and demo replication activities. Part of the woody biomass fuels production will be allocated for these uses, and it is estimated that the projected annual sales of woody biomass fuels (100 tons/year) for use as replacement for energy end-use activities that are currently using fossil fuels will translate to about 128 tCO_2 per year.

Annex 12A.3: Demo Component 3
Application of Energy Efficient Woody Biomass-fired Industrial Equipment

This component of the demo is showcasing the application of energy efficient woody biomass based thermal energy technologies in rural industries. This is an integral component of the demonstration of an integrated system of woody biomass-based energy (power and non-power) generation and utilization to support rural electrification and rural development.

Currently in Pakistan's rural areas, the use of biomass such as wood is remarkably high and mostly because the wood fuel is used inefficiently, and the wood-fired equipment used are typically based on old inefficient technologies. There are also few rural industries that are already using biomass, in particular woody biomass for their thermal energy use e.g., boilers, kilns, etc. Building on ongoing and planned socio-economic development programs for the rural areas by the provincial governments, this demo component is intended to supplement the ongoing activities for improving the rural industries through the introduction of improved woody biomass based combustion equipment in selected rural industries to demonstrate the application of such equipment that can be used to replace fossil fuel-fired industrial equipment used in rural industries thereby lessening the dependency on fossil fuel and reducing GHG emissions from the use of fossil fuels.

Demo Component Description: For the industrial applications, selected rural industries in each province that are currently using fuel (apart from woody biomass) and has the potential to convert its production equipment to use woody biomass will provide an energy efficient woody biomass combustion system. The before and after fuel consumption will be monitored and evaluated and determine the energy savings and corresponding GHG emission reduction.

Technology Application: This demo component will feature the application of energy efficient industrial wood-fired combustion equipment. The type will depend on the kind and size of the equipment. Nonetheless, typical wood-fired industrial equipment is estimated to consume about 2 tons woody biomass daily or about 730 tons annually. At this rate of usage, and for the demo that will involve the deployment of woody biomass fuel combustion system for industrial equipment, the estimated GHG emissions that can be reduced is about 936 tCO_2 annually.

ANNEX 12B: Estimation of Potential GHG Emission Reduction from the PASBET Project

The estimation of the GHG emission reductions that are attributable to the PASBET Project are based on the most recent version of the Calculation Guide for GEF Project Greenhouse Gas Emission Reduction, as well as the 2008 Manual for Calculating GHG Benefits of GEF Projects: Energy Efficiency and Renewable Energy Projects.

A. Data and Assumptions

Energy Parameters of Biomass Gasifier Plant	Quantity	Unit
Specific energy consumption (Combustion):	1.6	kg/kWh
Specific energy consumption (Gasification):	1.16	$kg_{Biomass}$/kWh
Currency Conversion Factor	175	PKR/USD
Estimated cost of gasifier	2.2	USD/Watt
Cost of Biomass	38	PKR/kg
Cost of land lease	28,000	PKR/acre
GCV of Wood (On received basis)	3,800	KCal/kg
NCV of Wood (On received basis)	14.32	GJ/Ton
Calorific value of Coal	26.17	GJ/ton
Emission Factor of Coal	0.0895	tCO_2/GJ
Grid Emission Factor	0.586	tCO_2/MWh
Size of Power Plant	310	kW
Annual operating days	300	days
Daily operating hours	18	Hr.
Average plant loading factor	70%	%
Annual electricity generation from bio-gasifier	1,171,800	kWh/yr.
Size of Solar PV Power Plant	250	kW
Solar PV Net Capacity Factor	0.167	
Electricity from PV Solar plant (back up)	365,730	kWh/yr.
Total electricity generation	1,537,530	kWh/yr.
Woody biomass requirement	2,326373	Ton/yr.
GCV of Wood (On received basis)	3,800	KCal/kg
NCV of Wood (On received basis)	14.32	GJ/Ton
Energy generated from woody biomass	5,337	GJ_{th}
Conversion factor to Ton of Oil Equivalent (toe)	41.87	GJ/toe
Ton of Oil Equivalent	127	toe
Grid Emission Factor	0.586	tCO_2/MWh
Emission Reduction	905	tCO_2/yr.

Energy Comparison with NG-fired Genset	Quantity	Unit
Specific Energy Consumption of NG based genset	7.3	kWh/m^3
Annual Energy generation	1,171,800	kWh
Annual NG Consumption	160,521	m^3/yr.

NCV of NG	0.033	GJ/m3
Energy generated from NG	851	GJ
Conversion factor to 1 Ton of Oil Equivalent (toe)	41.87	GJ/toe
Ton of Oil Equivalent	20	toe

Biomass Treatment	Value	Unit
Cost of woodchipper for biomass gasifier	4,270	USD
Cost of pelleting plant	2,850	USD

Biomass Supply for Industrial Use	Value	Unit
Cost of Woody biomass burners for industrial use	3,429	USD
Woody Biomass required for operation of industrial applications	2	Tons/day
Woody Biomass required for operation of industrial applications	730	Tons/yr.

Total Woody Biomass Requirement	Value	Unit
For 310 kW Biomass Gasifier	2,326	Tons/yr.
For industrial biomass usage	730	Tons/yr.
Estimated Woody biomass generated from the TBTTP	11,173	Tons/yr.
Total	3,056	Tons/yr.

Assumptions:

1. Demo implementation schedule (cumulative): Year 1 = 1 demo installed and operational; Year 2 = 3 demos installed and operational; Year 3 = All 4 demos installed and operational.
2. There will be 150 demo replications; 40 supported by PASBET Project; and 110 influenced (assisted during the PASBET Project implementation and implemented after the PASBET Project completion (influence period).
3. Demo replication implementation schedule (cumulative):

Year	Directly Supported	Assisted and Influenced	Total Replications
PASBET Year 3	20	40	60
PASBET Year 4	30	50	80
PASBET Year 5	40	60	100
Influence Period Year 1	40	70	110
Influence Period Year 2	40	80	120
Influence Period Year 3	40	90	130
Influence Period Year 4	40	100	140
Influence Period Year 5	40	110	150

4. Lifetime of each demo = 20 years
5. Woody biomass accounts for about 5% of the annual biomass energy (all forms) consumption in the country during the influence period.

B. GHG Emissions from Demonstrations

There are four (4) sets of demonstrations, one set for each province, and all of them are on an integrated system of woody biomass-based energy (power and non-power) generation and utilization to support rural electrification and rural development. Under each demo design, the following are the major items that collectively make up an integrated inclusions to be integrated as a system for woody biomass-based power generation and utilization in the selected rural sites in each of the four provinces:

- Demo Component 1
  - Woody biomass-fed gasifier and gas engine for electricity generation (310 kW)
  - Solar PV power (250 kW) for back up or augmentation to meet the village rural electricity demand
  - Mini/micro grid power distribution for rural electrification
- Demo Component 2
  - Fuel tree plantation, nursery, and harvesting/gathering system
  - Woody biomass fuel processing, storage, handling, and transportation system
- Demo Component 3
  - Woody biomass combustion systems for industrial thermal (non-power) applications

The expected GHG emission reductions from one demo are the following:

Demo Sub-Component	Annual GHG Emission Reduction, tCO_2
Woody biomass gasification-based power generation and distribution (mini/micro-grid system)	691
Back-up solar PV power system	214
Woody biomass fuel production	N/A
Woody biomass combustion systems for industrial thermal (non-power) applications	936
Total Annual GHG Emission Reduction per Demonstration	1,840
Total Annual GHG Emission Reduction from 4 Demos (1 per province)	7,360

C. Project CO_2 Emissions Reduction Estimates

Direct CO_2 Emission Reductions (DER)

During the PASBET Project implementation period, specific woody biomass, production, processing, and utilization technologies will be demonstrated in 4 provinces. These will showcase the utilization of woody biomass fuel as energy source for the power (i.e., woody biomass-based power generation and distribution) and non-power (e.g., woody biomass-fired industrial equipment) applications. The demonstrations involve the design, installation, operation, and maintenance of woody biomass-based energy system applications in selected rural areas where there are available potential sustainable woody biomass resources.

The estimated cumulative direct emission reductions during the 5-year PASBET Project implementation period are about 195,051 tons CO_2. This amount comes from the implementation of the demonstrations (29,442 tons CO_2) and also from the supported demo replications (165,609 tons CO_2). This is from a total of 44 projects (inclusive of the 4 demos). This amount of GHG emission reductions will all be realized during the implementation period of the PASBET Project. The installation of all demos is expected to be

completed by end Year 2 and start to be all operational by Year 3, while the anticipated replications will be operational starting Year 4. These GHG emission reductions are directly attributable to the PASBET Project because the activities that brought about and enabled the generation of these emission reductions are all part and parcel of the project.

DER_{EOP} = Direct CO_2 emission reductions due to PASBET Project intervention (cumulative by end-of-project) = 195,050 tons CO_2.
$DER_{LIFETIME}$ = Direct CO2 emission reductions due to PASBET Project intervention (cumulative during the lifetime) = 1,619,293 tons CO_2. This is based on a 20-year lifetime.

Direct Post-Project CO2 Emission Reductions (DPPER)

The interventions that will be carried out under the PASBET Project are intended to showcase the cost-effective applications of woody biomass-based energy generation technologies for power and non-power applications in the rural areas of 4 provinces (Punjab, KP, Sindh and Balochistan). It is intended to influence change or bring about changes in the attitude of the project stakeholders (i.e., the rural villages that are currently unelectrified) towards sustainable low carbon rural development. Based on discussions with provincial governments, there is high level of interest in these sustainable and low carbon technologies using indigenous woody biomass resources. The project will provide technical assistance to proponents of the demo replications, for example in the design and sourcing financing of the required investments for the demo replication. The expectation is that the investment activities and the demonstrations that will be implemented will consequentially be replicated or where feasible, scaled up, by the project stakeholders using public and/or private funds. Such replications and other follow-up initiatives will be carried out by the provincial governments and rural communities, and by the private sector in these rural areas after the completion of the PASBET Project. In this case, it is expected that the same level of replications that are attributable to the PASBET Project will be implemented. In this case, the lifetime DPPER is about 3,238,586 tons CO_2

$DPPER_{LIFETIME} = DER_{LIFETIME}$ x 1 = 1,619,293 tons CO_2 x 1 = 1,619,293 tons CO_2

Total Lifetime DER ($LDER_{TOTAL}$) = $DER_{LIFETIME}$ + $DPPER_{LIFETIME}$ = 1,619,293 + 1,619,293 = 3,238,586 tons CO_2

Consequential CO_2 Emission Reductions (CER)

It is expected that the implemented investment and demonstration activities will influence the stakeholders from the provincial governments, rural communities, and the private sector in the provinces to replicate the technology applications that will be demonstrated (conceptualization, design, engineering, securing of financing, installation, commissioning, operation, and maintenance). The best practices and lessons learned from these activities/demos will help a lot in the replications, and the resulting energy savings and GHG emission reductions from such replications can be considered as consequentially or indirectly attributable to the PASBET Project.

Bottom-up Approach (BUA)

In estimating the Lifetime Consequential Emission Reductions using the bottom-up approach (CER_{BUA}), the Lifetime DER is multiplied by a replication factor (RF). As the investment activities and demonstrations are contributing to market transformation, the GEF prescribed RF value is 2 to 3. Considering that there are

still many unelectrified villages, particularly in provinces with sustainable woody biomass resources, it can be conservatively estimated that the value of RF for indirect GHG emissions reductions is about 2.0.

CER_{BUA} = Total Lifetime DER ($LDER_{TOTAL}$) x RF = 3,238,586 x 2.0 = 6,477,172 tons CO_2

Summary of CO_2 Emission Reductions

CO_2 Emission Reduction Type	Quantity, tCO_2
End-of-Project Direct CO_2 Emission Reduction (DER_{EOP})	195,050
Lifetime Direct CO_2 Emission Reduction ($DER_{LIFETIME}$)	1,619,293
Lifetime Direct Post Project CO_2 Emission Reduction ($DPPER_{LIFETIME}$)	1,619,293
Total Lifetime Direct CO_2 Emission Reduction ($LDER_{TOTAL}$)	3,238,586
Lifetime Consequential CO_2 Emission Reduction - BU Approach (CER_{BUA})	6,477,172
Total Lifetime Direct and Consequential (CER_{BUA}) CO_2 Emission	9,715,757

ANNEX 13: ADDITIONAL AGREEMENTS: CO-FINANCING LETTERS

(separate file)

ANNEX 14 - GEF and/or LDCF/SCCF Core indicators

GEF 7 Core Indicator Worksheet

Core Indicator 6	**Greenhouse gas emission mitigated**				***(Metric Tons)***
		Tons (6.1+6.2)			
		Entered		Entered	
		PIF stage	Endorsement	MTR	TE
	Expected CO_2e (direct)	3,124,000*	3,238,586*		
	Expected CO_2e (indirect)	0	6,477,172**		
Indicator 6.1	Carbon sequestered or emissions avoided in the AFOLU sector				
		Metric Tons			
		Entered		Entered	
		PIF stage	Endorsement	MTR	TE
	Expected CO_2e (direct)				
	Expected CO_2e (indirect)				
	Anticipated Year				
Indicator 6.2	Emissions avoided				
		Metric Tons			
		Expected		Achieved	
		PIF stage	Endorsement	MTR	TE
	Expected CO_2e (direct)	3,124,000*	3,238,586*		
	Expected CO_2e (indirect)	0	6,477,172**		
	Anticipated Year	*2023-2038*	*2023-2038*		
Indicator 6.3	Energy saved				
		Energy Savings, MJ			
		Expected		Achieved	
		PIF stage	Endorsement	MTR	TE
Indicator 6.4	Increase in installed renewable energy capacity per technology				
	Technology	Capacity (MW)			
		Expected		Achieved	
		PIF stage	Endorsement	MTR	TE
	Biomass (Power)	5.0	5.2		
	Solar (Power)	0	4.16		
	Biomass (Thermal)	N.E.	8.32		
Core Indicator 11	**Number of direct beneficiaries disaggregated by gender as co-benefit of GEF investment**				
				Number Achieved	
		PIF Stage	Endorsement	MTR	TE
	Female	2,000	160,380		
	Male	2,000	107,000		
	Total	*4,000*	*267,380*		

Estimated direct GHG ERs by end of 10-years influence period after end-of-project.

**Estimated indirect GHG ERs starting end-of-project till end of 10-years influence period.

ANNEX 15 - GEF Taxonomy

Level 1	Level 2	Level 3	Level 4
☒ **Influencing models**			
	☒ **Transform policy and regulatory environments**		
	☒ **Strengthen institutional capacity and decision-making**		
	☒ **Convene multi-stakeholder alliances**		
	☒ **Demonstrate innovative approaches**		
☒ **Stakeholders**			
	☒ **Private Sector**		
		☐ Large corporations	
		☒ SMEs	
		☒ Individuals/Entrepreneurs	
	☒ **Beneficiaries**		
	☒ **Local Communities**		
	☒ **Civil Society**		
		☒ Community Based Organization	
		☒ Non-Governmental Organization	
		☐ Academia	
	☒ **Type of Engagement**		
		☒ Information Dissemination	
		☒ Partnership	
		☒ Consultation	
		☒ Participation	
	☒ **Communications**		
		☒ Awareness Raising	
		☒ Public Campaigns	
☒ **Capacity, Knowledge, and Research**			
	☒ **Enabling Activities**		
	☒ **Capacity Development**		
	☒ **Learning**		
		☐ Theory of Change	
		☒ Adaptive Management	
		☐ Indicators to Measure Change	
	☒ **Innovation**		
	☒ **Knowledge and Learning**		
		☒ Knowledge Management	
		☒ Innovation	
		☒ Capacity Development	
		☒ Learning	
	☒ **Stakeholder Engagement Plan**		
☒ **Gender Equality**			
	☒ **Gender Mainstreaming**		
		☒ Beneficiaries	
		☒ Women groups	
		☒ Sex-disaggregated indicators	
		☐ Gender-sensitive indicators	
	☒ **Gender results areas**		
		☒ Access and control over natural resources	
		☒ Participation and leadership	
		☒ Access to benefits and services	
		☒ Capacity development	
		☒ Awareness raising	

		☒Knowledge generation	
☒**Focal Areas/Theme**			
	☒**Climate Change**		
		☒**Climate Change Mitigation**	
			☒Agriculture, Forestry, and other Land Use
			☒Energy Efficiency
			☐Sustainable Urban Systems and Transport
			☒Technology Transfer
			☒Renewable Energy
			☒Financing
			☐Enabling Activities
		☒**Technology Transfer**	
			☒Technology Needs Assessment
		☒**Rio Marker**	
			☒Climate Change Mitigation 1

ANNEX 16 – Letter of Agreement for CO Support to NIM
(separate file)

Annex 17: GEF CEO Endorsement
https://pims.undp.org/attachments/6542/217455/1762339/1817748/6542%20CEOEndorsementLetter_10720Pakistan%20-%20FSP%20CEO%20Endorsement%20Ltr.01.13.2023.pdf

Annex 18: Results of the Partners Capacity Assessment Tool (PCAT) and HACT Micro Assessment

- PASBET_MoCC&EC PCAT 12 Dec 2021.xlsm
- UNDP_Micro_Assessment_Report_-_Ministry_of_Climate_Change_&_Environmental_Coordination__Feb 2022_.pdf

Annex 19: UNDP Project Quality Assurance Report (to be completed in UNDP online corporate planning system)

https://undp-my.sharepoint.com/:w:/g/personal/anila_ajmal_undp_org/EaYI9P8BI-xAqHuJQuQt-IUBPci-3NWODhc5fe3oNqhvYQ?e=Z0KwIX